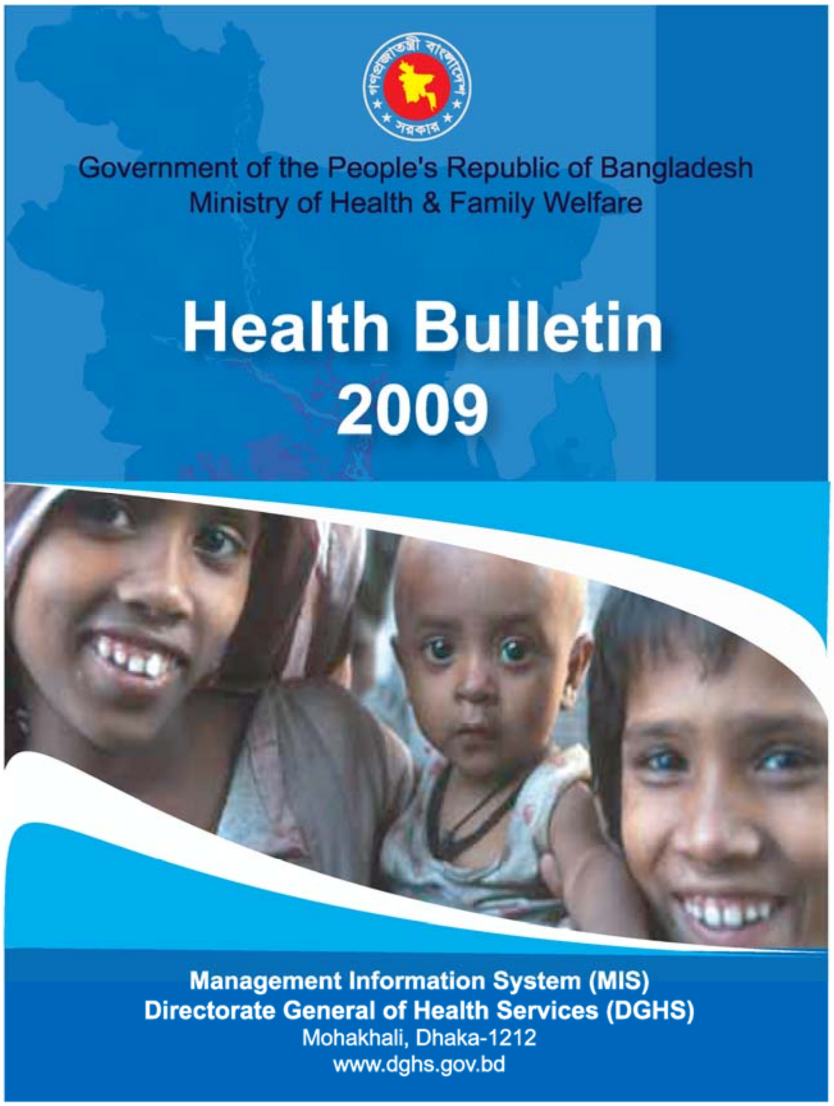
Government of the People's Republic of Bangladesh
Ministry of Health & Family Welfare

Health Bulletin
2009

Management Information System (MIS)
Directorate General of Health Services (DGHS)
Mohakhali, Dhaka-1212
www.dghs.gov.bd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Acknowledgement

Professor Dr AFM Ruhal Haque, MP
Honorable Minister for Health & Family Welfare
Professor Dr Syed Modasser Ali
Honorable Adviser to the Prime Minister for Health, Family Planning & Social Welfare
Dr Captain (Retd.) Mozibur Rahman Fakir, MP
Honorable State Minister for Health & Family Welfare
Shaikh Altaf Ali
Secretary, Ministry of Health & Family Welfare
Heads, managers and data providing staffs of all public and private health organizations

Message

Professor Dr A. F. M. Ruhal Haque

Honorable Minister

Ministry of Health and Family Welfare

Government of the People's Republic of Bangladesh

It is indeed my great pleasure to see that the **"Health Bulletin 2009"** of MIS, DGHS is going to be published. This Health Bulletin is a statistical year book that contains information on health situation of the country based principally on the information derived from health facilities and services run by the Directorate General of Health Services (DGHS).

It is blissful to note that MIS, DGHS has been able to make publication of this Health Bulletin as a yearly routine. I believe that information contained in the pages of this book will be much help to understand the health situation of the country and to assess needs and then make plans for health improvement.

Currently our country is heading through a wave of tech-savvy movement in the name of **"Digital Bangladesh"** under the leadership of our Honorable Prime Minister Sheikh Hasina. The Ministry of Health and Family Welfare has also tuned itself in this wave through initiative of implementing digital culture in its systems and processes. The ultimate aim of such exercises is to create an environment for evidence based decision making. The MIS, DGHS is in the forefront and leadership role in this effort. The information collection and processing systems have been much improved in the past few months and we will soon observe the benefits of the new systems soon. My ministry will always provide active support to all such developments towards evidence based decision making.

I am hopeful that "Health Bulletin 2009" will fulfill a large extent of information needs for our health planning and management. I thank Professor Dr Abul Kalam Azad, Director of MIS, DGHS and his team to publish this wonderful resource book.

Joy Bangla, Joy Bangobandhu.
Long live Bangladesh

Professor Dr A.F.M. Ruhal Haque

Message

Professor Dr Syed Modasser Ali

Honorable Adviser to the Prime Minister

Of the Government of the People's Republic of Bangladesh
For Health, Family Planning and Social Welfare Affairs

I am happy to see that the Management Information System (MIS) of the Directorate General of Health Services (DGHS) is going to publish its "Health Bulletin 2009". This publication has now become an important document revealing statistical information of different health programs, hospital services, academic and public health institutions, and of health workforce mainly under the DGHS of Ministry of Health and Family Welfare.

The contemporary age is regarded as information age and our Government welcomes this information age for the sake of evidence based decision making. We believe in in-depth information to explore the underlying key facts so that decisions can be specific and result oriented for the good of the people. Therefore, our present Government under the visionary and able leadership of Prime Minister Sheikh Hasina, the Daughter of the Father of the Nation Bangobandhu Sheikh Mujibur Rahman is implementing the Digital Bangladesh to harness an information society.

I have learnt that in the last few months since our Government took charge, the MIS of DGHS has made remarkable progress in extending the Internet connectivity across all the hospitals, health managers' offices and academic institutions under DGHS. Data and information communication has become much faster now and so the decision making process. This achievement is the important first step in the introduction of digital health and e-governance in our health sector. I firmly believe that this development will be further consolidated and strengthened in future.

I have been informed that Health Bulletin 2009 is improved than its predecessor as information could be gathered quickly through using the new electronic communications system and the future publications are expected to be even better.

I have noted this positive change and wish that the Health Bulletin 2009 will be appreciated by the learned readers.

Joy Bangla, Joy Bangobandhu.
Long live Bangladesh

Professor Dr Syed Modasser Ali

Message

Dr Captain (Retd.) Mozibur Rahman Fakir, MP

State Minister

Ministry of Health and Family Welfare

Government of the People's Republic of Bangladesh

I am pleased to learn that Health Bulletin 2009 containir information regarding different parameters of health status of tr people of Bangladesh and Health Service of the Government Bangladesh is being published. The Government is committe towards ensuring all the fundamental rights that includes heal of the people. For this we need to strengthen our system services delivery through efficient management. And to fulfill o objectives, we need to know concrete information on the heal profile and health service delivery mechanism. In the absence sound health information, the limited resources available f meeting the health needs of the people can not be used mo effectively.

In this information age, use of electronic medium for establishir networks and linkages with the hospitals and health centers the upazila, district and national levels can ensure eas accessibility of current data. Our Government has a clear focu on accomplishing digitalization at all tier of health care delive system. The Government puts priority to the expansion modern and well-functioning Management information systen for health to generate accurate information on a regular basis. hope that through effective digitization; soon it will be possib for the planners to proceed for an evidence-based decisic making leading towards better health of the people.

I would request all concerned to help not only to generate heal information but also make use of information at all tiers of tr health care delivery system. It will ensure improvement of tr health status of the people thus help us to bring in the chang and enable us to reach our vision of 2021.

I extend my sincere appreciation the Management Informatic System for Health (MIS-Health) Department of the Directora General of Health Bulletin 2009. I hope this endeavor of MI Health, DGHS will be of use to all concerned.

Joy Bangla, Joy Bangobandhu.
Long live Bangladesh

Dr Captain (Retd.) Mozibur Rahman Fakir, MP

Message
Shaikh Altaf Ali
Secretary
Ministry of Health and Family Welfare
Government of the People's Republic of Bangladesh

Health data and statistics are important because they measure a wide range of health indicators of our life, society and governance no matter what is its context. The context may either be a small community or an entire country. Statistical evidences help us to understand treatment efficacies of drugs, cost-effectiveness of health care, right kind of prevention measures, and outcomes of public health interventions.

The Health Bulletin 2009, which is a yearly publication of MIS, DGHS provides us such statistical evidences about our health situation in the country. Although there remains scope to improve the publication covering data from more stakeholders, more areas and more in-depth, obvious limitations prevented the concerned team members to restrict temptations. Nonetheless, Health Bulletin 2009 is the only comprehensive book that provides recent health data in a comprehensive format. One of the limitations of data scarcity is absence of culture in keeping records, limitation in data communication and absence of tools for data analysis. The recent development of ICT and digital communication systems in public health sectors of Bangladesh will remove such barriers in future and the future publications will definitely see the benefits.

I thank Director of MIS, DGHS Professor Dr Abul Kalam Azad and his able team for taking initiatives for publishing this Health Bulletin.

Shaikh Altaf Ali

Message

Professor Dr Shah Monir Hossain

Director General of Health Services

Government of the People's Republic of Bangladesh

We always work under extreme pressures of routine health service management and unexpected health crisis arising out from natural disasters and global pandemics of communicable diseases. All of our directorates, hospitals, institutes, branches, sub-branches, managers and staffs have to confront these pressures. I am delighted to see that despite such pressures, the Management Information System of DGHS did not stop its regular endeavor of publishing the yearly Health Bulletin.

I have learnt that Health Bulletin 2009 furnishes better and more information than Health Bulletin 2008. This has been possible due to recent deployment of digital systems across our health care network which improved our data communication systems in terms of availability, quicker transmission and processing. Thanks to "Digital Bangladesh" vision of the present Government and thanks to our Honorable Minister for Health and Family Welfare to show strong support to establish our digital network systems.

I firmly believe that there is enough space for improvement in the upcoming Health Bulletins. With this I would like to laud Professor Dr Abul Kalam Azad, Director of MIS, DGHS, who brought the "Health Bulletin" at this stage. My thanks are also for the editorial board members, and concerned officers and staffs who were associated with this publication.

Professor Dr Shah Monir Hossain

Message

Professor Dr Abul Kalam Azad

Director & Line-Director,MIS-Health, DGHS
Government of the People's Republic of Bangladesh

I am delighted to see that the Health Bulletin 2009 is ready for our valued readers. This issue of Health Bulletin is blessed with benefit of our recent development of information and communication technology (ICT) backbone created across the health sector of the country as low as up to the upazila level. Through this network we could receive the information more rapidly than earlier through email or direct Internet based central database, in soft copy rather than in hard copy, and from more health facilities satisfying reasonably wider coverage. Therefore, we had some opportunity to check quality of the data. However, as the ICT backbone was created about the end of April 2009, and we had a time frame to publish this Health Bulletin by June 2009, we certainly could not give enough attention for ensuring its quality and content. We apologize to our readers for this shortcoming.

The experience of our current working environment, which is a combination of ICT backbone, improved staff skill and motivation, and monitoring and supervision system, shows that the future issues of Health Bulletins could be published as soon as the year will be ended and would be richer in quality, content, writing style and design.

While this issue of Health Bulletin 2009 is better in quality than our previous issues, we still caution our readers to consider the limitation of the data sources, specially those related to morbidity and mortality. The existing health information system allows us to get data only from the public health facilities. Estimates say that about one quintile to one quarter of health care seekers of the country avail services from public health facilities. Therefore, our data may not represent the entire population of the country. Again, diagnoses are often made in situation of limited availability of laboratory, equipment, and skilled staff facilities. This situation cannot exclude possibility of mistake in ascertaining causes of morbidity or death of a patient. However, we are in relentless efforts to improve the data quality situation to gift to you the benefits in future.

Whatever improvements we have made, are the results of supports and inspirations you and your organizations have given to us continuously over the years. I am deeply indebted to our Honorable Minister for Health & Family Welfare Professor AFM Ruhal Haque for continuous support to our work. Likewise, we acknowlegdge the cordial support and continuous guidance of Professor Syed Modasser Ali. Honorable Adviser to the Prime Minister for Health, Family Welfare and Social Welfare Affairs. The Honorable State Minister for Health and Family Welfare Dr. Captain (Retd.) Mujibur Rahman was kind enough to give attention to this work even it was just after his taking change of office. The Secretary, MOHFW Mr. Shaikh Altaf Ali and Director General of Health services Professor Shah Monir Hossain were always beside us to improve the quality of work. I express my thanks to them. While all the health managers and data related staffs all over the country and in MIS-Health office were in close relationship and engaged with us in publication of this Health Bulletin, I feel special urge to mention the names of Dr. Nasreen Khan, Md. Nezam Uddin Biswas and Kaniz Fatema who were actually at the hearts of key activities of this publication. I wish healthy active and successful life of everybody.

Professor Abul Kalam Azad

ACRONYMS

ALS	Average Length of Stay
ARI	Acute Respiratory Tract Infection
BBS	Bangladesh Bureau of Statistics
BCC	Behavior Change Communication
BDHS	Bangladesh Demographic and Health Survey
BEOC	Basic Emergency Obstetric Care
BGC	Bangladesh Geographic Survey
BMI	Body Mass Index
BMRC	Bangladesh Medical Research Council
BOR	Bed Occupancy Rate
BSMMU	Bangabandhu Sheikh Mujib Medical University
CABG	Coronary Artery Bypass Grafting
CC	Community Clinic
CDC	Communicable Disease Control
CIDD	Control of Iodine Deficiency Disorder
CMMU	Construction, Maintenance and Management Unit
CMNS	Child and Mother Nutrition Survey
CNS	Child Nutrition Survey
CRF	Chronic Renal Failure
CS	Civil Surgeon
DAB	Diabetic Association of Bangladesh
DDA	Directorate of Drug Administration
DF	Dengue Fever
DGFP	Directorate General of Family Planning
DGHS	Directorate General of Health Services
DHF	Dengue Hemorrhagic Fever
DNS	Directorate of Nursing Services
DOTS	Directly Observed Treatment Strategy
DPHE	Department of Public Health Engineering
DSF	Demand Side Financing
EmOC	Emergency Obstetric Care
ESD	Essential Service Delivery
ESP	Essential Service Packages
ETT	Exercise Tolerance Test
FEP	Filariasis Elimination Program
GAVI	Global Alliance for Vaccine and Immunization
GFTAM	Global Fund Tuberculosis, Aids & Malaria
GO	Government
GOB	Government of Bangladesh
HIV	Human Immunodeficiency virus/Acquired Immunodeficiency Syndrome
HNPSP	Health, Nutrition and Population Sector Program
HPSP	Health and Population Sector Program
IAPB	International Association for Prevention of Blindness
ICT	Information and Communication Technology
IDA	Iron Deficiency Anemia
IDD	Iodine Deficiency Disorder
IDH	Infectious Diseases Hospital

IEC	Information, Education and Communication
IEDCR	Institute of Epidemiology, Disease Control and Research
IHSM	Improved Hospital Services Management
IMHR	Institute of Mental Health and Research
IMR	Infant Mortality Rate
IOL	Intraocular Lens
IPH	Institute of Public Health
IPHN	Institute of Public Health Nutrition
IRS	Indoor residual spraying
IST	In-Service Training
IVM	Integrated Vector Management
IYCF	Infant and Young Child Feeding
LAN	Local Area Network
LLIN	Long lasting Insecticidal Net
MATS	Medical Assistants' Training School
MDA	Mass Drug administration
MDG	Millennium Development Goal
MIS	Management Information System
MNC	Maternal, Neonatal and Child
MNHC	Maternal and Neonatal Health Care
MO	Medical Officer
MOHFW	Ministry of Health and Family Welfare
MOU	Memorandum of Understanding
MWM	Medical Waste Management
NCD	Non-communicable Diseases
NEMEW	National Equipment Maintenance and Engineering Workshop
NGO	Non-government Organization
NICRH	National Institute of Cancer Research and Hospital
NICVD	National Institute of Cardiovascular Diseases
NID	National Immunization Day
NIDCH	National Institute of Chest Diseases and Hospital
NIKDU	National Institute of Kidney Diseases and Urology
NIMHR	National Institute of Mental Health and Research
NIO	National Institute of Ophthalmology
NIPORT	National Institute of Population Research and Training
NIPSOM	National Institute of Preventive and Social Medicine
NNP	National Nutrition Program
NITOR	National Institute of Traumatology, Orthopedics and Rehabilitation
OP	Operational Plan
OPD	Outpatient Department
ORS	Oral Rehydration Salt
ORT	Oral Rehydration Therapy
OT	Operation Theater
PMIS	Personnel Management Information System
PRSP	Poverty Reduction Strategy Paper
PSM	Preventive and Social Medicine
RDU	Research and Development Unit
RHC	Rural Health Center
SBTP	Safe Blood Transfusion Program
SEARO	South East Asian Regional Office

SVRS	Sample Vital Registration Survey
TB	Tuberculosis
TT	Tetanus Toxoid
TTU	Technical Training Unit
UHC	Upazila Health Complex
UHFPO	Upazila Health and Family Planning Officer
UHFWC	Union Health and Family Welfare Center
UNICEF	United Nations Children's Educational Fund
USC	Union Sub Center
USI	Universal Salt Iodization
VAC	Vitamin A Capsule
VAD	Vitamin A Deficiency
WAZ	Weight of Age Z score
WCBA	Women of Child Bearing Age
WHO	World Health Organization

CONTENTS

Bangladesh - Basic Information

Location and Geography

Bangladesh was emerged as an independent and sovereign country in 1971 following a nine months war of liberation. The country is one of the largest deltas of the world with a total area of 147,570 sq km. Being a low-lying country it stretches latitudinally between 20º34' and 26º38' north and longitudinally between 88º01' and 92º41' east. It is mostlysurrounded by Indian Territory (West Bengal, Tripura, Assam and Meghalaya), except for a small strip in the southeast by Myanmar. Bay of Bengal lies on the south. The standard time of the country is GMT +6 hrs.

History

Bangladesh has a glorious history and rich heritage. Once it was known as 'Sonar Bangla' or the 'Golden Bengal'. The territory now constituting Bangladesh was under the Muslim rule for over five and a half centuries from 1201 to 1757 AD. Subsequently, it came under the British rule following the defeat of the sovereign ruler, Nawab Sirajuddaula, at the battle of Palassey on 23 June 1757. The British ruled over the Indian sub-continent including this land for nearly 190 years from 1757 to 1947. During that period, Bangladesh was a part of the British Indian provinces of Bengal and Assam. With the termination of British rule in August 1947, the sub-continent was partitioned into India and Pakistan.

Bangladesh was a part of Pakistan and was called 'East Pakistan'. It remained so for about 24 years from August 14, 1947 to March 25, 1971.

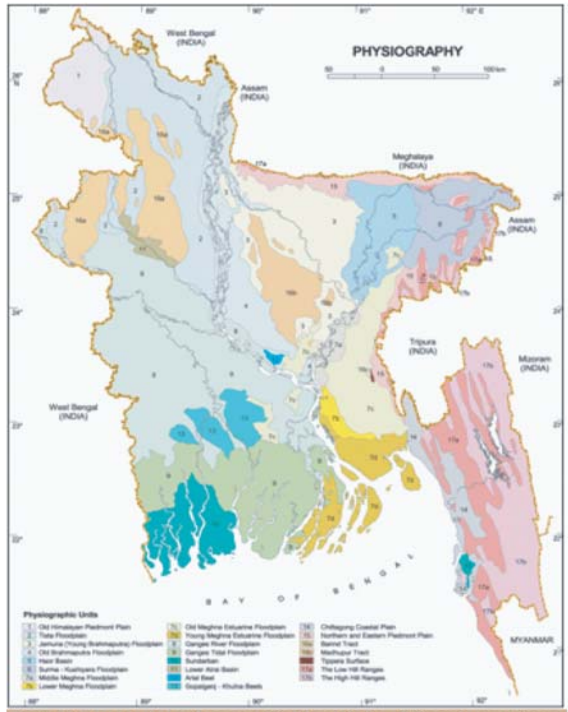
Physiography

With about half of its surface below the 10 m contour line, Bangladesh is located at the lowermost reaches of three mighty river systems -the Ganges-Padma river system, Brahmaputra-Jamuna river system and Surma-Meghna river system. Coinciding with the division of the country based on altitude the land can be divided into three major categories of physical units: Tertiary hills, Pleistocene uplands and Recent plains (formed in recent epoch). The heavy monsoon rainfall coupled with the low altitude of major parts of the country makes floods an annual phenomenon in Bangladesh. Quaternary (began about 2 million years ago and extends to the present) sediments, deposited mainly by the Ganges, Brahmaputra (Jamuna) and Meghna rivers and their numerous distributaries, cover about three-quarters of Bangladesh. The physiography and the drainage pattern of the vast alluvial plains in the central, northern and western regions have gone under considerable alterations in recent times. In the context of physiography, Bangladesh may be classified into three distinct regions: (a) floodplains, (b) terraces and (c) hills, each having distinguishing characteristics of its own. The physiography of the country has been divided into 24 sub-regions and 54 units.

Climate

Bangladesh has a tropical monsoon-type climate, with a hot and rainy summer and a dry winter. January is the coolest month with temperatures averaging near 260 C (780 F) and April is the warmest with temperatures from 330 to 360 C (910 to 960 F). The climate is one of the wettest in the world. Most places receive more than 1,525 mm of rain a year, and areas near the hills receive 5,080 mm). Most rains occur during the monsoon (June-September) and little in winter (November-February).

Bangladesh has warm temperatures throughout the year, with relatively little variation from month to month. January tends to be the coolest month and May the warmest. In Dhaka, the average January temperature is about 19°C (about 66°F), and the average May temperature is about 29°C (about 84°F).



Bangladesh has been divided into 24 sub-regions based on surface.

Administration

From the administrative point of view, Bangladesh is divided into 6 Divisions, 64 Districts, 6 City Corporations, 308 Municipalities, 482 Upazilas and 4498 Unions. The six administrative division's are namely, Dhaka, Chittagong, Rajshahi, Khulna, Barisal and Sylhet. The country is governed by the Parliamentary Democracy and it has a unitary National Parliament, named Bangladesh Jatiya Sangsad. There are 40 Ministries and 12 Divisions. The Ministry of Health & Family Welfare is one of largest ministries in the country. At the national level, the Ministry of Health & Family Welfare (MOHFW) is responsible for policy, planning and decision making at macro level. Under MOHFW, there are four Directorates, viz., Directorate General of Health Services, Directorate General of Family Planning, Directorate of Nursing Services and Directorate of Drug Administration. Beside, there are a separate National Nutrition Project (NNP)and Construction, Maintanance and Management Unit (CMMU).

Economy

Bangladesh has an agrarian economy, although the share of agriculture to GDP has been decreasing over the last few years. Yet it dominates the economy accommodating major rural labour force. From marketing point of view, Bangladesh has been following a mixed economy that operates on free market principles. The GDP of Bangladesh is 6.21% and per capita income is US$ 599. The principal industries of the country include readymade garments, textiles, chemical fertilizers, pharmaceuticals, tea processing, sugar, leather goods etc. The principal mineral includes Natural gas, Coal, white clay, glass sand etc.

Communication

The transport system of Bangladesh consists of roads, railways, inland waterways, two sea ports, maritime shipping and civil aviation catering for both domestic and international traffic. Presentlythere are about 21,000 km of paved roads; 2,706 route-kilometres of railways (BG-884 km and MG -1,822 km); 3,800 km of perennial waterways which increases to 6,000 km during the monsoon, 2 seaports (Chittagong and Chalna) and 3 international (Dhaka, Chittagong and

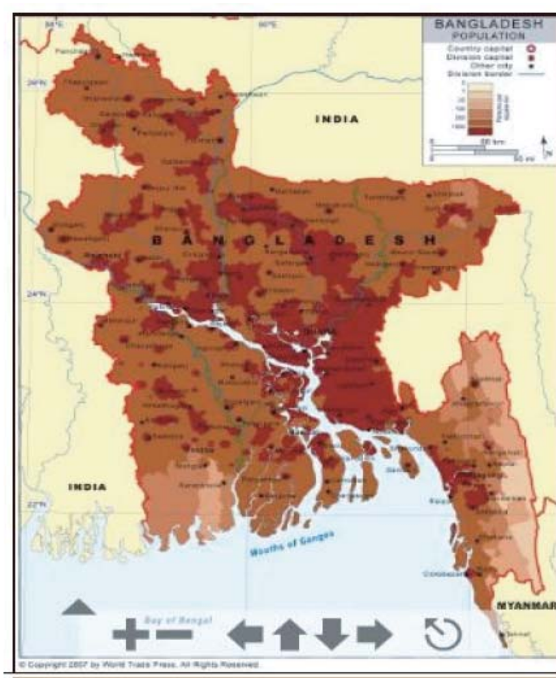
Sylhet) and 8 domestic airports. Along with the development of road transport, efforts are under way to develop the water transport system. The country is covered with a network of rivers and canals forming a maze of interconnecting channels. Rivers are still the lifeline of the nation and are the cheapest mean of transport. Dhaka is connected by air with many global cities by the national airline (Biman Bangladesh Airlines). A number of foreign airlines operate their international services with a link to Dhaka, Chittagong and Sylhet.

Religion and Culture

The majority (about 88%) of the people are Muslim. Over 98% of the people speak in Bangla. English, however is widely spoken. Bangladesh is heir to a rich cultural legacy. In two thousand or more years of its chequered history, many illustrious dynasties of kings and Sultans ruled the country and have left their mark in the shape of magnificent cities and monuments. Apart from this, the century old cultural traditions can be viewed in innumerable tangible and intangible heritages -in archaeological sites, in sculptures, in stones and terracotta, in architectures, museums, archives, libraries, classical music, songs and dance, paintings, dramas, folk arts, festivals, games as well as ethnic cultural activities. The people of Bangladesh are very simple and friendly. A beautiful communal harmony among the different religions has ensured a very congenial atmosphere. More than 75% of the population lives in rural areas. Urbanisation has, however, been rapid in the last few decades.

Population and Demograph

Bangladesh is now Asia's fifth and world's eighth populous country with an estimated population of about 146 million. Density of population is around 979 per square kilometer, the highest in the world. Rural population comprises about 76% while urban constitutes about 24%. Adult literacy rate is 54% (2006). Census of 2001 reveals that 43 per cent of the population is below 15 years of age. This young age structure constitutes built-in population momentum. Also urban population is increasing quite fast. Though Bangladesh has made progress in reducing poverty and per capita income has been creeping up, a substantial number of population are poor. Progress made in improving Bangladesh's Human Development Index (HDI) has placed her among the medium-ranking HDI countries.



In this map the deep colours indicate the highest population concentration in Bangladesh.

Strong policy interventions led to continuous reduction in the annual growth rate of population from the level of 2.33 % in 1981 to 1.54 in 2001 and further to 1.48 (2007). The Total Fartility Rate (TFR) also went down from 3.4 in 1993-94 to 2.2 (2007). The CPR (any method) increased from 44.6% in 1993-94 to 58.1% in 2004, but again fell down to 55.8% in 2007. Life expectancy at birth has continuously been rising, and is now 65 years (2007) from the level of 58 (1994). Reversing past trends, women now live longer than men. The country, however, is over burdened with about two million new faces every year creating extra pressure on food, shelter, education, health, employment, etc., and thus making the anticipated economic growth difficult.

District-wise distribution of population in Bangladesh (estimated)

SL. NO	District	Adjusted	Estimated						
		2001	2002	2003	2004	2005	2006	2007	2008
Barisal Division									
1	Barisal	2,465,392	2,501,880	2,538,368	2,574,855	2,611,343	2,650,513	2,690,271	2,727,934
2	Barguna	887,144	900,274	913,403	926,533	939,663	953,758	968,064	981, 616
3	Bhola	1,788,019	1,814,482	1,840,944	1,867,407	1,893,870	1,922,278	1,951,112	1,978,427
4	Jhalakhati	727,175	737,937	748,699	759,462	770,224	781,777	793,504	804, 613
5	Patuakhali	1,537,747	1,560,506	1,583,264	1,606,023	1,628,782	1,653,214	1,678,012	1,701,504
6	Pirojpur	1,154,549	1,171,636	1,188,724	1,205,811	1,222,898	1,241,241	1,259,860	1,277,498
Division -Total		8,560,026	8,686,714	8,813,403	8,940,091	9,066,780	9,202,782	9,340,823	9, 471,594
Chittagong Division									
7	Bandarban	315,717	320,390	325,062	329,735	334,407	339,423	344,514	349,337
8	Brahman-baria	2,496,403	2,533,350	2,570,297	2,607,243	2,644,190	2,683,853	2,724,111	2,762,248
9	Chandpur	2,352,623	2,387,442	2,422,261	2,457,079	2,491,898	2,529,276	2,567,216	2, 603,157
10	Chittagong	6,869,744	6,971,416	7,073,088	7,174,761	7,276,433	7,385,579	7,496,363	7,601,312
11	Comilla	4,819,989	4,891,325	4,962,661	5,033,997	5,105,332	5,181,912	5,259,641	5,333,275
12	Cox's Bazar	1,847,186	1,874,524	1,901,863	1,929,201	1,956,539	1,985,887	2,015,675	2,043,894
13	Feni	1,266,038	1,284,775	1,303,513	1,322,250	1,340,987	1,361,102	1,381,518	1,400,859
14	Kharachari	542,642	550,673	558,704	566,735	574,766	583,387	592,138	600,427
15	Laxmipur	1,560,570	1,583,666	1,606,763	1,629,859	1,652,956	1,677,750	1,702,917	1,726,757
16	Noakhali	2,698,658	2,738,598	2,778,538	2,818,478	2,858,419	2,901,295	2,944,815	29,860,42
17	Rangamati	551,250	559,409	567,567	575,726	583,884	592,642	601,532	609, 953
Divisi on-Total		25,320,820	25,695,568	26,070,316	26,445,064	26,819,813	27,222,110	27,630,442	28,017,266

District-wise distribution of population in Bangladesh (estimated) (contd...)

SL. NO	District	Adjusted	Estimated						
		2001	2002	2003	2004	2005	2006	2007	2008
	Dhaka Division								
18	Dhaka	9,047,911	9,181,820	9,315,729	9,449,638	9,583,547	9,727,300	9,873,210	10,011,434
19	Faridpur	1,829,507	1,856,584	1,883,660	1,910, 737	1,937,814	1,966,881	1,996,384	2,024,333
20	Gazipur	2,124,018	2,155,453	2,186,889	2,218,324	2,249,760	2,283,506	2,317,759	2,350,207
21	Gopalgonj	1,209,160	1,227,056	1,244,951	1,262,847	1,280,742	1,299,953	1,319,452	1,337,924
22	Jamalpur	2,210,921	2,243,643	2,276,364	2,309,086	2,341,808	2,376,935	2,412,589	2,446,365
23	Keshore - gonj	2,684,591	2,724,323	2,764,055	2,803,787	2,843,519	2,886,172	2,929,464	2,970,476
24	Madaripur	1,186,211	1,203,767	1,221,323	1,238,879	1,256,435	1,275,282	1,294,411	1,312,532
25	Manikgonj	1,366,735	1,386,963	1,407,190	1,427,418	1,447,646	1,469,361	1,491,401	1,512,280
26	Munshigonj	1,353,297	1,373,326	1,393,355	1,413,383	1,433,412	1,454,913	1,476,737	1,497,411
27	M-singh	4,682,234	4,751,531	4,820,828	4,890,125	4,959,422	5,033,813	5,109,321	5,180,851
28	N-gonj	2,278,843	2,312,570	2,346,297	2,380,024	2,413,751	2,449,957	2,486,707	2,521,520
29	Narshingdi	1,996,552	2,026,101	2,055,650	2,085,199	2,114,748	2,146,469	2,178,666	2,209,167
30	Netrokona	2,069,408	2,100,03 5	2,130,662	2,161,290	2,191,917	2,224,796	2,258,168	2,289,782
31	Rajbari	999,704	1,014,500	1,029,295	1,044,091	1,058,886	1,074,769	1,090,891	1,106,163
32	Shariatpur	1,134,498	1,151,289	1,168,079	1,184,870	1,201,660	1,219,685	1,237,980	1,255,311
33	Sherpur	1,331,083	1,350,783	1,370,483	1,390,183	1,409,883	1,431,031	1,452,497	1,472,831
34	Tangail	3,424,028	3,474,704	3,525,379	3,576,055	3,626,730	3,681,131	3,736,348	3,788,656
Division - Total		40,928,701	41,534,446	42,140,191	42,745,935	43,351,680	44,001,955	44,661,985	45,287,252

District-wise distribution of population in Bangladesh (estimated) (contd...)

SL. NO	District	Adjusted	Estimated						
		2001	2002	2003	2004	2005	2006	2007	2008
	Khulna Division								
35	Bagerhat	1,592,358	1,615,925	1,639,492	1,663,059	1,686,626	1,711,925	1,737,604	1,761,930
36	Chuadanga	1,055,238	1,070,856	1,086,473	1,102,091	1,117,708	1,134,474	1,151,491	1,167,611
37	Jessore	2,592,670	2,631,042	2,669,413	2,707,785	2,746,156	2,787,348	2,829,159	2,868,767
38	Jhenardha	1,646,548	1,670,917	1,695,286	1,719,655	1,744,024	1,770,184	1,796,737	1,821,891
39	Khulna	2,475,365	2,512,000	2,548,636	2,585,271	2,621,907	2,661,236	2,701,154	2,738,970
40	Kustia	1,823,881	1,850,874	1,877,868	1,904,861	1,931,855	1,960,833	1,990,245	2,018,108
41	Magura	862,768	875,537	888,306	901,075	913,844	927,552	941,465	954,645
42	Meherpur	616,883	626,013	635,143	644,273	653,402	663,203	673,151	682,575
43	Narail	729,506	740,303	751,099	761,896	772,693	784,283	796,048	807,192
44	Shatkhira	1,937,007	1,965,675	1,994,342	2,023,010	2,051,678	2,082,453	2,113,690	2,143,281
Division Total		15,332,224	15,559,141	15,786,058	16,012,975	16,239,892	16,483,490	16,730,743	16,964,974
	Rajshahi Division								
45	Bogra	3,165,567	3,212,417	3,259,268	3,306,118	3,352,969	3,403,264	3,454,312	3,502,672
46	Dinajpur	2,772,459	2,813,491	2,854,524	2,895,556	2,936,589	2,980,638	3,025,347	3,067,701
47	Gaibandha	2,235,759	2,268,848	2,301,937	2,335,027	2,368,116	2,403,638	2,439,692	2473847.69
48	Joupurhat	899,217	912,525	925,834	939,142	952,451	966,738	981,239	994976.346
49	Kurigram	1,850,713	1,878,104	1,905,494	1,932,885	1,960,275	1,989,679	2,019,524	2,047,797
50	Lalmonir-hat	1,159,357	1,176,515	1,193,674	1,210,832	1,227,991	1,246,411	1,265,107	1,282,818
51	Naogoan	2,504,718	2,541,788	2,578,858	2,615,927	2,652,997	2,692,792	2,733,184	2,771,448
52	Natore	1,592,547	1,616,117	1,639,686	1,663,256	1,686,826	1,712,128	1,737,810	1,762,139
53	Nawabgonj	1,492,543	1,514,633	1,536,722	1,558,812	1,580,902	1,604,616	1,628,685	1,651,486
54	Nilphamary	1,639,956	1,664,227	1,688,499	1,712,770	1,737,041	1,763,097	1,789,543	1,814,596
55	Pabna	2,272,775	2,306,412	2,340,049	2,373,686	2,407,323	2,443,433	2,480,084	2,514,805
56	Panchgar	879,711	892,731	905,750	918,770	931,790	945,767	959,953	973,392
57	Rajshahi	2,387,602	2,422,939	2,458,275	2,493,612	2,528,948	2,566,882	2,605,385	2,641,860
58	Rangpur	2,652,908	2,692,171	2,731,434	2,770,697	2,809,960	2,852,109	2,894,891	2,935,419
59	Sherajgonj	2,806,178	2,847,709	2,889,241	2,930,772	2,972,304	3,016,889	3,062,142	3,105,011

District-wise distribution of population in Bangladesh (estimated) (contd...)

SL. NO	District	Adjusted	Estimated						
		2001	2002	2003	2004	2005	2006	2007	2008
					Rajshahi Division				
60	Thakurgoan	1,275,150	1294,022	1,312,894	1,331,767	1,350,639	1,370,899	1,391,462	1,410942
Division-Total		31,587,160	32,054,650	32,522,140	32,989,630	33,457,120	33,958,977	34,468,361	34,950917
					Sylhet Division				
61	Habigonj	1,837,339	1,864,532	1,891,724	1,918,917	1,946,109	1,975,301	2,004,930	2,032999
62	Moulavi Bazar	1,688,981	1,713,978	1,738,975	1,763,972	1,788,969	1,815,804	1,843,041	1,868843
63	Sunamgonj	2,089,480	2,120,404	2,151,329	2,182,253	2,213,177	2,246,375	2,280,070	2,311990
64	Sylhet	2,674,177	2,713,755	2,753,333	2,792,910	2,832,488	2,874,975	2,918,100	2,958953
Division-Total		8,289,977	8,412,669	8,535,360	8,658,052	8,780,744	8,912,455	9,046,142	9,172786
Country-Total		130,018,908	131,943,188	133,867,468	135,791,748	137,716,027	139,781,767	141,878,494	143864792

Source: BBS Census Population (adjusted). Note: We estimated population for 2008 based on 2007 estimated population and adding 1.4% average growth rate. We estimated population for 2006 and 2007 based on 2005 estimated population and adding 1.5% average growth rate. The BBS used different method of calculation. Therefore, our figures for 2006, 2007 and 2008 are different than BBS estimates.

Health Status

Since independence Bangladesh has made significant progress in health outcomes. Infant and Child mortality rates have been markedly reduced. The underfive mortality rate in Bangladesh declined from 151 deaths per thousand live births in 1991 to 65 deaths/1000 live births in 2007 and during the same period infant mortality rate reduced from 94 deaths per 1000 live births to 52. EPI coverage extended its reach from 54% in 1991 to 87.2% in 2006.The MMR reduced from 574/100,000 live births in 1991 to 290 in 2007. Deliveries attended by skilled birth attendants increased from only 5% in 1990 to 20% in 2006. The prevalence of malaria dropped from 42 cases /100,000 in 2001 to 34 in 2005. Bangladesh has also achieved significant success in halting and reversing the spread of tuberculosis (TB). Detection of TB by the Directly Observed Treatment Short-course (DOTS) has more than doubled between 2002 and 2007, from 34 to 92%. The successful treatment of tuberculosis has progressed from 84% in 2002 to 91% in 2007. Polio and leprosy are virtually eliminated. HIV prevalence is still very low. Development of countrywide network of health care infrastructure in public sector is remarkable. However, availability of drugs at the health facilities, deployment of adequate health professionals along with maintenance of the health care facilities remain as crucial issues, impacting on optimum utilization of public health facilities

Nutrition Status

There has been considerable progress in reducing malnutrition and micro nutrient deficiencies in Bangladesh. According to BDHS, percentage of U5 underweight (6-59 months) has reduced to 46.3 (2007) from 67 (1990) and that of U5 stunted (24-59 months) from 54.6 (1996) to 36.2 (2007). Percentage of children 1-5 years receiving vitamin-A supplements in last six months has increased from 73.3 (1999-00) to 88.3 (2007). The rate of night blindness has reduced to 0.04 per 1000 people (IPHN, HKI 2006). However, in spite of efforts taken by the government, high rates of malnutrition and micronutrient deficiencies along with gender discrimination remain common in Bangladesh.

Urban Health Service

The urban areas provide a contrasting picture of availability of different facilities and services for secondary and tertiary level health care, while primary health care facilities and services for the urban population at large and the urban poor in particular are inadequate.

Rapid influx of migrants and increased numbers of people living in urban slums in large cities are creating continuous pressure on urban health care service delivery. Since the launching of two urban primary health care projects, the services have been delivered by the city corporations and municipalities through contracted NGOs in the project's area. Rest of the urban areas and services are being covered by MOHFW's facilities. Moreover, 35 urban dispensaries under the DGHS are providing outdoor patient services including EPI and MCH to the urban population.

Organizational Setup of MOHFW

The Ministry of Health & Family Welfare is one of largest ministries in the country. At the national level, the ministry of Health & Family Welfare (MOH&FW) is responsible for policy, planning and decision making at macro level.

Executing Authorities of MOHFW:

Under MOHFW, there are four Directorates General or Directorates, e.g., Directorate General of Health Services, Directorate General of Family Planning, Directorate of Nursing Services and Directorate of Drug Administration.

Directorate General of health Services (DGHS)

The Directorate General of Health Services (DGHS) is entrusted for the implementation of the policy decisions of the Ministry of Health and Family Welfare (MOHFW) as regards health service delivery to all the people under the jurisdiction of the Government of the People's Republic of Bangladesh. It provides technical guidance to the ministry. DGHS carries out its activities through different directors, line directors, project directors, institution heads, district and upazila health managers and union health staffs.

Level	Designation of Manager	Responsibility
National	Director General	Looks after overall administration and implementation for health services
	Additional Directors General	Assist Director General
	Directors of DGHS	Assist Director General
	Line Directors	Implement respective Operational Plan of HNPSP
	Project Directors	Implement respective project under HNPSP
	Program Managers	Assist respective Line Director
	Deputy Program Managers	Assist respective Program Manager
	Directors of different National Institutions	Administer and manage respective national level institution
Regional	Principals of academic institutions	Administer and manage respective medical college, institute of health technology, medical assistants' training school, etc.
	Directors of medical college hospitals	Administer and manage respective medical college hospital
Division	Divisional Directors for Health	Administer and supervise activities of health managers at district and lower levels
District	Civil Surgeons (CS)	Implement, administer and manages health programs of district level. In some cases, looks after district hospital
	Superintendents	Administer and manage respective sadar hospital, general hospital, district hospital, mental hospital, chest hospital, etc.
Upazila	Upazila Health & Family Planning Officers (UHFPO)	Implement, administer and manage health programs of upazila level. Run respective upazila health complex
Union	Health Inspectors and Assistant Health Inspectors	Manage and supervise health programs at union level and below

Health care delivery systems of Bangladesh

Distribution of public health care services and facilities follows similar pattern of administrative tiers, viz. national (mostly capital-based in Dhaka), regional (in divisions), district, upazila, union and ward. The country has 6 divisions, 64 districts, 482 upazillas and 4,498 unions. As the Ministry of health and family Welfare deploys health workforce according to the older ward system, which divides each union into 3 wards. Therefore, number of MOHFW wards is 13,494.

Primary health care (PHC), which includes family planning services in the urban area (city corporations and municipalities), is provided by Ministry of Local Government; and in rest of the country by Ministry of Health and Family Welfare (MOHFW) provides health care service. Provision of secondary and tertiary care, in both urban divisional directorate with necessary staff. and rural areas, is the sole responsibility of MOHFW. The MOHFW delivers its services through two separate executing authorities, viz. Directorate General of Health Services (DGHS) and Directorate General of Family Planning (DGFP). The names explain their functions.

PHC services of both DGHS and DGFP begin at the ward level through a set of community health staffs, at least one in each ward (Table). To supervise these field staffs, there is one assistant health inspector (for DGHS) and one family planning inspector (for DGFP) at union level. There are several hundred non-bed community facilities to provide outpatient services (1466 for DGHS and 3500 for DGFP). Besides DGFP also operates additional 97 maternal and child welfare centres (MCWCs) (union: 23; upazila: 12; district: 62), 471 MCH-FP clinics (upazila: 407; district: 64), 177

NGO clinics (upazila: 68; district: 104; national: 05), 08 model clinics (national: 02; regional: 06) and organizes 30,000 makeshift satellite clinics per month.

The public sector hospital care in Bangladesh is mainly provided by DGHS.

Pimary level hospital care	Secondary level hospital care	Tertiary level hospital care
Begins through Upazila Health Complex (31 to 50 bed) existing in 418 upazilas. The other upazilas (64 Nos.) being their administrative offices in district head quarters, provide primary level hospital care from the respective district hospitals.	The district hospitals (50 to 375 bed), one in each district, provide secondary level hospital care in several specialty areas.	The regional hospitals are multidisciplinary tertiary care hospitals (250 to 1700 beds) mostly affiliated with teaching institutes. At the national level, there are postgraduate and specialized hospitals (100 to 600 beds).

Divisional level health organization

At the divisional level, there is a divisional Director for Health. S/he is the head of a Divisional Directors supervise the activities of the civil Surgeons.

District level health organization

At the district level, Civil Surgeon is the health manager. S/he has own administrative office supported by various categories of staff. There is either a Sadar Hospital or a General Hospital in each district head quarter. The Hospital provides services under the management of Civil Surgeon with a view to render out-patient, in-patient, emergency, laboratory and imaging services to the people. The in-patient services internal medicine, general surgery, obstetric and gynecology and other common specialist clinical services. It is the secondary level referral facility of health services of Bangladesh. Currently there are 59 Sadar district hospitals and 2 General hospitals in the country each having 100-250 bed.

Upazila level health organization

Upazila Health Complex (UHC) is another fixed service delivery point next to district level hospital. It provides the first level referral services to the population. In each UHC, there are posts for 9 (nine) doctors including one Upazila Health and Family Planning Officer (UHFPO). UHFPO is the Chief Health Officer of upazila and also Head of the UHC. Other doctors of UHC are Junior Consultants-4, Resident Medical Officer-1, Assistant Surgeons (MO)-2 and Dental Surgeon-1. There are 418 Upazila Health Complexes (UHC) in the country of which 153 are 50-bed and rests are 31-bed. UHC provides out-patient, in-patient and emergency services, limited diagnostic and imaging services, emergency obstetric care, contraceptive services and dental care.

Union level health organization

There are four types of static health facilities in the union level. These are Rural Health Centers (RHC, 10-bed hospital), Union Sub-centers (USC), Union Health and Family Welfare Centers (UHFWC) and Community Clinics (CC). There are 22 RHCs, in each of these, there are sanctioned posts of 20 staffs. RHC provides both out-patient and inpatient services. In an USC, there is sanctioned posts for one medical officer, one medical assistant, one pharmacist and one MLSS. Number of USC is 1,362; that for UHFWC is 87. Under HPSP, Government planned for establishing one Community Clinic for every 6000 rural populations. Number of CCs so far built is 11,883. But, these were not made functional. Recently Government has decided to start the CCs again. The total number of CCs will be 18000. The existing UHCs and Union level facilities will also provide services of CCs in the respective communities. So,13,500 additional CCs will be required.

The main health workforce in the union level is the domiciliary staff called health assistants. They are placed in each ward, which is the lowest and smallest administrative unit of the health sector. They visit the homes of the local people for providing primary health care services and collection of routine health data. The health assistants routinely organize satellite clinics for immunization services. Besides there are other small to large hospitals and special purpose hospitals spread across the country both in rural as well as in urban areas. Under the DGHS, there are altogether 40 teaching/training institutes and 589 small to large hospitals. In Family Planning sector, there are one national research-cum-training institute, two hospital-based training centres, and 32 other training centres (national: 12; regional: 20).

Nearly six hundred health managers under DGHS and a similar number under DGFP, from national to upazila levels, play roles in administering the health and family planning services (1,17). This figure does not include the institute and clinic/hospital heads.

Health, Nutrition & Population Sector Program (HNPSP)

The constitution Bangladesh mandates for basic health care services for its people as one of the fundamental responsibilities of the state. Towards this goal, the government has taken different endeavors to extend health facilities to the population. The broader policy document of the Government of Bangladesh that shapes direction of health care is the Proverty Reduction Strategy Paper (PRSP) although the current government has indicated that it will go for Five-Year Plan. The Government of Bangladesh is running a program through which the health care services are provided to the people from the grass root to the central level. The program is entitled Health, Nutrition and Population Sector Program for the period of July 2003 through June 2010 (HNPSP 2003-2010). The Ministry of Health and Family Welfare (MOHFW) designed the Program Implementation Plan (PIP) in accordance with the PRSP to implement its sector-wide program popularly known as Health, Nutrition and Population Sector Program (HNPSP).

The HNPSP covers 38 Operational Plans (OP) to be implemented by 38 Line Directors and 14 Projects/Programs. The Government has recently decided to continue HNPSP until 2011. The details of the program are well documented in the form of Program Implementation Plan (PIP) duly endorsed at the highest policy level of the government, the Excecutive Committee for National Economic Council (ECNEC).

The Implementing Agency of the program is Ministry of Health and Family Welfare (MOHFW) with its attached departments. The financial involvement is estimated to be around Taka 324,503 million which includes contributions for GOB (Government of Bangladesh) and DPs (Development Partners).

Priority Objectives and Goal

One of the important goals of PRSP and HNPSP is attainment of Millennium Development Goals (MDGs). The health sector is specially striving for attainment of health related MDGs. The priority objectives of HNPSP are: (i) reducing MMR; (ii) reducing TFR; (iii) reducing malnutrition; (iv)reducing infant and under-five mortality;
(v) reducing the burden of TB and other diseases; and (vi) prevention and control of non-communicable diseases including injuries. The commitment of the government targets towards

reaching the goal of sustainable improvement in health, nutrition and family planning status of the people by the end of the program period.

Distribution of Operational Plans and Projects/ Programs under HNPSP (2008-2011)

Implementing Authority	OP	ject / Pro Program
Ministry of Health & Family Welfare (includes NNP)	7	1
Directorate General of Health Services	19	9
Directorate General of Family Planning	9	1
Directorate of Drug Administration	1	0
Directorate of Nursing Services	1	2
National Institute of Population Research and Training	1	1
Total	38	14

It may be mentioned here that HNPSP deals with health care service delivery of the public sector. Nevertheless, it strives to maintain a strong cooperation and coordination with the efforts of the Private Sector as well so as to ensure the overall well-being of every citizen of the country.

Of the 38 OPs, 7 are under MOHFW, 19 under Directorate General of Health Services (DGHS), 9 under Directorate General of Family Planning (DGFP), 1 under Directorate of Nursing Services (DNS), 1 under Directorate of Drug Administration (DDA) and 1 under National Institute of Population Research and Training (NIPORT) and. Of the 14 projects/programs, 1 is under MOHFW, 9 under DGHS, 1 under DGFP, 2 under DNS and 1 under NIPORT.

The Health Bulletin 2009 is an attempt of Management Information System (MIS) of DGHS to provide an overview of the current health profiles of Bangladesh.

Basic Information and Indicators

Name				Source
A. GEOGRAPHY				
1	Location		Between 20°34' and 26° 38' north latitude and between 80°01' and 92° 41' east longitude	Staistical Pocket Book of Bangladesh 2008
2	Boundary		North: India West: India South: Bay of Bengal East: India & Myanmar	Staistical Pocket Book of Bangladesh 2008
3	Area (Sq.Km.)		147,570 Sq.Km. (56977 Sq. miles)	Staistical Pocket Book of Bangladesh 2008
4	Territorial Water		12 Nautical miles	Do
5	Standard Time		GMT + 7 hrs	
6	Rainfall		203mm/month	
B. ADMINISTRATION				
7	Division		6	Statistical Pocket Book Bangladesh 2008, BBS Health MIS, 2009 BBS, 2008
	City Corporation		6	
	Metropolitan City		4	
	Municipality		308	
8	Districts		64	
	Upazila		482	
9	Union		4,498	
10	Mouza		59,229	
11	Village (approximately)		87,310	
12	Household		2,54,90,822	
13	Average size of Household		4.7	SVRS, 2007
C. DEMOGRAPHY				
14	Population (2001 Census)	Total	124.355 million	Statistical Pocket Book Bangladesh 2008, BBS
		Male	64.091 million	
		Female	60.264 million	
15	Population Projected July 2007	Total	143.91 million	Statistical Pocket Book Bangladesh 2008, BBS
		Male	74.09 million	
		Female	69. 81 million	
16	Sex Ratio (Male per 100 Female)		106.0	Statistical Pocket Book Bangladesh 2008, BBS
17	Under 5 Population (in %)		11.7	SVRS, 2007, BBS
18	Under 00-14 Population (in %) Both sexes		35.1	SVRS, 2007, BBS
19	Female Population (15-49 yrs in %)		53.0	SVRS, 2007, BBS
20	Population (60 yrs + in %) Both sexes		6.6	SVRS, 2007, BBS
21	Population Density per sq.km.		966	SVRS, 2007, BBS

22	Crude Birth Rate (Per 1,000 pop.)		20.9	SVRS, 2007, BBS
23	Crude Death Rate (Per 1,000 pop.)		6.2	SVRS, 2007, BBS
24	Population Growth Rate (%)		1.40	SVRS, 2007, BBS
25	Total Fertility Rate (birth per women 15-49 yrs)		2.39 2.70	SVRS, 2007, BBS BDHS, 2007
26	Gross Reproduction rate		1.17	SVRS, 2007, BBS
27	Net Reproduction rate (NRR)		1.14	SVRS, 2007, BBS
28	Urban Population (in millions)		35.7	SVRS, 2007, BBS
29	Life Expectancy at Birth	Both sexes	66.6 years	SVRS, 2007, BBS
		Male	65.4 years	SVRS, 2007, BBS
		Female	67.9 years	SVRS, 2007, BBS
30	Mean Age at First Marriage	Male	23.6 years	SVRS, 2007, BBS
		Female	18.4 years	SVRS, 2007, BBS
D. HEALTH STATUS				
31	Infant Mortality Rate (per 1000 live birth)		52	SVRS, 2007, BBS BDHS, 2007
32	Maternal Mortality Ratio (per 1,000 live births)		3.37 3.20	SVRS, 2007, BBS BMMS, 2001
33	Neonatal Mortality Rate (per 1,000 live births)		31 37	SVRS, 2007, BBS BDHS, 2007
34	Under 5 Mortality Rate (per 1,000 live births)		60 65	SVRS, 2007 BDHS, 2007
35	Percent of population using safe drinking water (Tap & Tubewell)	National	98.8	SVRS, 2007, BBS
36	Percent of population using (water seal) latrines		54.24	SVRS, 2007, BBS
37	Prevalence of night blindness among pre school children (HKI 2005)		0.04	Director, IPHN, Mohakhali.
38	% of births attended by skilled personnel		17.8	BDHS , 2007
	% of women received at least one antenatal care		51.7	BDHS, 2007
	% of mother received PNC from a trained provider within 42 days of delivery		21.3	BDHS, 2007
39	Malaria slide positivity rate (positive per hundred slide examined)		8.47	DGHS, 2008
40	Malaria incidence rate per 1000 population		0.63	DGHS, 2008
41	TB incidence rate per 100000 population		99	DGHS, 2008
42	% of smear-positive pulmonary TB cases detected put under DOTS		72%	TB Annual Report 2008 NTCP, DGHS.
43	% of smear -positive pulmonary TB cases detected cured under DOTS		92%	
44	% urban population with access to improved sanitation		77.1	SVRS, 2007, BBS
45	% of <5 children with diarrhea treated with ORT (ORS or home made solution)		81.2	BDHS, 2007
46	% of <5 children with symptoms of ARI seeking care from trained provider		28.13	BDHS, 2007
E. EDUCATION and ECONOMY				
47	Per Capita GDP (in U.S.$) 2007-08		554	Statistical Pocket Book Bangladesh 2008, BBS
48	Per Capita Income (in US$) 2007-08		599	
49	Per capita public health expenditure on H&FP (Taka)		281	BBS, 2006
50	Adult Literacy Rate (Pop.15+), (Both sexes)		56.3	SVRS, 2007, BBS
F. HEALTH SERVICES PROVISION				
51	No. of Hospitals in Health Sector		589	DGHS (MIS) 2009
	No. of Non. Govt. Hospitals (Regtd. by DGHS)		2271	March 2009 DGHS

52	No. of Beds in Health Sector	Functioning	38,171	MIS, 2008 (Dec)
53	No. of Beds in Private Sector (Regtd by DGHS)		36,244	DGHS (Hospital) 2009
54	No. of Registered Physicians (April 2009)		49,994	BMDC, 2009
55	No. of Registered Dental Surgeon (April -2009)		3481	BMDC, 2009
56	No. of Government Medical colleges		18	DGHS (MIS) 2008
57	No. of Private Medical colleges		41	DGHS, (ME) 2008
	No. of Private Dental colleges		11	DGHS, 2008
58	No. of Private Institute of Health Technology (IHT)		39	DGHS, (MIS) 2008 (Dec)
59	No. of Personnel under DGHS (Sane-110144)	Existing	79,896	DGHS, 2009 (June)
60	No. of Doctors under Health Services (Sane-19243	Existing	12382	DGHS, (MIS) 2009 (June)
61	No. of Registered Nurses (as on April-2009)		23,729	DGHS, (MIS) 2009 (June)
62	No. of Nurses in Public sector	Existing	14,377	DNS, 2009
63	No. Registered Mid - wives		22,253	BNC, 2009 (April)
64	No. of Trained Skilled Birth Attendance		5000	UNFPA, 2009 (June)
G. HEALTH SERVICES				
65	EPI & Vit-A Coverage	DPT3	95%	Bangladesh EPI Coverage Evaluation Survey 2007 Expanded Programme of Immunization, DGHS & BDHS, 2007
		BCG	90% 96.8%	
		Measles	96%	
		TT2	94%	
		HAP-B3	95%	
		VIT-A	93% 88.3%	
		OPV3	92% 90.8%	
		Fully Immunized	75%	
66	Population per Physicians		2860	DGHS 2008
67	Population per Bed (Beds of Health Sector + Regtd.Privet Hospital)		1860	DGHS 2009
68	Physician to Nurse Ratio		2:1	DGHS 2008
69	Population per Nurse		5720	DGHS 2008
H. PLACE OF DELIVERY		**Age of birth**		
70	Public health facility	<20	7.6	BDHS 2007
		20-34	7.0	
		35+	4.0	
71	Private & NGO health facility	<20	6.5	BDHS 2007
		20-34	8.2	
		35+	6.9	
	In health facility		15.0	BDHS 2007

72	Home	<20	85.7	BDHS 2007
		20-34	84.3	
		35-49	88.3	
73	Other	<20	0.2	BDHS 2007
		20-34	0.3	
		35+	0.1	
I. PERCENT OF DELIVERY ATTENDED BY				
74	Qualified Doctor	<20	12.0	BDHS 2007
		20-34	13.2	
		35+	10.2	
75	Nurse/midwife/paramedic	<20	5.7	BDHS 2007
		20-34	5.2	
		35+	2.1	
76	Traditional birth attendant	<20	11.6	BDHS 2007
		20-34	10.4	
		35+	10.2	
77	Untrained birth attendant	<20	63.4	BDHS 2007
		20-34	62.1	
		35+	63.6	
78	Relatives and friends	<20	5.7	BDHS 2007
		20-34	6.2	

Sources

1. Directorate General of Health Services (DGHS)-MIS, Hospital,Medical Education.
2. Bangladesh Bureau of Statistics (BBS), Statistics Division, Ministry of Planning.
3. Bangladesh Demographic & Health Survey-2007 (BDHS), NIPORT.
4. Human Resources Development (HRD) Unit, 2008, Ministry of Health & Family Welfare, Bangladesh Secretariat, Dhaka.
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6. Bangladesh Medical and Dental Council (BMDC).
7. Bangladesh Nursing Council (BNC).

Health Related Millennium Development Goals (MDGs)

The MDGs mark a shift in the perception of development, away from GDP and per capita growth rates. Progress on MDG yardsticks can be assessed by ascertaining the extent to which the MDGs are mainstreamed in the development agenda; progress towards the MDGs themselves; progress relative to comparator countries; and the likelihood of the MDGs being reached. As MDGs encompass all aspects of social development, overall success will be due to the mainstreaming of the MDGs (into PRSPs, for example); public finance reforms; economic structure diversification; increasing rates of growth (currently at 6%); growing exports and remittances; and social protection programmes, including health and nutrition. The table in this page shows that Bangladesh is making success in achieving the health related MDGs. In child survival, Bangladesh is at the 8th position out of the 16 countries of the developing countries in the world.

Health related Millennium Development Goals (MDGs) and Bangladesh

Goal	Target	Indicator	Bangladesh Situation		
			Bench mark	Current	Target 2015
1. Reduced Child Mortality	Reduce by 2/3rds, between 1990 and 2015, the under five mortality rate	Under five mortality rate (per 1000 live births)	144.0 (1990)	65.0 (2007)	48.0
		Infant mortality rate (per 1000 live births)	94.0 (1990)	52.0 (2007)	31.3
		Proportion of infants immunized against measles	65.0 (1990)	83.1 (2007)	
2. Improved maternal health	Reduced by 3/4th, between 1990 and 2015, the maternal mortality ratio	Maternal mortality ratio (per 1000 live births)	4.8 (1990)	3.2 (2001)	1.2
		Proportion of births attended by skilled health personnel	7.0% (1990)	17.8% (2007)	50.0% (by 2010)
3. Combat HIV/ AIDS malaria and other diseases	Half halted by 2015 and begun to reverse the spread of HIV/AIDS	Prevalence of HIV (% among high risk groups)	-	0.57-0.75% (2007)	Halt
	Half halted by 2015 and begun to reverse the incidence of malaria and other major diseases	New TB cases detected%	-	72.2% (2007)	>70.0%
		TB cure rate under DOTS	29.2 (1993)	92.0% (2007)	>85.0%

Essential Service Delivery

Reproductive health

In spite of the fact that maternal mortality has declined from nearly 574 per 100,000 live births in the 1990 to between 320 and 400 in 2001, the maternal mortality ratio (MMR) in Bangladesh remains one of the highest in the world.

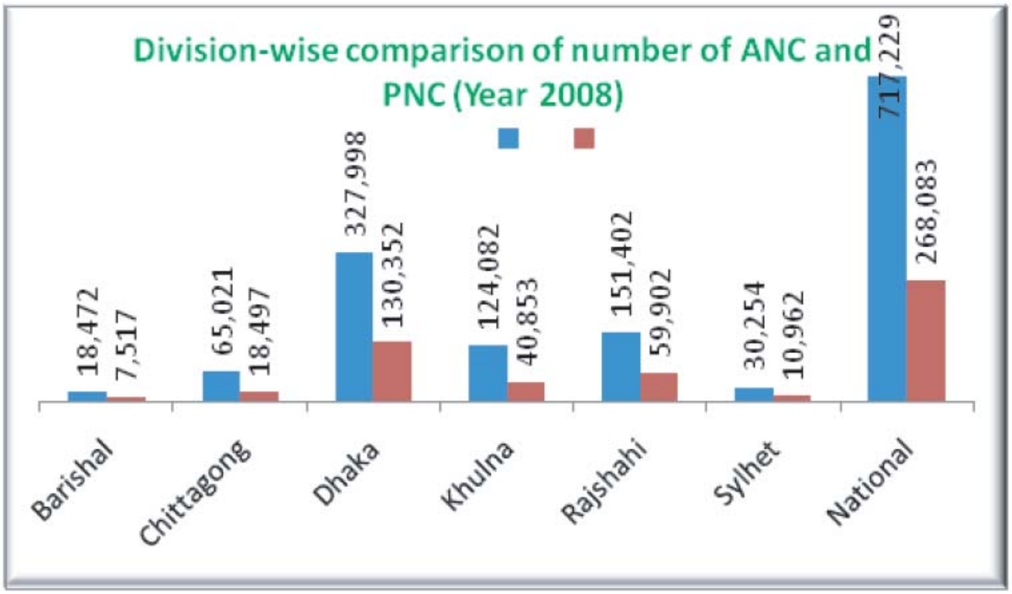
The chief causes of maternal deaths are haemorrhage, unsafe abortion, and the 'three delays dynamics'. The first delay, arising mainly from poverty, is in seeking professional care; the second delay is logistical as most of the health centres and private clinics are located in towns whereas 70% of the population are rural based; the third delay arises from the lack of adequate human recourses and trained personnel at the service centres.
It is estimated that 14% of maternal deaths are caused by violence against women, while 12,000 to 15,000 women die every year from maternal health complications. Some 45% of all mothers are malnourished.
The population of Bangladesh is relatively young, with a third falling within the age group of 10-24 years. Nearly half the adolescent girls (15-19 years) are married, 57% of them become mothers before the age of 19, and half these adolescent mothers are acutely malnourished. Thus MMR among adolescent mothers is 30-50% higher than the national rate.

Division-wise EmOC Services (Summary Report), January to December 2008

Division	No. of mothers			No. Referred		No. Received PNC	No. of deaths	
	Received ANC	Admitted	Had Complications	In	Out		Mothers	Newborns
Barisal	18,472	25,373	10,263	776	1,713	7,517	160	431
Chittagong	65,021	88,818	34,651	3,049	3,972	18,497	278	488
Dhaka	327,998	184,166	58,737	5,166	8,612	130,352	534	511
Rajshahi	151,402	174,042	48,430	2,670	9,250	59,902	382	578
Sylhet	30,254	35,870	17,197	985	2,167	10,962	190	271
Khulna	124,082	86,978	26,468	324	3,216	40,853	155	122
National	717,229	595,247	195,746	13,078	28,930	268,083	1,699	2,401

Division-wise EmOC Services (Summary Report), January to December 2008

Division	Type of delivery (No. of cases)					No. of births		Other Operations (No.)
	Normal	Forceps/ Vacuum	Vaginal Breech / Face	Caesarean	Total	Live Births	Still Births	
Barisal	10,725	39	28	7,894	18,686	18,404	769	247
Chittagong	38,897	867	941	25,955	66,660	64,610	3,542	2,191
Dhaka	66,830	1,119	759	69,480	138,188	134,007	5,178	3,768
Rajshahi	81,979	1,570	1,183	40,971	125,703	121,367	5,297	4,874
Sylhet	14,633	537	436	8,677	24,283	22,401	2,155	2,109
Khulna	**35,163**	**380**	**401**	**24,102**	**60,046**	**58,471**	**1,870**	**1,400**
National	248,227	4,512	3,748	177,079	433,566	419,260	18,811	14,589



Demand side financing: Maternal Health Voucher Scheme

To address the situation of high maternal mortality rate and enable pregnant women receive the quality antenatal, intra-natal and postnatal cares, demand side financing through Maternal Health Voucher Scheme was introduced in Bangladesh in functional form from 2006.

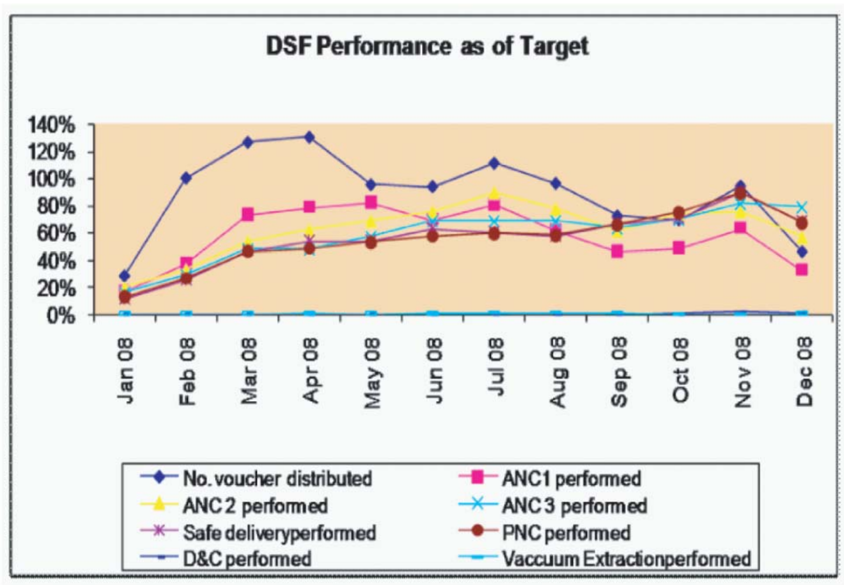
On consideration of literacy rate, population density, poverty levels, and presence of skilled birth attendant (SBA) program, the scheme has been gradually extended to cover 33 upazilas. Programs in 3 of the upazilas are financed by UNFPA and remaining 30 from pool fund. Technical assistance is given by WHO. The objective of the program is to create awareness of the poor pregnant mothers about obstetric care and increase their demand for maternal health services contributing to reducing the MMR. The service is provided from the respective upazila health complexes.

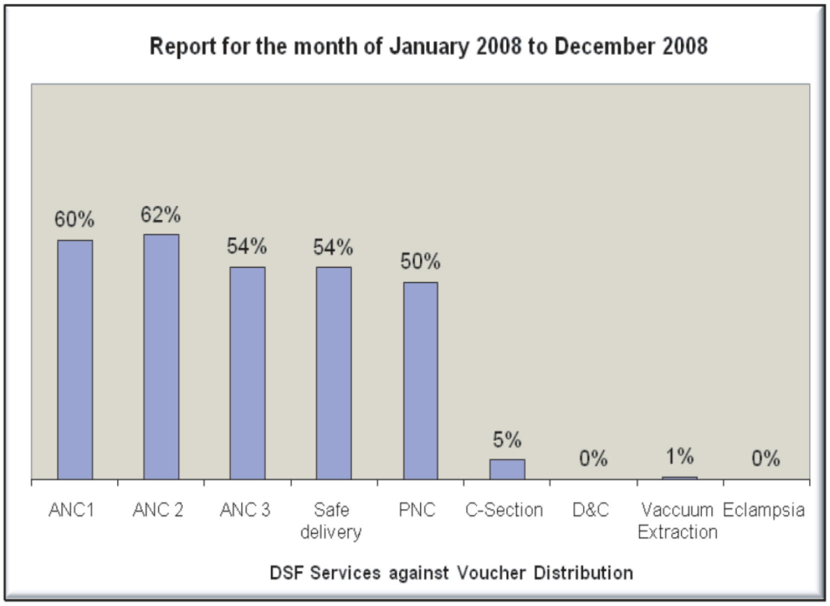
List of 33 upazilla covered by DSF

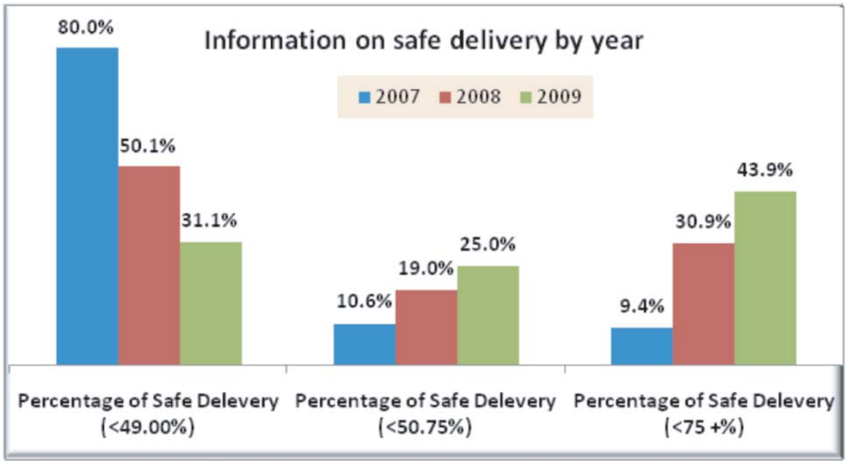
District	Upazila	District	Upazilla	Disrtict	Upazila
Barisal	Banaripara	Tangail	Sakhipur	Cox'sbazar	Ukhia
Patuakhali	Kalapara	Jessore	Chaugacha	Cox'sbazar	Teknaf
Chittagong	Mirsharai	Khulna	Paikgacha	Shariatpur	Naria
Comilla	Daudkandi	Kustia	Daulatpur	Shunamgonj	Sulla
Noakhali	Chatkhil	Bogra	Khetlal	Kurigram	Ulipur
Chittagong	Ramu	Dinajpur	Khanshama	Chandpur	Matlab-North
Dhaka	Harirampur	Pabna	Shahjadpur	Panchagar	Debigangnj
Faridpur	Bhanga	Rajshahi	Shibganj	Naogaon	Atrai
Jamalpur	Sarishabari	Rangpur	Gobindaganj	Laximpur	Raipur
Mymensingh	Tarail	Sylhet	Baniachong	Pirojpur	Nazipur
Narayanganj	Raipura	Madaripur	Shibchar	Meherpur	Gangni

Assistance in monetary form for DSF services

Type of Service	Taka
Registration	10.00
Lab tests for 3 ANCs (2 Blood and 2 Urine tests)	140.00
Consultati on fees for 3 ANCs and 1 PNC	200.00
Safe Delivery	300.00
Medicines	100.00
Total	750
Type of Complications	Taka
Forceps/ Manual Removal of Placenta/ DE&C/ Vacuum extraction	1000
Management of Eclampsia	1000
C-Section	6000







The community skilled birth attendant program in Bangladesh

In 2003, the Ministry of Health and Family Welfare initiated a 6-month basic community skilled birth attendants'(C-SBA) training program, with the goal to reduce the ratio of maternal and infant mortality and morbidity in Bangladesh. After a positive evaluation in 2004, the government decided to scale up the program nation-wide and the training was extended to a total of 18 months (6 months basic + 9 months field work under supervision + 3 months additional course).

Under the overall supervision and direction of the Directorate General of Health Services (DGHS) and Directorate General of Family Planning (DGFP) and with financial and technical support from UNFPA and WHO, the program is being implemented by the Obstetric and Gynecological Society of Bangladesh (OGSB). The OGSB is entrusted with the overall responsibility for the coordination of the training program.

The purpose of the training is to equip Family Welfare Assistants (FWA) and Female Health Assistants (FeHA) with the basic skills to function as C-SBAs in an effort to contribute to the achievement of MDGs 4 and 5 on child and maternal health and to further guarantee safe motherhood for pregnant women in Bangladesh.

To improve their performance and to monitor their activities in a supportive way, a supervisory system was established. The Ministry of Health and Family Welfare (MoHFW) has identified Family Welfare Visitor (FWVs) to provide supportive supervision to the C-SBAs and monthly submit their performance reports. A total of 140 FWVs have already been trained on this Pro-Active Supervision training.

Up to June 2009, nearly 5000 C-SBAs have been trained and are serving in their own area of living. In order to create a better and safer service for the mothers in the communities, there is an urge to update their midwifery competencies through providing a refresher training, to keep up the quality of their services. Under the leadership of DGHS, a 2-week refresher training is in progress with the aim to improve and strengthened the C-SBAs with further technical skills and knowledge based on the 6 months basic course.

Joint Government-UN Maternal and Newborn Health Program (MNH)

MNH program is being implemented in 4 districts (Thakurgaon, Jamalpur, Narail and Moulavi bazar) covering all areas of the respective districts. The program has been based on local level planning (LLP) with decentralization.

The activities are being implemented through Director, PHC under Operational Plan of Essential Service Delivery (ESD) with special emphasis on need based demand and priority setup. Civil surgeons are the focal persons in the district level and performance is monitored and evaluated by national and district MNH committees. If the program is successfullyimplemented, then the MNH program will be expanded in another 16 districts. The program is technically supported by UNFPA, UNICEF and WHO and is funded by EC and DFID.

National Fistula Program

In Bangladesh, fistula and other maternal morbidities affect 400,000 women (BIRPERHT study 1996). The first ever need assessment done in the country estimated that approximately 71,000 women are currently living with fistula in Bangladesh. This gives a figure of 1.69 women living with fistula per 1,000 ever Twelve centers in Bangladesh, 9 in Government Medical College Hospitals and 3 in Private Medical College Hospitals treat obstetric fistula since October 2003. From the same time, a National Fistula Center has been working in Dhaka Medical College Hospital (DMCH) with the financial assistance of UNFPA. A full-fledged National Fistula Centre is being established at DMCH under the IDB fund and with the technical assistance of UNFPA. This center would probably be the only Fistula Hospital in South-East Asia.

IMCI

Integrated Management of Childhood Illness (IMCI) is a strategy as well as a program developed in mid-1990s by WHO, UNICEF and other partners to unify existing vertical child health programs (e.g., Control of Diarrheal Diseases and Acute Respiratory Infections). IMCI addresses morbidities which are responsible for almost 75% of under-5 deaths. The Ministry of Health and FamilyWelfare (MOHFW) of Bangladesh introduced IMCI in 2002. It is estimated that neonatal death rate makes the highest married women (GOB, UNFPA & EngenderHealth Study 2003). Obstetric fistula is a devastating pregnancy related disability. Prolonged obstructed labour and obstetric fistula are vivid examples of poor maternal and reproductive health care and the unfortunate ending of motherhood leading to unacceptably high maternal mortality and morbidity.

contributions to the under-5 deaths of Bangladesh, which is 71% of infant mortality rate and 57% of under-5 mortality rate. IMCI targets to reduce both neonatal as well as all under-5 deaths. Therefore, the Health, Nutrition and Population Sector Program (HNPSP 2003-11) of MOHFW, Bangladesh also identified IMCI program as one of the major program areas of

Essential Service Delivery. To simplify case management in the primary health care settings by the health workers and paramedics, IMCI program in Bangladesh classified childhood diseases into 9 broad categories, viz. Very Severe Disease, Pneumonia, Cough and cold-not pneumonia, Diarrhea, Dysentery, Malaria, Non-malarial fever, Measles and others. IMCI is provided through facility-based treatment as well as through home care. UNICEF and WHO provide technical and financial assistance while various other development partners and NGOs help in implementation.

List of Upazilas under coverage of IMCI Facilities in Bangladesh by Division and District

Rajshahi Division						
Kurigram (9)	Thakurgaon (5)	Rangpur (8)	Nilphamari (6)	Gaibandha (7)	Bogra (11)	Dinajpur (13)
Sadar	Sadar	Sadar	Jaldhaka	Sadar	Kahaloo	Sadar
Rajarhat	Pirganj	Mithapukur	Sadar	Gobindagonj	Gabtali	Chirirbandar
Nageshwari	Baliadangi	Pirgonj	Kishoregonj	Shadullapur	Sadar	Khansama
Rowmari	Ranishankail	Pirgacha	Saidpur	Sundargonj	Adamdighi	Bochaganj
Char Rajibpur	Haripur	Kaunia	Domar	Fulchari	Sherpur	Kaharool
Phulbari	Sirajgonj (9)	Taragonj	Dimla	Polashabri	Sonatola	Birganj
Bhurungamari	Shahajadpur	Badargonj	C'nawabganj (5)	Shaghata	Shariakandi	Biroil
Chilmari Ulipur	Sadar	Gongachara	Sadar	Naogaon (11)	Dhunat	Parbotipur
	Kamarkhand	Joypurhat (5)	Gomostapur	Sapahar	Dhupchachia	Fulbaria
Natore(6)	Ullapara	Sadar	Nachole	Dhamairhat	Nandigram	
Sadar	Raigonj	Akkelpur	Bholahat	Patnitala	Shibgonj	
Gurudaspur	Belkuchi	Kalai	Shibganj	Porsha		
Singra	Kazipur	Khetlal	Panchagarh (5)	Badalgachi		
Baraigram	Tarash	Panchbibi	Sadar	Mohadebpur		
Bagatipara	Chowhali		Tetulia	Niamatpur		
Lalpur			Boda	Manda		
Lalmonirhat (5)			Atwari	Sadar		
Hatibanda			Debigonj	Raninagar		
				Atrai		

Barisal Division			Sylhet Division			
Barisal (10)	Bhola (7)	Patuakhali (6)	Sunamgonj (10)	Habigonj (8)	Sylhet (11)	Moulvibazar (6)
Bakergonj	Sadar	Sadar	Sadar	Madhabpur	Balagonj	Rajnagar
Babugonj	Borhanuddin	Golachipa	Chatak	Sadar	Bianibazar	Sadar
Muladi	Daulatkhan	Bauphol	Dharmapasa	Baniachong	Golapgonj	Sreemongol
Sadar	Tajumudd in		Doarabazar	Chunarughat	Zokigonj	Barolekha
Banaripara	Lalmohan		Salla	Azmirigonj	Bishwnath	Kulaura
Mehendiganj	Monpura		Tahirpur	Lakhai	Sadar	Kamalgonj
Wazirpur	Charfashon		Jamalgonj	Bahubol	Companigonj	
Gouranadi			Bishawmbharpur	Nabigonj	Goainghat	
Agailjhara			Dirai		Jointapur	
Hizla			Jagannathpur		Kanaighat	

.

List of Upazilas under coverage of IMCI Facilities in Bangladesh by Division and District (contd...)

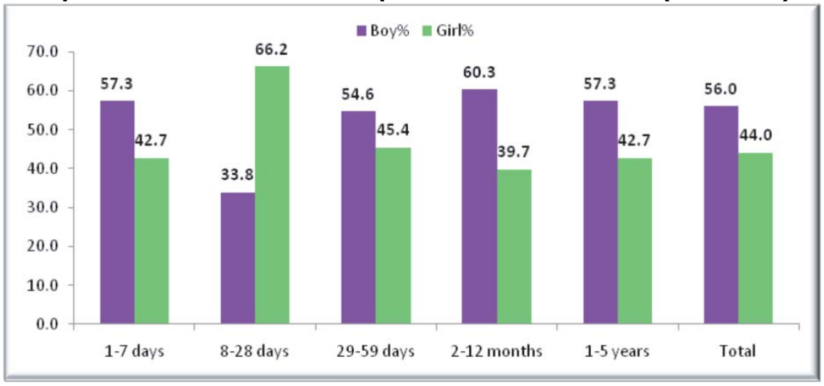
Chittagong Division				Khulna Division	
B. Baria (7)	Chittagong (14)	Bandarban (7)	Cox's Bazar (7)	Khulna (9)	Jessor (8)
Sarail	Hathazari	Sadar	Sadar	Fultala	Sharsha
Nasirnagar	Sandhip	Lama	Ukhia	Terokhada	Sadar
Sadar	Fatikchari	Roangachari	Teknaf	Daulatpur	Chowgacha
Nabinagar	Raujan	Thanchi	Chokoria	Rupsha	Bagerpara
Kashba	Rangunia	Alikadam	Kutubdia	Dumuria	Jhikorgacha
Akhaura	Mirehwarai	Ruma	Moheshkhali	Batiaghata	Monirampur
Bancharampur	Patia	Comilla (12)	Ramu	Paikgacha	Avoynagar
Laxmipur (4)	Lohagara	Daudkandi	Chandpur (7)	Dacope	Keshobpur
Sadar	Anowara		Maltab	Koyra	
Ramgati	Banshkhali				
Ramganj	Chandanaish				
Raypur	Shatkania				

Dhaka Division						
Sherpur (5)	Mymesingh (12)	Netrokona (10)	Kishoregonj (13)	Narshingdhi (6)	Shariatpur (6)	Dhaka (5)
Sadar	Gaforgaon	Sadar	Bhoirob	Shibpur	Goshairhat	Dhamrai
Nalitabari	Sadar	Mohangonj	Sadar	Manohardi	Janjira	Savar
Nakla	Trishal	Durgapur	Pakundia	Sadar	Sadar	Keranigonj
Jhenaigati	Muktagacha	Kalmkanda	Nikli	Polash	Naria	Dohar
Sreebordi	Phulbaria	Kendua	Katiadi	Raipura	Bhederganj	Nawabgonj
Jamalpur (7)	Gouripur	Madan	Kuliarchar	Belabo	Damudda	Gazipur(5)
Sharishabari	Bhaluka	Khaliajuri	Karimgonj	Gopalgonj (5)	Madaripur (4)	Sreepur
Sadar	Haluaghat	Atpara	Tarail	Sadar	Sadar	Sadar
Melandah	Dhubaura	Barhatta	Mitamoin	Kashiani	Rajor	Kaligonj
Islampur	Phulpur	Purba Dhala	Hossainpur	Moksedpur	Kalkini	Kapasia
Dewangonj	Ishwargonj		Itna	Kotalipara	Shibchar	Kaliakoir
Madargonj	Nandail		Bajitpur	Tungipara		
Bakshigonj			Austogram			

In-patient children in the IMCI Upazila health facilities by age group

Age group	Boy		Girl		Both sex	
	No.	%within boys	No.	%within girls	No.	%within both sex
1-7 days	342	3.5	255	3.4	597	3.5
8-28 days	538	5.6	1,054	13.9	1,592	9.2
29 -59 days	867	9.0	721	9.5	1,588	9.2
2-12 months	3,723	38.5	2,454	32.3	6,177	35.8
1-5 years	4,190	43.4	3,119	41.0	7,309	42.3
Total	9,660	100.0	7,603	100.0	17,263	100.0

In-patient children in IMCI upazila health facilities (n=17,263)



Sex distribution of morbidities in each age cluster of children who attended out-patient and emergency departments of IMCI facilities

Diseases	0-7d age cluster					
	GIRL	%	BOY	%	TOTAL	%
Very severe disease	139	9.45	152	11.41	291	10.4
Pneumonia	131	8.91	174	13.06	305	10.9
Cough & cold	462	31.41	405	30.41	867	30.9
Diarrhea	92	6.25	79	5.93	171	6.1
Dysentery	88	5.98	70	5.26	158	5.6
Malaria fever	4	0.27	4	0.30	8	0.3
Non-malaria fever	84	5.71	91	6.83	175	6.2
Measles	8	0.54	7	0.53	15	0.5
Eare problem	58	3.94	41	3.08	99	3.5
Malnutrition	71	4.83	59	4.43	130	4.6
Drowning	4	0.27	4	0.30	8	0.3
Injury	38	2.58	29	2.18	67	2.4
Others	292	19.85	217	16.29	509	18.2
Total	1471	100.00	1332	100.00	2803	100.0

Disease	8-28d age cluster					
	GIRL	%	BOY	%	TOTAL	%
Very severe disease	272	6.9	283	7.9	555	7.4
Pneumonia	300	7.6	315	8.8	615	8.2
Cough & cold	1279	32.4	1205	33.5	2484	32.9
Diarrhea	346	8.8	293	8.1	639	8.5
Dysentery	163	4.1	152	4.2	315	4.2
Malaria fever	5	0.1	8	0.2	13	0.2
Non-malaria fever	346	8.8	266	7.4	612	8.1
Measles	10	0.3	12	0.3	22	0.3
Eare problem	130	3.3	118	3.3	248	3.3
Malnutrition	119	3.0	109	3.0	228	3.0
Drowning	4	0.1	4	0.1	8	0.1
Injury	148	3.7	142	3.9	290	3.8
Others	827	20.9	690	19.2	1517	20.1
Total	**3949**	**100.0**	**3597**	**100.0**	**7546**	**100.0**

Disease	29-59d age cluster					
	GIRL	%	BOY	%	TOTAL	%
Very severe disease	324	3.0	477	4.7	801	3.8
Pneumonia	816	7.5	784	7.7	1600	7.6
Cough & cold	3934	36.4	3985	39.3	7919	37.8
Diarrhea	968	9.0	870	8.6	1838	8.8

Sex distribution of morbidities in each age cluster of children who attended outpatient and emergency departments of IMCI facilities (contd...)

Disease	8 28d age cluster					
Dysentery	497	4.6	413	4.1	910	4.3
Malaria fever	11	0.1	10	0.1	21	0.1
Non -malaria fever	975	9.0	869	8.6	1844	8.8
Measles	17	0.2	34	0.3	51	0.2
Eare problem	377	3.5	284	2.8	661	3.2
Malnutrition	512	4.7	449	4.4	961	4.6
Drowning	7	0.1	4	0.0	11	0.1
Injury	233	2.2	212	2.1	445	2.1
Others	2142	19.8	1744	17.2	3886	18.6
Total	10813	100.0	10135	100.0	20948	100.0

Disease	2-12m age cluster					
	GIRL	%	BOY	%	TOTAL	%
Very severe disease	593	1.0	830	1.3	1423	1.2
Pneumonia	6066	9.9	6926	11.2	12992	10.5
Cough & cold	18468	30.0	19244	31.0	37712	30.5
Diarrhea	7248	11.8	7258	11.7	14506	11.7
Dysentery	3489	5.7	3500	5.6	6989	5.7
Malaria fever	58	0.1	72	0.1	130	0.1
Non-malaria fever	6488	10.5	7113	11.5	13601	11.0
Measles	41	0.1	37	0.1	78	0.1
Eare problem	2122	3.4	1984	3.2	4106	3.3
Malnutrition	2643	4.3	2317	3.7	4960	4.0
Drowning	40	0.1	43	0.1	83	0.1
Injury	928	1.5	904	1.5	1832	1.5
Others	13392	21.7	11781	19.0	25173	20.4
Total	61576	100.0	62009	100.0	123585	100.0

Disease	1-5y age cluster					
	GIRL	%	BOY	%	TOTAL	%
Very severe disease	677	0.6	781	0.7	1458	0.6
Pneumonia	8480	7.5	9480	8.0	17960	7.8
Cough & cold	31507	27.8	32746	27.7	64253	27.8
Diarrhea	12370	10.9	13472	11.4	25842	11.2
Dysentery	7658	6.8	7650	6.5	15308	6.6
Malaria fever	149	0.1	194	0.2	343	0.1
Non-malaria fever	12703	11.2	14217	12.0	26920	11.6
Measles	71	0.1	56	0.0	127	0.1
Eare problem	4087	3.6	4502	3.8	8589	3.7
Malnutrition	5564	4.9	5159	4.4	10723	4.6
Drowning	165	0.1	193	0.2	358	0.2
Injury	2525	2.2	3030	2.6	5555	2.4
Others	27204	24.0	26553	22.5	53757	23.3
Total	113160	100.0	118033	100.0	231193	100.0

Sex distribution of causes of death in each age cluster of children who attended out-patient and emergency departments of IMCI facilities

Disease	0-7d age cluster				
	GIRL	%	BOY	%	TOTAL
Very severe disease	27	1227	25	11.36	52
Pneumonia	26	11.82	16	7.27	42
Cough & cold	2	0.91	15	6.82	17
Diarrhea	18	8.18	4	1.82	22
Dysentery	8	3.64	5	2.27	13
Malaria fever	1	0.45	8	3.64	9
Nonmalaria fever	1	0.45	5	2.27	6
Measles	3	1.36	3	1.36	6
Ear problem	0	0.00	0	0.00	0
Malnutrition	1	0.45	2	0.91	3
Drowning	0	0.00	4	1.82	4
Injury	14	6.36	11	5.00	25
Others	8	3.64	13	5.91	21
Total	109	49.55	111	50.45	220

Disease	8-28d age cluster				
	GIRL	%	BOY	%	TOTAL
Very severe disease	7	8.24	15	17.65	22
Pneumonia	2	2.35	8	9.41	10
Cough & cold	2	2.35	5	5.88	7
Diarrhea	7	8.24	3	3.53	10
Dysentery	0	0.00	0	0.00	0
Malaria fever	0	0.00	4	4.71	4
Nonmalaria fever	0	0.00	4	4.71	4
Measles	3	3.53	3	3.53	6
Ear problem	0	0.00	0	0.00	0
Malnutrition	0	0.00	0	0.00	0
Drowning	4	4.71	4	4.71	8
Injury	4	4.71	0	0.00	4
Others	5	5.88	5	5.88	10
Total	34	40.00	51	60.00	85

Disease	25-59d age cluster				
	GIRL	%	BOY	%	TOTAL
Very severe disease	5	8.77	8	14.04	13
Pneumonia	2	3.51	5	8.77	7
Cough & cold	4	7.02	0	0.00	4
Diarrhea	0	0.00	3	5.26	3
Dysentery	0	0.00	0	0.00	0
Malaria fever	0	0.00	4	7.02	4
Nonmalaria fever	0	0.00	0	0.00	0
Measles	0	0.00	3	5.26	3
Ear problem	0	0.00	0	0.00	0
Malnutrition	0	0.00	0	0.00	0
Drowning	4	7.02	4	7.02	8
Injury	4	7.02	5	8.77	9
Others	0	0.00	6	10.53	6
Total	19	33.33	38	66.67	57

Sex distribution of causes of death in each age cluster of children who attended out-patient and emergency departments of IMCI facilities (contd...)

Disease	2-12m age cluster				
	GIRL	%	BOY	%	TOTAL
Very severe disease	9	7.83	28	24.35	37
Pneumonia	14	12.17	10	8.70	24
Cough & cold	6	5.22	1	0.87	7
Diarrhea	0	0.00	0	0.00	0
Dysentery	0	0.00	0	0.00	0
Malaria fever	0	0.00	4	3.48	4
Nonmalaria fever	0	0.00	4	3.48	4
Measles	3	2.61	3	2.61	6
Ear problem	2	1.74	0	0.00	2
Malnutrition	0	0.00	0	0.00	0
Drowning	4	3.48	4	3.48	8
Injury	4	3.48	4	3.48	8
Others	9	7.83	6	5.22	15
Total	51	44.35	64	55.65	115

Disease	1-5y age cluster				
	GIRL	%	BOY	%	TOTAL
Very severe disease	10	14.71	6	8.82	16
Pneumonia	2	2.94	2	2.94	4
Cough & cold	1	1.47	1	1.47	2
Diarrhea	2	2.94	0	0.00	2
Dysentery	0	0.00	0	0.00	0
Malaria fever	0	0.00	0	0.00	0
Nonmalaria fever	1	1.47	0	0.00	1
Measles	0	0.00	3	4.41	3
Ear problem	2	2.94	0	0.00	2
Malnutrition	0	0.00	0	0.00	0
Drowning	9	13.24	12	17.65	21
Injury	0	0.00	0	0.00	0
Others	6	8.82	11	16.18	17
Total	33	48.53	35	51.47	68

Distribution of causes of death within each sex and age cluster of children who attended out-patient and emergency departments of IMCI facilities

Disease	0-7d age cluster					
	GIRL	%	BOY	%	TOTAL	%
Very severe disease	27	24.77	25	22.52	52	23.64
Pneumonia	26	23.85	16	14.41	42	19.09
Cough & cold	2	1.83	15	13.51	17	7.73
Diarrhea	18	16.51	4	3.60	22	10.00
Dysentery	8	7.34	5	4.50	13	5.91
Malaria fever	1	0.92	8	7.21	9	4.09
Non-malaria fever	1	0.92	5	4.50	6	2.73
Measles	3	2.75	3	2.70	6	2.73
Ear problem	0	0.00	0	0.00	0	0.00
Malnutrition	1	0.92	2	1.80	3	1.36
Drowning	0	0.0	4	3.60	4	1.82
Injury	14	12.84	11	4.91	25	11.36
Others	8	7.34	13	11.71	21	9.55
Total	109	100.0	111	100.0	220	100.0

Distribution of causes of death within each sex and age cluster of children who attended out-patient and emergency departments of IMCI facilities (contd...)

Disease	8-28d age cluster					
	GIRL	%	BOY	%	TOTAL	%
Very severe disease	7	2059	15	29.41	22	25.88
Pneumonia	2	5.88	8	15.69	10	11.76
Cough & cold	2	5.88	5	9.80	7	8.24
Diarrhea	7	20.59	3	5.88	10	11.76
Dysentery	0	0.00	0	0.00	0	0.00
Malaria fever	0	0.00	4	7.84	4	4.71
Non-malaria fever	0	0.00	4	7.84	4	4.71
Measles	3	8.82	3	5.88	6	7.06
Earproblem	0	0.00	0	0.00	0	0.00
Malnutrition	0	0.00	0	0.00	0	0.00
Drowning	4	11.76	4	7.84	8	9.41
Injury	4	11.76	0	0.00	4	4.71
Others	5	14.71	5	9.80	10	11.76
Total	34	100.0	51	100.0	85	100.0

Disease	29-59d age cluster					
	GIRL	%	BOY	%	TOTAL	%
Very severe disease	5	26.32	8	21.05	13	22.81
Pneumonia	2	10.53	5	13.16	7	12.28
Cough & cold	4	21.05	0	0.00	4	7.02
Diarrhea	0	0.00	3	7.89	3	5.26
Dysentery	0	0.00	0	0.00	0	0.00
Malaria fever	0	0.00	4	10.53	4	7.02
Non-malaria fever	0	0.00	0	0.00	0	0.00
Measles	0	0.00	3	7.89	3	5.26
Ear problem	0	0.00	0	0.00	0	0.00
Malnutrition	0	0.00	0	0.00	0	0.00
Drowning	4	21.05	4	10.53	8	14.04
Injury	4	21.05	5	13.16	9	15.79
Others	0	0.00	6	15.79	6	10.53
Total	19	100.0	38	100.0	57	100.0

Disease	2-12m age cluster					
	GIRL	%	BOY	%	TOTAL	%
Very severe disease	9	17.65	28	43.75	37	32.17
Pneumonia	14	27.45	10	15.63	24	20.87
Cough & cold	6	11.76	1	1.56	7	6.09
Diarrhea	0	0.00	0	0.00	0	0.00
Dysentery	0	0.00	0	0.00	0	0.00
Malaria fever	0	0.00	4	6.25	4	3.48
Non-malaria fever	0	0.00	4	6.25	4	3.48
Measles	3	5.88	3	4.69	6	5.22
Ear problem	2	3.92	0	0.00	2	1.74
Malnutrition	0	0.00	0	0.00	0	0.00
Drowning	4	7.84	4	6.25	8	6.96
Injury	4	7.84	4	6.25	8	6.96
Others	9	17.65	6	9.38	15	13.04
Total	51	100.0	64	100.0	115	100.0

Distribution of causes of death within each sex and age cluster of children who attended out-patient and emergency departments of IMCI facilities (contd...)

Disease	1-5y age cluster					
	GIRL	%	BOY	%	TOTAL	%
Very severe disease	10	30.30	6	17.14	16	23.53
Pneumonia	2	6.06	2	5.71	4	5.88
Cough & cold	1	3.03	1	2.86	2	2.94
Diarrhea	2	6.06	0	0.00	2	2.94
Dysentery	0	0.00	0	0.00	0	0.00
Malaria fever	0	0.00	0	0.00	0	0.00
Non-malaria fever	1	3.03	0	0.00	1	1.47
Measles	0	0.00	3	8.57	3	4.41
Ear problem	2	6.06	0	0.00	2	2.94
Malnutrition	0	0.00	0	0.00	0	0.00
Drowning	9	27.27	12	34.29	21	30.88
Injury	0	0.00	0	0.00	0	0.00
Others	6	18.18	11	31.43	17	25.00
Total	33	100.0	35	100.0	68	100.0

EPI

Expanded Programme on Immunization (EPI) was officially launched in Bangladesh on 7th April 1979 aiming to reduce child mortality and morbidity by providing vaccines against 6 vaccine preventable diseases (BCG, OPV, DPT, and Measles). Initially EPI services were limited in districts and major municipalities. In the year 1985, intensification was started to cover all the target population through out the country under UCI (Universal Child Immunization) initiative. Intensification was completed by the year 1990. The same service delivery was used for TT vaccination for pregnant women. In the year 1993 GoB endorsed TT5 dose schedule for women of child bearing age initially from 15 to 45 years age and later 15 to 49 years age. Hepatitis-B vaccine was incorporated in the programme in 2003 with GAVI (Global Alliance for Vaccines and Immunization) phase I support bundle with injection safety supply. Since 2007, government is procuring injection safety material from local manufacturer using own fund.

Target Population

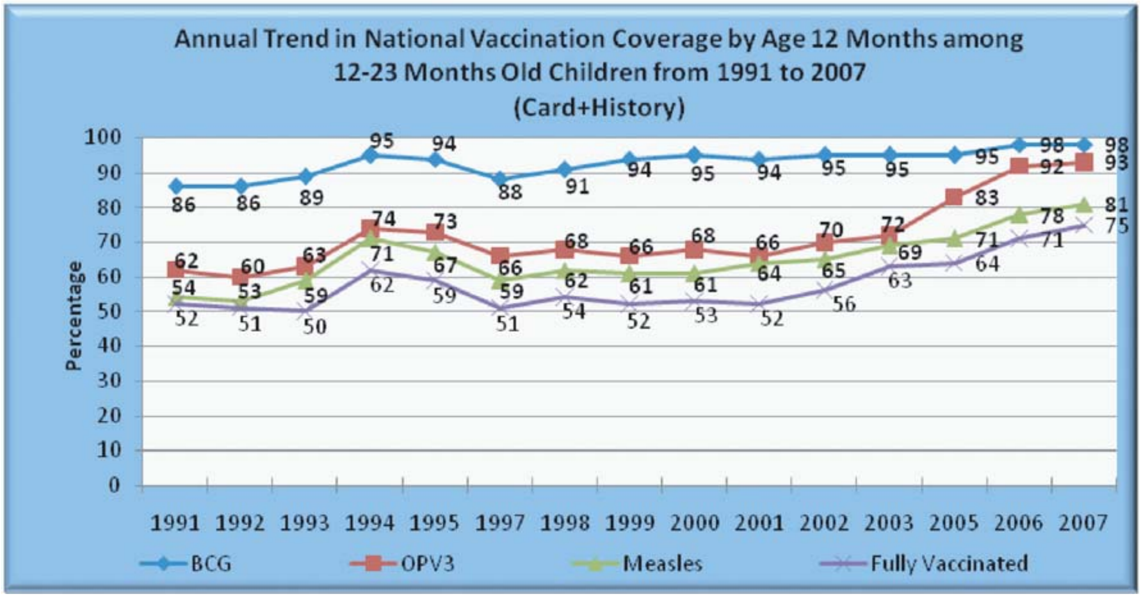
- Children under 1 year (4 million)
- Women of Child Bearing Age (35 million)

After intensification of EPI, following UCI GoB has made commendable progress in the last two decades. Fully Vaccination Coverage rate for the children has increased from 2 % in 1985 to 75 % in 2007 according to CES-2007.

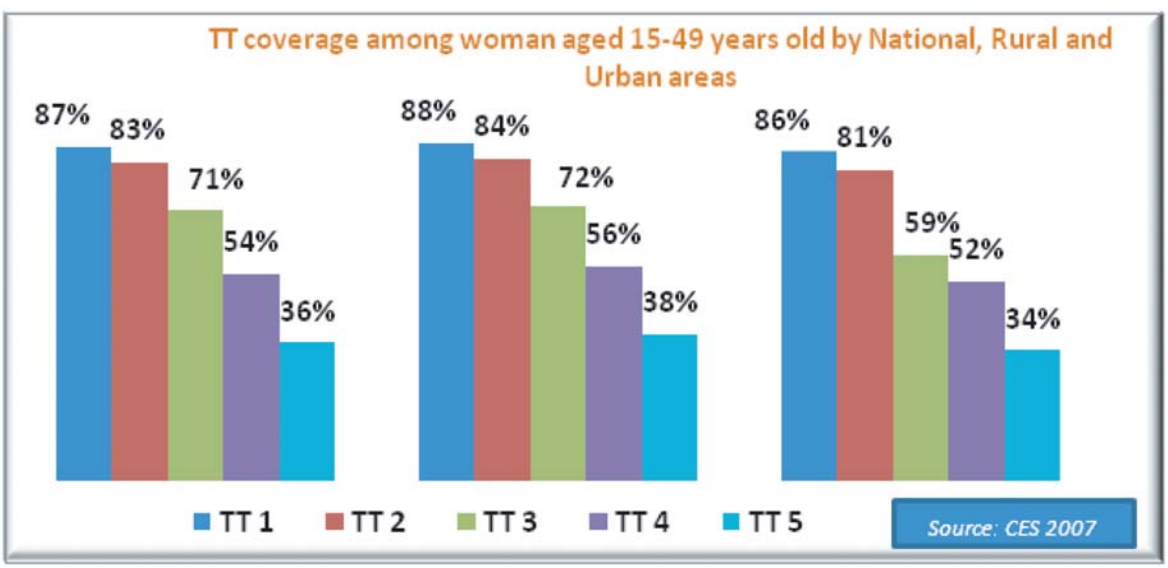
In year 2008, Bangladesh has successfully conducted the 2nd round of 17th NIDs. Coverage in the 1st round was 97.5% nationally. Analysis of data from the Independent Observers Checklists showed that high coverage in this 2nd round has also been achieved. Independent observers visited 2888 sites in 62 Districts including 6 City Corporations and among these sites they visited 1905 fixed sites on the day of the NID and evaluated 9570 children during the Child to Child search were completed. According to the findings of Independent Observers checklists 97.9% of the targeted 20.4 million children of 0-59 months old were given OPV. Of those children 90.3% were vaccinated at the fixed sites and 7.6% during house to house child search. Division wise coverage in the 2nd round is ranging from 96.5% in Chittagong to 99.6% in

Khulna. 2.1% of the children were missed which represents 504,000 children.

Bangladesh is maintaining polio free status for more than 2 years and the real challenge for the future is to sustain the high quality surveillance and increasing the coverage of routine EPI. It is also essential to routine EPI services to further minimize the number of susceptible children. A lower number of susceptible will minimize the chance of outbreak due to imported wild polio virus. In 2005 and 2006, EPI conducted phase by phase country wide Measles Catch-Up Campaign. From 15 July 2008, EPI has introduced Measles case based surveillance in all Health facilities. EPI has taken initiative for providing proper management of measles cases during outbreak investigation. EPI is currently planning for Measles Follow-Up Campaign in 2010.



TT coverage among child bearing age women: TT1-87% and TT5-36% nationally (CES-2007)



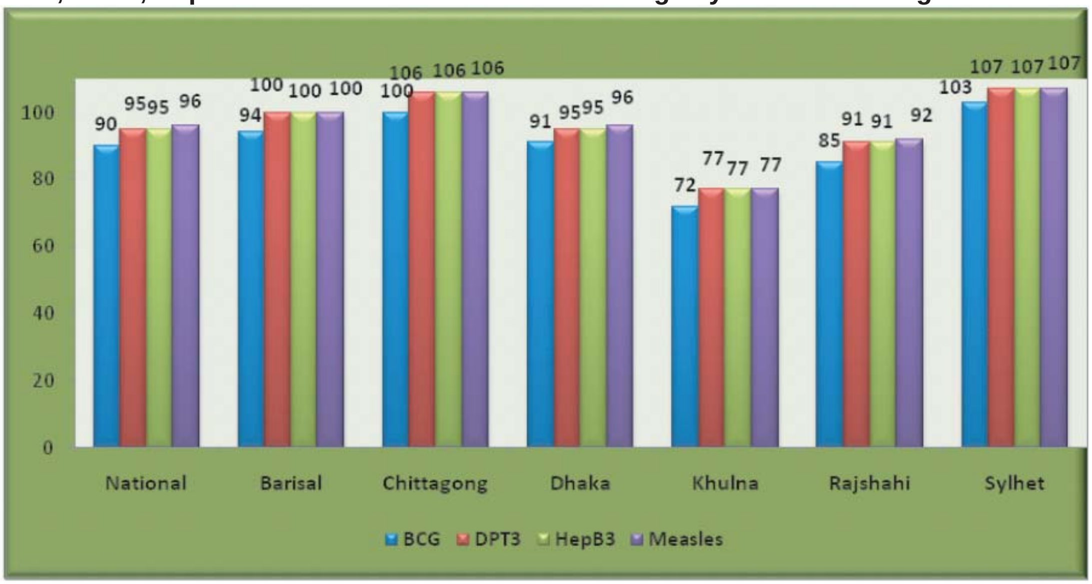
Routine Immunization Coverage by District Level for 2008

Division	District Name	2008 Number of Live Births	Routine Immunization Coverage Percent							
			BCG %	OPV3 %	HepB1 %	HepB3 %	DTP1 %	DTP3 %	MCV1* %	TT2+ %
Barisal	Barguna	27,103	96%	102%	102%	102%	102%	102%	102%	52%
	Barisal	74,033	97%	102%	103%	102%	103%	102%	103%	51%
	Bhola	56,872	106%	113%	113%	113%	113%	113%	113%	53%
	Jhalakati	21,453	73%	77%	77%	77%	77%	77%	77%	31%
	Patuakhali	48,856	97%	102%	103%	102%	103%	102%	101%	59%
	Perojpur	35,008	77%	82%	82%	82%	82%	82%	82%	39%
Chittagong	Bandarban	10,776	98%	102%	103%	102%	103%	102%	98%	29%
	Brahmanbaria	77,548	115%	126%	122%	126%	122%	126%	127%	59%
	Chandpur	81,662	85%	90%	91%	90%	91%	90%	92%	38%
	Chittagong	224,330	93%	97%	99%	97%	99%	97%	98%	48%
	Comilla	152,070	104%	110%	110%	111%	111%	111%	109%	53%
	Cox's Bazar	61,981	116%	125%	125%	125%	125%	125%	126%	60%
	Feni	40,040	94%	100%	101%	100%	101%	100%	98%	33%
	Khagrachari	21,300	73%	79%	78%	79%	78%	79%	79%	27%
	Lakshmipur	48,999	106%	111%	113%	111%	113%	111%	114%	57%
	Noakhali	86,265	112%	114%	120%	114%	120%	114%	112%	61%
	Rangamati	19,284	74%	81%	80%	81%	80%	81%	79%	36%
Dhaka	Dhaka	361,070	70%	71%	69%	71%	69%	71%	71%	37%
	Faridpur	58,374	94%	100%	99%	100%	99%	100%	102%	41%
	Gazipur	67,543	120%	127%	127%	127%	127%	127%	128%	72%
	Gopalganj	36,910	85%	90%	91%	90%	91%	90%	89%	39%
	Jamalpur	70,132	94%	99%	99%	99%	99%	99%	100%	34%
	Kishoreganj	83,144	111%	116%	116%	116%	116%	116%	117%	38%
	Madaripur	35,732	97%	101%	103%	101%	103%	101%	101%	37%
	Manikganj	42,228	85%	89%	90%	89%	90%	89%	89%	28%
	Munshiganj	41,129	89%	97%	95%	97%	95%	97%	96%	28%
	Mymensingh	146,602	102%	108%	107%	108%	107%	108%	109%	46%
	Narayanganj	74,660	96%	105%	102%	105%	102%	105%	103%	50%
	Narsingdhi	65,976	95%	101%	101%	101%	100%	101%	102%	40%
	Netrokona	65,345	106%	111%	111%	112%	111%	111%	112%	52%
	Rajbari	31,500	91%	95%	97%	95%	97%	95%	95%	47%
	Sariatpur	35,781	87%	91%	92%	91%	92%	91%	92%	18%
	Sherpur	41,276	95%	101%	101%	101%	101%	101%	102%	38%
	Tangail	104,448	96%	97%	98%	97%	98%	97%	99%	33%
Khulna	Bagerhat	47,741	67%	72%	71%	72%	71%	72%	70%	21%
	Chuadanga	35,467	75%	78%	79%	78%	79%	78%	79%	27%
	Jessore	83,275	73%	81%	77%	82%	77%	82%	81%	51%
	Jhenaidah	52,369	76%	78%	81%	78%	80%	78%	81%	39%
	Khulna	77,864	60%	66%	64%	66%	64%	66%	64%	20%
	Kushtia	58,441	84%	87%	89%	87%	89%	87%	86%	37%
	Magura	27,813	76%	80%	80%	80%	80%	80%	80%	30%
	Meherpur	20,133	76%	77%	79%	77%	79%	78%	80%	34%
	Narail	21,841	80%	83%	86%	83%	86%	83%	87%	28%
	Satkhira	61,599	71%	75%	76%	75%	76%	75%	74%	37%
Rajshahi	Bogra	99,981	75%	82%	81%	82%	81%	82%	87%	31%
	Dinajpur	89,286	82%	89%	86%	89%	86%	89%	89%	32%
	Gaibandha	68,475	96%	106%	104%	106%	104%	106%	104%	52%
	Joypurhat	27,805	71%	76%	75%	76%	75%	76%	76%	32%
	Kurigram	57,122	110%	117%	118%	117%	118%	117%	117%	61%
	Lalmonirhat	36,977	89%	97%	95%	97%	95%	97%	100%	45%
	Natore	48,930	83%	89%	87%	89%	87%	89%	89%	40%
	Nilphamari	52,481	91%	102%	98%	102%	98%	102%	101%	71%
	Noagoan	77,551	71%	75%	75%	75%	75%	75%	75%	32%
	Nowabganj	49,150	83%	87%	88%	87%	88%	87%	88%	23%
	Pabna	71,224	93%	98%	98%	98%	98%	98%	100%	45%
	Panchagharh	28,382	85%	93%	91%	93%	91%	93%	93%	35%
	Rajshahi	77,623	72%	72%	72%	72%	72%	72%	74%	33%
	Rangpur	85,317	85%	93%	89%	93%	89%	93%	95%	38%
	Sirajganj	90,795	94%	100%	98%	100%	98%	100%	100%	44%
	Thakurgoan	41,178	84%	92%	89%	92%	89%	92%	94%	50%

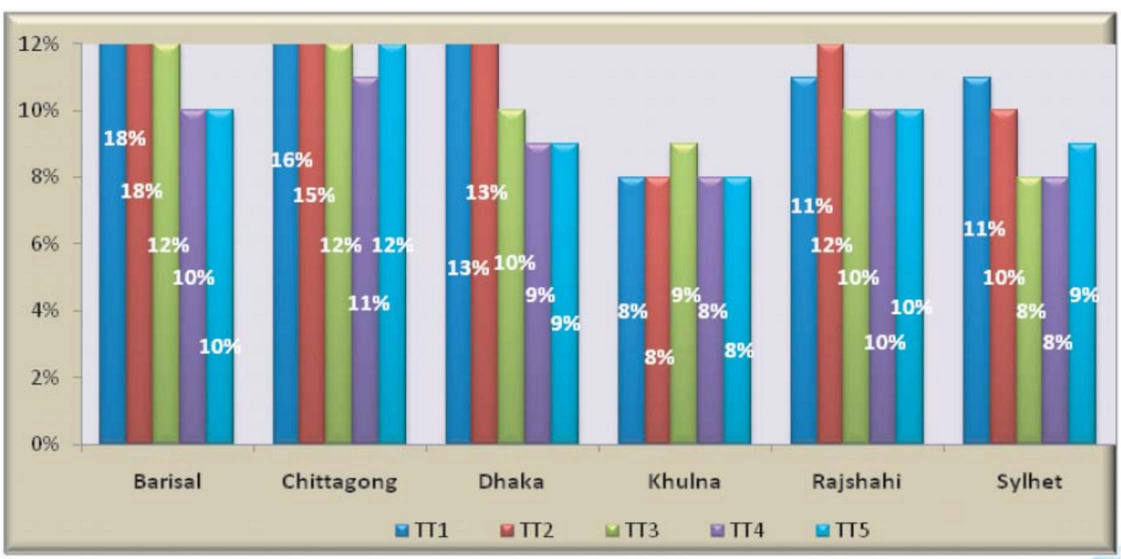
Routine Immunization Coverage by District Level for 2008 (contd...)

Division	District Name	2008 Number of Live Births	Routine Immunization Coverage in Percent							
			BCG %	OPV3 %	HepB1 %	HepB3 %	DTP1 %	DTP3 %	MCV1* %	TT2+ %
Sylhet	Habiganj	58,208	102%	108%	109%	108%	109%	108%	107%	33%
	Moulvi Bazar	54,727	90%	95%	95%	95%	95%	95%	95%	32%
	Sunamganj	67,015	112%	114%	119%	115%	119%	115%	116%	45%
	Sylhet	92,572	105%	109%	110%	109%	111%	109%	107%	33%
Total		4,210,773	90%	95%	95%	95%	95%	95%	96%	42%

BCG, DPT3, HepB3 and Measles Cumulative Coverage by Division% Bangladesh in 2008



Pregnant Women (Cumulative) TT Coverage by Division Bangladesh in 2008



Support service and coordination for Essential Service Delivery

In the Health, Nutrition and Population Sector Program, Line Director for Essential Service Delivery has been given special responsibility to support the health facilities at upazila and below. A program called "Support service and coordination" has been introduced to provide inputs to all newly constructed and upgraded health facilities at the level of upazila and below with a view to (a) improve coordination among all the components of ESD and others; (b) establish referral linkage among public-public and public-private health facilities; and (c) monitor and accelerate maximum utilization of beds of the health facilities.

Ambulance, X-Ray machines and Generators at upazila level and below

Name of the Item	Existing	Functioning	%	Non -functioning	%
Ambulance	490	306	62.44	184	37.56
X-Ray	467	266	57.00	179	43.00
Generator	349	242	69.34	107	30.66

Over all service

It is reported that overall service of the health facilities at upazila level and below has been improved. There is local level health care improvement committee for each facility which provides regular monitoring and supervision with regard to Health Facility Citizen Charters. Improvement has been claimed in attendance of doctors and staffs, diet quality, availability of medicines, cleanliness, etc.

Community clinics (CCs)

The Ministry of Health and Family Welfare initiated Community Clinics program under Health and Population Sector Program 1998-2003. It aimed to provide essential service package (ESP) for health, family planning and nutritional to rural people from a single-stop demand-based static center for every 6000 population. In April 1999, MOHFW issued guidelines for operation of CC, where there was provision of involving local people in the management of CC. A Community Group comprised of 9 to 11 members was proposed to function as executive board of the CC. It was planned to construct 13,695 CCs, of which 11,883 were completed. Community Clinic functioning was declared abandoned in 2001 and the facilities were sitting idle until 2008. In June 2008, the MOHFW again issued a circular to resume operation of CCs phase by phase. The newly elected Government in December, 2008 has an election manifesto of restarting the community clinics. This fresh attempt has created much hope among the community people with regard to getting health services right at their door steps. CCs are being resuming quickly.

Division	Built	Hand over completed
Dhaka	2951	2822
Chittagong	1663	1621
Rajshahi	3255	3171
Khulna	1385	1380
Barisal	801	737
Sylhet	668	618
Total =	**10723**	**10349**

Medical Waste Management at Upazila level -Progress in 2008

Health care wastes are generated as a by-product of health care activities and its generation is unavoidable. The medical waste is capable of transmitting diseases either through direct contact or by contaminating soil, air and water. If not properly handled, medical waste is a risk to service providers, other individuals, community and the environment. As there was no policy of waste management, the generated wastes were all collected together resulting in making the whole bulk as hazardous waste and thus exposing the population to a highly risky situation. Furthermore, safety of the waste handlers, generators, community and environment are not always properly addressed.

Under HPSP and later HNPSP, Government of Bangladesh included medical waste management as a priority sector as an activity under improved hospital services component. The upazila and below level health centers were not considered in that initiative as they fall under the jurisdiction of primary health care. Considering the situation government incorporated the waste management initiative for hospitals at the upazila and below as a component of essential services delivery (ESD) in line with the national goal to ensure safe, environment friendly, cost-effective and sustainable management of medical wastes derived from curative, diagnostic, preventive and rehabilitative health care services both in public and private sector.

To implement proper medical waste management system at primary health care level the main components in implementing the strategies are:

1. Construction of pits (for infectious, sharps, general and recyclable waste) in UHC
2. Procurement and regular supply of logistics for collection and transportation of waste and the safety materials for the waste handlers
3. Training/Orientation of personnel on proper waste management
4. Community awareness on waste, its management and individual responsibility.

Target is to establish a sustainable medical waste management system in all the UHC's by 2010-11 fiscal year.

Achievements in 2008

1. Pits were constructed in 133 upazila's (included in the fiscal year 2007-08).
2. Necessary logistics (for safety and managing waste) were procured and distributed to those 133 upazila.
3. Hands on training were imparted to 133 upazila waste related personnel's and concerned district officials to establish proper waste management system in those UHC's.
4. Orientation on MWM of key officials of CS office and all the upazila staffs (except field staffs) of 18 districts.
5. Construction of pits in 76 more Upazila health complexes has started and will be completed by July 2009
6. Procurement process through CMSD has already started for procuring logistics for the 2008-09 fiscal year.
7. For behaviour change communication one poster is developed on medical waste management to use at UHCs.

Facility Based Health Services

Hospital service is one of the important activities of health sector, which is the most visible health service also. This chapter of the Health Bulletin 2009 will provide an overview of the hospitals and their bed capacity as well as utilization based on the information from January through December of 2008.

No. of hospitals by bed capacity

There are 585 hospitals ranging from 10 beds to 1,700 beds under DGHS currently. All of these hospitals provide a total of 37,090 beds. The table below gives a detail profile.

No. of hospitals by bed capacity and total beds under DGHS (Year 2008)

Sl. No.	Bed capacity	No. of hospitals in this type	Total beds
1	1700 beds	1	1700
2	1010 beds	1	1010
3	900 beds	1	900
4	800 beds	1	800
5	600 beds	5	3000
6	500 beds	3	1500
7	414 beds	1	414
8	375 beds	1	375
9	250 beds	19	4750
10	200 beds	2	400
11	150 beds	3	450
12	100 beds	53	5300
13	80 beds	1	80
14	56 beds	1	56
15	50 beds	158	7900
16	31 beds	271	8401
17	30 beds	1	30
18	25 beds	1	25
19	20 beds	43	860
20	10 beds	22	220
	Total =	589	38171

Type of hospitals

Following list gives an overview of the type of hospitals currently in operation under DGHS:

Type of hospitals	No. of hospitals	Total bed capacity
Postgraduate institute hospital	7	2014
Dental college hospital	1	20
Hospital for alternative medicine	2	200
Medical college hospital	14	8685
Mental hospital, Pabna	1	500
Shekh Abu Naser Specialized Hospital	1	250
Narayanganj 200 bed Hospital	1	200
Specialized Health center (Asthma & Burn unit)	2	150
Sarkari karmochari hospital	1	100
Chest hospital	12	566
Infectious disease hospital	5	180
Leprosy hospital	3	130
District Level Hospital	60	8100
50 bed hospital(Tongi, Saidpur)	2	100
100 bed hospital (Narsingdi)	1	100
25 bed hospital (Jhenidah)	1	25
Bangladesh korea moitree hospital	1	20
Upazila health complex	421	15958
Health complex (31 bed)	3	93
20 bed hospital	28	560
10 bed hospital	22	220
Total	589	38171

The table below provides further detail of the different types of hospitals.

No. of different types of hospitals under DGHS with bed capacity as of December 2008

Type of hospital	No. of hospitals	No. of beds				
		Total	Revenue	Develop.	Proposed	Bed will Increase
Postgraduate Institute Hospital	7	2014	1700	314	250	200
Medical College Hospital	14	8685	8685	0	7500	2150
Pabna Mental Hospital	1	500	400	100	0	0
Dental College and Hospital,	1	20	20	0	200	180
Unani & Ayurvadic College	1	100	100	0	0	0
Homoeopathic Degree College	1	100	100	0	0	0
Shekh Abu Naser Specialized	1	250	0	250	0	0
National Ashma Center	1	100	0	100	0	0
Burn Unit at DMCH	1	50	0	50	200	150
Sarkari Karmochari Hospital	1	100	100	0	0	0
Narayanganj 200 bed Hospital	1	200	200	0	250	50
District Level Hospital	60	8100	6620	1480	3000	1200
Chest Hospital	12	566	566	0	0	0
Infectious Diseases Hospital,	5	180	180	0	0	0
Leprosy Hospital	3	130	130	0	0	0
Tongi 50 beded Hospital	1	50	50	0	0	0
Saidpur 50 beded Hospital,	1	50	50	0	0	0
Narsingdi 100 Bed	1	100	0	100	0	0
Bangladesh Korea Moitree	1	20	0	20	0	0
Upazila Health Complex (50 bed)	153	7650	7650	0	0	0
Upazila Health Complex (31 bed)	268	8308	8308	0	0	0
Health complex (1 development)	3	93	62	31	0	0
Jhenaidah 25 Bed Sishu Hospital	1	25	0	25	0	0
20 Baded Hospital (2	28	560	520	40	0	0
10 Beded Hospital(RHC)(1	22	220	220	0	0	0
National TB Control Project	1	0	0	0	250	250
General Hospital (Proposed)	6	0	0	0	2000	2000
Total Institute =	596	38171	35661	2510	13650	6180

Postgraduate Institute Hospitals (all are national level hospitals and are located in Dhaka)

Total = 7		No. of beds				
		Total	Revenue	Develop.	Proposed	Beds will Increase
1.	National Institute of Chest Disease and Hospital (NIDCH)	600	600	0	0	0
2.	National Institute of Cardiovascular Disease (NICVD)	414	250	164	0	0
3.	National Institute of Traumatology and Rehabilitation (NITOR)	500	500	0	0	0
4.	National Institute of Cancer Research and Hospital (NICR&H)	50	50	0	250	200
5.	National Institute of Ophthalmology (NIO)	250	250	0	0	
6.	National Institute of Kidney Disease and Hospital (NIKDU)	100	0	100	0	0
7.	National Institute of Mental Health (NIMHR)	100	50	50	0	
	Total =	**2014**	**1700**	**314**	**250**	**200**

Medical College Hospitals of Teaching Hospitals of equivalent level (Regional hospitals and are used as undergraduate and postgraduate teaching hospitals)

Division	District	Name of hospital (Total = 17)	No. of beds				
			Beds	Revenue	Develop.	Proposed	Bed will increase
Barisal	Barisal	Sher-e-Bangla Medical College Hospital	600	600	0	1000	400
Chittagong	Chittagong	Chittagong Medical College Hospital	1010	1010	0	0	0
	Comilla	Comilla Medical College Hospital	250	250	0	500	250
Dhaka	Dhaka	Dhaka Medical College Hospital	1700	1700	0	2000	300
		Sir Salimullh Medical College Hospital	600	600	0	0	0
		Shahid Suhrawardy Hospital, Dhaka	375	375	0	0	0
		Homoeopathic Degree College & Hospital	100	100	0	0	0
		Unani & Ayurvadic College & Hospital	100	100	0	0	0
		Dental College and Hospital, Dhaka	20	20	0	200	180
	Faridpur	Faridpur Medical College Hospital	250	250	0	0	0
	Mymensingh	Mymensingh Medical College Hospital	800	800	0	1000	200
Khulna		Khulna Medical College Hospital	250	250	0	500	250
Rajshahi	Bogra	SZR Medical College Hospital	500	500	0	0	0
	Dinajpur	Dinajpur Medical College Hospital	250	250	0	500	250
	Rajshahi	Rajshahi Medical College Hospital	600	600	0	0	0
	Rangpur	Rangpur Medical College Hospital	600	600	0	1000	400
Sylhet	Sylhet	MAG Osmani Medical College Hospital	900	900	0	1000	100
		Total	8905	8905	0	7700	2330

Specialized Centers under DGHS with bed capacity (Year 2008)

Division	District	Name of hospital (Total = 2)	No. of beds				
			Beds	Revenue	Develop.	Proposed	Bed will increase
Dhaka	Dhaka	1. National Asthma Center at NIDCH	100	0	100	0	0
		2. Burn Unit	50	0	50	200	150
		Total	150	0	150	200	150

Other Special Purpose Hospitals under DGHS with bed capacity (Year 2008)

Division	District	Name of hospital (Total = 4)	No. of beds				
			Beds	Revenue	Develop.	Proposed	Bed will increase
Rajshahi	Pabna	1. Mental Hospital	500	400	100	0	0
Khulna	Khulna	2. Shekh Abu Naser Specialized	250	0	250	0	0
Dhaka	Narayanganj	3. 200 bed Hospital	200	200	0	0	0
	Dhaka	4. Sarkari Karmochari Hospital	100	100	0	0	0
		Total =	**1050**	**700**	**350**	**0**	**0**

TB Center and Hospitals under DGHS with bed capacity (Year 2008)

Division	District	Name of hospital (Total = 13)	No. of beds				
			Beds	Revenue	Develop.	Proposed	Bed will increase
Dhaka	Dhaka	1. National TB Control Project	0	0	0	250	250
	Faridpur	2. Chest Hospital	20	20	0	0	0
Barisal	Barisal	3. Chest Hospital	20	20	0	0	0
Chittagong	Brahmanbaria	4. Chest Hospital	20	20	0	0	0
	Chittagong	5. Chest Hospital	150	150	0	0	0
	Feni	6. Chest Hospital	20	20	0	0	0
Khulna	Khulna	7. Chest Hospital	100	100	0	0	0
	Jessore	8. Chest Hospital	20	20	0	0	0
Rajshahi	Bogra	9. Chest Hospital	20	20	0	0	0
	Pabna	10. Chest Hospital	20	20	0	0	0
	Rajshahi	11. Chest Hospital	100	100	0	0	0
	Rangpur	12. Chest Hospital	20	20	0	0	0
Sylhet	Sylhet	13. Chest Hospital	56	56	0	0	0
		Total =	566	566	0	250	250

Infectious Disease Hospital under DGHS with bed capacity (Year 2008)

Divisi	District town where located (Total = 5)	No. of beds				
		Bed	Revenue	Develop.	Proposed	Bed will Increase
Chittagong	1. Chittagong	20	20	0	0	0
Dhaka	2. Dhaka	100	100	0	0	0
Khulna	3. Khulna	20	20	0	0	0
Rajshahi	4. Rajshahi	20	20	0	0	0
Sylhet	5. Sylhet	20	20	0	0	0
	Total =	180	180	0	0	0

Leprosy Hospital under DGHS with bed capacity (Year 2008)

Division	District town where located (Total = 3)	No. of beds				
		Bed	Revenue	Develop.	Proposed	Bed will Increase
Dhaka	1. Dhaka	30	30	0	0	0
Rajshahi	2. Nilphamari	80	80	0	0	0
Sylhet	3. Sylhet	20	20	0	0	0
	Total	130	130	0	0	0

District Level Hospital under DGHS with bed capacity (Year 2008)

Division	Sl. No.	District	Total	Revenue	Devlop.	Proposed	Bed will il
Barisal	1	Barguna	100	100	0	0	0
Barisal	2	Barisal	100	100	0	0	0
Barisal	3	Bhola	100	100	0	150	50
Barisal	4	Jhalokathi	100	100	0	150	50
Barisal	5	Patuakhali	250	150	100	0	0
Barisal	6	Pirojpur	100	100	0	150	50
Chittagon	7	Bandarban	100	100	0	0	0
Chittagon	8	Brahmanbaria	100	100	0	250	150
Chittagon	9	Chandpur	200	100	100	0	0
Chittagon	10	Chittagong	150	150	0	0	0
Chittagon	11	Comilla	100	100	0	0	0
Chittagon	12	Cox's Bazar	250	100	150	0	0
Chittagon	13	Feni	100	100	0	250	150
Chittagon	14	Khagrachari	50	50	0	100	50
Chit tagon	15	Lakshmipur	100	100	0	0	0
Chittagon	16	Noakhali	250	150	100	0	0
Chittagon	17	Rangamati	100	100	0	0	0
Dhaka	18	Faridpur	100	100	0	0	0
Dhaka	19	Gazipur	100	100	0	0	0
Dhaka	20	Gopalganj	250	100	150	0	0
Dhaka	21	Jamalpur	250	100	150	0	0
Dhaka	22	Kishoreganj	250	100	150	0	0
Dhaka	23	Madaripur	100	100	0	0	0
Dhaka	24	Manikganj	100	100	0	0	0
Dhaka	25	Munshiganj	100	100	0	250	150
Dhaka	26	Narayanganj	100	100	0	0	0
Dhaka	27	Narsingdi	100	100	0	0	0
Dhaka	28	Netrokona	100	100	0	0	0
Dhaka	29	Rajbari	100	100	0	0	0
Dhaka	30	Shariatpur	100	100	0	0	0
Dhaka	31	Sherpur	100	50	50	0	0

Dhaka	32	Tangail	250	100	150	0	0
Khulna	33	Bagerhat	100	100	0	150	50
Khulna	34	Chuadanga	100	100	0	150	50
Khulna	35	Jessore	250	250	0	0	0
Khulna	36	Jhenaidah	100	100	0	0	0
Khulna	37	Khulna	150	150	0	0	0
Khulna	38	Kustia	250	150	100	0	0
Khulna	39	Magura	100	100	0	150	50
Khulna	40	Meherpur	100	100	0	0	0
Khulna	41	Narail	100	100	0	0	0
Khulna	42	Satkhira	100	100	0	150	50
Rajshahi	43	Bogra	250	250	0	0	0

District level Hospital under DGHS with bed capacity (Year 2008) (contd...)

Division	Sl. No.	District	Total	Revenue	Devlop.	Proposed	Bed will ..
Rajshahi	44	Chapai Nawabganj	100	100	0	0	0
Rajshahi	45	Dinajpur	250	250	0	0	0
Rajshahi	46	Gaibandha	100	100	0	0	0
Rajshahi	47	Joypurhat	100	100	0	150	50
Rajshahi	48	Kurigram	100	100	0	150	50
Rajshahi	49	Lalmonirhat	100	100	0	0	0
Rajshahi	50	Naogaon	100	100	0	0	0
Rajshahi	51	Natore	100	100	0	0	0
Rajshahi	52	Nilphamari	100	100	0	150	50
Rajshahi	53	Pabna	250	120	130	250	0
Rajshahi	54	Panchagarh	100	100	0	0	0
Rajshahi	55	Serajganj	100	100	0	0	0
Rajshahi	56	Thakurgaon	100	100	0	0	0
Sylhet	57	Habiganj	100	100	0	150	50
Sylhet	58	Moulavibazar	100	100	0	250	150
Sylhet	59	Sunamganj	250	100	150	0	0
Sylhet	60	Sylhet	100	100	0	0	0
		Total	**8100**	**6620**	**1480**	**3000**	**1200**

Miscellaneous type of hospital under DGHS with bed capacity (Year 2008)

Division	District	Name of hospital (Total = 10)	No. of beds				
			Beds	Revenue	Develop.	Proposed	Bed will increase
Chittagong	Chittagong	1. General Hospital (proposed), Port Area	0	0	0	100	100
		2. Institute of Tropical Medicine	0	0	0	100	100
Dhaka	Dhaka	3. Bangladesh Korea Moitree Hospital	20	0	20	0	0
		4. General Hospital (proposed), Khilgaon	0	0	0	500	500
		5. General Hospital (proposed), Kurmitolla	0	0	0	500	500
		6. General Hospital (proposed), Mirpur	0	0	0	500	500
		7. Institute of Neuro Science	0	0	0	300	300
	Gazipur	8. Tongi 50 bed Hospital	50	50	0	0	0
	Narsingdi	9. Narsingdi 100 bed Hospital (Devlop.)	100	0	100	0	0
Khulna	Jhenaidah	10. Jhenaidah 25 bed Sishu Hospital	25	0	25	0	0
Rajshahi	Nilphamari	11. Saidpur 50 bed Hospital, Nilphamari	50	50	0	0	0
		Total =	245	100	145	2000	2000

31-bed hospital under DGHS with bed capacity (Year 2008)

Division	District	Name of hospital (Total = 10)	No. of beds				
			Beds	Revenue	Develop.	Proposed	Bed will increase
Chittagong	Chittagong	1. Sultanpur 31 Bed Hospital, Rawjan	31	0	31	0	0
Rajshahi	Rajshahi	2. Godagari 31 Bed Hospital	31	31	0	0	0
	Rangpur	3. Haragach 31 Bed Hospital	31	31	0	0	0
		Total	93	62	31	0	0

The Directorate General of Family Planning under the Ministry of Health and FamilyWelfare also runs several facilities as shown below:

Number of hospitals with bed capacities and health facilities under DGFP

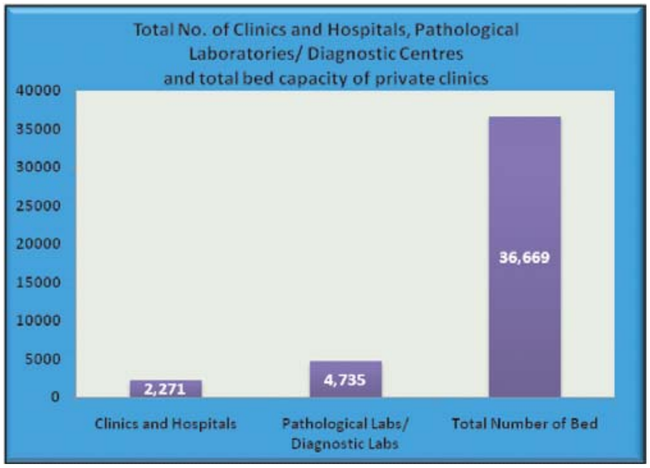
Type of facility	No. of facility	No. of bed in each	Total bed (No.)	No. of hospitals in this category	No. of non-bed facility in this category	No. of beds in this category
Community health facility	3500	0	0	0	3500	0
Maternal and Child Welfare Center	97	0	0	0	97	0
MCH-FP Clinic	471	0	0	0	471	0
NGO clinic supported by DGFP	177	0	0	0	177	0
Model cl inic	8	0	0	0	8	0
Hospital	2	0	0	2	0	0
Total	4255				4253	

Source: Website of DGFP

City Corporations in the six metropolitan cities run hospitals and health clinics through their own management and through Urban Primary Health Care Projects (UPHCP). UPHCP clinics are the good examples of Public-Private Partnership (PPP). Armed Forces of Bangladesh, Bangladesh Police, Bangladesh Rifles, Bangladesh Ansars, Bangladesh Railway, and Ministry of Social Welfare also run some hospitals and clinics mainly for their service men and families.

Private health facilities

From 1982, the Director of Hospital of DGHS has given registration to a total of 7,006 hospitals, clinics, pathological laboratories and diagnostic centers in the private sector. The number of hospitals and clinics is 2,271 and that of pathological laboratories and diagnostic clinics is 4,735. The total number of beds in the registered private hospitals and clinics is 36,669.



Government rules for licensing of private hospital or clinic

To start a private hospital or clinic, the interested party has to take license from DGHS through submission of application in prescribed forms. The conditions for issuing and continuing a license are as follows:

1. The hospital or clinic will require to have adequate space and healthy environment
2. It should have at minimum 80 square feet space per patient
3. The operation theater should be air-conditioned
4. There should be appropriate instruments as per prescribed guidelines
5. The hospital or clinic will have to store adequate life saving drugs and other medicines

6. There should be full time doctors, nurses and other staffs as per prescribed guidelines (for every 10 beds 3 doctors, 6 nurses, 3 cleaners, specialist doctors for surgery and for follow-up).

In violation of any of the clauses, the accused clinic owner will be liable to 6 months jail or fine worth Tk. 5,000 or both or cease of all removable assets in favor of state.

Basic Information of Some Private and autonomous Hospital in Bangladesh

Private Hospital	Sanc- Bed	Free Bed	Department	Wards	Cabins	No. of OT	Doctors	Trainee	Physician	Surgeons	Gynecolgst	Nurses	Medi.Asstt	Medi.Tech	Technician
Lions Eye Hospital	84	10	6	6	10	4	29	3	0	19	0	14	13	5	2
Monowara General Hospital	60	4	6	16	44	3	21	0	15	15	20	60	0	2	7
Islami Bank Hospital	160	0	4(9)	8	76	5	68	0	62	28	8	143	0	12	9
Aysha Memorial Hospital	50	0	6	20	30	2	47	0	22	9	3	86	0	6	13
Meditech General Hospital	10	0	10	2	6	1	6	0	8	2	1	13	0	3	1
Metropolitn Hospital	70	0	8	40	30	5	32	0	15		2	79	3	4	19
Samarita Hospital	200	5	12	76	75	4	45	0	30	25	7	7	4	10	9
Sikder Women's MCH	100	0	9	3		2	31	0	14	1	2	32	5	13	2
Uttara Adhunik MCH	500		12	12		11	153		14	22		164		11	48
Panpacific Hospital	62	0	10	20	42	3	20	0	34	34	6	38	0	4	5
Aichi Hospital Lt	50	5	7	25	25	2	15	0	14	9	3	40	0	2	3
Bhasani MCH	500	200	19	13	20	6	175	71	36	19	10	58	15	8	2
Apollo Hospital	304		31	15	49	8	170		27	17	3	284	82	46	31
BSMMU Hospital	1212	452	48	48	105	18	1033	328	205	18 7	45	536		62	52
East West MCH	400	80	14	14	29	6	105		45	18	5	98	8	10	8

Hospital Statistics

For the reporting period of January to December 2008, the MIS, DGHS received information from 7 postgraduate institute hospitals, 14 medical college hospitals and 58 districts (district and other hospitals and the health facilities below the district level) in the public sector. Number of patients attending out-patient and emergency departments was estimated to be 65,437,876; numbers of patients who were admitted were 3,221,642. The hospitals reported 78,152 deaths among the admitted patients giving a death rate of 2.4%.

Indoor and outdoor statistics of different types of hospitals under DGHS (Year 2008)

Type of hospital	No.of Hospital	No.							
		Total bed	Beds com	Admission	Death	Bed Occupancy Rate	Hospital Death Rate	Average Length of Stay	OPD pts
Post Graduate Institutes	7	2075	1950	72,759	3,486	83.49	5.24	9	508,584
Medical College Hospital	14	9555	8935	807,455	38,439	103.87	5.22	5	6,029,039
Mental Hospital	1	500	500	1,412	4	80.22	0.24	86	21,410
Chest Clinics & Chest Hospital (TB Hospital)	4	406	406	1,833	56	57.82	3.29	50	0
Chest Clinics & Chest Hospital (TB Seg. Hospital)	7	140	140	864	41	113.84	4.54	64	0
Leprosy Hospital	3	130	130	601	0	75.52	0.00	61	16,080
I.D. Hospital	5	180	160	2,338	194	67.58	8.73	18	51,312
District Hospital	58	6870	6870	853,453	19,728	109.81	2.42	3	7,396,097
Upazila Health Complex	421	14797	14368	1,429,832	16,204	71.37	1.14	3	25,563,324
Rheumatic Fever and heart disease	1	-	-	-	-	-	-	-	48,167
School Health Clinics	23	-	-	-	-	-	-	-	61,249
Urban Dispensarie	35	-	-	-	-	-	-	-	136,080
Union Sub-centre	1362	-	-	-	-	-	-	-	25,606,534
Type	520	34,653	33,509	3170547	78152	86.37	2.57	4	65437876

No. of OPD patients seen by different out-patient only health facilities under DGHS (Year 2008)

Type of health facility	No. of OPD patients
Rheumatic Fever and heart disease (n=1)	4,8167
School Health Clinics (n=23)	61,249
Urban Dispensaries (n=35)	136,080
Union sub-centers (n=1362)	25,606,534
Total (n=1421) =	25,852,030

Indoor & outdoor Statistics of postgraduate institute hospitals (Year 2008)

Name of hospital	Bed (N)	Admission (N)			Death (N)			Outdoor (N)				ALS	BOR	HDR	ADA	ADOP
		M	F	T	M	F	T	M	F	Child	T					
NICVD	414	23214	10432	33646	1640	772	2412	91147	47889	8534	147570	5	117.85	7.06	92	404
NIDCH	670	6971	2608	9579	669	181	850	45408	29428	0	74816	24	92.35	8.93	26	205
NITOR	500	17535	3317	20852	137	87	224	77378	39216	26703	143297	8	89.15	1.12	57	393
NIO	100	1252	1368	2620	0	0	0	30756	21948	11742	64446	10	75.63	0.00	7	177
NIKDU	116	3056	1830	4886	126	84	210	34884	17830	3840	56554	0	90.97	0.00	13	155
IMHR	150	749	427	1176	0	0	0	12692	9209	0	21901	42	79.11	0.00	3	60
Total	2050	48972	17725	72759	2446	1040	3486	244689	138481	46979	508584	9	83.49	5.24	33	232

Note: M- Male; F- Female; T- Total; ALS- Average Length of Stay; BOR- Bed Occupancy Rate; HDR- Hospital Death Rate; ADA- Average Daily Admission; ADOP- Average Daily Outdoor Patients

Indoor & outdoor Statistics of medical college hospitals (Year 2008)

Name of hospital	Bed (N)	Admission (N)			Death (N)			Outdoor (N)				ALS	BOR	HDR	ADA	ADOP
		M	F	T	M	F	T	M	F	Child	T					
Rangpur	600	40996	34336	75332	1880	1200	3080	150532	147664	35158	333354	2	55.18	4.34	206	913
Rajshahi	600	35056	36228	71284	1292	920	2212	69164	173232	246132	488528	2	47.54	3.35	195	1338
Barisal	600	36780	29106	65886	1504	1360	2864	184802	167728	159464	511994	6	187.57	4.45	181	1403
Dhaka	1700	37197	51778	88975	3264	3550	6814	241691	241389	206988	690068	0	0.00	9.57	244	1891
Mitford	600	23948	18233	42181	1148	475	1623	95004	100795	68350	264149	7	130.64	3.76	116	724
Mymensingh	800	45024	47958	92982	2244	2164	4408	202262	280184	197658	680104	6	185.68	4.53	255	1863
Sylhet	900	40318	38876	79194	1744	1442	3186	232462	205684	47312	485458	4	100.06	4.24	217	1330
Chittagong	1010			115573			6614				926239	8	172.72	8.48	317	2538
Dinajpur	250	16736	22490	39226	430	850	1280	115534	110414	58684	284632	3	150.61	3.07	107	780
Khulna	500	15662	22454	38116	938	608	1546	98438	122292	47460	268190	7	134.35	4.56	104	735
Faridpur	250	11290	16084	27374	574	672	1246	99942	90608	35094	225644	5	137.31	4.63	75	618
Comilla	250	12387	9938	22325	655	572	1227	88631	77943	36062	202636	6	119.76	7.13	61	555
Bogra	500	17908	21757	39665	950	1147	2097	139110	136391	25692	301193	6	123.76	5.54	109	825
Shuhrowardi	375	3724	5618	9342	190	52	242	158116	189172	19562	366850	9	83.98	1.88	26	1005
Total	8935	337026	354856	807455	16813	15012	38439	1875688	2043496	1183616	6029039	5	103.87	5.22	158	1179

Note: M- Male; F- Female; T- Total; ALS- Average Length of Stay; BOR- Bed Occupancy Rate; HDR- Hospital Death Rate; ADA- Average Daily Admission; ADOP- Average Daily Outdoor Patients

Indoor & outdoor Statistics of General Hospitals (Year 2008)

Name of hospital	Bed (N)	Admission (N)			Death (N)			Outdoor (N)				ALS	BOR	HDR	ADA	ADOP
		M	F	T	M	F	T	M	F	Child	T					
Mental	500	1182	230	1412	4	0	4	9662	11748	0	21410	86	80.22	0.24	4	59
Tongi labour	50	2735	2486	5221	19	9	28	29587	39128	36876	105591	5	85.38	0.94	14	289
Total	**550**	**3917**	**2716**	**6633**	**23**	**9**	**4**	**39249**	**50876**	**36876**	**21410**	86	72.93	0.24	9	174

Note: M- Male; F- Female; T- Total; ALS- Average Length of Stay; BOR- Bed Occupancy Rate; HDR- Hospital Death Rate; ADA- Average Daily Admission; ADOP- Average Daily Outdoor Patients

Indoor & outdoor Statistics of Chest Clinics & Hospitals (TB Hospital) (Year 2008)

Name of hospital	Bed (N)	Admission (N)			Death (N)			Outdoor (N)				ALS	BOR	HDR	ADA	ADOP
		M	F	T	M	F	T	M	F	Child	T					
Rajshahi	150	280	70	350	0	0	0					60	38.72	0.00	1	0
Khulna	100	346	168	514	10	4	14					56	73.78	2.93	1	0
Sylhet	56	275	82	357	20	3	23					44	70.11	7.01	1	0
Chittagong	100	372	240	612	18	1	19					43	63.63	3.52	2	0
Total	406	1273	560	1833	48	8	56	na	na	na	na	50	57.82	3.29	1	na

Indoor & outdoor Statistics of Chest Clinics & Hospitals (TB Hospital) (Year 2008)

Name of hospital	Bed (N)	Admission (N)			Death (N)			Outdoor (N)				ALS	BOR	HDR	ADA	ADOP
		M	F	T	M	F	T	M	F	Child	T					
Bogra	20	156	26	182	6	0	6	na	na	na	na	46	99.32	3.85	0	
Pabna	20	52	8	60	3	0	3	na	na	na	na	160	135.86	4.84	0	0
Jessore	20	53	19	72	0	0	0	na	na	na	na	97	85.29	0.00	0	0
Barisal	20	34	32	66	0	0	0	na	na	na	na	80	92.00	0.00	0	0
Faridpur	20	36	54	90	0	0	0	na	na	na	na	65	90.79	0.00	0	0
B. Baria	20	108	72	180	11	0	11	na	na	na	na	-	81.95	0.00	0	0
Rangpur	20	138	36	174	21	0	21	na	na	na	na	-	146.44	0.00	0	0
Total	**140**	**577**	**247**	**824**	**41**	**0**	**41**	na	**na**	**na**	**na**	62	104.52	**4.78**	**na**	**na**

Note: M- Male; F- Female; T- Total; ALS- Average Length of Stay; BOR- Bed Occupancy Rate; HDR- Hospital Death Rate; ADA- Average Daily Admission; ADOP- Average Daily Outdoor Patients

Indoor & outdoor Statistics of Leprosy Hospitals (Year 2008)

Name of hospital	Bed (N)	Admission (N)			Death (N)			Outdoor (N)				ALS	BOR	HDR	ADA	ADOP
		M	F	T	M	F	T	M	F	Child	T					
Nilphamari	20	70	0	70	0	0	0	4224	4864	508	9596	41	28.93	0.00	0	26
Dhaka	30	196	56	252	0	0	0	3420	1092	12	4524	32	73.40	0.00	1	12
Sylhet	80	246	33	279	0	0	0	1564	386	10	1960	91	87.96	0.00	1	5
Total	**130**	**512**	**89**	**601**	**0**	**0**	**0**	**9208**	**6342**	**530**	**16080**	61	75.52	**0.00**	**1**	**14**

Note: M- Male; F- Female; T- Total; ALS- Average Length of Stay; BOR- Bed Occupancy Rate; HDR- Hospital Death Rate; ADA- Average Daily Admission; ADOP- Average Daily Outdoor Patients

Indoor & outdoor Statistics of Infectious Disease Hospitals (Year 2008)

Name of hospital	Bed (N)	Admission (N)			Death (N)			Outdoor (N)				ALS	BOR	HDR	ADA	ADOP
		M	F	T	M	F	T	M	F	Child	T					
IDH																
Rajshahi	20	230	84	314	38	2	40	168	96	26	290	31	112.11	15.27	1	1
Khulna	20	326	266	592	16	12	28				0	7	53.53	4.75	2	0
Dhaka	100	654	516	1170	74	30	104	24782	10010	16230	51022	16	46.09	9.63	3	140
Sylhet	20			0			0				0	0	0.00		0	0
Chittagong	20	192	70	262	10	12	22				0	37	144.56	7.61	1	0
Total	**180**	**1402**	**936**	**2338**	**138**	**56**	**194**	**24950**	**10106**	**16256**	**51312**	18	60.07	**8.73**	**2**	**35**

Note: M- Male; F- Female; T- Total; ALS- Average Length of Stay; BOR- Bed Occupancy Rate; HDR- Hospital Death Rate; ADA- Average Daily Admission; ADOP- Average Daily Outdoor Patients

Indoor & outdoor Statistics of district level Hospitals (Year 2008)

Name of Dist.	Bed (N)	Admission			Death (N)			Outdoor				ALS	BOR	HDR	ADA	ADOP
		M	F	T	M	F	T	M	F	child	T					
Jessore	250	17496	20326	37822	766	480	1246	93132	105856	40376	239364	3	144.5	3.3	104	656
Tangail	250	9080	22676	31756	292	594	886	29128	64452	44462	138042	3	110.1	2.9	87	378
Jamalpur	250	10282	10456	20738	216	124	340	54482	63408	47480	165370	3	71.9	1.7	57	453
Noakhali	250	9960	9198	19158	336	222	558	8914	9896	9846	28656	4	87.2	2.9	52	79
Bogra	250	7054	9234	16288	101	65	166	71638	82931	18426	172995	4	100.2	0.8	45	474
Narsingdi	200	6314	10586	16900	158	214	372	100936	158128	93650	352714	6	114.5	2.6	46	966
Chandpur	200	3790	7152	10942	92	142	234	28618	64396	63160	156174	5	61.9	2.8	30	428
Kushtia	150	11762	12732	24494	570	272	842	50786	63722	43614	158122	3	154.8	2.3	67	433
Khulna	150	3626	4922	8548	94	52	146	64400	98172	24440	187012	6	93.8	1.8	23	512
Chittagong	150	1838	2410	4248	25	22	47	29604	30856	25312	85772	9	69.6	1.1	12	235
Patuakhali	150	7406	7296	14702	208	102	310	26820	37772	34778	99370	5	137.6	2.1	40	272
Pabna	120	13260	16898	30158	518	368	886	57324	55244	54582	167150	2	177.9	2.3	83	458
Panchagarh	100	5390	5908	11298	98	64	162	23958	39078	44900	107936	3	85.3	1.6	31	296
Thakurgaon	100	5648	18628	24276	156	174	330	50760	56432	60820	168012	4	120.5	2.7	67	460
Nilphmari	100	9196	18478	27674	72	96	168	51078	70214	58570	179862	2	140.5	0.6	76	493
Lalmonirhat	100	3407	6266	9673	84	118	202	39187	31282	70469	140938	4	102.5	2.4	27	386
Kurigram	100	8316	6846	15162	324	132	456	47468	37822	48816	134106	3	136.9	3.1	42	367
Gaibandha	100	5192	6880	12072	134	228	362	28000	36956	29796	94752	6	113.2	5.3	33	260
Joypurhat	100	7618	10088	17706	224	172	396	48860	77432	47002	173294	2	102.4	2.2	49	475
Serajgonj	100	15744	20437	36181	159	211	370	21270	26518	15404	63192	7	181.5	3.9	99	173
Natore	100	7222	7394	14616	208	132	340	25572	35860	30380	91812	3	100.7	2.4	40	252
Naogan	100	9020	7914	16934	282	133	415	42341	60544	32759	135644	3	129.8	2.4	46	372
Ch.Nawabganj	100	7970	10520	18490	214	150	364	31860	42328	31618	105806	3	133.2	1.9	51	290
Meherpur	100	3766	7574	11340	118	76	194	34880	56276	35504	126660	3	79.2	1.7	31	347
Chuadanga	100	4664	8408	13072	152	184	336	16240	36784	18326	71350	2	86.8	2.6	36	195
Jhenaidah	100	6989	14262	21251	215	322	537	57447	76389	47153	180989	3	161.6	2.5	58	496
Magura	100	9640	8972	18612	250	154	404	36872	39266	34584	110722	3	142.4	2.2	51	303
Narail	100	4376	4996	9372	98	78	176	24884	38268	26414	89566	3	84.6	1.9	26	245
Satkhira	100	6116	6914	13030	246	294	540	66416	75240	40932	182588	5	134.9	5.7	36	500
Bagerhat	100	4322	6446	10768	152	100	252	26192	29090	14258	69540	4	131.9	2.3	30	191
Perojpur	100	9464	4892	14356	108	60	168	25712	30308	33944	89964	2	62.9	1.5	39	246
Jhalakathi	100	4602	4870	9472	40	26	66	33590	52964	26712	113266	4	91.3	0.7	26	310

Barguna	100	12932	13084	26016	224	176	400	81388	87788	43052	212228	2	140.9	1.6	71	581

Indoor & outdoor Statistics of district level Hospitals (Year 2008) (contd…)

Name of Dist.	Bed (N)	Admission			Death (N)			Outdoor				ALS	BOR	HDR	ADA	ADOP
		M	F	T	M	F	T	M	F	Child	T					
Bhola	100	4800	7718	12518	92	156	248	20124	29098	24492	73714	4	106.1	2.3	34	202
Shariatpur	100	3022	4634	7656	82	90	172	20806	26206	26570	73582	4	83.7	2.2	21	202
Madaripur	100	4116	8102	12218	48	68	116	21776	18666	18676	59118	3	91.5	0	33	162
Gopalganj	100	6312	5422	11734	218	148	366	26682	27568	27496	81746	3	101.8	3.2	32	224
Faridpur	100	7570	8480	16050	200	128	328	33202	38128	38450	109780	4	161.0	2.1	44	301
Rajbari	100	3146	6900	10046	134	58	192	28600	28844	21414	78858	4	105.5	0	28	216
Manikganj	100	4676	5736	10412	192	226	418	58604	72214	72234	203052	4	113.7	4.3	29	556
Munshignj	100	3641	6110	9751	53	29	82	46086	52778	42288	141152	4	100.7	0.8	27	387
Narayangonj	100	3294	4578	7872	14	22	36	100574	71556	40280	212410	3	73.4	0.4	22	582
Gazipur	100	3472	7276	10748	52	96	148	23414	41000	35494	99908	3	71.9	1.6	29	274
Kishoreganj	100	18484	15552	34036	398	256	654	47240	63038	45290	155568	4	234.8	2.8	93	426
Netrokona	100	6858	8614	15472	132	44	176	29918	38154	33442	101514	3	109.5	1.1	42	278
Sunamganj	100	3067	9509	12576	55	194	249	27876	37584	40518	105978	3	88.6	2.1	34	290
M.Bazar	100	10308	9510	19818	186	150	336	66030	58140	64410	188580	2	126.8	1.7	54	517
Hobiganj	100	6734	14214	20948	138	580	718	44118	45216	33942	123276	3	144.1	3.7	57	338
B. Baria	100	8418	7118	15536	320	246	566	65672	72080	60222	197974	4	175.2	3.8	43	542
Comilla	100	5366	6148	11514	160	134	294	54154	66708	41944	162806	5	150.3	2.7	32	446
Laximpur	100	8024	7180	15204	234	196	430	45570	44494	38780	128844	4	152.27	3.1	42	353
Feni	100	7744	18054	25798	200	332	532	86384	66858	60472	213714	3	214.7	2.1	71	586
Rangamati	100	3978	3686	7664	78	84	162	17216	16554	19574	53344	4	81.5	2.1	21	146
Bandarban	100	1650	2232	3882	10	36	46	9266	10718	7672	27656	4	43.6	1.3	11	76
Cox's Bazar	100	12354	12478	24832	376	342	718	85406	90632	157936	333974	3	183.8	3.4	68	915
Sherpur	100	12728	4698	17426	154	82	236	17614	18110	13852	49576	3	80.9	2.8	48	136
Khagrachhari	100	3770	3440	7210	106	94	200	22110	25538	21370	69018	5	73.1	3.6	20	189
Total	4500	301356	383726	685082	7084	6777	13861	1790307	2071183	1756861	5618351	3	116.9	2.32	1877	15393

Note: M -Male; F -Female; T -Total; ALS -Average Length of Stay; BOR -Bed Occupancy Rate; HDR -Hospital Death Rate; ADA -Average Daily Admission;

ADOP -Average Daily Outdoor Patients

Bed occupancy rate of Upazila Health Complexes-2004-2008

Bed Occupancy Level (in Percentage)	Years									
	2004		2005		2006		2007		2008	
	No.	%	No.	%	No.	%	No.	%	No.	%
Below 40%	4	1.10	8	2.33	14	3.92	11	3.05	10	2.48
40.01 – 50.00	7	1.92	9	2.62	20	5.60	14	3.88	42	10.40
50.01 – 60.00	25	6.87	17	4.96	26	7.28	27	7.48	59	14.60
60.01 – 70.00	27	7.42	45	13.12	36	10.08	48	13.30	74	18.32
70.01 – 80.00	82	22.53	58	16.91	62	17.37	80	22.16	76	18.81
80.01 – 90.00	77	21.15	97	28.28	75	21.01	69	19.11	61	15.10
90.01 – 100.00	89	24.45	73	21.28	70	19.61	67	18.56	48	11.88
100.01 – 110.00	40	10.99	27	7.87	36	10.08	37	10.25	16	3.96
110.01- 120.00	19	5.22	22	6.41	15	4.20	7	1.94	16	3.96
120.01- 130.00	1	0.27	3	0.87	2	0.56	1	0.28	0	0.00
130.01 – 140.00	2	0.55	1	0.29	1	0.28	0	0.00	0	0.00
Above 140%	1	0.27	0	0.00	0	0.00	0	0.00	2	0.50
Total No. of Upazila Health Complexes	364	100.00	343	100.00	357	100.00	361	100.00	404	100.00

BSMMU

Bangabandhu Sheikh Mujib Medical University (BSMMU) is the premier Postgraduate Medical Institution of the country. It bears the heritage to Institute of Postgraduate Medical Research (IPGMR)which was established in December 1965. In the year 1998 the Government converted IPGMR into a Medical University for expanding the facilities for higher medical education and research in the country. It has an enviable reputation for providing high quality postgraduate education in different specialties. The university has strong link with other professional bodies at home and abroad. The university is expanding rapidly and at present, the university has many departments equipped with modern technology for service, teaching and research. Besides education, the university plays the vital role of promoting research activities in various discipline of medicine. Since its inception, the university has also been delivering general and specialized clinical service as a tertiary level healthcare center. The university provides patient care services on various disciplines like Psychiatry, Physical medicine, Pediatrics, Neonatology, Pediatric neurology, Pediatric surgery, Clinical pathology, Dermatology, Colorectal surgery, Nephrology, Urology, Neurology, Neuro-Surgery, Internal Medicine, Ga stroenterology, Hepat ology, Ophthalmology, ENT, Obstetrics & gynecology, Surgery, Hepatobiliary Surgery, dentistry, and blood transfusion services. It provides different treatment services like Intensive Care, Lithotripsy, Pain management and diagnostic services like radiology, endoscopy, CT scan & MRI and a one-stop laboratory service. BSMMU runs Institute of Nuclear Medicine (INM). INM is a joint project of Bangladesh Atomic Energy Commission and BSMMU. The INM has modern diagnostic and therapeutic facilities including computerized ultrasonography, gamma camera and a well equipped radioimmunoassay (RIA) laboratory. This is considered to be the best center for noninvasive diagnoses.

Hospital Statistics of BSMMU (Year -2008)

Month	No. of Outdoor patient			No. of Emergency patient			No. of Admitted patient		
	M	F	Total	M	F	Total	M	F	Total
January	35721	28785	64506	120	95	215	1354	1000	2354
February	36604	25782	62386	104	106	210	1140	1028	2168
March	38242	30202	68444	149	108	257	1536	1200	2736
April	36774	31213	67987	135	139	274	1410	907	2317
May	37645	32253	69898	209	158	367	1210	1009	2219
June	39270	28982	68252	180	125	305	1480	905	2385
July	36392	33352	69744	174	130	304	1401	950	2351
August	34972	26720	61692	137	121	258	1260	1008	2268
September	36962	29892	66854	152	127	279	1303	804	2107
October	35891	31290	67181	151	136	287	1410	821	2231
November	36780	32120	68900	158	135	293	1070	615	1685
December	37202	35320	72522	170	145	315	1170	902	2072
Total of the year	442455	365911	808366	1839	1525	3364	15744	11149	26893

Note: M- Male; F- Female; T- Total; ALS- Average Length of Stay; BOR- Bed Occupancy Rate;

Hospital Statistics of BSMMU (Year - 2008)

Month	No. of patient death			ALS (days)	BOR (%)	No. of major operation
	M	F	Total			
January	52	31	83	10.03	2:11	853
February	60	28	88	9.06	0.1 1	743
March	55	38	93	10.04	2:46	900
April	45	36	81	8.05	2:13	865
May	56	33	89	10.02	2:13	827
June	49	36	85	9.03	2:21	926
July	57	35	92	8.06	2:19	915
August	61	29	90	10.02	0.14	785
Sept ember	52	33	85	10.05	0.1 2	802
October	57	37	94	9.07	2:07	647
November	53	34	87	10.01	0.0 9	892
December	56	30	86	8.07	0.1 1	486
Total of the year	653	400	1053	9.29	1.2	9641

Note: M- Male; F- Female; T- Total; ALS- Average Length of Stay; BOR- Bed Occupancy Rate;

Smiling Sun Franchise Program (SSFP)

The Smiling Sun Franchise Program is a project funded by the United States Agency for International Development (USAID). It is intended to complement the wide network of health-care facilities set up by the Government of Bangladesh resorting to an innovative approach to health care franchising. SSFP is committed to improve the quality of life of all Bangladeshis by providing superior, friendly and affordable health services in a sustainable manner.

To achieve relevant health outcomes, SSFP is jointly working with partnering NGOs to convert the existing network into a viable social health system. SSFP objective is to strengthen partnering organization's quality of care while helping them to enhance their financial sustainability, thus enabling them to continue serving an important segment of the Bangladeshi society, including the poorest of the poor.

Currently 29 NGOs are providing health care services to women, children and through 319 static and 8,500 satellite clinics in 61 districts of Bangladesh. 34 clinics of this network are providing Emergency Obstetric Care (EmOC) services. This network will continue to expand the volume and types of quality health care under ESD provided to the able-to-pay customers as well as underserved and poor clients.

By the fourth year of this project SSFP aims to generate sufficient income to support approximately 70% of the operational cost while maintaining access to those who cannot afford to pay for services.

During 2008, three hundred nineteen (319) smiling sun clinics treated 21,832,000 outdoor patients while 11, 277 patients were admitted and discharged in 34 EmOC (ultra) clinics. No patient died in smiling sun clinics during 2008. In 34 ultra clinics patient stayed on an average 3 days while their bed occupancy rate was 35%. In 34 EmOC clinics on an average 30 patients were admitted per day while SSFP network treated on an average 72,773 outdoor patients per day. During 2008 twenty comprehensive EmOC clinics conducted 3,575 major surgeries (C-section).

Patient statistics of SSFP during January-December 2008

Month	Outdoor Pt.	Admitted Pt.	No. of patient days	ALS	BOR %	Ave. Daily Admn.	ADOP	Total No. of Surgery
January	1841397	901	2703	3	32	29	73656	294
February	1843190	919	2757	3	36	33	76800	301
March	1894555	910	2730	3	33	29	75782	282
April	1866097	926	2778	3	34	31	74644	285
May	1866459	865	2595	3	31	28	71787	276
June	1801721	874	2622	3	32	29	69297	268
July	1848260	982	2946	3	35	32	71087	314
August	1802036	1004	3012	3	36	32	69309	340
September	1764061	1002	3006	3	37	33	80185	301
October	1760685	995	2985	3	36	32	67719	309
November	1833725	1017	3051	3	38	34	73349	332
December	1709700	882	2646	3	32	28	68388	273
Total	**21,831,886**	**11,277**	**33,831**	**3**	**418**	**30**	72,773	**3575**

ALS- Average Length of Stay; BOR- Bed Occupancy Rate; ADA- Average Daily Admission; ADOP- Average Daily Outdoor Patient

Urban Primary Health Care Project (UPHCP-II):

About 35 million people representing almost 25 percent of the population of Bangladesh live in urban areas, a large proportion of whom are slum dwellers. The health knowledge of the urban slum dwellers and their access to essential basic health services are low. Children living in urban slums are deprived of education and health care, and vulnerable to violence, abuse and exploitation. On the other hand, high rate of mortality and morbidity exists among women who remain neglected in terms of meeting their basic health needs and ensuring their rights.

The Government of Bangladesh is committed to put in place strategies to address the issues of improving the health status of the urban population. This is to be done through improved access to and utilization of efficient, effective and sustainable Primary Health Care Services. The provision of public health services in urban areas is the responsibility of Local Government Bodies by dint of City Corporation Ordinance of 1983 and Pouroshova Ordinance of 1977. For primary health care services delivery, the public sector works in partnership with NGOs and the local government institutions such as the City Corporations and Pouroshovas. The health service delivery mechanism in urban areas involves diverse roles of the government (MOLGRD&C and MOHFW), NGOs and the private sector.

Urban Primary Health Care Project, a Public-Private Partnership is an innovative initiative with the goal to improve the health status of the urban population, speciallythe poor, particularly focusing on women and children. These population segments are usually undeserved by the health care facilities due to many reasons. UPHCP-II is committed to provide all essential health services and reproductive health services to them for improvement of their livelihood. With the aim to contribute to achieving the national goals and targets of the Millennium Development

Goals (MDGs), the First Urban Primary Health Care Project (UPHCP-I) and Second Urban Primary Health Care Project (UPHCP-II) were initiated in 1998 and 2005 respectively which are milestones in urban health care services. Local Government Division, MOLGRD&C is the responsible executing agency for UPHCP-II. This project is funded by GOB, ADB, DFID, SIDA, UNFPA and ORBIS International. Project Area of UPHCP-II includes City Corporations of Dhaka, Chittagong, Khulna, Rajshahi, Barisal & Sylhet and Pouroshovas of Bogra, Comilla, Sirajgong, Madhobdi & Savar.

Patient statistics of UPHCP-II during January-December 2008

Outdoor Pt.			Emergency Pt.			Admitted Pt.			No. of pt. death			No.of Surgery		
M	F	Total	M	F	Total	M	F	Total	M	F	Total	Major	Minor	Total
1,976,674	5,832,917	7,809,591	4,544	9,959	14,503	0	38,678	38,678	1	2	3	10,468	11,731	22,199

Private Hospital Statistics Report (Jan - Dec 2008)

Hospital Name	No.of Beds	District	No. of Outdoor Pt.			No. of Emergency Pt.			No. Of Admitted Pt.		
			M	F	T	M	F	T	M	F	T
Lions Eye Hospital Private	84		39952	37471	77423	330	174	504	2879	3026	5905
United Hospital Ltd.	350	Dhaka	46745	28090	74835	7435	4140	11575	5095	2679	7774
Monowara General Hospital	60	Dhaka	1421	1927	3348	Included in Emergency		1153	3289	4442	
Islami Bank Hospital	160	Dhaka			137963			31448			9890
Aysha Memorial Hospital	50	Dhaka	7093	5088	12181	502	566	1068	1744	1503	3247
Meditech General Hospital	10	Dhaka	4400	5367	9767	2700	3668	6368	431	699	1130
Metropolitan Hospital	70	Dhaka				2332	945	3277	1953	1584	3537
Samorita Hospitl	200	Dhaka	3656	1786	5442	6023	3173	9196	3323	3040	6363
Dhaka Hospital ICDDR,B		Dhaka	17278	10916	28194	Included in OPD			57833	41253	99086
Christian Hospital		Rangamati	8933	14224	2315	951	1507	2458	2050	2562	4612
Bhasani MCH	500	Dhaka1230	39158	29045	68203	28798	21182	49980	4332	3244	7576
Apollo Hospital	304	Dhaka	78031	79664	157695	2682	2077	4759	5404	4756	10160

Note: M- Male; F- Female; T- Total; ALS- Average Length of Stay; BOR- Bed Occupancy Rate

Private Hospital Statistics Report (Jan - Dec 2008) (contd...)

Hospital Name	No. of Patient Death			No. of Surgery done			ALS	BOR
	M	F	T	Major	Minor	T	(In days)	(%)
Lions Eye Hospital Private	0	0	0	5410	1273	6683	2 days/ Pt.	16 - 63 %
United Hospital Lt	147	50	197				4d - 5d	34 - 49 %
Monowara General Hospital	32	9	41	2496	935	3431	3-4d,5-6d	82%
Islami Bank Hospital			157	2807	769	3576		52 - 94 %
Aysha Memorial Hospital	41	27	68	525	245	770		
Meditech General Hospital	0	1	1	257	27	284	2 d / Pt	39 - 87 %
Metropolitan Hospital	119	66	185	720	480	1200	4.8d - 6.2d	66 - 82 %
Samorita Hospitl	23	17	40	4602				
Dhaka Hospital ICDDR,B	90	69	159				0.9	63 - 83 %
Christian Hospital								
Bhasani MCH	17	7	24	365	899	1264	3.2d/perPt.	54.06%
Apollo Hospital	158	125	283			4578	5.71-6.38d	45 - 53 %

Note: M- Male; F- Female; T- Total; ALS- Average Length of Stay; BOR- Bed Occupancy Rate

Morbidity Profile

National level

The MIS-health, DGHS received morbidity status on the indoor patients of hospitals in the public sector for the year 2008 (January-December). A total of 344 hospitals sent reports on distribution of admitted patients against 156 selected diseases which are common in Bangladesh. The hospitals include 301 upazila health complexes (87.4%), 27 district level hospitals (7.9%) (22 district hospitals and 5 general hospitals), 6 medical college hospitals (1.7%), 3 postgraduate institute hospitals (0.9%) and 7 other hospitals (2.1%). The latter group of hospitals includes 100-bed hospital (1 No.), 50-bed hospital (1 No.), 31-bed hospital (1 No.), 10-bed hospital (1 No.), leprosy hospital (1 No.), and specialized center-National Asthma Center and TB hospital (1 No.) Following table give distribution of top-10 diseases by age group irrespective of sex counting all the patients together reported from the above 344 hospitals. Diagnoses for 19, 96,541 patients were received. Number of patients in each group for which diagnoses were received has been shown at the bottom row of the table. The table shows that diarrhea topped the disease burden (15.1%), followed by assault (9.8%) and peptic ulcer (5.9%). The other seven diseases that top the list are viral fever (2.3%), poisoning (2.0%), road traffic accident (2.0%), enteric fever (2.0%), bronchial asthma (1.8%) and anemia (1.8%).

It may be mentioned here that the public hospitals of Bangladesh are attended by only about one fifth of the patients who seek health care. There is variation in the type of patients who attend each kind of public health facility. Therefore, the data presented here may not represent the overall population morbidities of the country.

Percent distribution of top-10 diseases among indoor patients of different public health facilities of Bangladesh in 2008

Position	0-28 d	29d-11m	1-4y	5-14y	15-24y	25-49 y	50+	All age
1	Pneumonia 34.1%	Pneumonia 38.3%	Diarrhea 30.4%	Diarrhea 21.3%	Assault 14.7%	Assault 15.1%	Diarrhea 11.9%	Diarrhea 15.1%
2	Diarrhea 8.6%	Diarrhea 26.5%	Pneumonia 20.9%	Assault 5.9%	Diarrhea 11.0%	Diarrhea 9.3%	Peptic Ulcer 8.6%	Assault 9.8%
3	Septicemia 6.2%	Bronchiolitis 2.9%	Dysentery 5%	Pneumonia 5.8%	Peptic Ulcer 6.0%	Peptic Ulcer 8.6%	Assault 8.5%	Pneumonia 7.4%
4	Anemia 2.7%	Viral fever 1.4%	Viral fever 2.4%	Peptic Ulcer 3.9%	Poisoning 3.3%	Obstructed Labor 3.6%	Hypertension 5.5%	Peptic Ulcer 5.9%
5	Bronchiolitis 0.9%	Enteric Fever 1.1%	Enteric Fever 1.7%	Viral fever 3.0%	Road Traffic Accident 2.8%	Poisoning 3.0%	Bronchial Asthma 3.5%	Viral fever 2.3%
6	Allergic Reaction 0.9%	Anemia 1.1%	Worm Infestation (Intestinal) 1.4%	Enteric Fever 2.9%	Enteric Fever 2.3%	Road Traffic Accident 2.8%	COPD 3.4%	Poisoning 2.0%
7	Bacillary Dysentery 0.6%	Protein Energy Malnutrition 1.0%	Assault 1.3%	Dysentery 2.6%	Viral fever 2.0%	Viral fever 2.6%	Road Traffic Accident 2.4%	Road Traffic Accident 2.0%
8	Congenital Heart Disease 0.5%	Suppurative Otitis Media 0.9%	Anemia 1.3%	Anemia 2.3%	Anemia 1.9%	Enteric Fever 1.8%	Anemia 2.2%	Enteric Fever 2.0%
9	Viral fever 0.4%	Dysentery 0.9%	Bronchial Asthma 1.2%	Worm Infestation (Intestinal) 2.3%	Obstructed Labor 1.8%	Hypertension 1.8%	Dysentery 2.1%	Bronchial Asthma 1.8%
10	Meningitis 0.4%	Bacillary Dysentery 0.8%	Suppurative Otitis Media 1.2%	Bronchial Asthma 2.0%	Head Injury 1.8%	Bronchial Asthma 1.7%	Enteric Fever 1.9%	Anemia 1.8%
No. of patients	**42,808**	**1, 52, 491**	**2,18,293**	**2,38,638**	**3,88,734**	**6,12,762**	**158,240**	**19,96,541**

District Level Hospitals

The distribution of indoor patients by disease diagnosis in the district level hospitals of Bangladesh is shown in the following table. MIS-health received report from 27 district level hospitals (22 sadar and 5 general hospitals). In Bangladesh, sadar/ district and general hospital provide similar type of treatment facilities to the patients. Irrespective of age, diarrhea (12.4%) tops the top-ten disease with assault (9.4%) and pneumonia (6.4%) in the next orders. Other diseases that are included in the top-10 diseases are peptic ulcer (3.3%), hypertension (2.8%), viral fever (2.5%), poisoning (2.3%), obstructed labor (2.1%), anemia (1.6%) and chronic obstructive pulmonary disease (1.6%). The distribution of top ten diseases according to different age group is shown in the table.

Percent distribution of top-10 diseases among indoor patients of different public health facilities of Bangladesh in 2008

Position	0-28 d	29d-11m	1-4y	5-14y	15-24y	25-49 y	50+	All age
1	Pneumonia 34.1%	Pneumonia 38.3%	Diarrhea 30.4%	Diarrhea 21.3%	Assault 14.7%	Assault 15.1%	Diarrhea 11.9%	Diarrhea 15.1%
2	Diarrhea 8.6%	Diarrhea 26.5%	Pneumonia 20.9%	Assault 5.9%	Diarrhea 11.0%	Diarrhea 9.3%	Peptic Ulcer 8.6%	Assault 9.8%
3	Septicemia 6.2%	Bronchiolitis 2.9%	Dysentery 5%	Pneumonia 5.8%	Peptic Ulcer 6.0%	Peptic Ulcer 8.6%	Assault 8.5%	Pneumonia 7.4%
4	Anemia 2.7%	Viral fever 1.4%	Viral fever 2.4%	Peptic Ulcer 3.9%	Poisoning 3.3%	Obstructed Labor 3.6%	Hypertension 5.5%	Peptic Ulcer 5.9%
5	Bronchiolitis 0.9%	Enteric Fever 1.1%	Enteric Fever 1.7%	Viral fever 3.0%	Road Traffic Accident 2.8%	Poisoning 3.0%	Bronchial Asthma 3.5%	Viral fever 2.3%
6	Allergic Reaction 0.9%	Anemia 1.1%	Worm Infestation (Intestinal) 1.4%	Enteric Fever 2.9%	Enteric Fever 2.3%	Road Traffic Accident 2.8%	COPD 3.4%	Poisoning 2.0%
7	Bacillary Dysentery 0.6%	Protein Energy Malnutrition 1.0%	Assault 1.3%	Dysentery 2.6%	Viral fever 2.0%	Viral fever 2.6%	Road Traffic Accident 2.4%	Road Traffic Accident 2.0%
8	Congenital Heart Disease 0.5%	Suppurative Otitis Media 0.9%	Anemia 1.3%	Anemia 2.3%	Anemia 1.9%	Enteric Fever 1.8%	Anemia 2.2%	Enteric Fever 2.0%
9	Viral fever 0.4%	Dysentery 0.9%	Bronchial Asthma 1.2%	Worm Infestation (Intestinal) 2.3%	Obstructed Labor 1.8%	Hypertension 1.8%	Dysentery 2.1%	Bronchial Asthma 1.8%
10	Meningitis 0.4%	Bacillary Dysentery 0.8%	Suppurative Otitis Media 1.2%	Bronchial Asthma 2.0%	Head Injury 1.8%	Bronchial Asthma 1.7%	Enteric Fever 1.9%	Anemia 1.8%
No. of patients	**42,808**	**1, 52, 491**	**2,18,293**	**2,38,638**	**3,88,734**	**6,12,762**	**158,240**	**19,96,541**

Medical College Hospitals

In Bangladesh, medical college hospitals are the teaching hospital for medical students of undergraduate and postgraduate level and usually provide tertiary healthcare to the patients. The following table shows distribution of patients of the government medical college hospitals of Bangladesh. MIS-health received data from indoors of 6 medical college hospitals in year 2008 and the top-10 diseases in all age category include diarrhea (12.2%), suppurative otitis media (5.3%), assault (4.6%), head injury (3.9%), pneumonia (3.4%), COPD (3.0%), arthritis (2.9%), bronchiolitis (2.8%), tonsillitis (2.7%), bronchial asthma (2.4%). The distribution is seen I the following table.

Percent distribution of top-10 diseases among indoor patients of government medical college hospitals in 2008

Position	0-28 d	29d-11m	1-4y	5-14y	15-24y	25-49 y	50+	All age
1	Septicemia 31.1%	Diarrhea 30.5%	Diarrhea 23.1%	Diarrhea 27.3%	Head Injury 11.9%	Assault 8.4%	CVA 7.7%	Diarrhea 12.2%
2	Anemia 14.1%	Bronchiolitis 18.2%	Congenital Heart Disease 7.3%	COPD 7.3%	Suppurative Otitis Media 9.4%	Head Injury 5.3%	Assault 7.3%	Suppurative Otitis Media 5.3%
3	Pneumonia 6.1%	Pneumonia 11.7%	Suppurative Otitis Media 6.9%	Suppurative Otitis Media 5.9%	Assault 8.2%	Arthritis 5.2%	Road Traffic Accident 6.8%	Assault 4.6%
4	Allergic Reaction 5.1%	Suppurative Otitis Media 5.9%	Bronchiolitis 6.5%	Tonsillitis 5.1%	Arthritis 4.4%	Heart failure 4.1%	Arthritis 5.6%	Head Injury 3.9%
5	Congenital Heart Disease 3.4%	Anemia 4.1%	Tonsillitis 6.5%	Bronchial Asthma 4.9%	Poisoning 4.0%	Abortion 3.0%	Hypertension 5.0%	Pneumonia 3.4%
6	Valvular Heart Disease 1.7%	Septicemia 3.9%	Pneumonia 6.3%	Hypertension 3.1%	Tonsillitis 3.9%	Suppurative Otitis Media 3.0%	COPD 3.9%	COPD 3.0%
7	Bronchiolitis 0.8%	Protein Energy Malnutrition 2.9%	Bronchial Asthma 3.6%	Pneumonia 2.7%	Abortion 3.1%	Peptic Ulcer 2.8%	Bronchial Asthma 2.6%	Arthritis 2.9%
8	Diarrhoea 0.6%	Allergic Reaction 2.4%	Tuberculosis (Pulmonary) 3.5%	Enteric Fever 2.3%	Road Traffic Accident 2.8%	Poisoning 2.7%	Peptic Ulcer 2.5%	Bronchiolitis 2.8%
9	Dysentery 0.6%	Bacillary Dysentery 2.1%	Protein Energy Malnutrition 2.9%	Heart failure 2.0%	COPD 2.1%	COPD 2.6%	C.C.F 2.1%	Tonsillitis 2.7%
10	Hemolytic Jaundice 0.5%	Congenital Heart Disease 1.8%	Dysentery 2.4%	Thalassemia 1.9%	Peptic Ulcer 2.0%	Diabetes Mellitus 2.4%	Heart failure 1.9%	Bronchial Asthma 2.4%
No. of Patients	6,580	20,863	28,611	41,226	44,216	43,057	29,180	2,13,733

Upazila Health Complex

The MIS-health, DGHS received morbidity status on the indoor patients of 301 upazila health complexes out of 322 for the year 2008 (January - December). The top-10 disease profiles of the different age group of patients admitted in the upazilla health complexes are shown in the following table. Irrespective of age, the top-ten diseases diarrhea (16.1%), assault (13.4%), peptic ulcer (5.4%), pneumonia (4.6%), bronchial asthma (3.8%), poisoning (2.6%), enteric fever (1.8%), road traffic accident (1.1%), hypertension (1.1%), anxiety & depressive disorders (0.8%).

Percent distribution of top-10 diseases among indoor patients of upazilla health complexes in 2008

Position	0d-11m	1-4y	5-14y	15-24y	25-49 y	50+	All age
1	Pneumonia 25.8%	Diarrhea 32.9%	Diarrhea 24.5%	Diarrhea 23.0%	Assault 24.0%	Diarrhea 12.6%	Diarrhea 16.1%
2	Diarrhea 16.7%	Pneumonia 13.3%	Assault 7.2%	Assault 15.2%	Diarrhea 8.3%	Assault 10.1%	Assault 13.4%
3	Septicemia 2.9%	Enteric Fever 1.3%	Peptic Ulcer 4.0%	Peptic Ulcer 6.2%	Peptic Ulcer 7.6%	Bronchial Asthma 7.6%	Peptic Ulcer 5.4%
4	Bronchial Asthma .9%	Bronchial Asthma 1.2%	Enteric Fever 3.7%	Enteric Fever 4.0%	Poisoning 5.3%	Peptic Ulcer 4.8%	Pneumonia 4.6%
5	Enteric Fever 0.2%	Assault 1.2%	Bronchial Asthma 2.4%	Road Traffic Accident 1.9%	Bronchial Asthma 5.2%	Hypertension 3.7%	Bronchial Asthma 3.8%
6	Glomerulo-nephritis 0.17%	Burn 0.8%	Road Traffic Accident 1.5%	Anxiety & Depressive Disorders 1.9%	Enteric Fever 1.6%	COPD 2.5%	Poisoning 2.6%
7	Poisoning 0.17%	Viral fever 0.4%	Pneumonia 1.4%	Anemia 1.4%	Road Traffic Accident 1.3%	CVA 1.7%	Enteric Fever 1.8%
8	Protein Energy Malnutrition 0.1.7%	Anemia 0.2%	Poisoning 1.4%	Poisoning 0.9%	Hypertension 1.1%	Road Traffic Accident 1.2%	Road Traffic Accident 1.1%
9	Meningitis 0.08%	Urinary Tract Infection .2%	Anxiety & Depressive Disorders 1.0%	Urinary Tract Infection 0.6%	Anxiety & Depressive Disorders 1.0%	Enteric Fever 1.0%	Hypertension 1.1%
10	Renal failure 0.08%	Encephalitis 0.2%	Anemia 0.7%	Viral fever 0.3%	COPD 0.7%	Poisoning 0.9%	Anxiety & Depressive Disorders 0.8%
No. of patients	**1,428**	**1,221**	**1,093**	**1,537**	**4,324**	**2,230**	**11,833**

Causes of Death

The MIS-health, DGHS regularly gather data on causes of deaths from the inpatient departments of public health facilities of Bangladesh. For the year 2008 (January to December), reports from 406 hospitals on causes of deaths were received. These hospitals include 337 upazila health complexes (83.0%), 50 district level hospitals (12.32%) (district and general hospitals), 5 medical college hospitals (1.23%), 5 postgraduate institute hospitals and 9 (3.4%) hospitals of other category. This latter group of hospitals include chest hospital (5 Nos.), infectious disease hospitals (3 Nos.) and labor hospital (1 No.)

Following table gives distribution of top-10 causes of death by level of health care delivery, irrespective of sex counting all the deaths from various causes together for which causes were reported has been reported from all the above hospitals. Causes of deaths for 27,789 cases were received. Number of deaths in each group shown at the top of each group. The table shows that respiratory failure topped the causes of death (8.8%), followed by pneumonia (7.8%) and birth asphyxia (7.6%). The other 7 leading causes of deaths are heart disease other than M.I. (7.2%), cerebrovascular diseases (6.4%), acute myocardial infarction (5.9%), complications of pre-term low birth weight (4.6%), septicemia (4.3%), , asthma (4.0%)and food poisoning (2.6%).

As mentioned earlier, only about one-fifth of our population seek health care from the public health facilities. So, the causes of deaths in our data represent only those deaths occurred in public health facilities and not the whole population of the country.

Top 10 causes of death at different level of health care delivery

Top 10 causes of death									
1	2	3	4	5	6	7	8	9	10
Upazila Health Complexes (report on 4,543 deaths)									
Pneumonia 16.1%	Asthma 8.6%	CVD[1] 6.4%	Respiratory infection 6.1%	Respiratory failure 5.4%	Food Poisoning 3.9%	Acute MI[2] 3.8%	Birth asphyxia 3.6%	Pesticide poisoning 3.6%	PUO[4] 3.1%
District Level Hospitals (report on 13,284 deaths)									
Birth asphyxia 12.6%	CVD[1] 9.6%	Pneumonia 8.6%	Respiratory failure 6.8%	Heart disease other than Acute M.I. 6.2%	Septicemia 5.3%	MI[2] 4.6%	Pre-term LBW[3] 4.4%	Asthma 4.1%	Pesticide poisoning 2.6%
Medical College Hospitals (report on 8,334 deaths)									
Respiratory failure 14.3%	Pre term LBW[3] 8.3%	Heart disease other than Acute M.I. 7.2%	Septicemia 5.8%	CVD[1] 5.7%	Shock 4.2%	MI[2] 4.2%	Pneumonia 4.0%	Birth asphyxia 3.6%	Injury of head 3.2%
National (report on 27,789* deaths)									
Respiratory failure 8.8%	Pneumonia 7.8%	Birth asphyxia 7.6%	Heart disease other than Acute M.I. 7.3%	CVD[1] 6.4%	Acute MI[2] 5.9%	Pre term LBW[3] 4.6%	Septicemia 4.3%	Asthma 4.0%	Food poisoning 2.6%
Note: 1. CVD: Cerebro-vascular disease; 2. MI: Myccardial Infarction; 3. LBW: Complications of Pre-term Low Birth Weight; 4. PUO: Pyrexia of unknown origin. *Includes number of deaths from 5 postgraduate institute hospitals and 9 specialized hospitals.									

Top 10 causes of death by age group

The causes of deaths for 27,789 cases, as we received, were grouped according to age. In age group, distribution of the causes of deaths was made to find the top 10 causes of deaths. The result is shown in following table.

It is found that birth asphyxia (44.9%) and complications of pre-term low birth weight (23.8%) are the two top causes of deaths in the newborns aged 0-to 7-days; septicemia (25.2%) and pneumonia rank first and second position in top 10 causes of deaths respectively in 7-to 28-days' age group. Causes of deaths like complications of pre-term low birth weight (12.5%) and birth asphyxia (11.3%) descend to third and fourth position in this age group. Pneumonia is at the top of the causes of death in 1 month-to 1 year's (42.4%) and 1-to 4-years' (28.1%) age group respectively. Among the 5-to 14-years' age group, encephalitis (11.1%) tops the causes of deaths list followed by pneumonia (8.4%). Among the 15-to 49years' age group, respiratory failure (12.3%) and pesticide poisoning (8.0%)rank first and second position in the list of causes of deaths. Acute myocardial infarction (14.5%) tops the causes of deaths in the age group of 50-to 59-years, indicating the rise in prevalence of cardiac problems in elderly population of Bangladesh. Similarly, another noncommunicable disease the cerebrovascular disease (17.4%) is at the top of the causes of deaths in the 60+ years' age group.

Top 10 causes of death by age group

Top 10 causes of death by age group

1	2	3	4	5	6	7	8	9	10
0-7days (Total patients:4,556)									
Birth asphyxia 44.9%	Pre-term LBW[3] 23.8%	Pneumonia 9.2%	Septicemia 7.6%	Respiratory failure 3.1%	PUO[4] 1.9%	Bacterial sepsis of newborn 1.1%	Neonatal jaundice 0.8%	Tetanus 0.4%	Diarrhea 0.4%
7- 28days (Total patients:947									
Septicemia 25.2%	Pneumonia 24.6%	Pre-term LBW[3] 12.5%	Birth asphyxia 11.3%	Acute lower RTI[5] 8.2%	Respiratory failure 4.0%	Bacterial sepsis of newborn 3.6%	PUO[4] 2.5%	Neonatal jaundice 1.8%	Meningitis I%
1m-1yrs (Total patients:2,493)									
Pneumonia 42.4%	Acute lower RTI[5] 12.5%	Septicemia 11.9%	Respiratory failure 5.2%	Diarrhea 3.1%	Meningitis 2.9%	Encephalitis 2.4%	PUO[4] 2.2%	Birth asphyxia 2.2%	Anemia 1.2%
1-4yrs (Total patients: 1,118)									
Pneumonia 28.1%	Encephalitis 13.5%	Acute lower RTI[5] 8.6%	Septicemia 6.4%	Diarrhea 4.7%	Meningitis 4.4%	Respiratory failure 4.1%	PUO[4] 3.6%	Malaria 3.0%	Convulsions, not elsewhere classified 2.9%
5-14yrs (Total patients: 1,063)									
Encephalitis 11.1%	Pneumonia 8.4%	Meningitis 7.1%	Respiratory failure 6.7%	Tetanus 5.1%	Injury 4.1%	RTA[6] 4.0%	PUO[4] 3.8%	Septicemia 3.5%	Pesticide Poisoning 2.9%
15-49yrs (Total patients: 7,438)									
Respiratory failure 12.3%	Pesticide Poisoning 8.0%	Heart disease other than Acute M.I. 6.7%	MI[2] 4.9%	CVD[1] 4.5%	Injury 3.5%	RTA[6] 3.1%	Food poisoning 3.0%	Asthma 2.9%	Eclampsia 2.2%
50-59yrs (Total patients: 2,871)									
Acute MI 14.5%	CVD[1] 13.1%	Respiratory failure 11.3%	CIHD[7] 9.7%	Asthma 6.1%	Heart failure 5.3%	Shock 4.0%	Injury 2.4%	RTA[6] 2.1%	COPD[8] 1.7%
60yrs > (Total patients: 7,303)									
CVD[1] 17.4%	Acute MI[2] 11.2%	CIHD[7] 12.9%	Respiratory failure 10.6%	Asthma 8.8%	Heart failure 6.2%	Other COPD[8] 2.0%	RTA[6] 1.9%	Injury 1.6%	Abdominal/ pelvic pain 1.6%
Total (Total patients: 27,789)									
Respiratory failure 8.8%	Pneumonia 7.8%	Birth asphyxia 7.6%	Heart disease other than Acute M.I. 7.3%	CVD[1] 6.4%	Acute MI[2] 5.9%	Pre term LBW[3] 4.6%	Septicemia 4.3%	Asthma 4.0%	Food poisoning 2.6%
Note: 1. CVD: Cerebrovascular disease; 2. MI: Myocardial Infarction; 3. LBW: Complications of Pre-term Low Birth Weight; 4. PUO: Pyrexia of unknown origin; 5. RTI: Respiratory tract infection; 6. RTA: Road traffic accident; 7. CIHD: Chronic Ischemic Heart Disease									

Improved Hospital Services Management (IHSM)

The Operational Plan of IHSM includes following objectives:

1. To equip secondary and tertiary level hospitals for provision of the expected range of services with quality of care
2. To introduce standard waste management (phase wise) for the reduction of the diseases amongst the service providers and community people and also to improve the hospital environment
3. To reduce the maternal mortality by strengthening existing EOC activity
4. To introduce hospital referral linkage for the improvement of the patient care
5. To improve the quality of care for the private sector and NGOs hospitals / clinics / laboratories through monitoring / supervision and also to strengthen the regulatory framework
6. To provide access for the poor to specialize clinical service, i.e., reconstructive surgery
7. To strengthen the workshop of the NITOR for providing better patients care
8. To develop the capacity of NEMEW for facilitating repair and maintenance of hospital electro-medical equipment
9. To strengthen the activities of TEMO by providing extra fund for repair maintenance of vehicle and other machineries, TEMO authority repaired a good number of motor vehicles and other machineries with the utilization of provided fund.

Medical waste management activity Inter-Ministrial decision

According to decision of the Ministry of Health and Family Welfare (MOHFW) all hospitals of the country will be brought under Medical Waste Management Program by the year 2011. As per Interministrial decision "In-house" management of the medical waste is the responsibility of MOH&FW and "Out-house" management of MW is the responsibility of City Corporation or Municipality under Ministry of Local Government. The government has adopted the pit method as standard for safe disposal of medical wastes at the Upa-zilla level hospitals.

The government's guidelines for medical waste management are as follows:

The responsibility for hospitals' in-house medical waste management lies with MOHFW.

That for out-house waste management lies with Ministry of Local Government and Rural Development (MOLGRD).

The hospitals will group medical wastes in specific categories and store them locallyin separate color coded containers.

The city corporations will collect the wastes for safe disposal. The city corporations will outsource the activities to Prism Bangladesh or to other NGOs.

Committees will be formed in national, district and upa-zila levels for collection, transportation and safe disposal of medical wastes.

MOHFW will allocate fund to each hospital on per bed basis for medical waste management. All the Public & Private health care facilities in Dhaka City will be bought under MWM program by December, 2009. All the Public & Private health care facilities in the country will be under MWM program phase-wise and priority basis by 2011.

Control of communicable diseases

The prevention and control of communicable diseases represent a significant challenge to those providing health-care services in Bangladesh. Sound knowledge on the disease epidemiology is a must for the health service providers in various levels.

The Bangladesh population is namely affected by diarrheal diseases, cholera, hepatitis A & E, Malaria, Mycobacterial Disease like Tuberculosis and Leprosy, Dengue, Japanese encephalitis, Nipah virus infection, etc.

Crowding, poor access to safe water, inadequate hygiene and toilet facilities, and unsafe food preparation and handling practices are associated with transmission. Cholera is endemic Bangladesh, between 800 and 1000 cases are usually being recorded daily at the hospital of the ICCDR, B in Dhaka. Hepatitis A and E levels are usually high in the country.

Malaria risk exists throughout the year in Bangladesh. Thirteen out of 64 administrative districts are high malaria endemic areas. 98% of all malaria cases reported are from these districts, which are mainly located in the border areas of India and Myanmar.

Tuberculosis still remains as a major public health problem, which ranks Bangladesh fifth among the high-TB burden countries in the world. The present revised National Tuberculosis Programme (NTP) was launched and field implementation of DOTS (Directly Observed Treatment short course) was started in 1993.

Leishmaniasis or Kala Azar is endemic in Bangladesh and has an incidence of 175 per 100,000 per annum. It is caused by a protozoa which is transmitted from the bite of infected sandfly and may present in cutaneous or visceral forms (particularly common in Bangladesh).

Filariasis is a mosquito borne parasitic disease causillili
urogenital organs, breast, etc. with long arm disability. In Bangladesh, it is endemic in 23 districts, mostly the bordering ones. About 20 million people are already infected, most of whom are incapacitated.

Leprosy has been a major health problem in Bangladesh for a long time. Bangladesh was considered a high endemic country and was listed among ten countries with high case load (1992). Leprosy situation has changed globally after 1981 when the Multi Drugs Treatment (MDT) were introduced.

Hepatitis A virus infection is common in Bangladesh with a prevalence of about 2% to 7%. Prevalence of hepatitis C virus infection is less than 1%. Sporadic outbreak is often seen caused by hepatitis E virus infection; but presence of hepatitis D infection is not exactly known.

Polio free status prevailed from 2001 until now (June 2009) except a small window period in 2006 when 18 cases of child polio were seen in boarder areas of Bangladesh. it is assumed that these cases were imported from India.

Dengue fever/Dengue haemorrhagic fever (DF/ DHF) is a viral disease transmitted by the Aedes aegypty mosquito. It is on the increase in South East Asia. Bangladesh reported 100, 000 cases in 2005. However case fatality rate (CFR) remained <1% up to 2006.

Kala azar

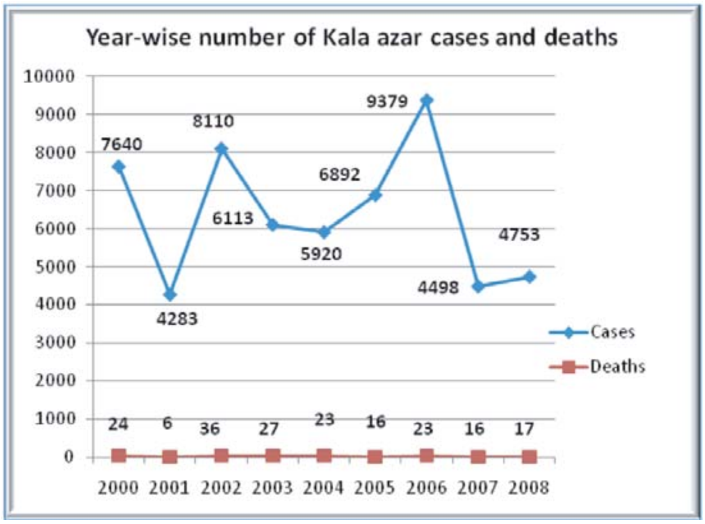
Visceral leishmaniasis (kala-azar), if left untreated, has a high mortality rate. In the territory that corresponds to present-day Bangladesh, the disease was an important public health problem during the pre-malaria eradication period in the 1950s, but, during the 1960s, it almost disappeared as a result of malaria control activities involving the widespread use of DDT residual spraying. A resurgence of the disease, including post-kala-azar dermal leishmaniasis (PKDL), occurred, however, in several parts of the country during the late 1970s when large-scale use of DDT ceased.

Kala-azar or Visceral Leishmaniasis is one of the complex of diseases, called leishmaniasis and is caused by the trypanosomatid parasite Leishmania donovani. In the Indian sub-continent it is transmitted by the sand fly, Phlebotomus argentipes. The disease presents with fever of long duration (more than two weeks) with splenomegaly, anemia and progressive weight loss. In endemic areas, children and young adults are its principal victims. Without timely treatment the disease is fatal. Kala azar and HIV coinfection has emerged as a health problem in recent years.

kala-azar has resurged in endemic regions of Bangladesh since the 1990s and has been reported in 45 districts, with the highest rates in the districts of Mymensingh, Pabna, and Tangail. In Mymensingh specifically, the average annual incidence rate between 1994 and 2004 was 5.8/10,000, and currently is as high as 300/10,000 in the most affected communities. During the last few years, kala azar situation assumed epidemic proportion with the number of reported cases increasing from 3,978 in 1993 to 8,505 in 2005. Present surveillance is weak and the current estimated total cases are about 45,000. Annually 10,000 cases are treated by the control program; but the cases treated by the private clinics and practitioners are not reported.

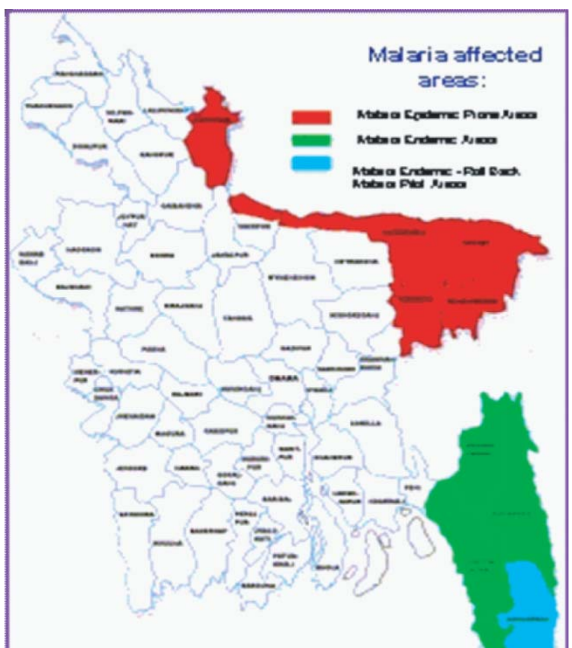
Bangladesh is now running a program against Kala azar with the objective to eliminate he disease by 2015 and with an aim to reduce the cases to less than one case per 10,000 populations in endemic upazila of Bangladesh.

India, Bangladesh and Nepal have expressed a commitment to eliminate Kala azar by 2015. In May 2005, the three countries signed a Memorandum of Understanding (MOU), in Geneva during the World Health Assembly. A Regional Strategic Plan has been prepared and endorsed by the WHO SEARO Regional Technical Advisory Group (RTAG) and partners supporting elimination.



Malaria

Malaria is one of the major public health problems in Bangladesh. Out of 64 districts in the country, malaria is highly endemic in 13 districts and 10.9 million people are at



Map of Bangladesh

Showing malaria areas

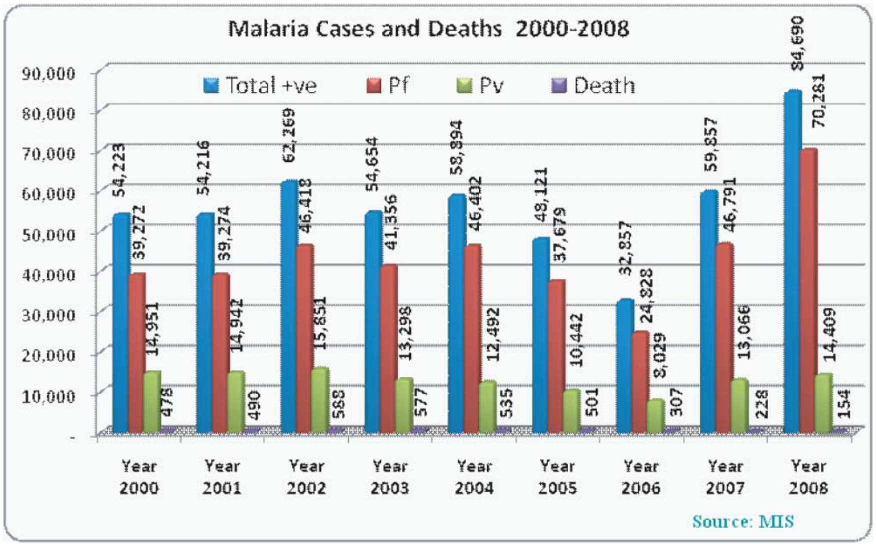
risk of the disease. More than 95% of the total malaria cases in the country are reported from these 13 high endemic districts. The three Hill Tract Districts (Bandarban, Khagrachari and Rangamati) and Cox's Bazar district report more than 80 percent of the malaria cases and deaths every year. These areas experience a perennial transmission of malaria with two peaks in pre-monsoon (March-May) and post-monsoon (September-November) periods. There is also reporting of outbreaks from bordering districts in the north and north-east. Both falciparum and vivax malaria are prevalent in the country of which the number of falciparum cases are 75% of the total cases in recent years due to increasing drug resistance. The first line drug chloroquin has been replaced by artemesinin based Combination Therapy (ACT) for treatment of falciparum malaria cases in 2004. *An. dirus, An. minimus* and *An.Philipinensis* are the principal vectors and all are susceptible to malathion and synthetic pyrethroid. Promotion and use of Insecticide Treated Nets, selective activities for containment of outbreaks and intensive campaigning for increasing awareness of the people are the main components for vector control. The malaria cases reported annually are an under estimate of the total disease burden because of shortcomings in surveillance and information management. The epidemiological data of 13 high endemic districts 2000-2008 is presented in the table below.

Malaria Epidemiological Data (2000-2008)

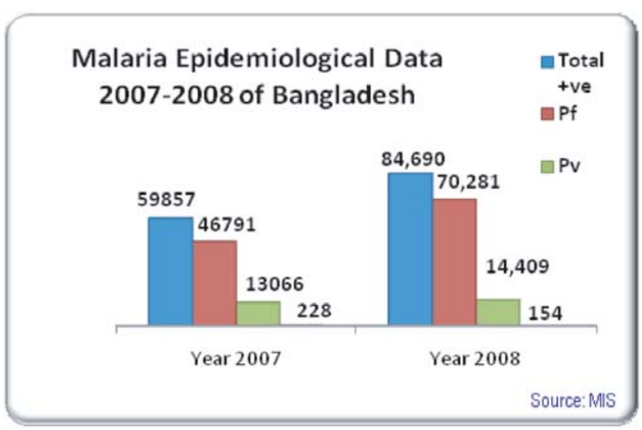
Year	Clinical Cases	Confirmed Cases			Pf %	Death
		Total +ve	Pf	Pv		
2000	294,358	54,223	39,272	14,951	72	478
2001	276,901	54,216	39,274	14,942	72	490
2002	305,738	62,269	46,418	15,851	75	588
2003	279,439	54,654	41,356	13,298	76	577
2004	224,003	58,894	46,402	12,492	79	535
2005	242,247	48,121	37,679	10,442	78	501
2006	313,794	32,857	24,828	8,029	76	307
2007	458,775	59,857	46,791	13,066	78	228
2008	526,478	84,690	70,281	14,409	83	154

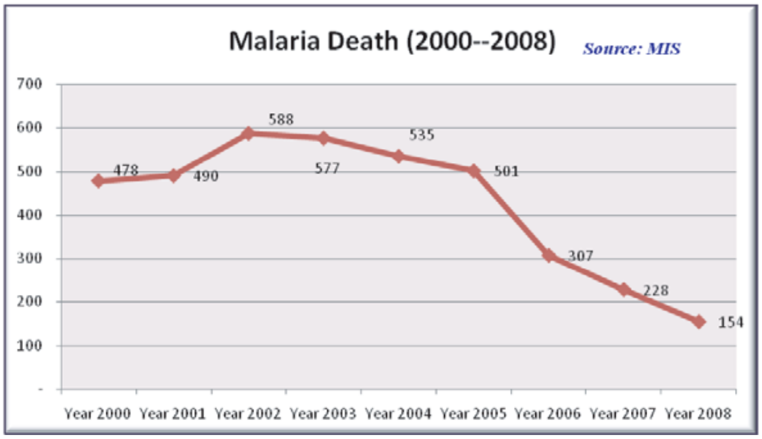
On an average, 60,000 confirmed cases are reported by the health institutions every year. This is, however, a gross under-reporting because of the fact that the medical college hospitals, specialized hospitals, NGO hospitals, private clinics and the private practitioners are not reporting malaria cases through the routine surveillance system except in few cases.

The surveillance system needs to add reports from all these service providers. Similarly those deaths are reported which are in hospitals and are diagnosed as confirmed cases during hospital admission only. Deaths are thus also grossly under reported. According to the expert opinion, the estimated burden of malaria may be three times higher than that of current reporting.

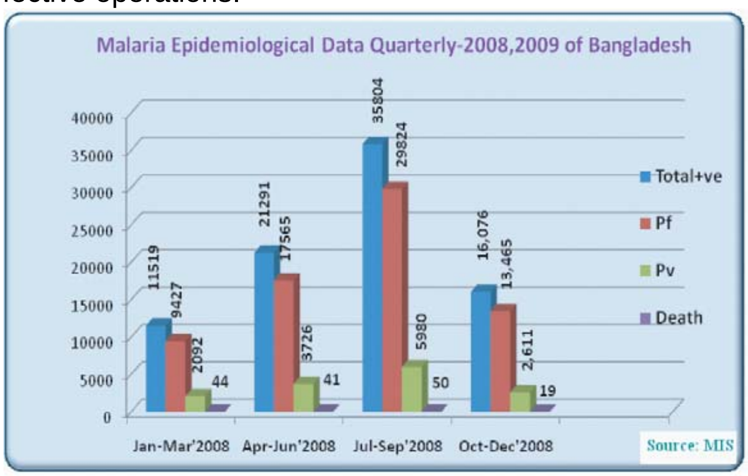


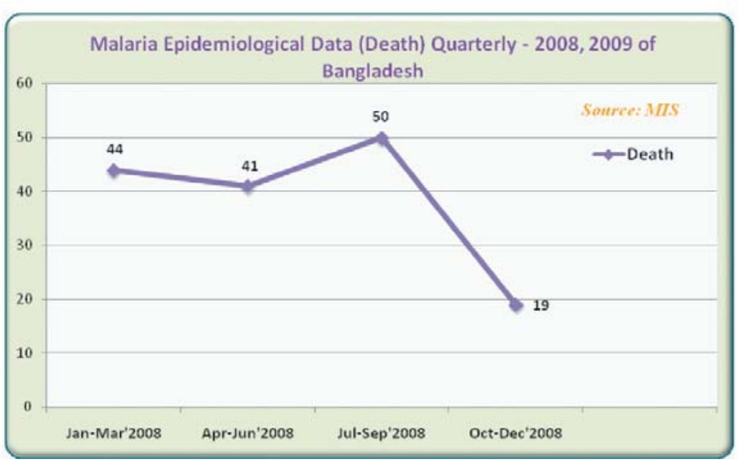
In Bangladesh, all age groups are at risk. In 2000, a total of 294,385 clinical cases were treated as against 54,430 confirmed cases. Of these, 72% were found P. falciparum and 460 deaths were reported. There are no remarkable changes in the total confirmed cases except some fluctuations year to year. Deaths are also around 500 each year with many variations. The remarkable feature is change in the P. falciparum proportion, which had been 72% in 2000 and increased up to 78% in 2005 and 2007. In 2008 the Pf% increased up to 83%. After inception of the Global Fund (GF) program the number of cases reported was increased in the year 2007 and 2008. But at the same time, number of deaths was reduced. This may be due to earlydiagnosis and treatment of uncomplicated malaria at the community level and improved referral and management of severe malaria in the hospital. The graph above shows the trend of malaria cases and deaths from 2000 to 2008. Eight districts and 34 upazilas having 7.5 million population in the east and northeast borders report focal outbreaks of malaria almost every year. A total of 37 outbreaks were reported to the Malaria and Parasitic Disease Control (M&PDC) unit in DGHS during the last 15 years from these 8 border districts. There is gross under reporting of outbreaks and sometimes reports are obtained from the mass media and print media and are delayed. The capacity to predict outbreaks, contain and deployment of appropriate preparedness and response programme is weak. Majority of the country's population lives in plain rural areas where only P. vivax malaria is reported in foci of low intensity. These sometimes lead to epidemic outbreaks.





During the 2004-2005 and 2006 -2007 biennium, with WHO support, Rapid Response Team (RRT) has been set up in each of these epidemic prone districts. Basic training of key staffs has been provided on effective and efficient management of malaria outbreaks. However a refreshers training is required for making the teams fully functional. It is expected that while these RRTs are functionally operational this will help in early detection, containment and reporting of outbreaks and reduce malaria morbidity and mortality. However response capacity at district and Upazila level needs to be strengthened and modern technology e.g. GIS needs to be introduced for effective operations.

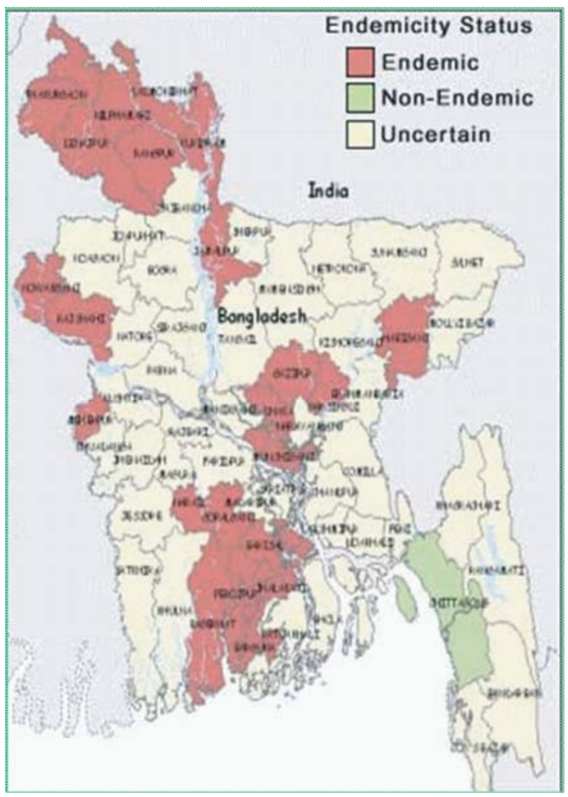




Filariasis

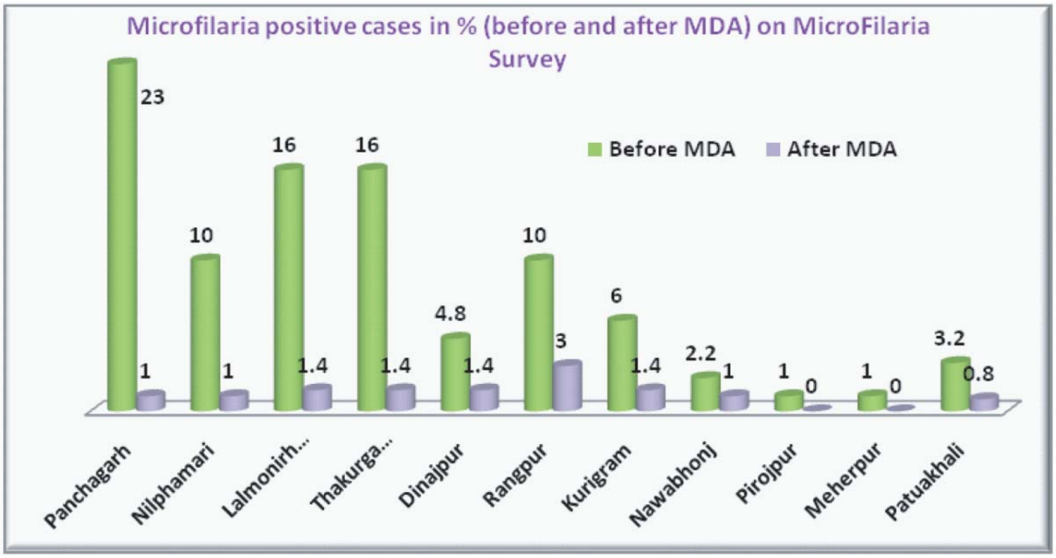
Filariasis is a mosquito borne parasitic disease causing swelling of the limbs, urogenital organs, breast, etc. with long-term disability. In Bangladesh, it is present now in 23 endemic districts, mostlybordering India. About 20 million are already infected, most of which are incapacitated and another 30 million are at risk of infection.

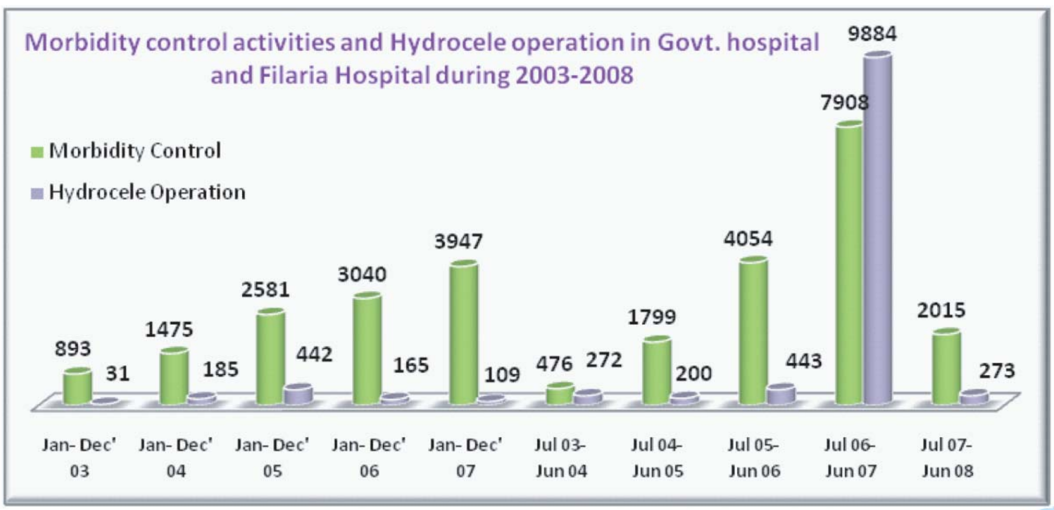
The disease can be eliminated by yearly single dose of Tab. Diethylcarbamazine and Tab. Albendazole for successive 4-6 years by Mass Drug Administration (MDA)as per WHO guideline. The morbidity can also be controlled by simple washing, hygiene, physiotherapy and surgery in case of hydrocele. Ministry of Health and Family Welfare is committed to eliminating filariasis by 2015 with global elimination of WHO commitment by 2020.



The target of Bangladesh is to eliminate the disease by 2015 through transmission and morbidity control. Accordingly Filariasis Elimination Program (FEP) was started from January 2001 as a new program under Director, Communicable Disease Control (CDC) of DGHS. The program has two principal goals: (a) to interrupt transmission of infection; and (b) to alleviate and prevent both the suffering and disability caused by the disease. The main strategy for filariasis elimination is mass drug administration (MDA) to the entire population at risk and morbidity control. The program was started in Panchagar District in October 2001 covering 0.91 million population with 93% coverage. In 2002, 5.1 million were under MDA with 87.32% coverage. In 2003 another 17.2 million, in 2004, 25 million and in 2008, 33.6 million were brought under MDA with >79.38% coverage found on post MDA coverage survey. Morbidity control activity will also continue side by side.Many lymphedema patients (7,908 cases) were trained on morbidity control management.

A Filaria Hospital has been established at Syedpur of Nilphamari in January 2003, which is the only center in the world. The disease mainly prevails in the northern parts of Bangladesh.





Division-wise distribution of filariasis patients by body parts affected (01 Jan-28 Dec 2008)

Division	Leg Swelling	Hand swelling	Breast Swelling	Scrotal Swelling	Hydrocele Operation
Rajshahi	48	4	3	74	114
Chittagong	0	0	0	0	0
Khulna	2	0	0	0	3
Barisal	0	0	0	0	4
Total	50	4	3	74	121

Source: Filariasis, Elimination Programme, Disease Control Unit, CDC,(2008) DGHS, Bangladesh. **Two Drugs**

Treatment Schedule (once in a year for 5 consecutive years)

Age	(100 mg) Tab Diethyl Carbamazine	(400 mg) Tab Albandazole
2-8 yrs	1 Tab	1 Tab
> 8 -12 yrs	2 Tab	1 Tab
>12yrs	3 Tab	1 Tab

Diarrhea

The Diarrheal disease is endemic in all countries of the South East Asia Region that includes Bangladesh. This disease constellation comprises of five major diseases, which causes the bulk of the under five mortality in Bangladesh. The diseases are associated with unsafe water and poor sanitation coupled with poor food handling practices and bottle-feeding of infants during the first few months of life. The main cause of death from acute diarrhoea is dehydration resulting from loss of fluids and electrolytes. Other causes of death are dysentery and malnutrition resulting from incorrect management of diarrhoea.Globally there are an estimated 1.8 billion episodes of childhood diarrhoea each year and diarrhoeal diseases, including dysentery, claim the lives of three million children annually. A regional programme for the control of diarrhoeal disease has been in operation since 1979.

The diarrhoeal diseases are of two types eatery diarrhoea and bloody diarrhoea (dysentery). Several microorganisms cause the disease, but major organisms responsible for the disease are V. cholera and Shigella. V.cholerae is responsible for cholera which is endemic in all countries of the South East Asia Region.

Diarrhoeal Report by Division and by Year (2001-2008)

Division	Year 2001		Year 2002		Year 2003		Year 2004		Year 2005		Year 2006		Year 2007		Year 2008	
	Attack	Death	Attack	Death	Attack	Death	Attack	Death	Attack	Death	Attack	Death	Attack	Death	Attack	Death
Barisal	25202	33	19869	19	14412	11	17986	19	15078	12	29072	5	31695	5	42584	12
Chittagong	337838	166	512215	296	379276	265	432829	277	405446	162	363710	84	446965	148	410195	123
Dhaka	518951	101	667406	140	610181	221	606782	172	606302	165	654172	46	710972	180	808390	160
Khulna	349735	56	528556	60	455683	82	401339	98	428502	81	413268	32	445631	37	476231	26
Rajshahi	412531	65	519503	121	528211	285	474848	382	441132	247	349203	49	661969	88	372203	38
Sylhet	213207	100	251296	96	209156	168	198650	119	144467	27	152425	23	178094	79	185376	34
Total	1857464	521	2498845	732	2196919	1032	2132434	1067	2040927	694	1961850	239	2335326	537	2294979	393

Division-wise case fatality rate of diarrhea in Bangladesh (2001-2008)

Division	Y2001	Y2002	Y2003	Y2004	Y2005	Y2006	Y2007	Y2008
Barisal	0.13	0.10	0.08	0.11	0.08	0.02	0.02	0.03
Chittagong	0.05	0.06	0.07	0.06	0.04	0.02	0.03	0.03
Dhaka	0.02	0.02	0.04	0.03	0.03	0.01	0.02	0.02
Khulna	0.02	0.01	0.02	0.02	0.02	0.01	0.01	0.01
Rajshahi	0.02	0.02	0.05	0.08	0.06	0.01	0.02	0.01
Sylhet	0.05	0.04	0.08	0.06	0.02	0.02	0.04	0.02
Average	0.03	0.03	0.05	0.05	0.03	0.01	0.02	0.02

Source: Director, Disease Control, DGHS, Mohakhali, Dhaka

Dengue

The fact surfaced from the collected data worldwide is that the incidence of D Dengue (fever) / Dengue Haemorrhagic (fever) has increased dramatically in all major tropical areas of the world in recent years. The frequency of epidemic activity is increasing with a trend toward larger epidemics and more severe cases. For most of the countries, the number of cases reported in the period 1981-1990 equaled or exceeded the total number reported in the previous 25 years.

Dengue (fever) is a reemerging vector borne communicable disease in Bangladesh established after its outbreak in the year 2000. Before which the disease was fairly unfamiliar though its presence was evident by a well organized scientific study in 1996-1997 by the National Control Program. The outbreak started in summer 2000 as acute febrile illness involving mainly three major cities of Bangladesh Dhaka, Chittagong and Khulna with the highest incidence rate in Dhaka.

Division	Year 2001		Year 2002		Year 2003		Year 2004		Year 2005		Year 2006		Year 2007		Year 2008	
	Attack	Death	Attack	Death	Attack	Death	Attack	Attack	Death	Death	Attack	Death	Attack	Death	Attack	Death
Barisal	3	0	54	0	0	0	8	0	0	0	1	0	0	0	0	0
Chittagong	9	0	92	0	21	1	9	0	2	0	0	0	0	0	0	0
Dhaka	2344	44	5859	49	450	9	3875	13	1033	4	2144	11	465	0	1151	0
Khulna	74	0	122	9	15	0	41	0	11	0	53	0	1	0	2	0
Rajshahi	0	0	4	0	0	0	1	0	2	0	2	0	0	0	0	0
Sylhet	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
Total	2430	44	6132	58	486	10	3934	13	1048	4	2200	11	466	0	1153	0

Source: Dengue Program, DGHS, Mohakhali, Dhaka

Nipah Encephalitis

Nipah is a zoonotic viral disease first identified in Nipah village of Malaysia in 1998-1999. The virus has been isolated from healthy fruit bats (Pteropus) which is being often found to cause infections in some animals including humans. In Bangladesh, Nipah emerged as a new killer disease from 2001.It may be manifested as a mild form of viral fever to severe form causing encephalitis or severe respiratory distress syndrome. Nipah experience in Bangladesh shows that the disease is a highly fatal one. So far 7 outbreaks have been recorded in Bangladesh.

Global Suspected Nipah Outbreaks

Year	Country	(N) Suspected Cases	Death (N)	Case Fatality Rate
1998 -1999	Malaysia	265	106	40%
1999	Singapore	11	1	9.1%
2001 -2008	Bangladesh	132	97	73.48 %
2002	India	6	-	-

Source: Nipah in short, 2006, Dr. Be-Nazir Ahmed, IEDCR, Mohakhali, Dhaka

Suspected Nipah Outbreak in Bangladesh

Year	Month	District	Suspected Cases (N)	Death (N)
2001	April-May	Meherpur	13	9
2003	January	Naogaon	12	8
2004	January	Rajbari	12	10
2004	January - March	Rajbari Dhaka, Natore, Gopalganj, Rajshahi, Manikganj, Joypurhat, Naogaon Faridpur	24	15
2004	Feb-April	Faridpur	36	27
2005	January	Tangail	12	11
2007	January	Thakurgaon	5	3
2007	April	Kushtia	8	5
2008	January - December	Manikganj & Rajbari	10	9
Total			132	97

Sources: 2009, Dr. Be-Nazir Ahmed, IEDCR, Mohakhali, Dhaka

No single measure is found to be effective in controll of the disease. Medical management depends on early diagnoses and symptomatic management including quick management of unconscious patient and respiratory distress conditions. For prevention, primary steps of creating awareness of people are the most important

HIV/AIDS

The first case of HIV/AIDS in Bangladesh was detected in 1989. Since then a total cumulative of 1495 cases of HIV/AIDS have been confirmed and reported as end of November 2008. Of these 476 have developed AIDS out of whom 165 have since died. However during the period December 2007 to November 2008 a total of 288 new HIV infection recorded and reported, of which 111 new AIDS cases identified of whom 42 died. The estimated total number of people living with HIV/AIDS is around 7,500 as of December 2006.

HIV/AIDS in Bangladesh (Since 1989)

Identified Cases	2008	Total (Cumulative)
HIV Cases	288	1495
AIDS Cases	111	476
AIDS Death	42	165

HIV prevalence in Bangladesh is low (< 1%) among the general population, even within the vulnerable population it continued to be low other than certain sections of injecting drug users. There are a number of riskfactors for the spread of HIV in Bangladesh such as: formal and informal commercial sex trade, low levels of condom use, increasing injecting drug use, and rising prevalence levels among injecting drug users.

Yearly Cases of HIV-AIDS (2003 to 2008)

Year	Cumulative Total	AIDS	Death	New cases
2003	363	34	10	115
2004	465	57	30	102
2005	658	134	74	193
2006	874	240	109	216
2007	1207	365	123	333
2008	1495	476	165	288

Over the period of 1999 -2008, HIV prevalence in central Dhaka showed rapid increase of HIV prevalence. The 8th Serological surveillance shows that the HIV rate has crossed the concentrated epidemic among IDUs. Rates in Central Bangladesh rose from 1.4% to 7% since 1999, up to as high as 11% in one neighborhood of Dhaka.

In Bangladesh, there are significant population of sex workers based in brothel, street, hotel and residence. They are composed of both male and female. An estimated 40,000 -90,000 female sex workers and 40,000 -150,000 MSM and male sex workers and 10,000 -15,000 are involved in high risk sexual behaviour and sex trade. Average client turn-over of these population ranges from 16-60/ week. The consistent condom use by female sex workers with new client past week improved significantly but still it less than 50% for all population across the country, further low by men having sex with men (MSM).

National AIDS/STD Programme:

NASP is the main coordinating and implementing body for the prevention of HIV/AIDS in the country through a coalition of 3 functionaries: National AIDS committee (NAC), The Ministry of Health and Family Welfare (MOHF&W) & the Directorate General of Health Services (DGHS). Apart from its stewardship role NASP is continuously organizing multiple awareness raising campaign and a number of advocacy workshops with different stakeholders all over the country. Besides, NASP has developed several national guidelines, manual and strategy document during the period 2005-2006 those are approved by government to augment the HIV/AIDS intervention activities.

National HIV Serological and Behavioural Surveillance

Following the UNAIDS/WHO guidelines for a revised "second generation surveillance" in a low prevalence situation, in 1998 the Government of Bangladesh set up a national HIV surveillance system. The system annually monitors HIV, syphilis, STIs and other behavioral factors that carry a risk of HIV infection.

HIV Prevalence over the Rounds

Surveillance Round	Numbers Tested	HIV (%)
1st Round	3886	<1% (0.4)
2nd Round	4634	<1% (0.2)
3rd Round	7063	<1% (0.2)
4th Round	7877	<1% (0.3)
5th Round	10445	<1% (0.3)
6th Round	11029	<1% (0.6)
7th Round	10368	<1% (0.9)
8th Round	12786	<1% (0.7)

Salient Features

- HIV Prevalence over the Rounds (1st to 8th) shows less than 1% among MARP Groups in Bangladesh
- MARP Groups are identified as IDUs, BSWs, SSWs, HSWs, MSM, Hijra, Clients of SW and Returnee Migrants.
- Size Estimation show a wide range of population among these groups.
- IDUs in Central region shows Concentrated Epidemic with HIV prevalence of 7%.

Ongoing HIV/AIDS Programs

MSA – UNICEF			
Divisions	6	Consortium	12
Districts	44	Associate NGOs	41
		DIC (Drop-in-Center)	142
MSA - Save the Children-USA			
Divisions	6	Consortium	5
Districts	45	NGOs (including 5 lead agencies)	16
		LSE (Life-Skills Education) Club	197
		YFHS (Youth Friendly Health Service) points	184
4. GFATM Round 6: HIV Prevention and Control among High-Risk Population and Vulnerable Young People In Bangladesh. - (2007-2012)			
MSA - Save the Children-USA			
Divisions	6	Consortium	8
Districts	55	NGOs (including 8 lead agencies)	46
		DIC (Drop-in-Center)	114
		IDUs	34
		SW	80

Major Achievements of HIV/AIDS Program

- Awareness creation among youth through extensive media campaign -34 million covered
- 184 YFHS points developed, LSE being given to 6,11,250 youths through 197 youth clubs

- Nearly 12 million students from Grade VI to XII exposed to HIV information in more than 14,000 educational institutions. Over 72,000 teachers trained
- 50,000 gate keepers (influential people) sensitized through advocacy
- 193,550 booklets distributed among 4 types of religious leaders.
- Baseline, operational and rapid assessments have been conducted on HIV/AIDS
- Mainstreaming done with 5 key ministries and 186 Community Based NGOs (CBOs)
- National mapping of IDUs, FSW, PLHIV and garment workers completed in 55 districts
- 34 DICs established for IDUs in 21 districts, 80 DICs for FSWs in 44 districts
- Counseling, condom distribution, STI management services provided to 32,375 FSWs
- 1,302,321 condoms distributed among IDUs and FSWs for safe sex promotion.
- ARV distributed to 200 listed PLHIV (People Living with HIV)
- Master Trainers developed for providing LSE to 660,000 garment workers
- DRE (Drug Resistance Education) provided to children of 6 City Corporation
- District AIDS Committees functionalized for coordinating an effective response to HIV.

Youth Friendly Health Services (YFHS) for Prevention of HIV/AIDS among Young People in Bangladesh

Youth Friendly Health Services (YFHS) intends to resolve the health related issues (physical, mental, Psycho-social, sexual) encountered by the 15-24 year old young people of Bangladesh, in a friendly manner, through the government, NGO and private Health Service Delivery Points (HSDPs). Since 2004, YFHS is implemented in selected 184 HSDPs of 32 districts. Technical Support of YFHS component is assigned to Ad-din Welfare Center. Other components are Life Skill Education (LSE) and Accessing Condom for Young (ACY). For mainstreaming and institutionalization of this service, a separate age-segregated line has been introduced in the relevant MIS forms of DGHS. MIS officials from the government and private facilities have received a one day orientation on the required record keeping and reporting procedures. Complete reports kept coming from October 2008. The number of 15-24 year old patients receiving services from HSDPs under YFHS from October to December 2008 is presented in Table.

Number of 15-24 year old patients received services from Health Service Delivery Points under Youth Friendly Health Service Program: "October to December 2008"

October			November			December			Cumulative		
Male	Female	Total	Male	Female	Total	Male	Female	Total	Male	Female	Total
31921	46193	78114	34337	53064	87401	28691	41909	70600	94949	141166	236115

Avian Influenza (Bird flu)

Avian Influenza or "Bird flu" is a contagious disease of animals caused by viruses that normally infect only birds and less commonly pigs. Avian influenza viruses are highly species-specific, but on rare occasions cross the species barrier to infect humans. It spreads very rapidly though poultry flocks, causes disease affecting multiple internal organs, and has a morality that can approach 100% often within 48 hours. The disease is first identified in Italy in 1878. In Asia Avian Influenza caused by H5N1 was initially recognized in Hong Kong in 1997. H5N1 Avian Influenza resurfaced in December 2003 initially in South Korea, with Additional outbreaks reported in January 2004 in Vietnam, Japan, Thailand, Cambodia and China. Despite extensive control efforts, new outbreaks of H5N1 Avian Influenza continued to be reported. In 2007, Bangladesh witnessed affection of poultry population by Avian Influenza. An epidemic is going on in poultry of Bangladesh, India and few neighbouring countries.

So far one Avian Influenza case has been identified in Bangladesh on 22 May 2008, but the person is still surviving. Total case reported to WHO till 11 February 2009 is 407. Total numbers of deaths are 254. Case Fatality is calculated as 63%. Since the detection of the first case in our country, Bangladesh is kept in Pandemic Alert Period.

IEDCR has been declared as the National Influenza Centre (NIC) of Bangladesh in 2007 by WHO. It is involved in regular and emergency activities for preparedness, prevention and control of Avian and Pandemic influenza.

Tuberculosis

Tuberculosis (TB) is a major public health burden of Bangladesh. In 2006, Bangladesh ranked sixth on the list of 22 highest burden TB countries in the world. The WHO estimated that in 2006 there were approximately 391 TB cases (all forms) per 100 000 population. It is estimated that, per 100 000 people, 225 new cases occur each year. Of these, approximately 101 per 100 000 were infectious cases, i.e. able to transmit TB in the community. It is further estimated that about 45 per 100 000 people die of TB every year. Applying these most recent WHO estimates for 2007, this translates to the following absolute numbers: 559 000 prevalent cases (all forms), 321 675 new cases (all forms), 144 397 new smear-positive cases and 64 335 people dying from TB. Although the HIV prevalence is still low, HIV poses a threat to TB control. The HIV prevalence in adult TB patient was about 0.1% as revealed in three limited surveys conducted in 1999, 2001 and 2006-07. The MDR-TB rate among new cases of TB was estimated to be 3.6% cases and 19% among re-treatment cases.

Basic TB indicators

In the absence of recent epidemiological studies, it is difficult to assess the current burden of TB in Bangladesh. Two nationwide prevalence surveys (1964-66 and 1987-88) were conducted. The first survey included a tuberculin survey to estimate the Annual Risk of Tuberculosis Infection (ARTI). There has been no new information about the ARTI trend over the past 40 years. NTP has commissioned a prevalence survey, funded by the United States Agency for International Development. Plans are also made to conduct nationwide representative drug resistance and TB/ HIV surveys. The increased service coverage was made possible by grants from the global fund to fight AIDS, Tuberculosis and Malaria (GFATM) and the Canadian International Development Agency (CIDA). Anti tuberculosis drugs needed to the treatment of the increased number of cases were provided with support of the Global Drug Facility (GDF)

Basic TB indicators	
Population (2008)	145279166
Rank among 22 high burden countries	6
Estimated incidence	all forms 223/100 000 pop./ year
New smear-positive cases	100/100 000 pop./ year
Estimated mortality	45/ 100 000 pop./ year
Estimated prevalence (all cases)	387/100 000 pop.
Proportion of MDR-TB Among new cases	3.6%
Proportion of MDR-TB Among re-treatment cases	19%
DOTS population coverage	100%
Case detection rate (2008)- New smear-positive cases	73.22%
New smear-positive cases (2007 cohort)	92%

Source: Bangladesh Bureau of Statistics (population) and WHO Global Tuberculosis Report (2009) (other indicators)

YEARLY SUMMARY OF TB CASE FINDING OF THE YEAR (2006-2008) Leprosy

	2006				
Area	Sub-Total				Grand Total
	Positive		Negative	Extra	
	New	Relap	New	New	
RURAL/ UPAZILA	89704	2645	16717	9707	118773
URBAN/METRO	9255	1279	5409	3499	19442
CDC	2806	287	2375	1155	6623
Grand Total:	101765	4211	24501	14361	144838

	2007				
Area	Sub-Total				Grand Total
	Positive		Negative	Extra	
	New	Relap	New	New	
RURAL/ UPAZILA	91606	2517	15852	10861	120836
URBAN/METRO	10264	1049	5449	4164	20926
CDC	2437	222	1934	1093	5686
Grand Total:	104307	3788	23235	16118	147448

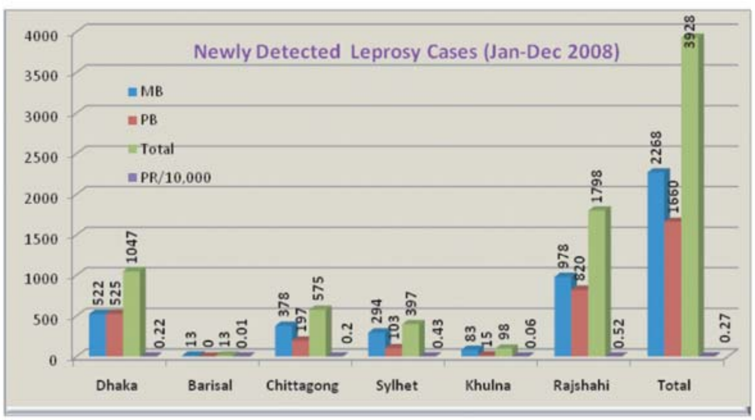
	2008				
Area	Sub-Total				Grand Total
	Positive		Negative	Extra	
	New	Relap	New	New	
RURAL/ UPAZILA	93659	2753	15069	12825	124306
URBAN/METRO	10289	1165	5660	4486	21600
CDC	2425	220	1463	1048	5156
Grand Total:	106373	4138	22192	18359	151062

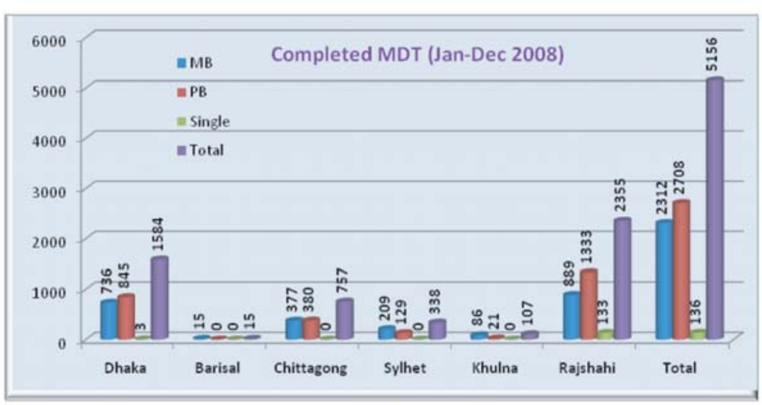
Year	Population (In Million)	Estimated +ve case	Detected +ve case	Case Detection Rate	Estimated Incidence per 100,000+00
2006	140.7	143,514	101,765	70.9	102
2007	142.96	144,390	104,307	72.24	101
2008	145.28	145280	106373	73.22	100

Leprosy

Elimination of leprosy as a public health problem, defined by a registered prevalence of less than one case per 10 000 population, was achieved by Bangladesh in 1998, and steady reduction in prevalence is ongoing. It is less certain whether a sustained reduction in case detection is occurring, with little overall change in some longstanding programme areas, though the overall annual new case detection rate has fallen by overone-third between 1996 and 2004, from 9.8 to 6.1 per 100 000. National Leprosy Elimination Campaign of 1999 detected countries, and mainly in low endemic areas. relatively fewer new cases than in other Further challenges remain in the area of urban leprosy control, where leprosy case finding represents 30% of the whole country, but public health infrastructure and community organization is weakest. Sustaining of leprosy services in the long term is a significant concern, and new modes of collaboration, with a more technical, supportive role for NGOs in some areas is being piloted.Major activities done by National Leprosy Elimination Programme (NLEP) during 2008 under WHO Biennium 2008-2009

Name of Activities	No. of Batches	Total numbers of Participants
Orientation on Leprosy for Scouts unit leaders	07	420
Orientation on Leprosy for Religious leaders	08	320
Orientation on Leprosy for Village Doctors	16	640
Total		**1380**

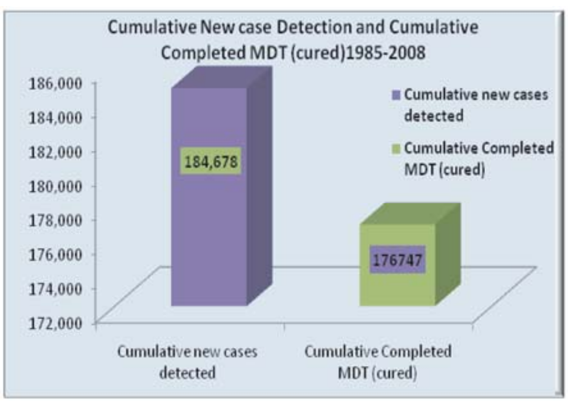


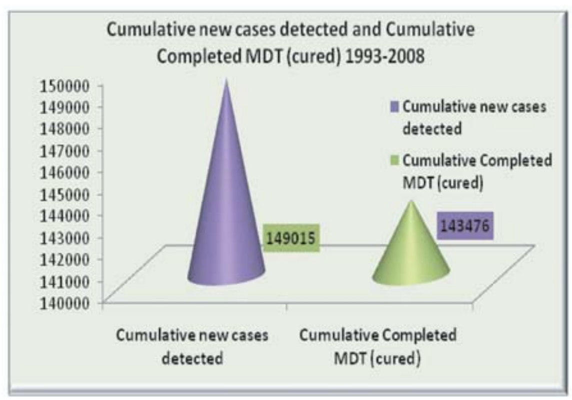


Newly detected cases of Leprosy (January -December 2008)

Division	Population	MB	PB	Total	PR/10, 000
Dhaka	46919988	522	525	1047	0.22
Barisal	9087073	13	0	13	0.01
Chittagong	28424321	378	197	575	0.20
Sylhet	9207106	294	103	397	0.43
Khulna	16920166	83	15	98	0.06
Rajshahi	34720511	978	820	1798	0.52
Total	145279165	2268	1660	3928	0.27
%		577393	42.2	100	

	Completed MDT (Jan -Dec 2008)			
Division	MB	PB	Single	Total
Dhaka	736	845	3	1584
Barisal	15	0	0	15
Chittagong	377	380	0	757
Sylhet	209	129	0	338
Khulna	86	21	0	107
Rajshahi	889	1333	133	2355
Total	2312	2708	136	5156
%	44.84	52.52	2.64	100





Emergency Preparedness & Response (EPR) Program

Bangladesh is a low lying deltaic country in the Indian subcontinent with a land of 147, 570 sq. km and 145 million inhabitants. Because of the geographical location, monsoonal influence and the impact of climate change, Bangladesh has been experiencing varieties of natural calamities such as cyclones, tornados, tidal surge, floods, flash floods, and land slide more frequently over time. Though significant earthquakes did not take place since over 100 years, potential threats of earthquakes and tsunamis are inevitable since the Asian fold crosses the eastern border of the country. Bangladesh has also experienced human induced disasters like fire, infrastructure collapse, water logging, road and river traffic accident, disease epidemics and various form of pollution. These events result in loss of valuable lives including injury, increased number of communicable diseases and huge economic loss, and thereby cause unbearable miseries to the affected population. Although we have no control over disaster, if we have enough skilled manpower, adequate logistics and appropriate policy at the right levels, then we can reduce the adverse health impact of disaster to a greater extent.

Current Activities

1. Training/workshop on psychosocial support, mass casualty management, preparedness and response in emergency, food & nutrition in emergency, vulnerability & capacity assessment for hospital staff & health personnel in various tiers.
2. Orientation course for field staff on disaster mitigation and for disaster focal points from divisional & district levels.
3. Training on emergency health information management for statistical assistants, emergency information management and emergency disease surveillance for medical officers.
4. Joint simulation exercises with Bangladesh Red Crescent Society (BDRCS) at most cyclone prone districts and training on search, rescue, evacuation and first aid during preparedness following disaster in collaboration with BDRCS and community (multi-sectoral approach).
5. Procurement of essential medicine, emergency response vehicle, equipments & other logistics for emergency response including bleaching powder and water purification tablets.
6. Consultative meetings on EHA bench marks, review & update of SOP, development of an earthquake/infrastructure collapse response plan, action plan on information and control regarding hospital preparedness plan for health sector.
7. Establishment of fully functional health sector disaster management institute

Recent capacity building activities at a glance (during Jan. 08 -Dec. 08)

Type	Activity Name	Source	Trainings	Trained
Training	Emergency Health Care for Health Care Providers	WHO	11	440
Training	Emergency Health Care for Community Level	WHO	31	1395
Training	Emergency Health Information System Mitigation.	WHO	21	415
Training	Emergency Preparedness & Response	HNPSP	23	858
Training	Mass Casualty Management	HNPSP	7	315
Training	Mass Casualty Management	WHO	6	198
Training	Psychosocial Support	WHO	3	120
Training	TOT on EPR	WHO	2	55
Training	TOT on Emergency Preparedness	HPNSP	4	120
Training	Vulnerability & Capacity Assessment	HPNSP	6	169
Workshop	Health Education & Personal Hygiene	WHO	22	820
Workshop	Vulnerability & Capacity Assessment	WHO	3	110

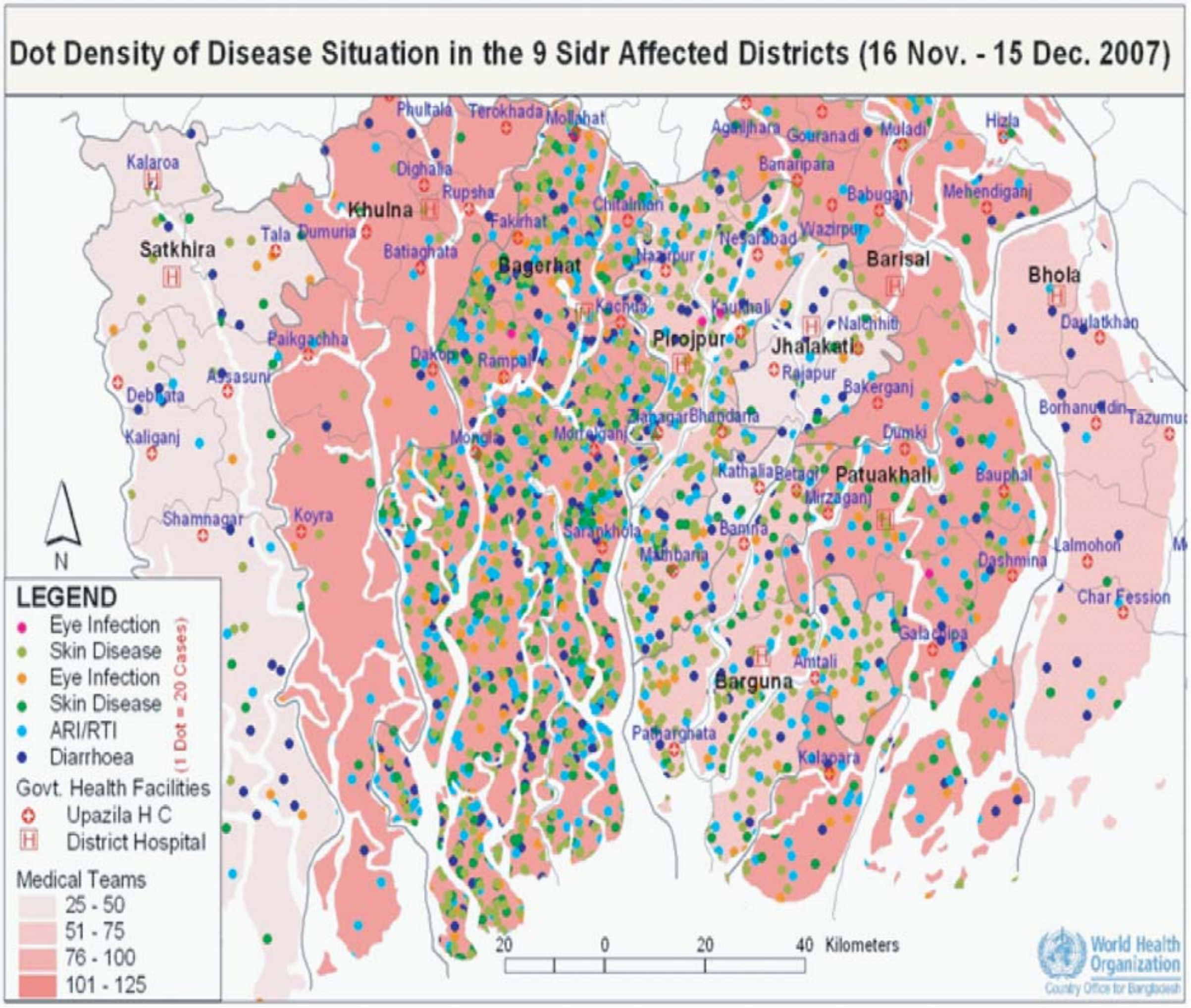
Response to Super Cyclone SIDR

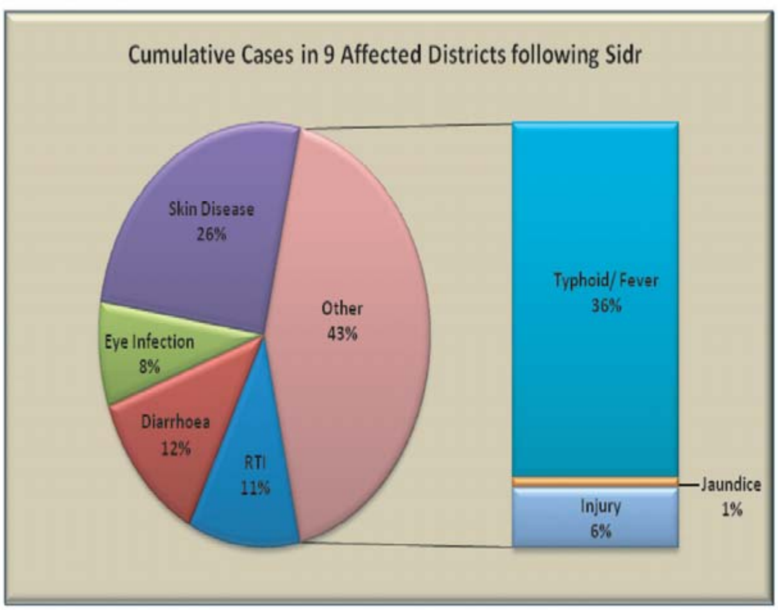
Super Cyclone SIDR hit the country on 15 November 2007 which caused a massive damage to the health infrastructures mainly in 9 districts of southern Bangladesh along with other 21 districts moderately and partially affected. Approximately 3500 lives were lost, 16 thousand people were injured and the safe drinking water sources and sanitary infrastructures were inundated by dirty water caused by dead animals, human bodies and other debris. The extent of casualty was quite minimum. The experts says its only because of better preparedness.

In response to SIDR consequence, the Directorate General of Health Services of MoHFW took necessary steps to mitigate the adverse effects. Thus monitoring the health situation was in place as ongoing basis and 690 medical teams were worked in 9 severely Sidr affected districts Health cluster coordination meetings (MoHFW, health related NGOs, UN & donor agencies) were continued in Dhaka and 9 districts; Members of the Armed Forces with local administration worked round the clock for rescuing assets and distributing relief goods, rice, safe drinking water, Water Purification Tablet (WPT) in the affected areas; Health education Bureau, (DGHS) had continued to further strengthen health promotional activities to build awareness among the affected population with technical support from WHO; emergency life-saving drugs had been supplied by MoHFW, WHO and other stakeholders and regular dispatch of medical supply was continuing to replenish emergency buffer stock; vacant posts of professors, associate professors, assistant professors and senior consultants were filed up in the medical colleges of affected areas by MoHFW; situational health assessment was completed by Marlin on behalf of DGHS, IEDCR & WHO existing disease surveillance form were redesigned for better reporting.

In response to Sidr, WHO also took necessary initiatives to avert the consequences of Sidr by deploying public heath specialists who were providing support for coordination among GOB, UN agencies, NGOs and other stake holders through district health cluster coordination meeting, initiating training on health education, capacity building for the health personnel and surveillance support.

WHO also provided 4 water purifying plants and 3 water pumps that are installed in Morelgonj and Sharankhola upazilas of Bagerhat district, Pathorghata upazila in Barguna district and Mothbaria upazila in Pirozpur district to ensure safe drinking water in health facilities and for affected population and therefore, no disease outbreak occurred in the Sidr affected areas.


Dot Density of Disease Situation in the 9 Sidr Affected Districts (16 Nov. - 15 Dec. 2007)
LEGEND
Eye Infection
Skin Disease
Eye Infection
Skin Disease
ARI/RTI
Diarrhoea
(1 Dot = 20 Cases)
Govt. Health Facilities
Upazila H C
District Hospital
Medical Teams
25 - 50
51 - 75
76 - 100
101 - 125
N
20 0 20 40 Kilometers
Khulna
Satkhira
Bagerhat
Pirojpur
Jhalakati
Barisal
Bhola
Patuakhali
Barguna
World Health Organization
Country Office for Bangladesh


Cumulative Cases in 9 Affected Districts following Sidr
Skin Disease 26%
Eye Infection 8%
Diarrhoea 12%
RTI 11%
Other 43%
Typhoid/ Fever 36%
Jaundice 1%
Injury 6%

Cyclone Sidr Recovery Activities

- Principles of Emergency Health Care training at all levels are ongoing for establishing Emergency Medical Services. 4 TOT training 'Principles on emergency health care in EPR' for Master Trainers on EMS system and subsequent implementation training on EPR in Barisal and Khulna divisions was held and at remaining district & UZ levels will held shortly.
- Video clips, manuals, booklets, placards & posters will be prepared for mass awareness on the three basic principles of Emergency Medical Services (EMS). These IEC materials will be made available in the TV channels, Radios, Cinema houses as trailers, in all levels.
- Procurement of Emergency Medical equipment and drugs is on process for replenishing buffer stock to reduce avoidable morbidity, mortality and disability due to disaster.
- More training on psychosocial support has been scheduled and preparation of basic messages for psychosocial support related IEC materials like pictorial poster, leaflet, audio cassette/CD and training manual in English and Bengali version is underway.
- Conduct training on caregivers on grief counseling, thus psychosocial support will provide for traumatizes people through counseling and proper referral system in Sidr affected areas.
- A hand book for identifying the vulnerable areas by category and to assess environmental impact on human health will be prepared for the Health Managers and health workers of other government, NGO and INGOs for preparedness plan and efficient resource management.

Non-communicable disease

There have been a number of demographic and lifestyle changes over the past two decades in Bangladesh. Improvement in health care delivery has increased the number in aging population. Similarly, industrialization has enhanced urbanization and change in life pattern. The rise in aging population and urbanization are accompanied by an increase in non-communicable diseases and mental health problems. The lifestyle changes associated with change in dietary pattern, lack of physical exercise and rest and recreation, use of tobacco etc., are all changing the epidemiology of morbidity and mortality in Bangladesh. Noncommunicable diseases such as cardio vascular disease, diabetes mellitus, cancer, chronic renal disease, mental problems are some of the important emerging non-communicable health problems in country like Bangladesh. Other conditions like injuries especially road traffic injuries, violence against women are on the rise. Data from some of the major health institutions who deals with noncommunicable disease are presented here. The data will give an indirect impression about the disease burden of NCDs in the country.

Cardiovascular Diseases

Cardiovascular diseases are the leading cause of death and disability in most of industrialized countries. They are also increasing in the developing world as well as in our country too. Major Cardiovascular diseases include Coronary Heart Disease, Hypertension,Rheumatic Heart disease etc. Statistics from National Institute of Cardiovascular Disease (NICVD) and National Center for Control of Rheumatic Fever and Heart Diseases indicates that numbers of patients suffering from cardiovascular diseases are rising over the years.

Statistics of National Institute of Cardiovascular Diseases (NICVD) 2002-2008

Year	Admissi on Total	Outdoor patients (No.)				Average daily admissi on (No.)	Average daily OPD patients (No.)	Ave rage length of stay (d)	Bed occupancy rate
		Male	Female	Child	Total				
2002	17081	52740	29532	4674	86944	47	238	6.91	129.63
2003	20083	54550	31939	5150	91639	55	251	7.07	157.76
2004	21522	56482	31250	4857	92589	59	253	6.90	164.03
2005	22419	59950	34608	5497	100055	62	274	6.46	160.39
2006	24376	61565	34861	6060	102486	67	281	6.47	175.80
2007	29147	76732	41792	7417	125941	80	345	5.48	174.80
2008	33946	91147	47889	8534	147570	93	403	5.21	147.70

Number of ETT performed at NICVD by year 2001-2008

Year	Male	Female	Total
2001	210	24	234
2002	454	55	509
2003	731	102	833
2004	828	167	995
2005	823	180	1003
2006	1233	321	1554
2007	1437	301	1738
2008	1798	339	2137

Cath Lab procedures done at NICVD by year (2001-2005)

Year	CAG	Cardiac cath			Renal Angiogram	Peripheral Angiogram	Interventional Procedure			Total
		Ped	Adult	Total			PCI	PTMC	Others	
2001	995	146	104	250	4	116	163	74	4	1654
2002	1378	111	95	206	6	97	228	92	11	2018
2003	2827	144	164	308	13	42	371	189	12	3762
2004	3210	55	170	225	69	93	599	273	13	4482
2005	2780	62	155	217	6	85	488	295	11	4109
Total	11190	518	688	1206	98	433	1849	923	51	16025

Cath Lab procedures done at NICVD by year (2006-2008)

Year	Interventional procedures											Cardiac Cath	Other	Total
	CAG	PCI	Device closure	Angiography		Angioplasty		PTMC	TPM	PPM	EPS			
				Renal	Peri-pheral	Renal	Peri-pheral							
2006	3105	584	1	-	106	-	7	280	675	321	161	229	4	5473
2007	3266	574	-	-	87	-	43	20	850	359	04	295	-	5698
2008	3980	889	0	-	112	-	23	130	741	414	113	380	18	6800
Total	10351	2047	1	-	305	-	73	430	2266	1094	478	904	22	17971

Open and closed heart and vascular surgeries performed at NICVD by year

Year	Open Heart Surgery					Closed Heart Surgery	Vascular Surgery	Total
	CABG	Valve	Congenital	Other	Total			
2001	60	134	133	3	330	157	-	487
2002	112	89	210	4	415	151	-	566
2003	170	142	162	22	496	140	-	606
2004	180	159	205	17	561	95	-	656
2005	267	102	237	20	626	93	-	719
2006	226	113	255	28	622	70	500	1192
2007	188	165	256	46	655	58	568	1281
2008	233	182	327	21	763	64	992	1819
Total	1436	1086	1785	161	828	764	2060	8120

Rheumatic heart disease (RHD)

Rheumatic heart disease (RHD) is a consequence of rheumatic fever and is the commonest heart ailment among the pediatric age-group and young adults of Bangladesh. In our country context, poverty, overcrowding, lack of nutrition and lack of health education concentrates the problem more. Statistics from the National Center for Control of Rheumatic Fever and Heart Diseases shows the increasing trend of rheumatic heart disease.

Statistics of National Center for Control of Rheumatic Fever and Heart Diseases

Number of Patients seen in OPD of National Center for Control of Rheumatic Fever and Heart Diseases year 2006-2008

Patients	2006		2007		2008	
	Male	Female	Male	Female	Male	Female
New patients in OPD	3964	5910	6744	10116	7620	11429
Old patients in OPD	11527	17290	12410	18614	11720	17398
Total	15491	23200	19154	28730	19340	28827

Distribution of patients attending OPD of National Center for Control of Rheumatic Fever and Heart Diseases by age group (2008)

Age group in years	No. of patients	Percentage
1 – 4	819	1.7
5 – 14	15799	32.8
15 – 49	29767	61.8
50 +	1782	3.7
Total	48167	100

Diabetes

Diabetes is a major public health problem for not only developed countries but also developing countries like us. The prevalence and the morbidity-mortality data due to diabetes are grossly under estimated all over the world. And the economic burden of the disease is also increasing. Without emphasis on prevention it will not be possible for Bangladesh to combat the disease epidemic. We are yet to develop large scale structured program on primary prevention of diabetes or secondary prevention of diabetes complication. Next to public sector, Diabetic Association of Bangladesh (BADAS) is playing a major role in treatment of diabetes in Bangladesh. Data from BIRDEM and affiliated associations under BADAS shows that diabetic patients are ever increasing.

Number of diabetic patients seen in BIRDEM OPD by Fiscal Year

Patients category	2001-2002	2002-2003	2003-2004	2004-2005	2005-06	2006-07	2007-2008
New Patients	20603	20883	21462	22324	22559	22363	22705
Total Patients	270190	291073	312535	334859	357418	37981	402468

Source: BIRDEM, Shahbagh, Dhaka

Discipline-wise number of patients attending different specialized OPDs of BIRDEM

Year	Endocrinology	Cardiology	Surgery	Ophthalmology	Obstetrics Gynecology	Dentistry	Skin diseases	Total
2003-2004	16096	20654	1015	69426	24192	19047	18833	169263
2004-2005	18298	23142	1103	69783	26631	21359	20326	180642
2005-2006	21269	29884	24023	69329	25979	24154	24755	219393
2006-2007	20977	28454	19110	75355	25760	20738	21704	212098
2007-2008	20483	25435	19248	70131	24625	17774	21662	394358

Discipline-wise number of patients attending different specialized OPDs of BIRDEM (contd...)

Year	Pulmonology	Pediatrics	Pediatric Endocrinology	Pediatric Neurology	ENT	Total
2007 -2008	1502	10298	1566	777	9534	23677

Source: Annual Report of Diabetes Association of Bangladesh (2003-2008)

Distribution of diabetic patients registered by BIRDEM

Fiscal Year	No. of registered diabetic patients		
	Male	Female	Total
2003 -2004	113982	88212	202194
2004 -2005	115427	94542	209969
2005 -2006	125149	116007	241156
2006-2007	82420	91777	174197
2007-2008	149890	89157	239047

Source: BIRDEM, Shahbagh, Dhaka; Note: Report were collected from 52 of diabetic affiliated institution

Cancer

Cancer is emerging as a public health concern worldwide. Though cancer occurs predominantly in elderly people of developed countries, developing country like Bangladesh is also having increased number of malignancies annually. As the health system is improving, mortality rate is gradually declining with the consequence of increased people in the elderly group. Rapid urbanization, environmental pollution, and change in lifestyle along with change in food habit are influencing the rise in number of cancer incidences of our country. The National Institute of Cancer Research and Hospital (NICRH) is the leading institute for cancer related hospital services and programs. Some of the Cancer related data provided by NICRH are included in the current health bulletin.

Number of Cancer patients treated at NICRH 2006-2008

Patients	2006		2007		2008	
	Male	Female	Male	Female	Male	Female
New patients in OPD	5130	2660	5712	2499	5924	3808
Old patients in OPD	3981	2261	4249	3667	8947	7583
Indoor patie nts	599	419	617	549	612	553

Source: Cancer Institute, Mohakhali, Dhaka

Discipline wise number of patients attending different specialized OPDs

Year	Surgery OPD	Medicine OPD	Gynae OPD	Radiotherapy OPD
2006	3319	10457	1932	14378
2007	3945	13778	2147	16879
2008	4078	15276	2472	22528

Distribution of patients attending the OPD by age group (2008)

Age group in years	No. of patients	Percentage
14	348	3.6
15-24	511	5.3
25-34	893	9.2
35-44	1633	16.8
45-54	2347	24.1
55-64	2033	20.9
65-74	1579	16.2
75-84	244	2.5
85-94	106	1.1
95	38	0.3
Total	9732	100.0

Distribution of Top ten diseases by gender among patients attending OPD of NICRH in 2008

ICD-10	Name of diseases	Male (%)	Female (%)	Total (%)
34	Lung	1446 (25.5)	227 (5.6)	1673 (16.5)
50	Breast	17 (0.3)	1039 (25.6)	1046 (12.0)
53	Cervix		872 (21.5)	872 (9.0)
77	Lymph node and lymphatic	420 (7.4)	167 (4.1)	587 (6.1)
15	Oesophagus	334 (5.9)	138 (3.4)	472 (4.7)
16	Stomach	272 (4.8)	101(2.5)	373 (3.6)
32	Larynx	306 (5.4)	40 (1.0)	346 (2.7)
22	Liver	187 (3.3)	65 (1.6)	252 (2.4)
2.9	Tongue	148 (2.6)	57 (1.4)	205 (2.0)
80	Unknown Primary	335 (5.9)	109 (2.7)	444 (3.1)

Renal Disease

Various type of renal disease is occurring in our country that includes not only acute but also chronic renal failure. Lack of optimum personal hygiene, inadequate health education and improper lifestyle influences the incidences of renal disease. Chronic renal failure not only increases disease burden but also creates huge economic burden. National Institute of Kidney Disease and Urology has taken a central role in treating the renal disease. Data from NIKDU shows that annual incidences of patients suffering from different type of renal diseases are constantly increasing.

Number of patients seen in OPD and Indoor of National Institute of Kidney Diseases & Urology (NIKDU) 2006-2008

No. of Patients	2006		2007		2008	
	Male	Female	Male	Female	Male	Female
New Patients in OPD	14876	7745	18951	9107	22445	11091
Old Patients OPD	5125	2750	6382	3169	7748	3830
Indoor Patients	2314	1219	2479	1343	1610	1583

Discipline wise number of patients attending different specialized OPDs of NIKDU

Year	Medicine OPD	Surgery OPD	Pediatrics OPD
2006	19076	14214	2828
2007	22958	17983	3318
2008	28265	20391	3515

Distribution of patients attending the OPD of NIKDU by age group in 2008

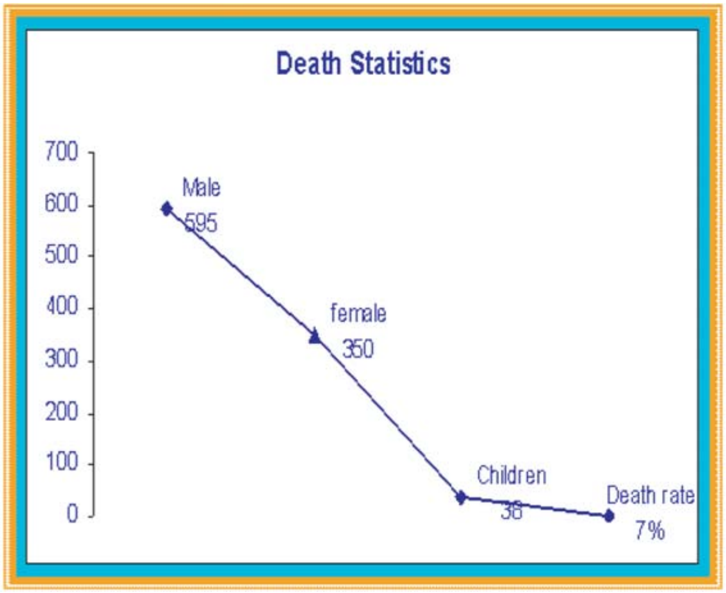
Age Group	No. of Patients	Percentage
1-4yrs	150	0.30%
5-14yrs	2205	4.50%
15-49yr s	30,875	63%
50+yrs	15437	32%
Total	48667	100%

Distribution of Top ten Diseases by gender among patients attending OPD of NIKDU in 2008

Name of Disease	Male	Female	Total
ARF	50500	25000	75500
CRF	5500	5000	10500
ESRD	4000	2200	6200
AGN	5000	1540	6540
Renal Stone	1000	500	1500
LUTS	8200	4000	12200
UTI	1000	2000	3000
Back Pain	500	1300	1800

Bladder Stone	100	110	210
BOO	200	20	220
Total	76000	41670	117670

Number of Total Deaths among the indoor patients in NIKDU (Jan.-Dec.)2008
Performance Records (Operation) Dept. of Urology -NIKDU:



Name of Operation	Jan.-2007 to Dec.-2007	Jan.-2008 to March,2008
Open Operation	523	121
Endoscopes procedure	662	93
ESWL	181	52
Other procedure's	602	179
Total	1968	445

Note: Open Surgery: Pyeloplasty., Pyelolithotomy, Nephrolithotomy, Anatrophic nephrolithotomy. Nephrostone : a)
Radical; b) Simple: Nephrostomy, Ureteroneocystostomy Cystectomy with diversion: a) Radical; b) Simple: Urethoplasty,Urethral dilation Penectomy, Cystolithotomy, SPC., Meatoplasty, Hypospadias repair,
2. Endoscopic
ı Upper Endourology: i) URS + ICPL, ii) PCNL, iii) ICPL, iv) Stenting
ı Lower Endourology: i) TURP, ii) TURBT, iii) OIU, iv) Cystoscopy,
v) Cystolitholapexy
3. Laparoscopic Urology: Nephrectomy, Pyeloplasty, Ureterolithotomy, Varicocele, Renal cystectomy, Urethrocystoscopy, Biopsy Optical Internal Urethrotomy , Stent Removal, Circumcision, Cystolitholapexy, Stenting
4. Transplant 2008 : 4 (Four) nos

Mental Health

Many people in our country suffer from mental illness and some of them are predicted to have serious and disabling mental disorders, and some to have psychosomatic disorders. Though epidemiological studies are not available, small scale studies suggest that prevalence of mental disorders is on increase in our country. National institute of Mental Health (NIMH) is one of the key institutes in treatment of mental disorders. To focus some light on the mental health related illness, this bulletin has included some of the patient related statistics of NIMH.

Number of patients seen in OPD and Indoor of NIMH

Patients	2006		2007		2008	
	Male	Female	Male	Female	Male	Female
New Patients in OPD	8001	4646	8959	5175	12692	9209
Indoor Patients	655	350	671	349	749	427

Distribution of patients attending the OPD by age group in year 2008

Age Group	NO. of Patients	Percentage
5- 14	1,039	8.19%
15- 49	3,529	67.20%
50 +	3,124	24.61%

Distribution of top 10 Diseases by gender among patients attending OPD of NIMH

Name of Disease	Male	Female	Total
Schizophrenia	4120	2138	6258
Bipolar Mood Disorder	2140	1139	3279
Depression	660	490	1150
GAD (Generalized Anxiety Disorder)	325	240	565
OCD	310	236	546
Conversion disorder	416	302	718
Phobic Disorder	272	110	337
MR	284	211	495
Epilepsy	275	203	478
Conduct Disorder	223	209	432

Top 10 cause of psychiatric morbidity among the indoor patients

Name of the Disease	Number of Cases	Percentage
Schizophrenia	327	32.80
Bipolar Mo od Disorder	215	21.57
Depression	112	11.24
Suicide and Para -suicide	40	4.12
Stupor	32	3.20
Mental Disorder	51	5.12
Extra -pyramidal Syndrome (EPS)	44	4.42

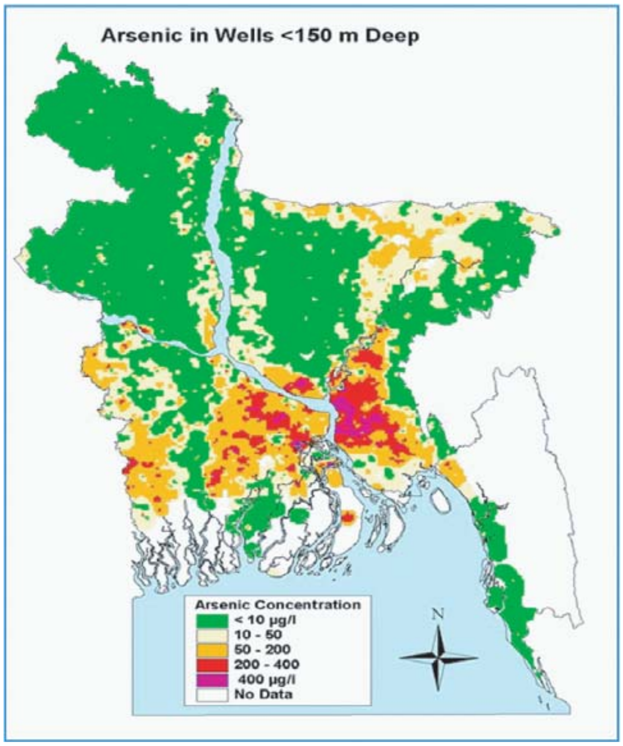
Substance abuse with Complication	56	5.62
Conversion and Dissociative Disorder (Hysteria)	100	10.03
Personality Disorder	20	2.00
Total	997	100%

Arsenicosis

In Bangladesh, arsenic contamination of water in tube-wells was confirmed in 1993 in the Nawabganj district. Department of Public Health Engineering (DPHE) at first identifies arsenic in 4 tubewells in Chamagram village of Chapainawabgong sadar Upazilla in Chapainawabgong district of Rajshahi Division in 1993. In 1994 Occupational and Environmental health Department of NIPSOM confirmed 8 patients with visible sign of skin lesion in the same village. In 1996 it was 23, in 1997 it was 42, in 1998 it was 86 and according to recent survey in 2008 the total number is 32,380.

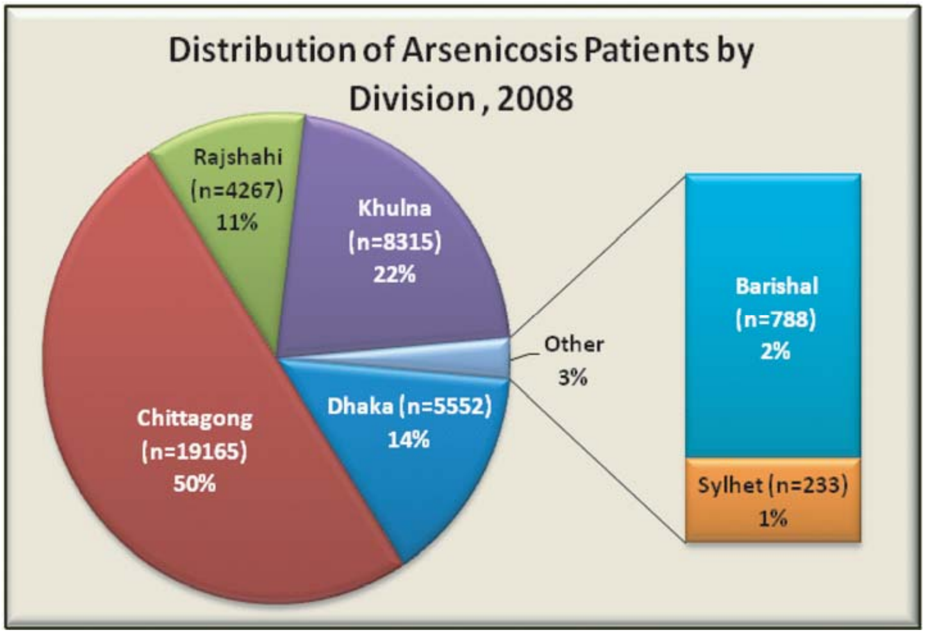
It is now clear that groundwater in the deltaic lands of Bangladesh is contaminated by significant levels of arsenic. It has been estimated that 30% of shallow tubewells in Bangladesh, that is, wells not deeper than 150 metres, are contaminated with arsenic, exceeding the allowable Bangladesh maximum of 50 mg/L

62 Districts out 64 districts are now with arsenic contamination in Tube-well water.



(50 parts per billion) and 46% of wells exceeded the World Health Organization's recommendation of 0.01 mg/L (10 parts per billion). The equivalent figures for deep wells exceeding 150 metres were 1% and 5% respectively. It was estimated that 66 million people drinks tube-well water exceeding the Bangladesh standard level for arsenic concentration (0.05 mg/) and the WHO's recommended level (0.01 mg/l). Non Communicable Disease and Other

Public Health Intervention (NCD&OPHI) is the 7th of 38th Operational Plan of Directorate General of Health Services. Arsenic related health problem mitigation program is implemented under this operational plan which is guided by National Policy for Arsenic Mitigation-2004. In 2008-2009 fiscal year, searching for patient of arsenicosis in 55 upazillas is completed. In 31 Upazillas searching is ongoing.



Injury

Injury is a major, but under-recognized, public health problem in the developing world including Bangladesh. Drowning, burns, falls, transport injury, poisoning, cuts, animal injury, suicide and violence are the most common mechanisms of injury. Bangladesh Health and Injury Survey (BHIS) conducted by Centre for Injury Prevention and Research, Bangladesh (CIPRB) revealed that injury is now the leading cause of children from the age of 1 to 17 years. BHIS revealed that as traditional causes of child death, communicable and non-communicable diseases, has declined the issue of child injury has emerged as a severely under-recognized issue.

Centre for Injury Prevention Launched the Prevention of child injuries through social-intervention and education (PRECISE) programme in 2005. PRECISE was developed to combat this silent epidemic, to reduce child and parental mortalities, morbidities and disabilities due to injury by developing cost effective injury prevention methods that can be replicated with ease in a low-income setting. This is the largest community based injury intervention programme ever implemented in the developing world. A population of more than 800,000 is covered by the project, three rural and one urban area, and is monitored by a unique injury surveillance system developed by CIPRB which records all changes in injury patterns. This monitoring tool is critical for all aspects of the project, to evaluate the effectiveness of the interventions and ascertain what can be replicated, up scaled and applied throughout Bangladesh and in other developing nations. The program has been extended to 2010 and the data collection is now done quarterly instead of previous monthly system. The programme consists of three main components:
1. Home Safety
2. School Safety
3. Community Safety

Since 2005, in Dhaka Medical College Hospital (DMCH) is providing one stop service to females suffering from domestic Patient statistics of National Institute of Traumatology, Orthopedics and Rehabilitation (NITOR) 2008 National Institute of Traumatology, Orthopedics and Rehabilitation deals with patients who suffer from orthopedic trauma due to various causes that includes Road Traffic Accidents. Patient statistics from NITOR encompasses the injuries due to various causes to some extent, as it is the central institute for trauma and orthopedics as well as rehabilitation.

Number of patients seen in OPD and Indoor of NITOR 2006-2008

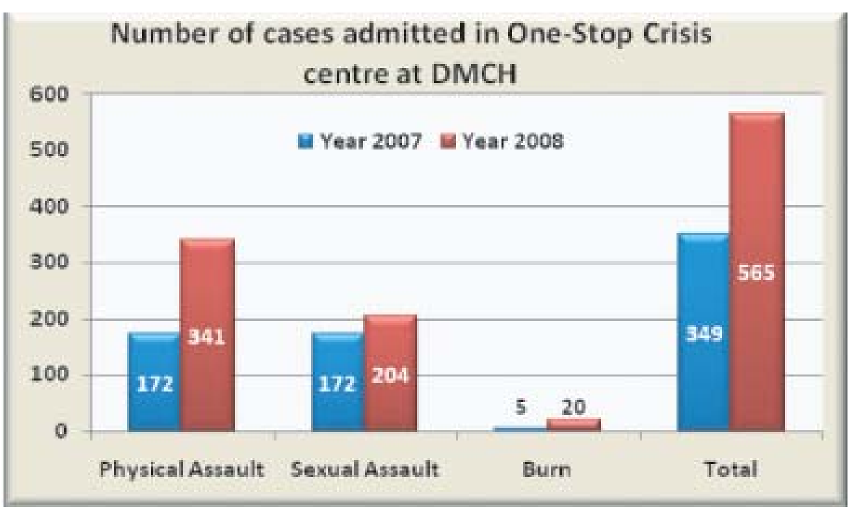
New Patients in OPD Old Patients in OPD Indoor Patients	Male Female 28347 11967 20451 7961 13985 3040	Male Female 30100 12300 21200 8000 13990 3050	Male Female 31500 13100 21100 8007 17535 3317

Distribution of Patients attending the OPD by age group (2008)

Age Group	No. of Patients	Percentage
0m – 1year	560	.5628%
1year -4years	1872	1.88%
5years – 14years	8180	8.221%
15years – 49years	61150	61.461%
50year+	27731	27.87%
Total	99493	100%

One-Stop Crisis Centre at DMCH

Since 2005, in Dhaka Medical College Hospital (DMCH) is providing one stop service to females suffering from domestic Patient statistics of National Institute of Traumatology, Orthopedics and Rehabilitation (NITOR) 2008 National Institute of Traumatology, Orthopedics and Rehabilitation deals with patients who suffer from orthopedic trauma due to various causes that includes Road Traffic Accidents. Patient statistics from NITOR encompasses the injuries due to various causes to some extent, as it is the central institute for trauma and orthopedics as well as rehabilitation. and other type of violence. Data of year 2007 and 2008 shows that the number of patients attending the OCC at DMCH is rising.



Safe Blood Transfusion

Though the need for safe blood is unanimous, millions of patients requiring transfusion do not get opportunity of transfusing safe blood and there is an obvious disproportion between developing and developed countries in access to safe blood. Of the estimated 80 million units of
blood donated annually worldwide, less than 45% is collected in developing countries. Acute haemorrhage due to accident and injuries and during post partum period, maternal and child anemia resulting from various cause are some of the leading cause of death in the developing countries. Timely access to safe blood transfusion is a life-saving measure in many of these clinical conditions and can also prevent serious illness in these patients. Besides, unsafe transfusion poses serious threat of transmitting infectious diseases that include HIV/AIDS, hepatitis-B, hepatitis-C, syphilis and malaria.

In our country the annual demand for blood transfusion is estimated to be 3, 00, 000 to 3,50,000 unit per year. But due to lack of voluntary donor and consciousness among people this demand is hardly met. South East Asia account for 25% of the world's population but collects only 9% of the world's blood supply as a result 7 million units of blood in a year, but there is need of a total 15 million units of blood.

Implementation of Safe Blood transfusion started in 2000 with the realization of judicial use and safety of blood and blood products in Bangladesh in order to ensure collection of blood not from commercial blood sellers and compulsory screening of blood for 5 blood-borne diseases viz. HIV/AIDS, hepatitis-B, hepatitis-C, syphilis and malaria before transfusion of blood. SBTP has 116 safe blood transfusion centers.

A national strategic plan for HIV/AIDS/STD prevention formulated incorporating blood transfusion sector. Since 2000, screening of Transfusion Transmitted Infection (WHO recommended) has been introduced in Bangladesh along with judicial act and national policies ruled by the Government of People's Republic of Bangladesh. Safe blood transfusion law 2002 has been approved by the parliament and published with an emphasis towards management and services of safe collection, processing, preservation and transfusion. The goal was to establish and operation of private blood transfusion centers and Upazilla blood transfusion centers by 2008. Official Gazette notification of the law has been published for implementation from 1st august 2004 and safe blood transfusion ruling order has been published on 7th may 2005. Emphasis is given to the outdoor and day care facilities provided by Transfusion Medicine Department. Safe blood Transfusion Programme was sponsored by United Nations Development Programme (UNDP) in 2000-2004 and from 2004-onwards is supporting by WORLD BANK, Development for International Development (DFID), World Health Organization(WHO) and Health, Nutrition and Population Sector Programme (HNPSP) .

The National Policy and Strategy on Blood Safety, adopted in 2007, defines minimum standards and requirements for health facilities to qualify and be authorized to screen blood for HIV before transfusion. A Reference Laboratory has been set up in Dhaka Medical College Hospital to conduct HIV confirmatory tests.

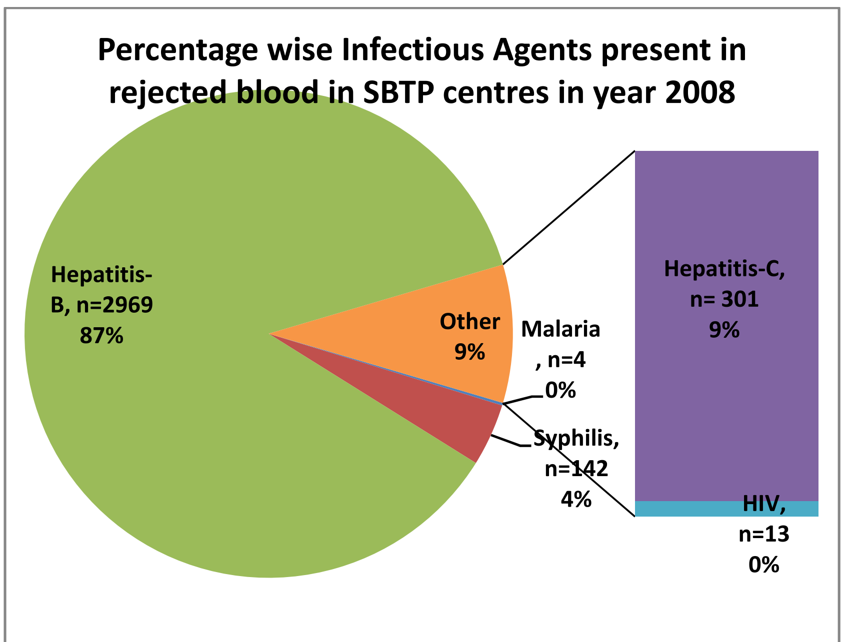
"Safe blood transfusion regulations 2008" was published in 17 June 2008. The regulations say that without taking license from a licensing authority, no person,
organization or institution will be allowed to establish and run a private blood bank.

The prerequisites for establishing a blood transfusion center: physical infrastructure, viz. specialist doctor, medical officer, staff nurse, technical supervisor, councilor, necessary equipment as per regulations, necessary furniture and reagents. The overall number of blood centers, however, is still inadequate. Efforts to promote voluntary blood donation and the mandatory screening of transfusion has reduced the practice of professional blood donation remarkably from 70 per cent in 2001 to 16 per cent in 2006. Over the same period, voluntary donation increased from 10 per cent to 24 per cent, and donations from relatives increased from 20 per cent to 27 per cent. Patients' family members, relatives, friends and acquaintances now donate 70% of the blood. Twenty five percent come from voluntary donors.

No. of SBTP centers (2007 and 2008)

Location of center	Year 2007	Year 2008
Medical College Hospitals	13	14
Specialized Hospitals	05	08
District Hospitals	53	59
Combined Military Hospital	13	14
Other Government and Non-Government Hospitals	11	11
Upazilla Health Complexes	00	02
BDR, Red Crescent , BIRDEM, Thalassaemia Center	04	08
Total	99	116

In 2007, SBTP screened 3, 58, 346 bags of blood and rejected 3,429 bags of blood due to presence of various infectious disease agents.

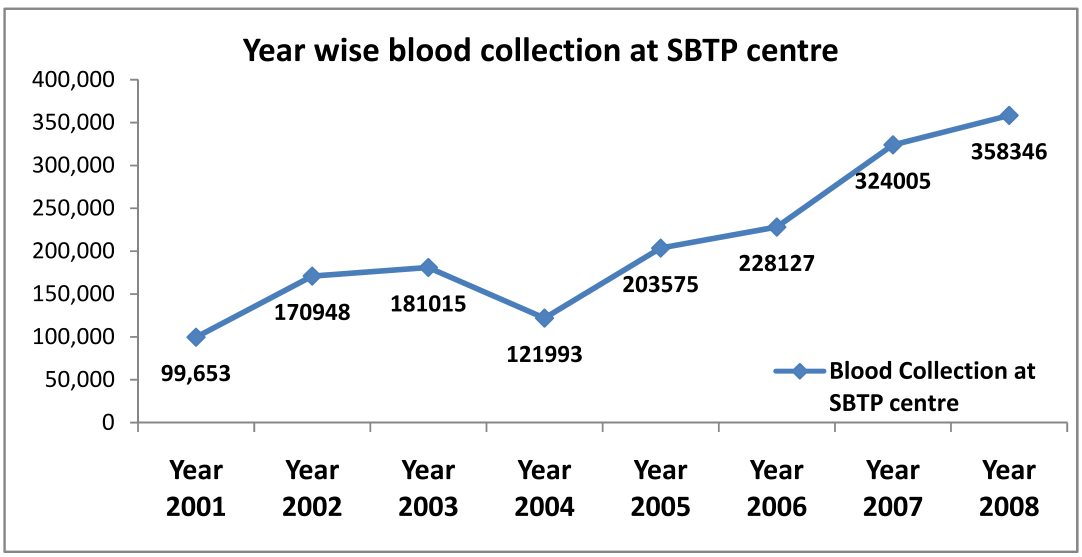


There are few philanthropic organizations who promote the cause of voluntary blood donation in the country. Sandhani, a wellknown medical and dental students' organization in the country pioneered the voluntary blood donation movement in the country in 1978. Since then the organization made significant contributions towards motivation of people for voluntary blood donation and safe blood transfusion. Later, other organizations also join in the efforts. These organizations are Bangladesh Red Crescent society, Quantum and Badhan.

Year-wise collection of safe blood by philanthropic organizations

Year	Sandhani	Red Crescent	Quantum	Badhan	Total
2001	43,702	19,300	1,781	8,785	73,568
2002	34,125	22,470	5,295	11,030	72,920
2003	35,223	22,810	6,720	13,000	77,753
2004	37,426	23,195	10,431	10,300	81,352
2005	38,989	24,842	15,063	17,000	95,894
2006	40,306	27,486	22,635	21,166	111,593
2007	29,312	14,612	33,712	28,000	107,643
2008	32,056	25,663	59,526	42,966	294,759
Total	**291,139**	**180,378**	**155,163**	**152,247**	**915,482**

Year-wise collection of Safe Blood by SBTP transfusion Centers (Including Red Crescent Society)



Nutrition

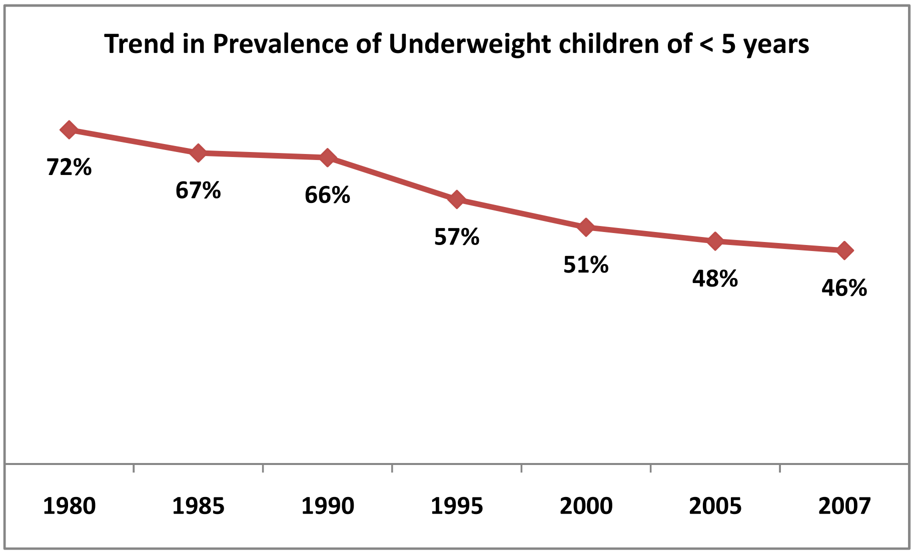
Nutritional status of people generally reflects health and welfare status of any country as poor nutritional status results from complex interactions between food intake or consumption, over-all health status, individual or community care practices at home and health care delivery centre. Numerous socioeconomic and cultural practices as well as individual access to food and feeding practice influence nutritional status of the country.

Young child and women of reproductive age are especially prone to nutritional deficits and micronutrient supplement. Like any other developing country, Bangladesh is highly populated country with small area of cultivable lands. Scores of people suffer from malnutrition in our country. Nutritional disorders like protein energy malnutrition (PEM), anemia, vitamin A deficiency (VAD), iodine deficiency, etc. have been recognized as major public health problems in our country. To achieve Millennium Development Goals (MDG) and indicators of Poverty Reduction Strategy Paper (PRSP), Bangladesh is committed to improve overall nutritional status especially the nutritional status of the women and children. Nutrition is included not only in the Health, Nutrition and Population Sector Program (HNPSP) of Ministry of health and Family Welfare but also being addressed by different ministries like Ministry of Agriculture. Two programs- National Nutrition Program (NNP) and Micronutrient Supplementation (MICS) are directly responsible for improving maternal and childhood nutrition. Under the National Nutrition Program, the intensified area based community nutrition approach (ABCN) through NGOs was maintained in the 109 upazilas. In ABCN, 2.9 million population, 1.1 million under 2 years children, 3 million pregnant women, 0.15 million newly-wed women and 1.5 million adolescent girls are brought under nutrition services. Data from Child Nutrition Surveys (CNS) 1995 and 2000, Child and Mother Nutrition Survey (CMNS) 2005 and Hellen Keller International (HKI) Survey 2007, UNICEF 2008 (State of the World Children 2008) and Bangladesh Health and Demography Survey (BDHS 2007), National Nutrition Program (NNP, 2008) are presented below to portray the current nutritional status of the children in Bangladesh.

Progress on relevant nutrition related indicators

Indicators	Status (2008)
Prevalence of women with BMI<18.5	28.4% (Oct-2008 MPR, NNP)
Prevalence of low birth weight (<2500)	7.5% (Oct-2008 MPR, NNP)
Under weight (<-2z) reduction in under 5 children	46.3% (2007)
Stunting (<-2z) reduction in under 5 children	36% (BDHS 2007)
Severe underweight in under 2 children (WAZ <-3)	10.3 (Oct-2008 MPR, NNP)
Moderate underweight in under 2 children (WAZ <-3)	22% (Oct-2008 MPR, NNP)
Prevalence of Anemia among adolescent girl	43.5% (HKI survey 2007)

The table above shows the overall progress of relevant part of result framework and some of the OP indicators of NNP.

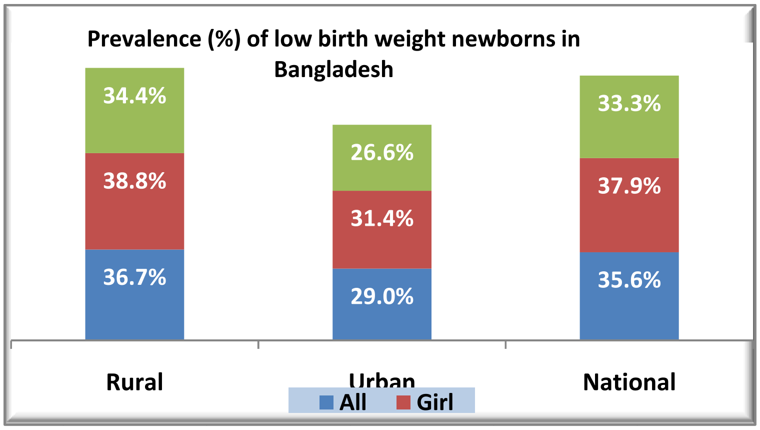


There is a downward trend in prevalence of childhood malnutrition during the last three decades, from 72% in 1980 to 46% in 2007. In Bangladesh, 17 percent of the children are considered to be underweight for their height, or wasted (BDHS 2007).

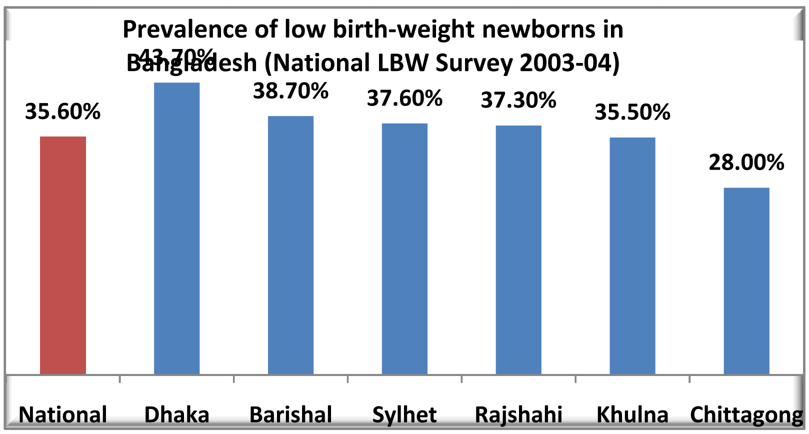
Prevalence (%) of malnourished <5 years children in Bangladesh 2000-2007

Year		Underweight	Stunted	Wasted
CNS 2000	Urban	41.80%	37.50%	10.90%
	Rural	52.60%	50.20%	12.20%
	National	51.00%	48.30%	12.00%
CMNS 2005	Urban	38.50%	32.50%	10.80%
	Rural	50.00%	44.90%	13.10%
	National	47.80%	42.40%	12.70%
2000-2006*	National	48.00%	43.00%	13.00%
BDHS 2007	Rural	43.00%	18.20%	45.00%
	Urban	33.40%	14.40%	36.40%
	National	41.00%	43.00%	17.00%
Source: CNS 1995-2000; CMNS 2005; *UNICEF 2008, BDHS 2007				

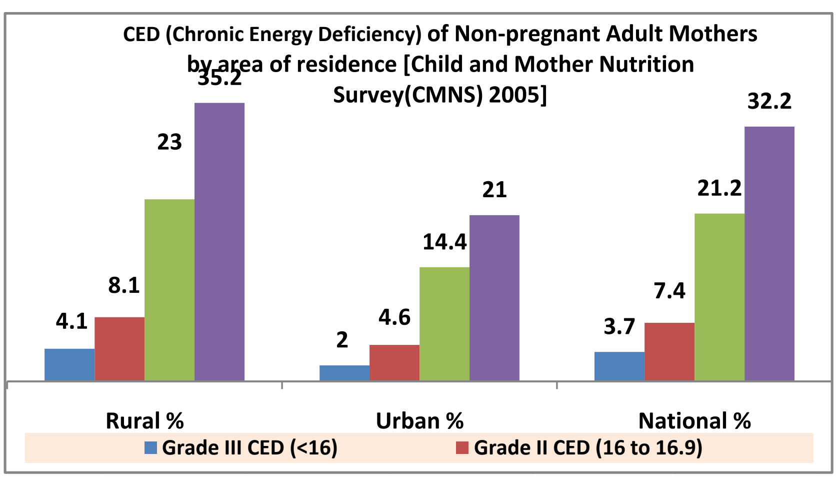
The National Low Birth Weight Survey of Bangladesh 2003-2004 provides data on percentage of low birth weight newborns in sex and urban-rural disaggregation. Prevalence of low birth weight newborns are more among the girls than among the boys both in rural as well as in urban areas. More rural newborns show low birth weight than urban newborns.

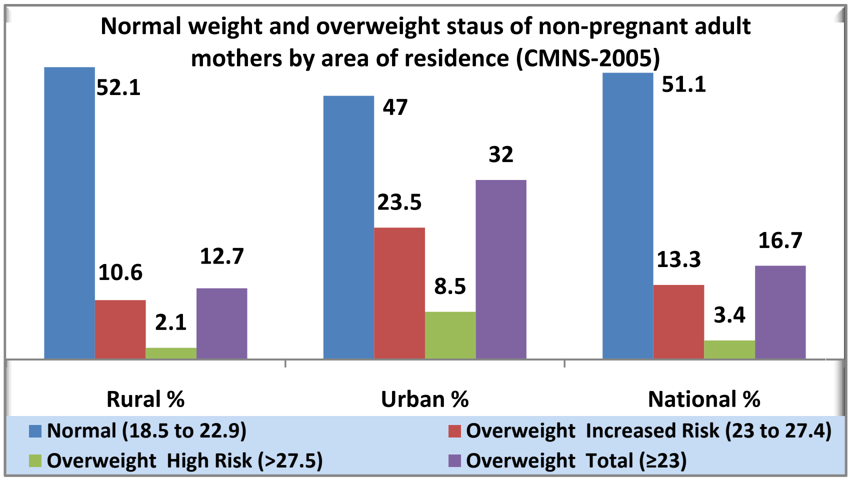


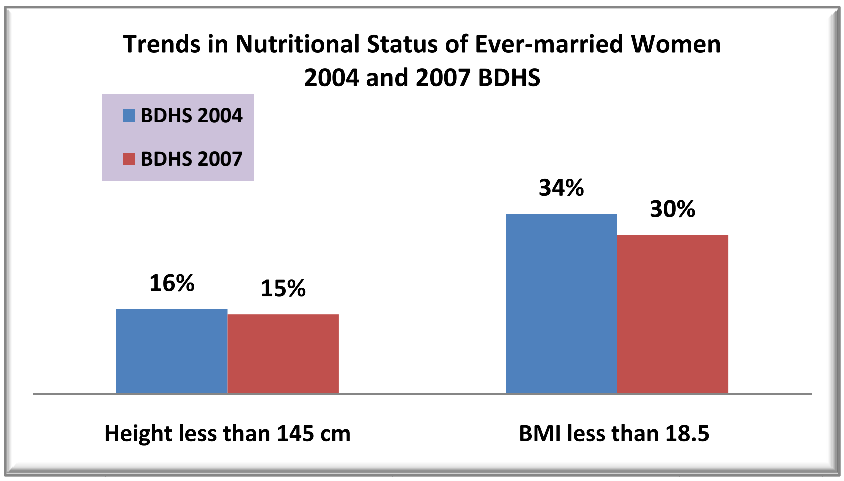
The division-wise distribution of rural newborns shows that Dhaka division has the highest prevalence of low birth weight newborns and the Chittagong division has the lowest prevalence. The prevalence of low birth weight newborns in other divisions is in intermediate levels.



The Child and Mother Nutrition Survey 2005 also collected data on non-pregnant adult mothers. Body mass index (BMI) was calculated and graded using Asian BMI criteria. Results are illustrated below:







Bangladesh Health and Demographic Survey 2007 (BDHS 2007) also indicate improvement in women's nutritional status as measured by BMI.

Institute of Public Health Nutrition

Institute of Public Health Nutrition (IPHN) is the focal institution for carrying out nutrition-related activities for Directorate General of Health Services. The institute is implementing number of activities intended for the improvement of the nutritional status of the people particularly of the mothers and <5 year children.

IPHN tries to improve following indices of nutrition:

i. Vitamin A deficiency disorders (VAD): % of night blindness among children and women of child bearing age (WCBA);
ii. Iodine deficiency disorder (IDD):
    a. Urinary iodide excretion rate;
    b. Quality control of iodized salt (estimation of iodine level of iodized salt)
iii. Iron deficiency anemia (IDA): % of anemia among WCBA and under 5 year children;
iv. Infant and young child feeding (IYCF) % of children consuming colostrums; % of children under exclusive breast feeding;
v. Weight for age Z score (WAZ): % of children under weight;

vi. Behavior change communication (BCC): % of population under direct contact of BCC activities.

Nutritional Blindness

It is estimated that 30,000 children's lives are saved in Bangladesh each year by Vitamin A supplementation. Very good progress has been made to reduce Vitamin A deficiency among children less than five years through Vitamin A supplementation. Night blindness in children under five years has reduced from 3.76% in 1983 to 0.04%9 in 2005 and is being maintained well below the WHO-recommended 1% threshold level. Vitamin A capsule distribution Coverage rate among 1 to 5 year age children has been increased to 99.7% (10 May 2008). De-worming tablet distribution (Tablet Albendazole) rate has been reached to 99.3% (10 May 2008). Postpartum coverage of vitamin A capsules has been raised to 35% (2008).

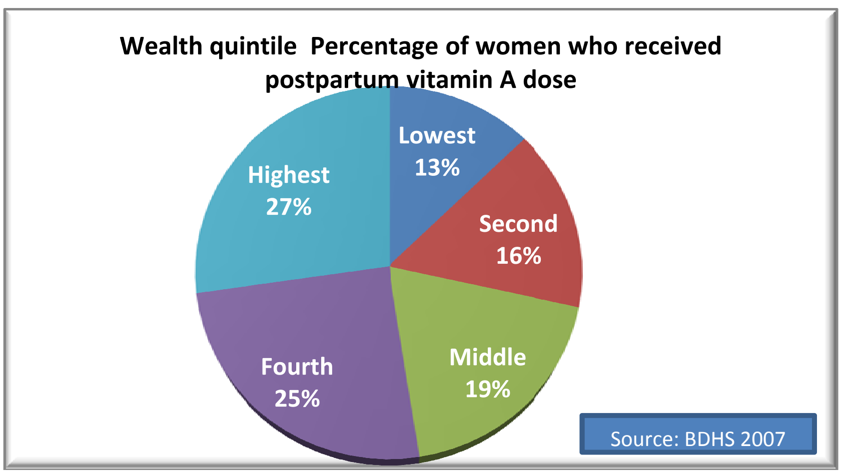
Seminar with eminent obstetricians and gynecologists of the country was arranged. They committed to advice vitamin A to pregnant mothers during their antenatal checkups and ask the mothers to have one red capsule (2 lac I.U.) within 6 weeks of delivery and according to their decisions, a guideline is sent to different hospitals to increase coverage. Vitamin A coverage for >1 year children has been raised to 73 % (2006).

Nutritional Blindness Prevention Program (NBPP)

Children under 1 year: High potency vitamin A capsules (1 lack international units) are supplemented during measles vaccination at EPI site.

Children 1 to 5 year: High potency vitamin A capsules (200,000 international units) are supplemented through two national events at 4 to 6 months Intervals every year.

Mothers: High potency vitamin A capsules (200,000 international units are supplemented during postpartum period within 6 weeks of child birth.



Status of nutritional blindness prevention program

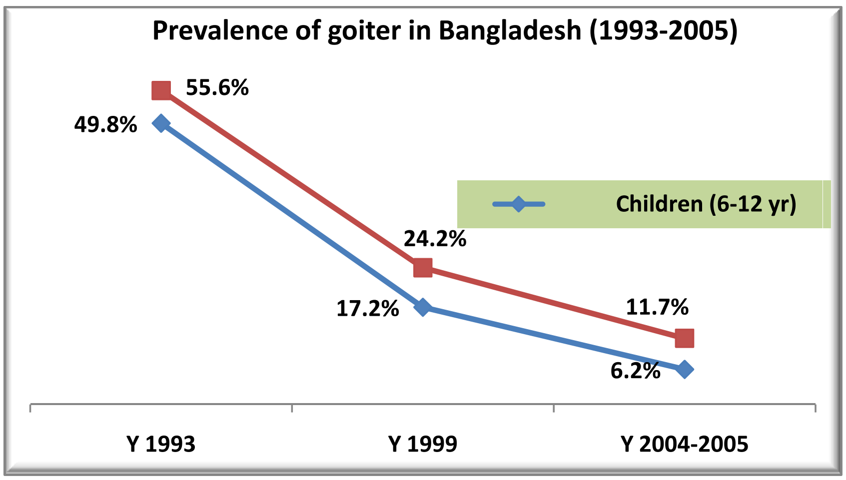
Name of the Program	Status
Night blindness prevalence (2005)	0.04%
Vitamin A capsule coverage among 1-5 year children (2008)	99.7%
Vitamin A capsule coverage among postpartum mothers (2006)	35.0%
De-worming (Tab. Albendazole) distribution rate (2008)	99.3%

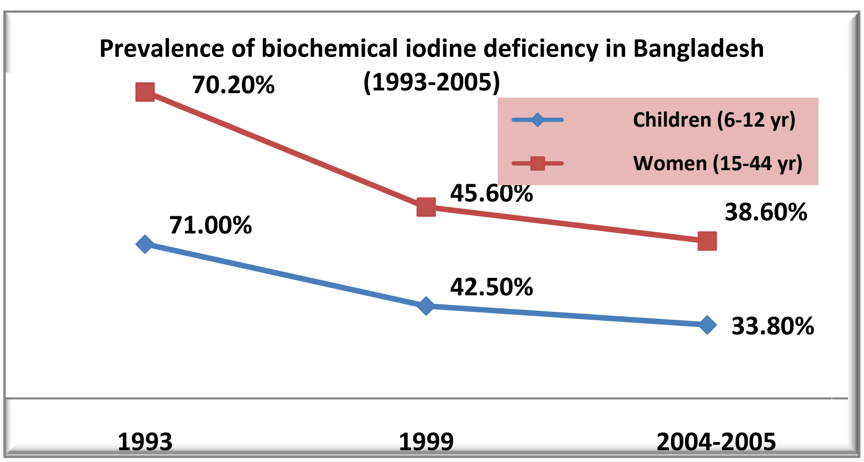
Reduction in the incidence of iodine deficiency diseases (IDD)

The program includes Iodized salt monitoring and awareness creation activities; training of field workers of health and family planning on control of iodine deficiency disorder (CIDD); training for testing Iodized salt, and surveillance of salt for iodization.

Bangladesh Iodine Deficiency Disorder / Universal Salt Iodization Survey, 2005 shows that 84% of all edible salt is now iodized, helping reduce iodine deficiency disorders. The prevalence of goiter in school-aged children decreased from 50% in 1993 to 6% in 2004/05 in the last decade as a direct result of salt iodization. Prevalence of severe iodine deficiencies in school aged children were reduced from 23.4% in 199313 to 4% in 2004/05.

Orientation on universal salt iodization (USI) law was distributed to health inspectors and sanitary inspectors of 63 districts. Universal salt iodization law is currently being updated and draft copy was sent to ministry for approval.





Reduction of Protein Energy Malnutrition (PEM) by training program

This program includes awareness program for PEM control; growth monitoring of 1 to 3 year old children; and communication on weaning. A national guideline for management of severe PEM is in process of publication.

Reduction of incidence of iron deficiency anemia

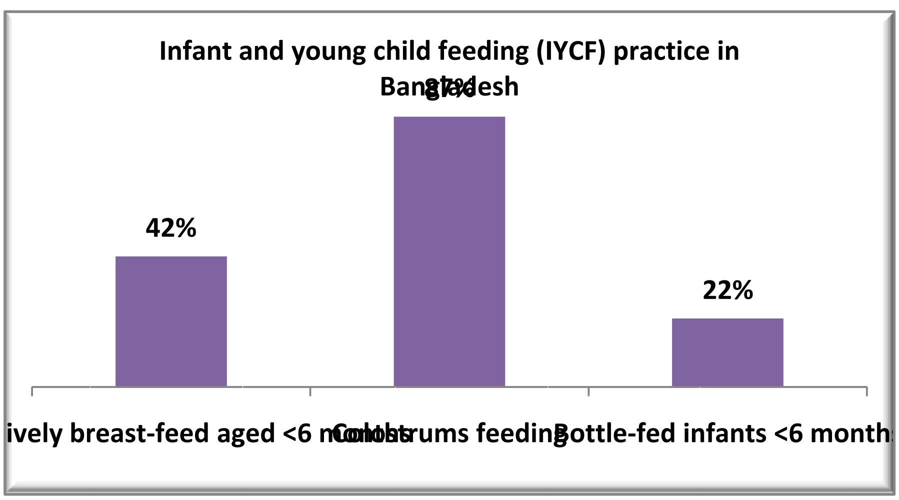
Micronutrient deficiencies especially iron, folic acid deficiencies that result in nutritional anemia, remain a public health in Bangladesh. Coverage of pre and postnatal iron and folic acid supplements is very low (only 15% of pregnant women in rural areas take at least 100 tablets during pregnancy).

The program for reduction of incidence of iron deficiency anemia includes awareness creating activities to control anemia and parasitic diseases; implementation of strategy to address the major causes of the malnutrition and anemia including ironfoliate supplementation, longtime food fortification and implementation. A guideline has been developed and distributed to all concerned in this respect for controlling anemia.

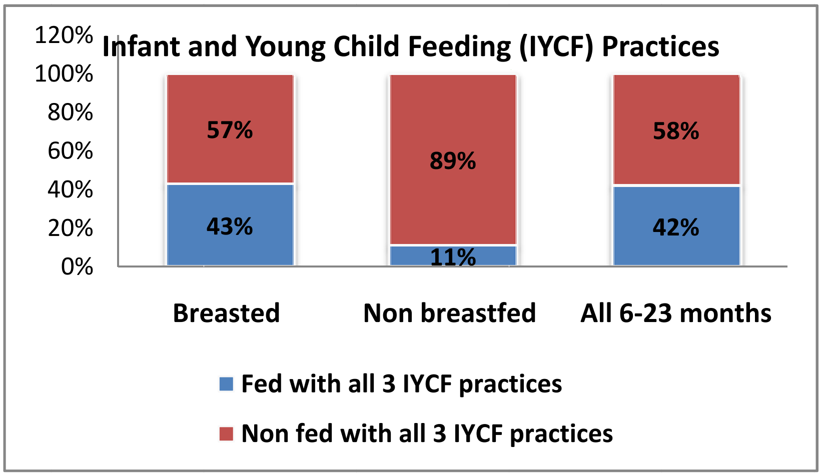
Prevalence of anemia in Bangladesh

Population group	Prevalence (%)
Pres-school aged children	49%
Pregnant women	46%
Non pregnant women	33%
Adolescents in country	23-29%
Adolescent in the Chittagong Hill Tracts	43%

Infant and Young Child Feeding (IYCF)



This program highlights promotion and protection of breast feeding through proper implementation of Breast Milk Substitute (BMS) code and proper child weaning Practice. In BDHS 2007 it is observed that appropriate infant and young children feeding practices increases sharply with the age of the children. Children in the urban areas and children of mothers who have completed primary education or higher are more likely to be fed according to IYCF recommendation. Feeding practices in the Rajshahi, Khulna, and Dhaka are better than other divisions.



Strengthening laboratory activities

The purpose of this activity is to develop the effective laboratory facilities of food and biochemical aspects of nutrition and to introduce serological tests for nutrition related diseases. IPHN is running its child nutrition units (CNUs) to full strength. All laboratory tests for identifying nutritional status of children and mothers are routinely done at IPHN laboratory. Iodine content of salt is also routinely tested.

Nutrition survey

To assess the nutritional situation surveys are now being carried out in selected districts. The survey will find answers to questions like impact of price hike on household food security, food quality and nutritional status of poor women and children in Bangladesh. A survey was done among 164 families in Bangalipoor union of Saidpur upazila under Nilphamari district.

Child Nutrition Unit (CNU)

The Institute of Public Health Nutrition is running 20 Child Nutrition Units, one located at IPHN and the others in 19 upazila health complexes in each of the greater 19 districts. The program has provided one nutritionist and a lady visitor to each center. The overall aim is to improve the nutritional status of the under- 5 children and mothers of the country. The CNUs provides services like growth monitoring; supplementation; complementary feeding; nutrition corner for mothers' education on nutrition; breast feeding corner; treatment of malnutrition and associated problem; referral center; and demonstration of home gardening. IPHN has taken steps to revitalize and provide more functional supports to CNUs.

National Rickets Survey , 2008

Rickets is a disease of children in which growing bones fail to calcify properly and become bent by the weight of the body and the pull of muscles. Rickets is most common due to a deficiency of vitamin D, an essential micronutrient obtained either from the diet or made in the body when the skin is exposed to sunlight. As vitamin D can be made in the skin, it is uncommon in the tropical or subtropical countries. When rickets does occur in sunny countries, it is usually due to other causes, such as calcium deficiency. Vitamin D deficiency rickets tends to be seen in very young children who have a soft and enlarged skull, swollen wrists and ankles and are prone to respiratory infections. In four different surveys conducted by the Institute of Child and Mother Health (ICMH), 1998; Bangladesh Rural Advancement

Committee (BRAC),2003 and Helen Keller International (HKI) in 2000 and 2004 identified that nutritional rickets was prevalent in different parts of the country specially in Chittagong and Sylhet division. National Rickets Survey in Bangladesh, 2008 is reported by Nutrition Program and members, Rickets Interest Group (RIG) and supported by National Nutrition Program (NNP), UNICEF and CARE Bangladesh. This study was conducted to measure the overall national prevalence of rickets among Bangladeshi children aged 1-15 years. Data were collected from 20,000 children in all six divisions in Bangladesh. Among these children, 197 (0.99%) had rickets. The prevalence was highest in children aged 1-5 years. Rickets was found in every division, with the highest prevalence reported in Chittagong division. According to radiological findings, 24% of the children with rickets had active disease.

Distribution of rachitic children by Division in 2008

Division	Total Number	Rickets (n)	Prevalence (%)
Barisal	4,449	7	0.16
Chittagong	6,884	151	2.19
Dhaka	12,070	14	0.12
Khulna	6,764	5	0.07
Rajshahi	12,813	9	0.07
Sylhet	3,912	11	0.28
Source: Health and Science Bulletin 7(1) 2009, ICDDR,B			

National Eye Care

Avoidable Blindness is one the major public health problem in Bangladesh. According to recently conducted National Blindness & Low vision survey, presently about 7.5 lakhs people aged 30 and above in the country are blind, besides about 40 thousand children are also blind. About 5 million people including children suffer from refractive errors while 250000 adultsare victims of low vision.

The number of blind population will go double by the year 2020 if no intervention is initiated immediately. In view of this critical situation, Bangladesh Government, being a signatory to Vision 2020, a global campaign for elimination of avoidable blindness by the year 2020, formulated a National Eye Care Plan under the leadership of the Bangladesh National Council for the Blind-an apex body under The Ministry of Health and Family Welfare in consultation with the stakeholders across the country and with the support from World Bank, International NGOs & WHO. To fulfill the Goal of the programme, DGHS through National Eye Care is working to eliminate avoidable blindness by the year 2020.

Objectives of the operational plan

- To develop /improve Eye Care infrastructure at secondary & primary level.
- To strengthen coordination among GONGO, private Eye Care providers.
- To increase awareness of mass population on eye care.
- To increase country cataract surgical rate through improving skill of Ophthalmologists.
- To prevent childhood Blindness.
- To increase affordability of eye care services by the poor patients particularly elderly, women and children through vouchering scheme.

Strategies

- Strengthening advocacy.
- Development of facilities & technology.
- Human resource development and management.
- Reducing the diseases burden
- Improving/expanding co-ordination and partnership.
- Developing/Strengthening eye health promotion system.
- Introducing/Strengthening in built supervision system.
- Supporting low vision patients with appropriate devices.
- Introducing in built MIS eye health system.
- Sustaining vouchering scheme.

Activities under taken during 2008

- Training, deployment and retaining of eye care providers.
- Procurement, distribution, installation and maintenance of eye care equipment.
- Procurement & distribution of MSR to District Hospital for SICS.
- Development of TV Spot, Radio spool.
- Development sharing and printing of treatment protocol.
- Development sharing and printing of training module for PHC workers.
- MSR support to outreach eye camps through district health administration.
- Sustaining of vouchering scheme for the poor and the marginalized.

- Development, field testing and use of MIS tools.

Achievements during the period 2008

- 20 (Twenty) Ophthalmologists from different district have been trained on micro surgery (SICS).
- 1500 primary health care workers have been trained on primary eye care
- Eye Care equipment procured, distributed and installed in 10 (Ten) Districts.
- One TV spot and one Radio spool developed and disseminated.
- Vision 2020 District committee formed in 6 (six) District e.g. ChapaiNawabganj, Nilphamari, B.Baria, Satkhira, Cox'sBazar and Narayanganj.
- MSR support to District hospitals B.baria, Satkhira, Narayanganj, Sariatpur, Madaripur, Bhola, Rajbari, Chandpur, Munshiganj, Netrokona, Pirojpur by GOB (Jamalpur, Manikganj, Chpai Nowabganj, Nilphamari, Gopalganj, Noakhali, Jenaidah, Jhalokati, Dinajpur) by GOB & NGOs.
- Vouchering scheme for free IOL (support to care providers) surgery in District of Manikganj sustained.
- Printing of 3000 copies of treatment protocol.
- Printing of 5000 copies of training module for PHC workers.
- Editing & reprinting of National Eye Care Plan
- Editing & reprinting of 3000 copies of National Eye Care Plan.
- Development, Sharing & introduction of monthly & annual reporting format for strengthening of MIS eye health.
- Country CSR rate (reported) increased to 1164.

Future plan of actions

- Improve co-operation & co-ordination among eye care providers.
- Introduction/strengthening of primary and secondary facilities to improve quality & expand coverage of eye care service delivery.
- Strengthening behavior change communication to increase awareness on primary eye care.
- Expansion of the coverage of vouchering scheme for IOL surgery to reach the poor.
- System development to monitor progress.
- Vision 2020 district committee formation in all 64 districts in phases.

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Health Education & Promotion Program

Background
Health promotion aims for improvement in health, preventing specific disease rather than treating illness alone. The major principles that underpin health promotion ideology are that health is essential for achieving a socially and economicallyproductive life. Successful implementation of health promotion strategies needs political and social actions to modify public policy. Without the policy support for creating enabling environment it will be difficult to change behaviour only through health education and promotion.

Initiatives
There are significant health promotion initiatives around the Globe. The Government of Bangladesh has initiated several health promotion policies such as Millennium Development Goals which include several health promoting objectives i.e. poverty reduction, water supply and sustaining the environment, reduction of maternal & infant mortality rate, improvement of health status and reduce disease burden. The Government of Bangladesh has ratified global framework convention of tobacco control adopted by World Health Organization. Bangladesh has been working on all these agenda with special emphasis to health promotion in a multisectoral approach.

Country Profile
Health education as health promotion initiative has started in Bangladesh in 1958 under the Directorate General of Health Services. It is considered as precondition for successful implementation of health care. The network of health promotion is extended up to the grass root level. The key strategic components of health education and promotion in Bangladesh are:

- Community Health Education
- School Health Education
- Industrial Health Education
- Hospital Health Education
- Environmental Health Education
- Education for prevention and control of communicable and non-communicable disease.
- Health Education for improvement of maternal and child health.
- Health Education for Diet and Physical activity.
- Education for improvement of nutritional status of the people.

Methods and Media for Health Promotion in Bangladesh

- Interpersonal communication and counseling.
- Group discussion and peer group education by the health care providers.
- Projection of documentary films, Videos on health issues in the community.
- Distribution and display of IEC materials on priority health issues.
- Social mobilization and advocacy at different levels.
- Use of Electronic and Print media to disseminate health messages.
- Dissemination of health messages through Audio-visual equipments in the hospitals and clinics.
- Health education and promotion campaign on different health problems.
- Health Education in the mosques and other religious institutions.

National Priority

Ministry of Health and Family Welfare has given priority on Health Education and Promotion program in the country. Health Promotion has been incorporated as an essential component in the Health, Nutrition and population sector program of the Government of Bangladesh with financial allocation of Taka 900 million for the period 2003-2010. This has been guided by health policy and aim to be contributed towards poverty alleviation, gender equity, violence against women, acid prevention, environment protection, disease prevention and control of drug abuse, maternal child health nutrition. The Country has been strengthening health promotion program at macro and micro level based on actual health needs of the community.

Vision 2015

- Improve knowledge, attitude and practices of the people towards prevention and control of communicable and non-communicable diseases.
- Strengthen multisectoral approach and stakeholders participation in the development of health promotion.
- Improve knowledge and skills of health service providers in communication.
- Establish community Support System in every village for health promotion program.
- Incorporate health promotion in all the ongoing health program.
- Strengthen Bureau of health Education and its network up to the sub-district levels.
- Share knowledge and experiences with regional countries to implement health promotion in a better way.
- Improve knowledge and skills of Health Education professionals through training, study tour in abroad.
- Conduct operational research and impact evaluation.
- Establish health promotion network within the region.

Commitment

- Increase investments in health promotion and to frame sound policies for health promotion as an essential component of equitable social and economic development.

- Establish effective mechanisms for a multisectoral approach in order to address effectively the social, economic, political and environmental determinants of health throughout the life-course.
- Support and foster the active engagement in health promotion of communities, civil society, the public and private sector and nongovernmental organizations, including associations of public health, while avoiding any possible conflict of interest.
- Monitor and evaluate systematically health promotion policies, programs, infrastructure and investment, on a regular basis, including consideration of the use of health-impact assessments.
- Close the gap between current practices and those functions based on the evidence of effective health promotion by the full use of evidence-based health promotion

National Level Health Education & Promotion Activities

No.	Activity	2008
1.	Training/Orientation (Local & Foreign)	
	(A) Training (Local)	
	· Scout leader orientation	1
	· School Health Education (Students & Teachers)	120
	· Training of Trainers (UH&FPO,s)	2
	· Trainers Training (Sr. Health Education Officers)	2
	(B) Foreign Training	
	· Training on IPC (Malaysia)	1 batch
	· Training on IPC (Nepal)	1 batch
	(C) Attend in training on different organization	10

No.	Activity	2008
2.	Workshop/Seminar/Conferences	
	· Press conference on 7th April World Health Day	1
3.	Advocacy & Orientation meeting.	
	· National level	10
	· Divisional level	25
	· District level	300
	· Upazila level	600
	· village level	2000
4.	Procurement of vehicles & motor cycles/AV Equipments/Office Equipments etc.	28 Jeep 92 motor cycle
5.	Health education & Promotion Model Village activities:	
	Health education & Promotion Model Village activities has already been started at 128 Upazillas. 3 guideline for model village were prepared and communicated to all districts. Necessary fund were placed at the disposal of Sr. Health Education Officers for implementation	
	· Develop messages on core health issues	11
	· Develop messages 33 behavioural issues	33
	· Develop messages for wall writing	120
6.	Production, Collection & Distribution of IEC Materials	
	· Leaflet on 16 issues (Diarrhea, Dengue, Water & Sanitation, Arsenic, EPI, & Iodine, Smoking, AIDS/World Health Day, HIV, Nutrition, ARI, MMR, IMR, Disaster, Vitamin "A", EOC management & Early Childhood, Development, Dowry, Cancer & Heart dieses.)	
	· Sticker on smoking	100000
	· Folder on smoking	10000
	· Calendar on Health Promotion messages	1200
	· Diary on Health Messages	1000
	· Book on Health Services in Bangladesh (English & Bangla)	5000+50000
	· Advocacy Materials for Imams	50000
	· Bangkok chartered	5000
	· Health Workers diary	30000
	· Monitoring checklist	30000
	· Billboards	13
	· Digital banner	200
	· Video on awareness program by Bangladesh Scouts	120
	· C.D Preparation on different Health issues	200
	· Reproduction of Films	200
	· Advertising materials on 25 types	10
7.	Advertisement through electronic media	
	TV Spot (Normal time TV program. - Total duration 2974 minutes. (Daily 25 minutes)	
	· EOC	1
	· Diarrhoea Prevention	2
	· ARI	1
	· Breast Cancer	1
	· Cervix, Cancer	1
	· Diabetic	1
	· Heart Diseases	1
	· Dowry	1
	· Newborn care	1
	· Dengue	2

No.	Activity	2008
7.	Advertisement through electronic media	
	TV program	
	· Develop TV drama serial on emerging and re-emerging health issues	26
	(II) Radio program	
	· On different Health and Population issues, Spots, Telop, Discussion etc. on Health issues were broadcasted.	
8.	Advertisement through Print media	
	· Advertisement through important national daily news papers, weekly & magazine (Both in Bangla & English) were implemented on the following Health issues · Diarrhea, Dengue, Water & Sanitation, Arsenic, EPI, Iodine, Smoking, HIV/AIDS, Nutrition, ARI, MMR, IMR, Disaster management & Early Childhood Development EOC. Diabetics, Heart diseases, dowry, Cancer (Total numbers of circulation)	500
	· Press note on disaster management every day average 5(five) national dailies during disaster.	
9.	Printing press Activities	
	(A) The following IEC materials were printed from BHE printing press	
	· Book lets/Gazette on anti Tobacco Acts	25,000
	· Poster (Mother & Child Health)	335000
	· Leaflet (Dengue)	6,00,000
	· Sticker (Diarrhea)	5000
	· Folder (Food safety)	3,000
	(B) Printing of Educational tools for Medical Education of DGHS & others programs:	
	1) Poster (Wood Frame)	150
	2) Cloth Banner	400
	3) Poster (Reduce maternal death rate)	12500
	4) Poster (Night blindness)	12500
	5) Poster (Child cough)	12500
	6) Poster (Smoking)	12500
	7) Poster (Diarrhoea)	25000
	8) Poster (gazette of smoking)	25000
	9) Leaflet (Disaster)	100000
	10) Poster (Dengue)	25000
	11) Leaflet (Health information during flood)	25000
	12) Leaflet of Diarrhoea, dysentery, Jandis etc	100000
	13) Duty of Disaster period (Leaflet)	100000
	14) Emergency message for affected area (‘’)	50000
	15) Leaflet (Dengue)	100000
	16) Leaflet (Diarrhoea)	100000
	17) Poster (Diarrhoea, Dysentery, Jaundice	25000
	18) Leaflet (Diarrhoea, Dysentery, Jaundice	100000
	19) Leaflet (Snake bite)	100000
	20) Breast & Cervical Cancer (Poster)	25000
	21) Diabetic (Poster)	25000
	22) Emergency Obstratic care (Poster)	25000
	23) Breast feeding	10000
	24) Adolescent health	10000
	25) Booklet on adolescent health	10000
	26) Poster on citizen charter	1200
	27) Diary (Including health message)	1500
	28) Diary cover	1500

No.	Activity	2008
9.	Printing press Activities	
	1) Calendar (Including health message)	2000
	2) Calendar Cover	2000
	(C) World Health Day	
	· Poster	70000
	· Leaflet	15000
	· Sticker	150000
	· Souvenir	15000
	· Folder	20000
	· Invitation card	3000
	· Envelop	3000
	· Certificate	200
10.	BHE participated and coordinated for observance the following International day's.	
	· World no Tobacco day	1 (One)
	· World Population day	1 (One)
	· World AIDS day	1 (One)
	· Breast-feeding day/weeks.	1 (One)
	· Safe mother hood day	1 (One)
11.	Health Education Exhibition	
	Health Education Exhibition were Organized in Dhaka city Corporation area and district level	4
	Painting Competition at Shishu Academy	1
	Essay Competition among the School Children	1
12.	Rally	
	Organized special anti smoking rally at Chapinawabgonj and Slyest District	1
	Rally on different health issues	10
13.	Folk songs, film shows & spot Announcement	
	Folk songs, film shows & spot announcement during observance of the world health day and other day's and events	1
	Organized Folk Songs	2
	Organized film shows	5
	Organized spot announcement on different health issue	10
	World Health Day	20 messages
	World Population Day	15 messages
	World No Tobacco Day	10 messages
	World Breastfeeding week	10 messages
14.	Meetings	
	· IEC Technical Committee	10
	· HEP Monthly Coordination Meeting	120
	· Partners & Stakeholders meeting (UNICEF, SNL, WHO &World Bank).	20
	· EPI communication sub committee meeting (Attend)	6
	· Meeting with Bangladesh scouts etc	2
15.	Monitoring and supervision:	
	BHE officers provided technical support and supportive supervision to the division HEO's, Sr./Jr. HEO's and Health Educators of the country	100
16.	Following activities done with the technical assistance of UNICEF	
	· Established community support system in 6 Upazilla	50
	· Organized planning meeting in 6 Upazilla	6
	· Developed community map	30
	· Conducted advocacy meeting at 30 villages	30
	· Conduct TOT for Sr. HEO's UHFPO's, UFPO's and others	1

No.	Activity	2008
16.	**Following activities done with the technical assistance of UNICEF**	
	• Conducted training for field workers in selected Upazilla	
	• Developed Upazilla action plan on EmOC held in BARD, Comilla	6
	• Conducted Interactive Popular theater in 30 selected villages	90
	• Popular theatre team formed and trained	1 batch
	• Organized IPT shows to disseminate messages on danger signs of pregnancy, importance of birth planning and importance of 3 visits on ANC.	
	• Developed community action plan on EmOC	6 + 2
	• Organized workshop on EmOC at National level (2 days)	
	• Hired consultant for development of community support group training manual	2
	• Develop Volunteers guideline	
	• Organized community support operational strategy development workshop	1
	• Organized messages and materials development workshop	2
	• Develop module for community support group	2
	• Implement safe motherhood activities	5
17.	**Package (2007-08)**	
	Health Education Intervention (IPC) towards Promotion of Community Health, Hospital Health, School Health, Environmental Health and Occupational Health.	HEP-S2
	Social Mobilization Cluster Group Education for the Promotion of Health in suport of poverty alleviation.	HEP-S3
	Country-wide Health Education Campaign in order to increase mass awareness on Prevention and Control of Communicable and Non-communicable Diseases and Mal-Nutrition.	HEP-S4
	Health education interventions towords prodution and distribution and display of IEC meterials for Prevention and Control of Emarging and Re-emarging Health Problems.	HEP-S5
	Need Assesment to design Intervention for Prevention and Control of Communicable and Non-Communicable Diseases.	HEP-S10
	Sensitization Workshop with Journalists and Stakeholders on Health Education and Promotion.	HEP-S12
	Development and Production of 16 mm films on communicable and non-communicable diseases.	HEP-S13
	Procurement of raw materials for the Utilization of printing press.	HEP-S15
	Production and Erection of Billboard near the Hihgways in 64 Districts, 6 Divesions and Dhaka city area in order to create mass awareness on Health Education and Promotion and also Dissemination Health Massages.	HEP-S18
19.	**Biennium program 2008-09 supported by WHO**	
	a. Develop Health Promotion Policy	
	b. Develop TOT manual on non-communicable disease	

No.	Activities		2008			Subject discussed
			Target	Achievement	Participant	
1.	Group discussion & counseling about Health Education at Health Education Resource Centre					
	a.	Group Discussion	20000	15325	251735	Personal hygiene
	b.	Personal Counseling	45560	35792	201140	Primary health care
	c.	Radio listening program	4056	2560	26895	AIDS, Acid, dowry prevention
2.	House hold visit at community level					
	a.	Personal Counseling	95966	70635	256000	EPI, Pregnant mother & child care
	b.	Group Discussion	15584	12224	254704	Reproductive health
3.	Community advocacy & Social mobilization					
	a.	Health Education Community meeting	4880	3052	62668	Newborn care, nutrition
	b.	Discussion with male person	35056	25300	152765	ARI, Communicable diseases
	c.	Discussion with local leaders	4516	2508	45527	Fisheries, livestock, etc.
	d.	Meeting with Union Parishad members	2000	1071	6936	Sanitation, violent against women
	e.	Film Show/Drama/Folk Song/Jatra etc.	5050	3044	35368	Manikar Sansar, Shapner Shuru, Environment
	f.	Others	4516	3004	408068	AIDS, Dengue, Heart disease, TV/Leprosy, etc.
4.	School Health Education Pro gram					
	a.	Big group meeting	9920	7016	208424	Adolescent health care
	b.	Classification discussion	12416	6400	209646	Sanitation & wastage disposal
	c.	Supports nutrition gardening	65728	38600	198440	Importance of nutrition
	d.	Film Show organize	9820	9020	802000	Diarrohea, Environment, Hook worm
	e.	Water & Sanitation	15976	7817	152150	Save water & importance of environment
5.	Rally/Kabadi/Jarry/Sari song etc.					
	a.	Rally	2020	2036	752061	World health day, World motherhood day, World No Tobacco day
	b.	Kabadi	-	-	-	
	c.	Jarry/Sari Song	2020	1836	957472	Unani medicine, pregnant, newborn, etc.
	d.	Comedy/ Path Nathak	-	-	-	
6.	Number of IEC/AV equipment materials					
	a.	Poster	75421	65294	-	5 danger sign, 3 delay sign, ARI, No Tobacco Day, World day etc
	b.	Leaflet	152691	152291		-Do-
	c.	Sticker	11521	6872		-Do-
	d.	Overhead Projector	64	64	-	Meeting, Workshop & Training
	e.	Film Show	-	-	-	
	f.	Miking etc.	4508	3596	-	NID, EPI, Dengue, Measles, health care regarding flood, Diarrohea prevention etc.

Alternative Medical Care (AMC)

Alternative Medicine popularly known as Unani, Ayurvedic and Homeopathic Medicine has been playing a significant role in the health care delivery system in the developing countries of this region including Bangladesh from time immemorial. Although tremendous progress has taken place in the field of modern medicine particularly in synthetic pharmaceuticals and antimicrobials, the practice and use of Alternative medicine is being continued through out the country even today. Because of unique geographical location and favorable climatic condition for cultivation and growth of a wide variety of flora and fauna having rich medicinal properties are intimately related and acceptable to our culture, diet and regimen.

Bangladesh being one of the few developing countries with a very large population living in the rural areas in the midst of extreme poverty can hardly afford the expensive diagnostic and treatment facilities of modern medicine. Due to lack of adequate support and patronization from state, the alternative medicine is being practiced mostly by unqualified persons in unscientific and unethical manner and the quality medicinal preparation are also scarce for lack of support in manufacturing process and industrial plants.

After the Drug Control Act (1982)Bangladesh Government has taken different steps for the development of Alternative Medicine. Govt. Unani and Ayurvedic Degree College established in 1990 and Homeopathic Degree College established in the same time. Both of the colleges are located at Mirpur and running smoothly. The admission criteria of both the colleges are similar to the MBBS course. After completion of 5 years course

BUMS(Bachelor of Unani Medicine and Surgery) and BAMS(Bachelor of Ayurvedic Medicine and Surgery) Degrees are offered.

There one year internship course is compulsory in the respective 100 bedded college Hospitals for those who have completed their five years course successfully. Both of the College Hospitals provide outdoor & indoor facilities Graduate Doctors are now available to provide quality services for the community at large.

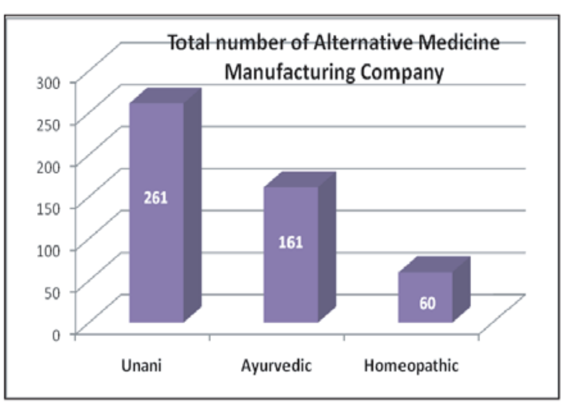
To provide quality services with Traditional Medicine, 45 Medical Officers (15 Unani, 15 Ayurvedic and 15 Homoeopathic) on Alternative Medicine have been appointed in the selected District level Hospitals under the work plan of HPSP so that the patients of these districts have the option to receive the types of treatment according to their own choice. To assist the Medical Officers 64 Support personnels (Compounder) have been appointed. To develop awareness on medicinal plants 467 herbal gardens have been established at 64 District Hospital premises and 403 Health Complex premises for demonstration of the community people. To look after the established herbal garden 467 herbal gardeners have been appointed. Now a total of 576 Alternative Medical Care (AMC) Health Personnel's have been working at different places. Beside this, one Government and different Non-Government Institutions is providing Diploma Certificates like DUMS, DAMS, DHMS under the Board of Unani, Ayurvedic and Homeopathic system of medicine. Apart from the Government services, some of the Graduates and Diploma certificate holders are working at different non-Government Organizations.

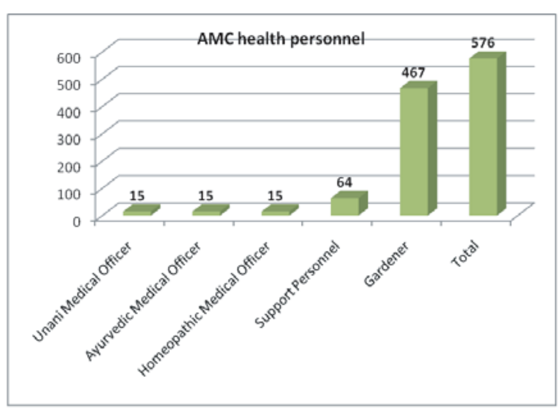
National Target with Indicators

Impact/Outcome Indicators(s)	Unit of Measurement	Achievement 2008	Target(2010)
Treatment coverage with AMC	% of population provided with AMC treatment.	About 10% (Average AMC Service Delivery)	About 32% (Total AMC Service Deliv ery)
1. Evaluation of AMC Service delivery	No. of Evaluation	3	9
2. Survey of AMC services	No. of survey to be conducted	3	9
3. Service providers skill Development	No. of orientation workshop	46	90
4. Fellowship for PG studies	No. of person provide d with fellowship.	01	24
5.Procurement of Medicine & Medical Requisites .	No. of Institutions provided with Medicine & MSR	49 Institutions	64
6. Overseas Training.	No. of Trainees providing Overseas Training.	-	60
7. Awareness development through BCC ac tivities i; e: Billboard, Poster, Sticker etc.	No. of Billboard No. &type of Poster No. &type of Sticker	22	64
8. Establishment of Graduate College.	No. of Graduate College	-	3
9. Establishment of Registration Council.	No. of Registration Council	-	3
10. Preparation of AMC Pharmacopoeia	No. of Pharmacopoeia	03(1st part)	3(Full)
11. Creation of herbal garden at central level	No. of herbal garden	-	01
12.Creation and maintenance of herbal garden at District Hospitals & UHC's	No. of herbal gardens	467	525
12. Establishment and functioning of research unit of GUADCH & GHDCH.	No. of institutions	-	3

Number of Qualified Personnel in AMC Year 2008

Qualified personnel in AMC	Number
Unani Graduates (BUMS)	364
Ayurv edic Graduates (BAMS)	297
Unani Diploma holders (DUMS)	1025
Ayurv edic Diploma holde rs (DAMS)	491
Homeopathic Graduates (BHMS)	616
Homeopathic Diploma holders (DHMS)	16222
Total	**19015**





Other Public Health Interventions

There are number of institutes carrying out public health activities like conducting disease surveillance and outbreak investigation; quality testing of drugs, food and water; production of vaccines, intravenous fluids, antisera and diagnostic reagents; diagnosis of infectious diseases and related research. Institute of Epidemiology, Disease Control and Research (IEDCR) and Institute of Public Health are two such important public health institutes under Directorate General of Health Services.

Institute of Epidemiology, Disease Control and Research (IEDCR)

Institute of Epidemiology, Disease Control and Research (IEDCR) is the national institute for conducting disease surveillance and outbreak investigation. It has been engaged in controlling disease and involved in research on diseases of public health importance. IEDCR was founded in 1976.

IEDCR has been nominated as National Influenza Centre (NIC), Bangladesh by WHO in 2007.

Main objectives of IEDCR are to conduct Disease Surveillance, Outbreak Investigation and Response, Research, and Training.

IEDCR have modern laboratories in the departments of Medical Entomology, Microbiology, Parasitology, Virology and Zoonosis. Among these, there are Biosafety Level (BSL) 2 and 3 labs and RTPCR laboratory.

The laboratories have wide range of diagnostic facilities, i.e., for parasitic and fungal diseases (Visceral Leishmaniasis (Kala-azar & PKDL), malaria, intestinal parasites, dermatophytes, candida etc), viral diseases (Nipah, hepatitis (HAV, HBV, HCV, HEV), HIV, influenza, Dengue etc), Bacterial Diseases (Enteric fever, Brucellosis, rickettsial diseases, other aerobic and anaerobic bacterial infections) and Biochemical tests. IEDCR also performs biological efficacy of insecticides regularly.

IEDCR is government mandated institute for conducting outbreak investigation of public health emergency of national concern in the country. In the year 2007, 17 outbreak investigations were conducted including Nipah outbreak, mass psychogenic Illness in 18 districts, toxic (Ghagra shak) outbreak in Sylhet, etc. In 2008, twenty five outbreak investigations were conducted including Nipah, puffer fish poisoning, mass psychogenic Illness, 1st human case of Avian influenza, Chikungunya outbreak etc.

IEDCR have some routine as well as disease specific surveillances.

Routine surveillances are :

- Priority Communicable Disease Surveillance
- Sentinel surveillance
- Institutional Disease Surveillance

Disease specific surveillances are -

- National HIV/AIDS Serological & Behavioral Surveillance (in collaboration with ICDDR,B)
- Nipah Surveillance (in collaboration with ICDDR,B)
- Community based Avian/Human Influenza Surveillance among poultry workers in H5 infected poultry farms
- Avian Influenza surveillance among the live bird handlers in wet markets of city corporations
- Acute Meningo-Encephalitis Surveillance (AMES) including Japanese B encephalitis (in collaboration with IPH and ICDDR,B with technical assistance from WHO)
- Hospital based influenza surveillance (in collaboration with ICDDR,B)
- Surveillance for hospital acquired respiratory infections in patients and health care workers in three tertiary care facilities (in collaboration with ICDDR,B)

IEDCR recently completed e-connection with 64 districts through wireless connectivity and established web based disease surveillance covering whole of Bangladesh.

IEDCR is routinely conducting training programmes specially focusing on epidemiology, surveillance, outbreak investigation, laboratory investigations, information technology etc. IEDCR also conducted certificate courses like:

- Three months Certificate course on Clinical Epidemiology
- Three months Certificate course on Medical Entomology

Continuous training activities are being conducted by IEDCR on Avian Influenza since 2007. IEDCR trained different tiers of health personnel including 64 District Rapid Response Teams, 471 Upazila Rapid Response Teams, 3700 Medical Personnel and 226100 Community Volunteers on Avian Influenza.

The main focus of researches conducted by IEDCR are:

- Parasitic Diseases: Kala-azar, Malaria, Filaria, Intestinal Parasites, etc.
- Viral Diseases: Hepatitis (HAV, HBV, HCV, HEV), HIV, Dengue, Influenza, Measles, Rabies, Nipah, Japanese B Encephalitis, Human Papilloma Viruses, etc.
- Bacterial: Enteric Fever, Rickettsia, Brucella, STDs, Anthrax, etc.
- Skin Diseases
- Entomological Surveys
- Non-Communicable Diseases (i.e.,Pattern of Obesity, CVD risk management)
- Mass Psychogenic Illness (MPI) and other socio-behavioural research

In the year 2008, IEDCR conducted a good number researches including:

- Phase IV clinical trial of oral Miltefosine for treatment of Kala-azar (2006-08)
- Dengue prevalence and entomological survey in Dhaka city
- Sero-prevalence of Hepatitis C with its Genotype (2007-08)
- Cure assessment of Kala-azar by detection of antigen in urine
- Gastro-enteritis situation in flood affected thanas in and around of Dhaka City (2008)
- National Health Account s (2007-08)
- Sero-prevalence of HIV among pregnant women (2007-08). 10470 HIV tests coducted in IEDCR Lab.

- Epidemiology of Influenza in Bangladesh (May 2008-09) in collaboration with ICDDR,B
- Oseltamavir Drug trial (May 2008-09)

IEDCR is also involved in developing many Standard Operating Procedures (SOPs),Training Manuals, Plan etc.

- Development of Standard Operating Procedures (SOPs) on; (a) Avian Influenza (11 SOPs); (b)Public Health Emergency of International Concern (PHEIC); (c) Outbreak Investigation
- Avian Influenza Pandemic Preparedness Plan (2006-2008, 2009-2011)
- International Health Regulation (IHR):National Strategy and Guideline
- Training Manuals on Outbreak investigations

Avian Influenza Activities

IEDCR has been declared as the National Influenza Centre (NIC) of Bangladesh in 2007 by WHO. IEDCR is involved in regular and emergency activities for preparedness, prevention and control of Avian and Pandemic influenza. IEDCR conducted community awareness programmes on Avian/Pandemic influenza encompassing population of 2,80,00,000. It has also conducted Table-top exercises at national and divisional level, TOT of the NRRT on Rapid Containment of Pandemic Influenza. IEDCR has accomplished Orientation on Avian and Pandemic Influenza of 64 districts Multisectoral Coordination Committee in 2008.

Institute of Public Health (IPH)

Institute of Public Health (IPH) established in 1953 is relentlessly carrying out public health activities since its inception. Testing for quality of drugs, food and water, production of vaccines, intravenous fluids, antisera and diagnostic reagents; diagnosis of infectious diseases and related research IPH also provides training to public health workers on laboratory quality, and food and water quality checking.

Activities of Institute of Public Health (IPH)

Production of intravenous fluid by year (No. of bags)

Fluid	Pack Size (ml)	Y2002	Y2003	Y2003	Y2004	Y2005	Y2006	Y2007	Y2008
Glucose Saline	1000	66780	39735	80904	81238	81238	6754	13242	130799
	500	3497	243610	241043	221026	221026	285145	217758	110179
Glucose Aqua	1000	56055	42569	84455	72429	72429	7823	11325	134416

	500	333213	248265	233086	211607	211607	277329	204345	110006
Normal Saline	1000	9291	17662	9783	17930	17930	5029	-	-
	500	77319	68492	50536	52518	52518	58338	67831	54379
Cholera Saline	1000	118519	129986	192907	10409	10409	1627	25304	108521
	500	308536	246718	472545	280402	280402	182789	240473	69401
P.D. Fluid	1000	93384	57657	68421	53666	53666	61391	38109	52481
	500	20278	-	-	-	-	-	10291	3640
3% Normal Saline	1000	-	-	-	-	-	-	-	-
	500	5022	5107	4578	6888	6888	6939	8456	7700
Baby Saline	1000	-	-	-	-	-	11000	12600	
	500	6717	4689	14307	8245	8245	500	-	
Haemodialysis Fluid	1000	33510	14200	21100	20650	20650	1000	-	10500
	500	-	-	-	-	-	500	8700	18680
Hartman's Solution	1000	-	-	-	-	-	1000	-	
	500	31694	42710	47520	70676	70676	500	21014	97752

Source: IPH, Mohakhali, Dhaka

Production of Blood Bags Infusion and Transfusion Sets by Year and by Size of Pack

Item	Pack type	Y2002	Y2003	Y2003	Y2004	Y2005	Y2006	Y2007	Y2008
CPD Blood Bag	Single	101844	107437	87586	59827	59827	65936	74435	55060
Baby Bag	150 ml	-	150	-	-	-	-	-	
Transfusion Set	-	37060	15650	51775	34775	34775	31860	24060	7925
Infusion Set	-	130200	107350	190300	188750	188750	86710	42200	30400

Source: IPH, Mohakhali, Dhaka

Production of anti-rabies vaccines (Unit in ml)

Fiscal Year	For man (5 ml)			For animal (10 ml)		
	ml	Ampoule	Course	ml	Ampoule	Course
2006	27,30,400	5,46,080	39,005	4,18,600	41,860	996

2007	24,46,900	4,89,380	34,955	4,83,750	48,375	1166
2008	30,17,125	28,48,440	83,793	6,19,620	51990	15,570

Source: IPH, Mohakhali, Dhaka

Production of Diagnostic Reagents to Support the Laboratories by Year (Unit in Litre)

Item	Y2001	Y2002	Y2003	Y2004	Y2005	Y2006	Y2007	Y2008
Benedict's Solution	420	600	555	470	460	294	480	560
ESR Fluid	60	160	160	150	160	110	237	380
20% Sulphuric Acid Solution	40	40	95	95	30	-	20	40
N/10 Hydrochloric Acid Solution	70	60	90	10	60	70	80	190
Acetone-alcohol	Nil	Nil	55	20	10	-	10	-
5% Acetic Acid Solution	60	60	80	100	60	20	70	170
WBC Fluid	60	50	80	40	60	20	40	100
RBC Fluid	20	50	80	70	30	-	20	80
30% Suplhosalicylic Acid	10	Nil	Nil	10	11.6	-	10	07
20% Sodium Hydroxide Solution	Nil	Nil	Nil	20	Nil	-	-	-
20% Potassium Hydroxide Solution	Nil	02	11.5	Nil	Nil	-	-	-
Semen Analysis Fluid	20	Nil	36.5	20	10	-	05	-
Normal Saline	60	100	90	70	40	30	80	60
Methylene Blue	20	35	57	30	10	10	05	25
Crystal Violet	15	15	30	10	10	-	-	05
Basic Fuchsin	05	10	33	10	05	32	-	22
Carbol Fuchsin	22	22	66	44	11	10	-	-
Gram Iodine	10.5	05	35	10	05	05	10	-
Lugol's Iodine	15.5	20	40	15	15	15	16	40
Leishman Stain	44	96	69	47	62	29	65.1	104
Giemsa Stain	29	48	39.5	Nil	36 l	16	51.8	60
Glucose Kits	100 kits	47 kits	78 kits	100 kits	48 kits	98	-	100

Bilirubin Kits	62 kits	68 kits	152 kits	97 kits	Nil	99	44	151
Creatinine Kits	Nil	Nil	54	51	Nil	-	-	69
Uric Acid Kits	Nil	Nil	27	60	Nil	-	-	-
EDTA Vial	Nil	Nil	Nil	Nil	Nil	500	-	-

Source: IPH, Mohakhali, Dhaka

Production of oral rehydration salt (ORS)

Year	Production (Packet)	Sale (Packet)
2001	32632350	30404625
2002	33713751	37000192
2003	34604700	3661741
2004	38094650	37942765
2005	39058284	38798539
2006	41050550	35472590
2007	41086825	44630025
2008	36287350	37466100

Source: IPH, Mohakhali, Dhaka

Food samples tested

Year	Total Samples	Genuine		Adulterated	
		No.	%	No.	%
2001	3280	1692	51.6%	1588	48.4%
2002	4300	2110	49.0%	2190	51.0%
2003	5120	2515	49.1%	2605	50.9%
2004	4413	2214	52.0%	2119	48.0%
2005	6337	3200	50.5%	3137	49.5%
2006	2779	1405	50.6%	1374	49.4%

2007	5992	3488	58.2%	2504	41.8%
2008	8734	5066	58%	3668	42%

Source: IPH, Mohakhali, Dhaka

Water samples tested by chemical method

Year	Total samples	Satisfactory		Unsatisfactory	
		No.	%	No.	%
2001	174	170	97.7%	4	2.3%
2002	248	240	96.8%	8	3.2%
2003	359	296	82.5%	63	17.5%
2004	319	315	98.7%	4	1.3%
2005	316	290	91.8%	26	8.2%
2006	301	278	92.4%	23	7.64%
2007	411	378	91.9%	33	8.03%
2008	418	407	97.37%	11	2.63%

Source: IPH, Mohakhali, Dhaka

Water Samples tested by bacteriological method

Year	Total samples	Satisfactory		Unsatisfactory	
		No.	%	No.	%
2001	386	332	86.0%	54	14.0%
2002	406	373	91.9%	33	8.1%
2003	492	426	86.6%	66	13.4%
2004	486	446	91.7%	40	8.3%
2005	290	248	85.5%	42	14.5%
2006	524	474	89.9%	53	10.11%
2007	725	580	80.0%	145	20.0%
2008	655	602	92%	53	08%

Source: IPH, Mohakhali, Dhaka

Number of drug samples received and tested

Year	Samples received	Satisfactory	Unsatisfactory	Not analyzed	Feed back to senders
2001	3625	3533	30	0	62
2002	3159	3017	26	0	113
2003	3842	3763	28	0	51
2004	3719	3641	45	0	33
2005	3472	3056	89	127	200
2006	2708	2664	44	-	-
2007	3097	2978	119	-	-
2008	4589	3639	100	-	-

Source: IPH, Mohakhali, Dhaka

Stool samples tested for polio virus

Item	Y2001	Y2002	Y2003	Y2004	Y2005	Y2006	Y2007	Y2008
Total Number of AFP Cases	1287	1365	1128	1301	1458	1619	1844	1809
Total Number of Samples	2728	2931	2388	2631	2910	3185	3611	4356
Total Polio Virus Isolates	74	93	91	118	59	253	181	80
Total Wild Polio Viruses	0	0	0	0	0	18	0	0
Total Vaccine (Sabin) Viruses	74	93	91	118	59	187	193	76
Total NPEV (Non Polio Entero Viruses)	804	815	565	517	574	473	553	1012
Total Negative Samples	1850	2023	1732	1996	2277	2492	2910	3264
Total	6817	7320	5995	6681	7337	8227	9292	10597

Source: IPH, Mohakhali, Dhaka

Measles and Rubella (IgM antibody) tested

Item	Y2003	Y2004	Y2005	Y2006	Y2007	Y2008
Total Blood Sample	71	616	1834	411	587	788
Measles Positive	59	404	769	170	06	16

Rubella Positive	0	55	609	164	432	243
Total Negative	12	157	453	77	149	529

Source: IPH, Mohakhali, Dhaka

Diagnostic services

Item	Y2001	Y2002	Y2003	Y2004	Y2005	Y2006	Y2007	Y2008
Biochemical tests (blood)	189	182	176	45	-	-	-	-
Serological tests	-	70	19	871	3333	923	2051	3293
Routine examination (Stool, Blood-CP, Urine, Sputum)	970	840	395	456	341	192	133	123
Culture & Sensitivity Test (Stool, Blood, Urine, Sputum, Throat Swab, Ear Swab)	222	231	381	146	121	161	108	98

Source: IPH, Mohakhali, Dhaka

No. of stool samples, throat swabs and rectal swabs **tests done for epidemiological purposes**

Y2001	Y2002	Y2003	Y2004	Y2005	Y2006	Y2007	Y2008
72	57	214	14	-	-	24	-

Source: IPH, Mohakhali, Dhaka

Number of 4th Year MBBS students visited for study

Y2001	Y2002	Y2003	Y2004	Y2005	Y2006	Y2007	Y2008
803	893	1158	924	1105	847	1476	1632

Source: IPH, Mohakhali, Dhaka

Manpower development under HNPSP at IPH by year

Sl. No.	Name of the Training/ workshop	Duration (in days)	Year							
			2007				2008			
			1st class	2nd class	3rd class	4th class	1st class	2nd class	3rd class	4th class
1.	Computer Training (Basic)	28	6	2	13	-	40	55	220	-

2.	GLP (Good Laboratory Practice)	10	-	4	110	100	-	50	140	-
3.	GLP (Good Laboratory Practice)	25	-	-	-	-	-	-	-	-
4.	GLP (Good Laboratory Practice)	20	-	-	-	-	-	-	-	-
5.	GLP (Good Laboratory Practice)	14	-	-	-	-	-	-	-	-
6.	GLP (Good Laboratory Practice)	5	-	20	60	40	-	-	-	240
7.	Computer Training (Refresher)	14	6	2	13	-	-	25	50	-
8.	Security Management Training	5	-	-	-	-	-	-	-	30
9.	English Language Training	28	-	-	-	-	40	51	-	-

Source: IPH, Mohakhali, Dhaka

Diagnosis of bronchiolitis may be masked under cover of pneumonia leading to improper management: findings of ICMH study

Bronchiolitis remains a major public health problem throughout the world exerting significant morbidity and mortality. Bronchiolitis due to Respiratory syncytial virus (RSV) remains a significant cause of respiratory disease all over the world including South-East Asian countries like India and Pakistan. It is estimated that proportional morbidity among the infants of Bangladesh due to respiratory diseases is 45%. In 2001-2002, epidemic of bronchiolitis was first reported in the country and high rate of this disease continued to prevail over the next five years. A recent study by Kabir, et al of Institute of Child and Mother Health shows 21% prevalence of bronchiolitis among the u-5 children who attended different

hospitals. It is found that nearly 95% of bronchiolitis cases are of viral origin, RSV being the commonest. The diagnosis of bronchiolitis is most often made on clinical grounds and the criteria may vary. Very simply, the first attack of wheezing in a previously healthy child of less than two years of age or for a diverse criteria with coryzal symptoms followed by rapid onset of wheeze, fever, tachypnea, chest retractions, crepitation, ronchi with radiographic evidence of chesthyperinflation. The study undertook middle line as the diagnostic criteria for

bronchiolitis (runny nose followed by breathing difficulty, chest indrawing and rhonchi (on auscultation) in children aged less than 2 years. Misdiagnosis of bronchiolitis as pneumonia has been observed, where antibiotics, have been used indiscriminately in 99% cases. Proportion of costly antibiotics, such as, ceftriaxone, a third generation cephalosporine, was in high proportion. Treatment for bronchiolitis require home management (in milder cases) and oxygen therapy or in severe cases measures such as nebulized salbutamol, adrenaline, corticosteroids, aerosolized ribavirin, hypertonic saline and in some cases critical management in pediatric intensive care unit (PICU). However, antibiotics remain a common practice in treating bronchiolitis despite the rare likelihood of bacterial infection. Use of antibiotics increases treatment cost and facilitates bacterial resistance. A recent multi-centre randomized control trial conducted by Kabir, et al have shown that managing acute bronchiolitis without antibiotics in adjunct to supportive measures remains preferable as clinical outcomes (recovery rates) were similar to those cases receiving antibiotics. It may be

concluded that bronchiolitis is one of the very common causes of respiratory distress for hospitalization of u-5 children in Bangladesh, the diagnosis of which requires only clinical skill and no laboratory investigation. Supportive measures are sufficient and use of antibiotics may only increase treatment cost and may lead to bacterial resistance without benefit in the recovery of illness.

Research and Development

Approximately 90 percent of the global disease burden exists in developing countries. It is apparent that priority action is needed to combat the health status and economic inequities faced by individuals and their families in these countries. Research is a mechanism that can contribute to health improvement, the implementation of health systems change and interventions to improve the overall health of our vulnerable populations. One of the critical roles of health research is to ensure that measures proposed to helpbreak the vicious cycle of ill health and poverty are based, as far as possible, on evidence, so that the resources available to finance them are used in the most efficient and effective way possible. A way to effectively utilize resources is through prioritizing health research.

The Directorate General of Health Services (DGHS) has a Line Director to oversee the priority areas that need health systems and other research and also support research agencies, organizations and individuals to carry out research. The principle that identifies priority research areas and research organizations focuses to the objectives of Health, Nutrition and Population Sector Program (HNPSP). It includes basic medical and bio-medical research, demographic, epidemiological, operational, and policy research, clinical research including research on reproductive health, impact and cost-effectiveness studies, behavioral and health systems research.

In 2008, the Research and Development Unit (RDU) of DGHS had given funds around two crore ninety five lac taka to different research projects on nutrition, poverty reduction, gender equity, child and maternal health, tuberculosis, and malaria. Among the major contenders, BMRC was funded for 14 projects, four researches of NIPSOM and IEDCR were funded and IPH got one research funding.

Bangladesh Medical Research Council (BMRC), Center for Medical Education (CME), National Institute of Preventive and Social Medicine (NIPSOM), Institute of Epidemiology, Disease Control and Research are among the major institutes under Directorate General of Health Services (DGHS) responsible for research and development.

Bangladesh Medical Research Council (BMRC)

Bangladesh Medical Research Council (BMRC) is the apex body of the Ministry of Health and Family Welfare for supporting health research in the Country. This organization was established in 1972 by order of the President as an Autonomous Body. As per resolution of the Government, BMRC is the focal point for Health Research. The objectives of BMRC are to identify problems and issues relating to medical and health sciences and to determine priority areas in research on the basis of health care needs, goals, policies and objectives. An account of research projects supported or done by BMRC is given below:

Institute-wise distribution of ongoing research projects during 2008-2009

Name of Institutions	No. of Projects (n = 29)		Total	%
	Regular Program	Development Program		
Research Cell, Medical College	04	05	09	31.0
Postgraduate Medical Institute	04	04	08	27.7
BSMMU	02	-	02	6.9
NGO	-	01	01	3.4
Other Institutes	03	06	09	31.0

Subject-wise distribution of ongoing research projects during 2008-2009 (contd...)

Subjects	No. of Projects (n = 29)		Total	%
	Regular Program	Development Program		
Maternal Health	01	01	02	6.9
Child Health	-	02	02	6.9
Infectious Diseases	03	04	07	24.2
Health System Research	03	03	06	20.8
Nutrition	01	01	02	6.9
Occupational and Environmental Health	-	01	01	3.4
Non-communicable Diseases				
Hypertension and Cardiovascular Diseases	01	02	03	10.3
Diabetes	-	-	-	-
Kidney Diseases	-	01	01	3.4
Mental Disorders	-	-	-	-
Others	04	01	05	17.2
Total	**13**	**16**	**29**	**100.0**

Centre for Medical Education (CME), Mohakhali, Dhaka

Centre for Medical Education (CME) since its establishment in the year 1983 is trying to improve the health professions education of Bangladesh as per it's TOR. CME has become a postgraduate institute by conducting two years MMEd course for the teachers of medical and allied health sciences institutes under the post graduate faculty of medical science and research, Dhaka university since 2004. Total number of seats for MMEd course is 15 out of which for Bangladeshi applicants 10 and for foreign applicants 05. Now CME is conducting it's MMEd course for 4th & 5th batches students as part I & part II respectively. 5 students acquired MMEd degree in the year 2008. Total six (6) WHO fellows from Nepal have received certificate after successful completion of short course from CME in 2008.

CME has conducted admission tests of Diploma in Nursing course and BSC in Nursing course in 2008. CME has also conducted admission tests of Govt. Homoeopathic and Govt. Unani & Ayurvedic Degree College in 2008. Total ten workshops on educational science and teaching methodology focusing on new assessment system of MBBS course have been conducted for

220 teachers of 11 subjects of different government and non government medical colleges. CME is acting as secretariat of national quality assurance body since 1998. A national consultative meeting was organized by CME with the support from PSE, DGSH for updating & enhancing the activities of quality assurance scheme in medical colleges in Bangladesh. CME developed the trainers guideline for medical assistants training, module on teaching health ethics, curriculum for training for health assistants with the support from WHO & curricula for IHT with the support from RTM International. CME also conducted five orientation workshops on developed module on teaching health ethics for the teachers of different government and non government medical colleges. Some of the important researches conducted by CME in year 2008 are as follows:

- Reviewing and Updating of MBBS Curriculum 2002: Teachers' views
- Present Status of Activities of Quality Assurance Scheme (QAS) in Medical and Dental Colleges of Bangladesh
- Perception of Teachers and Administrators of Health Sector on Accreditation in Medical Education in Bangladesh

The National Institute of Preventive and Social Medicine (NIPSOM)

The National Institute of Preventive and Social Medicine (NIPSOM), the only national level public health institute under the University of Dhaka, Bangladesh was established in 1978 with the aim to produce post-graduates capable of satisfying the needs of the community in promoting and restoring health. The institute is also supporting in the different health policy formulation of the government and community health programs through research, training and services. It conducts 8 (eight) Master of Public Health (MPH) courses of one-year duration each and 1 (one) M. Phil course of 2 years duration. The mission of this institute is to develop NIPSOM as a center of high credibility in academic, research, training and service delivery to support the government in the field of public health activities in order to improve the quality of health care. It also strives for continuous updating of the curriculum to keep pace with international standards so that students both local and foreign are attracted to be enrolled in the institute, which is intended to be a Collaborative Center of WHO/UNFPA. Some of the research activities conducted by NIPSOM in the year 2008 are enlisted below:

Parasitology

▪ Knowledge and attitude on avian Influenza with infections prevention practices among poultry workers in Dhaka

Medical Entomology

▪ Community based Visceral leishmaniasis vector control through insecticide treated bed nets: Feasibility, Cost and Coverage.

Community Medicine

▪ Hypertension and relevant risk factors among the secretariat employees of Bangladesh

Maternal and Child Health

- Reproductive Health Problems, their Determinants and Care Seeking Behaviour of Adolescent Girls in Rural Bangladesh
- Diabetes mellitus among arsenicsis affected mother and their reproductive outcome

- Adolescent pregnancy is an emergency research focus in Bangladesh
- Nutrition Education & Food Supplementation during Pregnancy & Their Influences on Birth Weight

Population Dynamics

- Effect of DOTS on sputum conversion after 2 months of treatment among pulmonary tuberculosis patients
- Ante Natal Care in selected Health Centers of Bangladesh

Health Education

- Pattern of Tobacco consumption among the diabetic patients
- Knowledge on cataract among slum dwellers
- Nutritional status of school children in a selected primary school in Dhaka city
- Periodontitis among the diabetic patient
- Periodontal condition among tobacco users
- Periodontal disease among pregnant women
- Knowledge about low birth weight among the pregnant women attending the Antenatal clinic
- Nutritional status of people living with HIV in Dhaka city
- Awareness among the nurses on HIV/AIDS infection in a selected hospital in Dhaka city
- Nutritional status of children suffering from rheumatic fever
- Attitude of the pregnant women towards antenatal care at Gazipur Sadar Hospital
- Knowledge on Visual Inspection of cervix with Acetic Acid (VIA) among Nursing personnel in a selected hospital
- Behavioral pattern about Household waste disposal among the urban slum dwellers.
- Knowledge and attitude of children about cigarette smoking in selected schools
- Prevalence and pattern of Tobacco consumption in a selected Rural Area
- Patients satisfaction on quality care nursing provided through model wards in Dhaka Medical College Hospital
- Effect of IEC materials on HIV/AIDS by the different segment of population in a selected community area
- Patient satisfaction on physiotherapy care in two selected hospitals
- Knowledge about smoking related health problem among the female prisoners of Nepal
- Knowledge and attitude about tuberculosis transmission, treatment and its control activities in peri-urban area

Epidemiology

- Risk factors of diabetic foot syndrome
- Risk factors associated with multi drug resistant tuberculosis

- Prevalence of overweight child of an English medium school and their fast food consumption pattern
- Influence of diabetes on physical function among the elderly persons
- Factors associated with hepatitis C infection
- Factors related to home delivery in urban slums despite receiving antenatal care
- Behavioral risk factors of noncommunicable diseases among house wives in a selected area of Dhaka
- Health risk behaviors among street adolescents of Dhaka city
- Human papilloma virus as a risk factors for carcinoma of cervix: Practitioners view
- HIV/ AIDS related stigma among health care personnel
- Pattern of poisoning in selected tertiary level hospitals
- Influence of host factors of tuberculosis patients on sputum conversion: A nested case control study
- Knowledge and attitude about tuberculosis transmission, treatment and its control activities in peri-urban area
- Rising prevalence of type 2 diabetes in rural Bangladesh: A population based study
- Metabolic syndrome in rural Bangladesh: Comparison of newlyproposed IDF, modified ATP III and WHO criteria and their agreements
- Global adult tobacco survey (GATS)

Nutrition & Biochemistry

- Food intake pattern and nutritional status of school children in a selected school in Dhaka city
- Assessment of nutritional status in early pregnancy in some selected hospitals of Dhaka city
- Nutritional status of the drug abusers in some selected rehabilitation centers in Dhaka city
- Nutritional status and class performance of children living in a selected orphanage in Dhaka city
- Personal Hygiene Practice among the food handlers of Dhaka University Hostels
- Knowledge attitude towards Avian influenza among the Medical College students in selected college in Bangladesh
- Intestinal parasitic infestation and Hygiene practices among the students of the specialized school for the street unchins in Dhaka city
- Treatment seeking behavior of filariasis patients in a selected endemic area of Bangladesh
- A study on awareness on sexually transmitted infections and its preventive measures among the medical Technologists of some selected institute in Dhaka city
- Awareness about the reproductive health issues among the unmarried adolescent girls in a selected slum
- Awareness about the reproductive health issue among the unmarried adolescent girls in a selected slum

- Awareness about otitis media among the parents of under five children attending a selected Hospital in Dhaka city
- Prescription pattern of diarrhoeal disease in selected upazilla health complexes
- Household food security and nutritional status among the under five children in a selected area of Kurigram District
- Job satisfaction among NGO health workers working in Dhaka city Corporation
- Job satisfaction among the primary health workers in a selected District
- Pattern of poisoning among commuters in selected hospital in Dhaka city
- Participation of women in household Decision making process in different issues in selected areas
- Low Backache among the Housewives

Public Health and Hospital Administration

- Public Health Workforce and Computerization of Information and Communication Technology (ICT) for Achieving Health related Millennium Development Goals (MDGs) in Bangladesh
- Analysis of current status of NTP activities in relation to its objectives and goals leading to achieving TB related millennium development goals in some urban and rural ladesh
- Maternal and child mortality and morbidity under the millennium development goals (MDGs) in two upazila of Bangladesh
- Patients safety care and hospital acquired infection.

Occupational and Environmental Health

- Assessment of impact of air pollution among school children in selected schools of Dhaka city.
- Quality of life of arsenicosis patients in an arsenic-affected area of Bangladesh
- Health problems among handloom workers
- Nutritional status and energy reserve of individuals exposed to arsenic contaminated water
- Status of blood pressure among individuals consuming saline water
- Respiratory problem among the workers of a selected furniture industry
- Depression among the garment workers in selected workplace
- Intellectual function of children of 5-10 years in arsenic exposed and non-exposed village in Bangladesh
- Noise induced hearing impairment among garment workers in Bangladesh
- Salinity in drinking water and pregnancy outcome in southern area of Bangladesh
- Health problems among deep sea fishermen
- Musculoskeletal disorderagricultural workers
- Respiratory problems among goldsmiths in selected workshops in Dhaka city
- Lung function of male individual having excess arsenic exposure in a selected rural area of Bangladesh
- Health Problems among the workers of Dhaka Export Processing Zone
- Pattern of injuries among workers of selected old and new textile mills in Dhaka

- Disease pattern and personal protective measure among the welding workers in a selected area in Dhaka city
- Developed Bangla manual for arsenicosis case detection and management
- Developed Lecture module on arsenicosis case diagnosis and management for undergraduate medical students

Institute of Child and Maternal Health

Institute of Child and Mother Health (ICMH) is a national level institute in Bangladesh committed to be a centre of excellence in the South East Asia. The institute is working for the improvement of health and nutrition of children and mothers in the country through its three objectives of human resource development, conducting research and patient care. This institute was made autonomous through an act in the parliament in 2002. ICMH is now administered through a Board of Governors. The institute has been recognized as a Lead Training Organization by the Government, Development Partners and others. ICMH conducts various short, mid and long term trainings on child and maternal health and nutrition. Training activities of ICMH is going in the full swing. So far 19036 health care providers have been trained till the end of 2008 in different short and long term courses in the field of child and mother health. ICMH offered Fellowship to PPD member states countries and successfullyconducted two International Fellowship training programme which were held on from15-24 February 2008 on International Fellowship Training Course on Reproductive Health and Safe Motherhood and International Fellowship Training Course on Essential Newborn Care and Infant & Young Child Feeding from 2-11 August'2008.

One of the mandates of ICMH is to conduct essential health service research. ICMH has so far conducted 119 research activities and the relevant articles have been published in different national and international scientific journals. Research work Conducted by faculty Members of ICMH in 2008:

- Magnitude of respiratory disorder in under five children in different hospitals of Bangladesh.
- Risk factors of recurrent wheeze with Iron Deficiency Anaemia (IDA)
- Sweet test in children with recurrent wheeze
- G-6 PD deficiency in icteric newborn with clinical course admitted in newborn unit of ICMH.
- Risk factors of Neonatal Pneumonia
- Double blind randomized trial of diazepam versus placebo for prevent of recurrence of febrile seizures
- Management practices of Epilepsy among various levels of health care providers: Optimizing the epilepsy management
- Effect of Chinese herbal medicine Goreisan (Wulingsan) on vomiting with acute watery diarrhea in children-a randomized, double blind, placebo controlled clinical trial.
- Lipid Profile of the Hospitalized Severely Malnourished Children.
- Study of blood lead level in school children of two schools of Dhaka city: a follow up study
- Perception and social determinants determining parental consanguineous marriage and health outcome in children: a hospital-based multi centre study.

- Socio-demographic profile and health problems of the adopted children: a hospital based multi centre study.
- Determination of socio-demographic risk factors for perinatal asphyxia: a hospital based, multi centre study.
- Neonatal seizure: evolution and outcome in infancy
- Socio economic impact of Childhood Disability on family Breast feeding and complementary feeding practices and morbidity among infants and young children with severe malnutrition.
- Effect of Zinc supplementation during pregnancy on birth weight in a periurban area of Bangladesh: A randomised, placebo controlled trial
- JiVitA -2 Newborn vitamin A dosing trial -for reduction of neonatal mortality in collaboration with Johns Hopkins University
- Community trial to determine impact of home based skilled birth attendant sevice on selected indications of sale motherhood
- Intra venous sucrose iron versus oral iron therapy in the treatment of iron therapy in the treatment of iron deficiency anamea.
- CRP in PROM A case control study
- Perinatal Death Audit
- Relationship of serum magnesium level with Eclampsia
- Intraumbilical oxytocin infection in the management of retained placenta
- An exploration of gender distribution, clinical profile, associated factors including pregnancy and health seeking behaviour of adult tuberculosis patient.
- Optimizing delivery care at the community: Acceptability and utilization of community based skilled birth attendant service in rural Bangladesh

Research work conducted by faculty members and students of different post graduate institutes and medical colleges in 2008:

National Institute of Cancer Research and Hospital (NICRH)

- Risk associated with breast cancer-by dept of Cancer Epidemiology
- Scoring of cancer related pain and management according to WHO guideline -by dept of Medical Oncology
- Effect on Lung in Breast Cancer Patients after local Radiation Treatment -by dept of Radiation Oncology
- Arsenic risk behavior of cancer patients attending NICRH, Dhaka -by dept of Cancer Epidemiology

National Institute of Mental Health (NIMH)

- Survey on prevalence of mental disorders in children: A community study in Bangladesh.

National Center for Control of Rheumatic Fever and Heart Diseases (NCCRFHD)

- Serum levels of Zinc and trace elements status in patients with Rheumatic Fever and Rheumatic Heart Diseases.

- Rheumatic Fever and Rheumatic Heart Disease.
- Development of mechanism for Rheumatic Fever and Rheumatic Heart Disease Surveillance.
- Changes in trace elements in patients with Rheumatic Fever and Rheumatic Heart Disease in Bangladesh.

National Institute of Traumatology, Orthopedics and Rehabilitation (NITOR)

- Comparative study between closed and open antegrade interlocking intramedullary nailing for the treatment of femoral shaft fracture in adult.
- Evaluation of replacement hemiarthroplasty by cemented bipolar prosthesis in femoral neck fracture in elderly but active patients.
- Evaluation and surgical management acetabular fracture.
- A prospective Randomized Comparative study for reverse oblique intertrochanteric fracture of femur treated by Dynamic Condylar Screw and proximal femoral nail in adult.
- Evaluation of the result of Management of prolapsed lumber intarvertebral disc by laminotomy and disectomy.
- Comparative analysis between proximal femoral nail and dynamic hip screw in the management of unstable trochanteric fracture neck of femur.
- The result of static interlocking Sign nailing in the treatment of femoral shaft fracture.
- Result of Treatment of Non-union tibial shaft fracture by intra-medullary interlocking nail with autogenous cnacellous bone graft.
- Evaluation of Replacement Hemiarthroplasty in femoral Neck Frafcture by bipolar prosthesis through lateral approach.
- Results of Retrograde interlocking intramedullary SIGN nailing in distal third femoral shaft fracture.
- A prospective study of the outcome of treatment of fracture neck of femur by Austin Moore prosthesis with autogenous cnacellous bone graft.
- Evaluation of treatment of intertrochanteric femoral fracture by PFN, Comparison with DCS.
- Evaluation of the results of treatment of intracapsular fracture neck of Femur by percutaneous cannulated cancellous hip screws
- Results of reconstruction of anterior cruciate ligament injury with Quadriple Hamstring graft by Miniarthrotomy
- Comparative study between early & late open reduction internal fixation by K-wire for the treatment of supracondylar fracture of the humerus in children.
- Evaluation of primary repair of tendo achilles injury caused by broken toilet pan
- Patellar tendon versus quadruple hamstring tendon graft for reconstruction of anterior cruciate ligament injury.
- Comparative study between conservative and operative management of traumatic unstable thoraco lumber spine injury with incomplete neurological lesion
- A study on exchange nailing with autogenous bone graft for aseptic nonunion of femoral shaft.
- Evaluation of results of open reduction and internal fixation of old Galeazzi fracture dislocation by LCDCP and stabilization of DRUI (Distal radio-ulna joint).

- Evaluation of results of open reduction and internal fixation of closed displaced fracture of medial malleolus by 4mm cannulated leg screws.
- Evaluation of Knee Stability following arthroscopically associated ACL reconstruction by BPTB interference screw.

Dhaka Medical College

Physiology

- Study of ventilatory function among the smokers
- Study of ECG changes in diabetes mellitus
- Association of Blood pressure and blood sugar among adult smoker and non-smoker stroke patients admitted in a tertiary level hospital
- Study of the effect of Mecobalamin on Electrophysiological changes of diabetic peripheral neuropathy
- A comparative study on serum Lipid Profile in the Hypertensive patients with & without insulin dependent diabetes mellitus.

Biochemistry

- Role of serum total and free prostate specific antigen (PSA) in the diagnosis of benign prostatic hyperplasia & prostate cancer
- Cystain C : Better Prodictor to assay the Renal Function in Diabetic Patients
- Comparison of Serum Calcium & Urinary excretion of Calcium in Preeclampsia normal pregnancy

Anatomy

- Evaluation of the in-course assessment system and first system based on anatomy module and curriculam (2002)

- Gross and histomorphological study of the parathyroid gland in Bangladeshi people
- A comparative anatomical study of spleen if human, cow & goat
- Gross and histomorphological study of the thyroid gland in Bangladeshi people
- Study of the ductal pattern of pancreas in Bangladeshi population
- Morphometry and histological study of the kidney in Bangladeshi People
- Morphological & histological study of the fallopian tube in Bangladeshi Female
- Gross & histomorphological study of Adrenal gland in Bangladeshi People
- Gross & histomorphological study of the Gallbladder in Bangladeshi People

- Histomorphological Study on Umbilical cord & fetal outcome Lipid Profile of cord blood in gestational diabetes mellitus

Pathology

- Histological scoring of nonalcoholic fatty liver disease and its correlation with risk factors in adult population in hospital patients

Microbiology

- Detection of chronic typhoid carrier state in patients Undergoing elective cholecystectomy
- Study on Prevalence of B. pertussis infection among children & adults
- Prevalence of Intestinal parasite in Bangladesh and Hookworm species differentiation by culture and microscopy

Pharmacology

- Study on the effect of Piper nighella saliva ground seed extract on Carrageenin & cotton Peller induced inflammation in rats
- Study of effect of Azadi Racta Indica on Fertility in Male Rats
- The Protective Effect of Extracts of cureuma Longa Rhizomes on Paracetamol induced hepatotoxicity in Rats

Community Medicine

- Risk factors with road traffic accident patients Attending tertiary level hospital of Bangladesh
- Treatment cost incurred by the patients treated at in-patient department of tertiary level hospital.
- Perception about prevention of bird flue among poultry retail sellers
- Factors influencing suicidal attempt
- Views of the elderly towards the behavior of his family members

Obstetrics and Gynaecology

- A lower dose of MgS04 for control of convulsion in Eclampsia Women in Bangladesh
- A study on cause and consequences of septic abortion in a tertiary hospital. Incidence & related risk factors for birth canal injury among obstetric patients admitted in DMCH
- Placenta Previa in scarred uterus versus unscarred uterus = a comparative study
- Medical induction of labor comparative study between intravaginal prostaglandin & intravenous oxytocin in primi patients in DMCH
- Cervical cytological changes associated with long term hormonal contraceptives.
- Study o risk factors, maternal & neonatal outcome in major placenta previa.
- Clinical evaluation of the Spondyloarthropathies

- Clinical Presentation of pregnancy with heart disease and its outcome
- Clinical analysis of fresh still birth cases in DMCH
- Elevated Plasma Homocysteine level as a predictor of severity of Preeclampsia and Eclampsia
- Study on liver function in severe preeclampsia
- Evaluation of dipstick Urinalysis & culture & sensitivity in detecting UTI in pregnancy.
- A Comparative study on maternal & fetal outcome in multiple & singleton pregnancy.
- A study on treatment outcome of patients admitted with Obstetrical fistula at national fistula center in DMCH
- Association of Toxoplasma gondii infection with Spontaneous Abortion
- Active Management of Third Stage of Labor: A study of 100 cases
- Risk factors for development of vesicovaginal fistula in Bangladesh

Medicine

Clinical Pattern and Profile of Hepatitis Viruses in a Tertiary Care Hospital.
Risk factors for Nephropathy in patients with diabetes mellitus
Modifiable risk factors among stroke patients -A hospital based study

Pediatrics

- Assessing students performance of Formative Assessment -does it need to improve
- Prevalence of Zinc deficiency in under the children in a slum of Dhaka Metropolitan City
- Assessment of child feeding practices by using Infant & Child feeding Index (ICFI) and its' association with nutritional status among children attending a tertiary level hospital in Bangladesh
- Study of immune response among preterm & low birth weight babies who completed 3 doses of hepatitis B vaccine in EPI schedule
- Medical & Social causes and Management of Obstructed Labor -50 cases analysis.
- The Pattern of childhood respiratory illness (1 month -12 yrs.) in different Hospital of Dhaka city providing health care facilities

Cardiology

- Stratification Risk Factors in Acute Coronary Syndrome (Bangladeshi) Patients
- ST Segment on initial Electrocardiogram (ECG) as a predictor of in Hospital outcome of ST elevated myocardial infarction (STEMI) patients

Radiology

- Barium swallow x-ray in evaluation of carcinoma of esophagus with histopathological correlation.
- Role of MRI in the Pre-operative diagnosis of lumber disc herniation: Correlation with

per-operative findings.

- Role of Barium meal follow through x-ray in the evaluation of ileo-caecal Tuberculosis with cytological & histopathological correlation.
- Role of CT scan in the evaluation of maxillary atrial mass with histopathological correlation
- Role of transvaginal ultrasonography in the evaluation of abnormal Uterine Bleeding with histopathological correlation
- Role of barium enema for the diagnosis of colorectal cancer and correlation with histopathological findings.
- CT evaluation of astrocytoma with histopathological correlation.

Neurology

- Study of clinical CSF & electrophysiological features of hospitalized patients with Guillain-Barré syndrome.
- Study of Association of small dense LDL with stroke.
- A Comparative study of Digital Subtraction Angiography (DSA) and Duplex

Gastroenterology

- Sequential therapy versus standard triple-drug therapy for Helicobacter pylori eradication in duodenal ulcer patients.
- Prospective Randomized, clinical trial comparing the efficiency of oral omeprazole

Intravenous

- Omeprazole and endoscopic therapy followed by intravenous omeprazole in patient with sever limb spasticity

Physical Medicine

- Use of 5% Phenol in Rehabilitation of Stroke Patients with severe limb spasticity.

Surgery

- Evaluation of Existing curriculum (2002)for Undergraduate Medical Students in Bangladesh
- Study of Acute Abdominal Presentation in Dengue and its outcome: A prospective observational study
- Study on Thyroid Malignancy
- Extrahepatic cases of Jaundice and outcome of surgical management -A study of 100 cases

Ophthalmology

- Kitotifen with Flurometh alone versus sodium cromoglycate with Fluromathalne in the management of Vernal Kerotoconjunctivitis (VKC) -A Comparative study of 100 cases

Urology

- Comparative study of early and conventional catheter removal following buccal mucasal graft urethroplasty
- Comparative study of interarectal lidocaine gel versus Intraprostatic lidocaine injection along with lidocaine gel as local anesthesia for TRUs Guided Prostate biopsy

Orthopedics Surgery

- Evaluation of the results of stabilization of talonaviecular and subtalar joints by K-wire in the surgical management of rigid variety of club foot

Otolaryngology

- Outcome of FESS in Chronic Maxillary Sinusitis

Neurosurgery

- Endoscopic third ventriculostomy for the treatment of Obstructive Hydrocephalus : A Comparative study on Outcome in newly Diagnosed & Shunt failure cases
- Study of the effects of pretreatment of Magnesium Sulphate on Suxamethonium induced complication during induction of General Anaesthesia
- Sevoflurane as an induction agent -A comparative study with Halothane in pediatric patients
- Addition of Clonidine or Fentanyl in supraclavicular block for upper limb surgery -A randomized comparative study

Anaesthesiology

- Bacteriological and sensitivity pattern of antibiotics of patients in Internsive Care Unit, DMCH -a system survey

Sir Salimullah Medical College

- Incidence of chips position carcinoma of prostate following TURP for clinicallyBEP patients
- Feeding practices in infant and young children upto 2 years of age
- Estimation of BMI and Its relation with self-perception and parents perceptions of the Nutritional Status in School Children of Dhaka City
- Prevalence of urinary tract infection in severely malnourished children
- Role of Tenonectomy in trabecutectomy
- Gross and histomorphological study of vermiform appendix in Bangladesh people
- Complications of ESRD in patients who are not in dialysis
- Abdominal Ultrasonogram in typhoid fever: An important diagnostic tool
- Evaluation of international prostate symptom score and quality of life following TURP in BPH
- Risk factor of stroke in young and old age group admitted in Sir Salimullah Medical College and Mitford Hospital Dhaka-a comparative study
- Study of causes of ileal perforation (s)

- Fournier's Gangrene-Approaches to the diagnosis & treatment
- Comparison between the post surgical corneal astigmatism in small incision cataract surgery (SICS) with straight and frown incision
- Prevalence of CTX-M gene in extended spectrum beta lactamase (ESBLs)producing Escherichia coli and klebsiella species, in public & private hospitals, Dhaka
- Detection of HBV DNA in HbsAg Negative cases by polymerase chain reaction
- Detection and prevalence of typical and atypical tubercular lymphadenitis in different hospital of Dhaka
- Study of serum ferritin level in preeclapmisia
- Study on some aspects of autonomic nerve function status in follicular luteal phase of menstrual cycle in Bangladeshi women
- The role of physical exercise and hormone replacement therapy in controlling dyslipidemia in postmenopausal women
- Relationship between homocysteine and carotid artery stenosis in ischemic stroke: A case control study
- A study on craniofacial anthropometry and their relationships with each other and with personal height of 8-12 years boys in Bangladesh
- Preterm delivery in pregnancy: Role of zinc and copper
- Cardio-pulmonary assessment after exercise in young adults
- Ultrasonographic evaluation of thyroid nodule with cytopathological correlation
- Prescribing patterns of Antibiotics in the Internal Medicine ward of a tertiary hospital in Bangladesh
- Effect of Ganoderma Lucidum on plasma lipid profile of hypercholesterolaemic rats
- Effect of honey on Blood Glucose Level of Alloxan Induced Diabetic Rats
- Hepatoprotective role of oyster mushroom against carbon tetrachloride induced liver damage in rats
- MRI evolution of tuberculous spondylitis with histopathological correlation
- Study on some aspects of Lung Function tests in Type-2 Diabetic patients in Bangladesh

Chittagong Medical College

Physiology

- Basal Gastric Acid Status in patients of Chronic Kidney Disease (CKD)

Biochemistry

- A Study of Plasma Fibrin D-Dimer and C-Reactive Protein in Patients with Ischemic Heart Disease (IHD)

Pharmacology and therapeutics

- Study of the safety profile of fenugreek seeds in rat.
- Study of effect of Boerhaavia diffusa plant extract on paracetamol induced hepatotoxicity in rat.

Pathology

- A comparative study of misoprostol versus manual vacuum aspiration (MVA) for the treatment of incomplete abortion.
- Last menstrual period method is a weak predictor in determination of expected date of delivety.
- Induction of labour versus expected management in women with premature rupture of membrane (PROM) between 34-37 weeks gestation.
- A comparative study of Laparoscopy assisted vaginal hysterectomy (LAVH) versus Total Abdominal Hysterectomy (TAH).
- Sonographic detection of high risk fetuses in a low risk antenatal population.
- Role of umbilical artery doppler velocity to assess fetal outcome in hypertensive disorder of pregnancy.
- Comparative study between colposcopic impression and histopathology for detection of cervical pre-cancerous lesion in VIA positive cases.
- A study on perinatal ourcome in term oligohydramnios pregnancy.

Microbiology

- Pattern of nosocomial agents in surgical wound infection in Chittagong Medical College.
- Pattern of bacteria causing urinary tract infections and their antibiotic susceptibility profile at Chittagong Medical College Hospital.
- Antimicrobial susceptibility pattern of Neisseria gonorrhoeae among Chittagong city dwellers and comparison of Gonorrhoea Rapid test with the conventional culture method for rapid diagnosis of Gonorrhoea.
- Isolation and Identification of Bacteria by Blood culture and estimation of IL-6 and CRP for the diagnosis of Septicemia in children.
- Study of drug resistance pattern of Mycobacterium tuberculosis in pulmonary and extra-pulmonary tuberculosis.
- Tumor necrosis factor-and soluble intercellular adhesion molecule-1 as prognostic marker of severe malaria cases.
- Isolation and detection of Extended spectrum
- B-lactamase producing bacteria from clinical isolates by disk diffusion test, disk approximation.
- Medicine (Internal Medicine & Tropical Medicine)
- Malarial retinopathy among the adult patient of Bangladesh.
- Study on Serum lipid, arterial blood pressure & body weight in post menopausal women.
- Risk factors of nom alcoholic fatty liver disease (NAFLD) in adult population of Bangladesh.
- Comparative study of PPI (Omeprazole)Versus H2 receptor antagonist (Ranitidine) in healing of peptic ulcer disease.
- A clinical trial of Nitazoxanide and Albendazole in Biliary Ascariasis.
- Sputum culture as a tool for diagnosing tubercular pleural effusion.
- Leucocytosis in management of OPC poisoning.
- Etiology and outcome of ARF in Nephrology ward of CMCH Study of 50 case.
- A review in lymphatic filariasis.

- Diagnosis and therapeutic strategy on viral encephalitis-A review.
- Role of Sputum cytology and CT guided FNAC of Bronchial tissue for diagnosing of Lung carcinoma-a comparative study.
- Recent update in the management of IBS. In collaboration with Welcome Trust Mahidul University, Bangkok
- RBC Deformity and sequestration of RBC in severe malaria.
- N-acetyle cysteine study in severe malaria.
- ECG Changes in severe malaria with artimisinine.
- The spectrum of retinopathy in adults with plasmodium falciparum malaria.
- Direct in vivo assessment of microcirculatory dysfunction in severe malaria.
- Levamisol therapy in Falciparum Malaria.
- Early enteral feeding in severe malaria.

Neuro-medicine Unit

- Pattern of presentation of different type of stroke patients.
- Study on neurological disorders during pregnency and puerperum.
- Trial of I/V Thiamin in suspected wet beriberi.
- Clinical presentation and short term outcome of patients of ATM (acute tranverse myelitis)

Nephrology

- Co-relation between urine analysis findings on renal biopsy in nephrotic Syndrome due to primary glomerulonephritis.
- Dialysis efficacy on ESAD patients on maintenance haemodialysis.
- Preliminary experience of kidney transplantation in Chittagong Medical College Hospital.

Dermatology

- Prevalence of Skin Cancer among the patients of chronic arsenicosis in Chittagong Medical College.
- Comparative study of clobestasol proprionate .05% cream & topical PUVASOL (8 methoxypsoralen .1% solution + sunlignt) in the treatment of Vitiligo.
- Efficacy of prednisolone alone and prednisolone plus azathioprine in the treatment of pemphigus vulgaris.
- A comperative study of intralesional & intramuscular Triamcenolone Acetonide on AlopeciaAreata.
- Mucocutaneons menifestation on systemic lupus erythematosus patient.
- Mucocutameons menifestation of chronic Arsenicosis.
- A comperative study of cryotherapy and topical Salicylic acid with lactic acid (Dhofilm) in the treatment of Verruca Vulgeris.
- Effects of topical taerolimus on vitiligo patient in children.

Pediatric Surgery

- Functional outcome of posterior urethral valve after cystoscopic fulguration.

- Role of heparin in the management of burn patients.
- Findings of Barium Enema in Hirschsprung's disease.
- Electrophysiological evaluation of GBS and CIDP patients.

Neuro-Surgery

- Surgical outcome of supratentorial meningioma a study of 25 cases in the department of Neurosurgery at Chittagong Medical College.
- Outcome of surgical management in compound depressed skull fracture with primary bone fragment replacement.
- Outcome of surgical management in extradural haemorrhage patient is directly related to the pre-operative Glasgow coma scale score
- Surgical outcome of extra-dural hematoma evacuated under local anesthesia-a prospective study
- A study of anterior cervical decompression with iliac bone-graft with screw-plate fixation for stabilization a prospective study

Otolaryngology and Head-Neck surgery

- Antibiotic resistance in neck space infection.

Gynecology & Obstetrics

- Efficacy of trans-thoracic FNAC in the Diagnosis of Bronchogenic carcinoma.

Anesthesiology

- Premedication with midazolam in children for smooth separation from parents: a comparison of two different routes of administration.
- Brachial plexus anesthesia (BPA): a comparative study on Supraclavicular Subclavian Perivascular Technique (SSPT) With the Axillary Transarterial Technique (ATT) with a Tourniquet.
- Smooth Insertion of Laryngeal Mask Airway: A comparison of propofol versus Thiopentone with Midazolam or Mini-dose Suceinylcholine.
- A single blinded randomized control trial of endotracheal and intravenous routes of lignocaine in attenuation of airway circulatory reflexes during emergence from general Anesthesia.

Physical Medicine & Rehabilitation

- Effects of Short Wave Diathermy in the patients with chronic low back pain.
- Study on referral of patients in the Department of Physical medicine & Rehabilitation in Chittagong medical College Hospital.
- Presentation of peri-partum paralysis attending in the department of PMR.
- Clinical profiles of patients with chronic low back pain.
- Study on low back Pain in Garments Workers in Chittagong.
- Awareness about disability among physicians in Chittagong.
- Physical functioning in patients with Rheumatoid Arthritis.
- Comparative study of supervised and unsupervised therapeutic exercise on pain and physical functioning from OA knee.

- Gross motor functional analysis of cerebral palsy patients
- Physical functioning of Spondyloarthropathy patients.

Pediatrics

- Burden of pneumococcal diseases in children in Bangladesh. A project to enhance laboratory capacity and create awareness and to prepare for introduction of a pneumococcal vaccine with ICDDR'B.
- Effect of severe protein energy malnutrition on circulating thyroid hormone
- Correlation of glycaemic & thermal status in severely malnourished children
- Protocolized management of severely malnourished children in Paediatric ward, CMCH. Partnership initiative with "CONCERN, Bangladesh"& ICDDR'B.
- "Acute Meningo Encephalitis Surveillance (AMES)"-a project which includes surveillance for Japanese.Encephalitis (JE) and Nipah Virus infection causing acute Meningoencephalitis, conducted by IEDCR.
- Timing of enteral feeding in cerebral malaria in the tropical setting-A randomized trial, collaborative study with Welcome.
- A hospital based survey of poisoning cases including snake bites admitted in CMCH-conducted by DGHS.
- Retinal changes amongst the children suffering from severe malaria in CMCH
- Effectiveness of Nestroft in detection of Beta-Thalassaemia Trait
- Efficacy of nebulized Ipratropium Bromide versus salbutamol in infants with acute bronchiolitis
- Diagnostic evaluation of protein creatinine index in children of Nephrotic Syndrome
- Clinical pattern of oedematous malnutrition-A hospital based study in infancy

Surgery

- Parietal fixation of mesh in case of TAPP laparoscopic hernia repair in a cheap & effective method
- Prevalence and risk factor of breast cancer: a population based study in Patiya, Chittagong
- Prevalence and risk factor of oral cancer: a population based study in Patiya, Chittagong
- Prevalence and risk factor of cervical cancer: a population based study in Patiya Chittagong
- Receptor status in breast cancer
- Establishing a population based breast cancer registry in chittagong
- Prevalence of tubercular mastitis kin patient present with breast lump
- Stapled hemorroidectomy in the treatment of hemorrhoid
- Closed lateral sphincterotomy under combined sedation and local anesthesia is preferred method over spinal anesthesia
- Smoking and dietary habits are risk factors for Gastric Carcinoma in Bangladesh

- Screening for risk factors of Diabetic foot
- Etiological relationship between acute pancreatitis and biliopancreatic ascariasis.
- Risk factors analysis for Ca. Breast
- Risk factor analysis for Colorectal Carcinoma
- Hospital based Cancer Registry
- Measurement of intra abdominal pressure as a criterion for Abdominal exploration

Urology

- Comparative Study of outcome of 1.5% glycine and 5% Dextrose in Aqua as an irrigation fluid intransurethral resection of prostate (TURP) for Benign Prostatic hyperplasia (BPH)
- Comparative Study between alfuzosin and tamsulosin in the management of Benign Prostatic hyperplasia (BPH)
- Comparative Study between alfuzosin monotherapy with combination of alfuzosin & finasteride in symptomatic benign hyperplasia of prostate patients.
- Clinical efficacy & Safety of oxybutynin IR & tolterodine ER in the treatment of symptomatic overactive bladder (OAB)
- Study for the safety and efficacy of adjunctive tamsulosin in enhancing the efficacy of renal and ureteral stone clearance when used with ESWL
- Assessment of renal functional deterioration & development of hypertension after ESWL
- Comparative Study of in situ ESWL versus push back, stenting and ESWL
- Incidence of intraabdominal hypertension in acute abdomen patient admitted in surgery word in Chittagong Medical College Hospital

Sylhet Osmani Medical College

Anatomy

- Comparative study on morphology of human placenta in normal and preclamptic pregnancy
- Gross & Histomorphology of spleen in autopsied human cadaver.
- Morphology of atrioventricular valves of the heart in autopsied human cadavar.
- Morphological evaluation of arch of aorta in autopsied human bodies.

Community Medicine

- Status of maternal health and family planning practice of some rural areas in Beanibazar Upazila, Sylhet.
- Socio-demographic characteristics and needs and demands of rural population of Beanibazar Upazila, Sylhet.
- Child rearing practice of village Maowa and Dobhag of Benaibazar Upazila Sylhet.

Microbiology

- Prevalence of IgM antibody against Measles, Mumps and Rubella in preschool children in Sylhet.

- Evaluation of the role of fluroscent microscopy and immunochromatography for rapid and efficient detection of malarial infection.
- Sero-prevalence of Hepatitis C virus antibody and their relationship with H.Pylori antibody among 1st year and final year student of SOMC.
- Study of sero-prevalence of CMV among children in Sylhet city, which is an indicator of humoral antibody and effect of nutrition on it.
- Evaluation of TNF-b level in malarial treatment.
- Correlation between inflammatory mediators

Pharmacology

- Comparative study on the effect of Olopatadine and Ketotifen in allergic conjunctivitis.
- Comparative study on the effect of Mirtazepine and Amitriptyline in major depression.
- Role of Omega-3 supplementation in suppression of Rheumatoid arthritis.

Forensic Medicine

- Various method of hanging.
- Study of Death due to poisoning.
- Domestic violence in Bangladesh.

Medicine

- Level of serum albumin and outcome of stroke
- Comparative study of lipid profile of newly detected uncontrolled type II diabetic patient and control subjects.
- Transthoracic FNAC in the diagnosis of lung cancer-A comparative study under guidance of ultrasound Vs CT.
- Higher level of calcium in the blood is associated with less severe ischemic stroke and better outcome.
- Comparative study of ischemic heart disease between patient with metabolic syndrome and normal individuals.
- Comparative study of bone mineral density between premenopausal and postmenopausal women

Orthopedics

- Outcome of early manipulation, serial plasters & percutaneous tanotomy of Achilles tendon in congenutal club foot.
- Evaluation of result of the treatment of remed interlocking intramedullary nailing of open fracture of tibia fibula without C arm.
- A comparison of I/V Ketorolac and Nalbuphine for post operative pain after orthepaedics surgery.

Ophthalmology

- Free conjunctival graft and amniotic membrane graft in pterygium surgery-reduces the recurrence rate of pterygium

Obstetrics and Gynecology

- Outcome of twin pregnancy among patient in SOMCH
- Fetal outcome of Preeclamptic mother in SOMCH
- Peripartum hysterectomy in SOMCH
- Obstetric haemorrhage related maternal death
- Outcome of local repair of VVF transvaginally in SOMCH
- Lipid profile in preeclampsia
- Thyroid function status in early pregnancy
- Adolescent pregnancy and low birth weight
- Control of convulsion by low dose magnesium sulphate in eclampsia
- Comparative study of efficacy of oral versus vaginal Misoprostol in management of incomplete abortion
- Study of serum homocystine concentration in pre-eclampsia
- A comparative study between pap smear and VIA in the diagnosis of cervical lesions
- Comparative study of colposcopy directed biopsy with colposcopic findings of VIA positive patient
- Role of imprint cytology in the diagnosis of Ovarian tumor

Paediatrics

- Efficacy of oral prednisolone in relieving hypoxia in children with acute Bronchiolitis: A randomized double blind placebo controlled trial.
- Role of maternal smokeless tobacco use in the delivery of preterm babies.
- Validity of red cell indices in Determination of β-thalassemia carrier-(Dissertation)
- Sensitivity, Specificity and predictive values of NESTROFT for detection of carriers of β-thalassemia
- Pregnancy outcome of Mothers who used smokeless tobacco for prolonged period
- Role of passive tobacco smoking in the development of IUGR babies
- Comparison of Nebulized Adrenaline versus salbutamol in bronchiolitis
- Otolaryngology and Head-Neck surgery
- Evaluation of topical povidone-Iodine versus ciprofloxacin in tubotympanic disease.

Shaheed Suhrawardy Medical College

Community Medicine

Assessment of the suitability of Practical Training sites providing Training to the students of

Health Technology Institute
Awareness of HIV/AIDS among the adolescent female garment workers
Recipients' perception of blood transfusion risk in some selected tertiary care hospitals
Characteristics of attendants assisting women at childbirth during domicillary delivery in rural community of Bangladesh

Gynecology and Obstetrics

- Assessment of knowledge and skill of health care providers in management of child abuse in different level Hospitals in Bangladesh
- Repair of incisional Herina-experience with a combined fascial & Prosthetic mesh repair
- Randomized controlled trial of Metformin & clomiphene on polycystic ovary syndrome Effect of Intravenous Iron in Postpartum Iron Deficiency Anemia.
- Symptoms Experienced by menopausal women
- Prevalence & risk factors of candidasis among women complaining vaginal discharge Risk factors & outcomes of Puerperal Sepsis
- Knowledge, Attitude & Perception of pregnant mothers on EOC patients
- Physical status of patients vaginal hysterectomy
- Prevalence of Candidiasis Tricomoniasis and Bacterial vaginosis among pregnant patients complaining vaginal discharge
- Accreditation of Women Friendly Hospital in Bangaldesh
- Facility and Community Based Maternal health review in Bangladesh
- Misoprostal and Combination if Misoprostal and Mefipristone in menstrual regualation : A randomised control study
- One stop crisis centre: Role in violence against women
- Rare congenital malformation of urogenital organs
- Colposcopic evaluation of clinicallly unhealthy cervix-A study of 500 cases
- Factors Associated with male infertility
- Study of Fibroid uterus-Clinical Feature and histological findings in SSH
- Female sexual Dysfunction Facts and factors among Gynae outpatients.
- Knowledge attitude and compliance of pregnant women for antenatal care
- Oral versus vaginal misoprostol in management of missed abortion
- Maternal outcome of vaginal delivery and caesarean section in eclamptic patients
- Dysteroscopic evaluation of intrauterine lesions
- Domestic Violence in antenatal period, fact and factors
- Effect of BMI on pregnancy outcome

- Prevalence and risk factor of urinary incontinence in women
- Effect of inter-pregnancy interval in fetal & maternal outcome

Pediatrics

- Study of Clinical Profile of 100 neonate
- Nosocomial Infection among the admitted patients in medicine & pediatrics department in collaboration with ICDDRB

Dhaka Dental College

- Scientific and Clinical aspect of Metal-Ceramic Fixed Partial Denture.
- Prosthetic Rehabilitation of Missing Teeth and oral health in the Adults
- Correlation between Pattern of Tooth loss and socio-deuographic status of the people in Bangladesh.

Human Resources

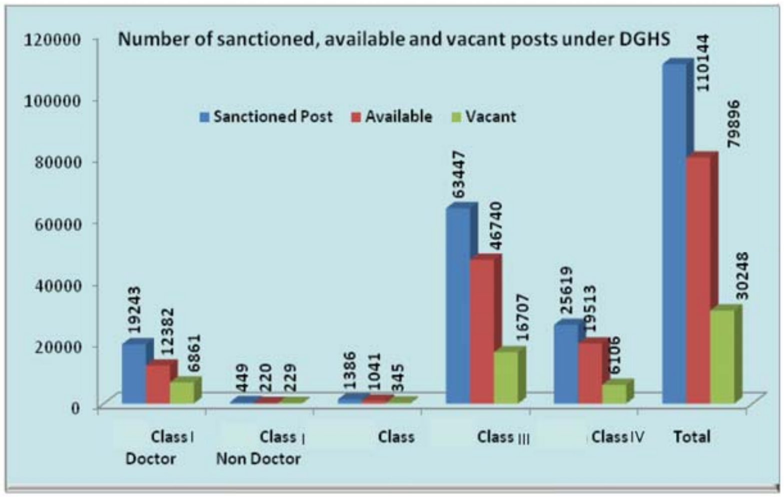
The Directorate General of Health Services is the key government organization, which is responsible for delivering health care to the people all over Bangladesh. In order to deliver the service over one lakh health care personnel and staffs are manned by DGHS throughout Bangladesh. The manpower deployment and redeployment process is very dynamic and at every point of time the picture is changing. Retirement, placement, transfer of manpower is constantly occurring and it influences the process of distribution of human resources. Besides this, depending upon the personnel category, transfer and posting may take place at different level viz., MOHFW, DGHS and offices of the Divisional Directors of Health and Civil Surgeons office. So limitations arise at any given point of time when we try to pick up accurate picture of human resource distribution. The information on distribution of human resources working under DGHS during the preparation of the current Health Bulletin are summarized in the following tables.

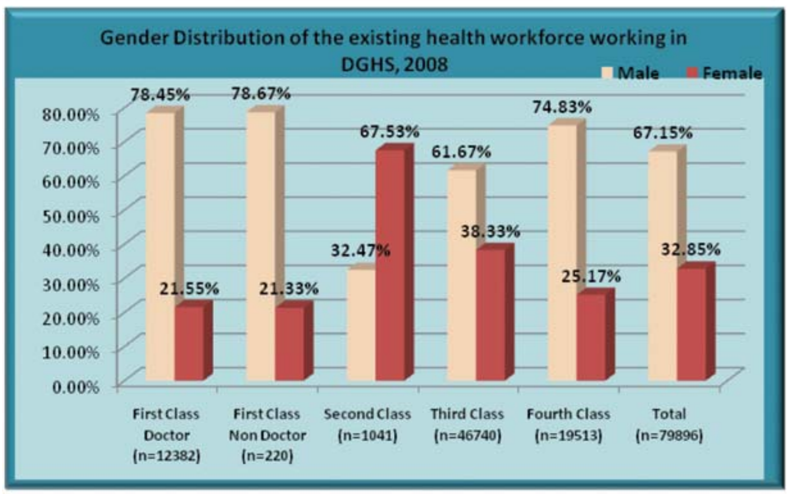
Class wise manpower summary under DGHS (No.)

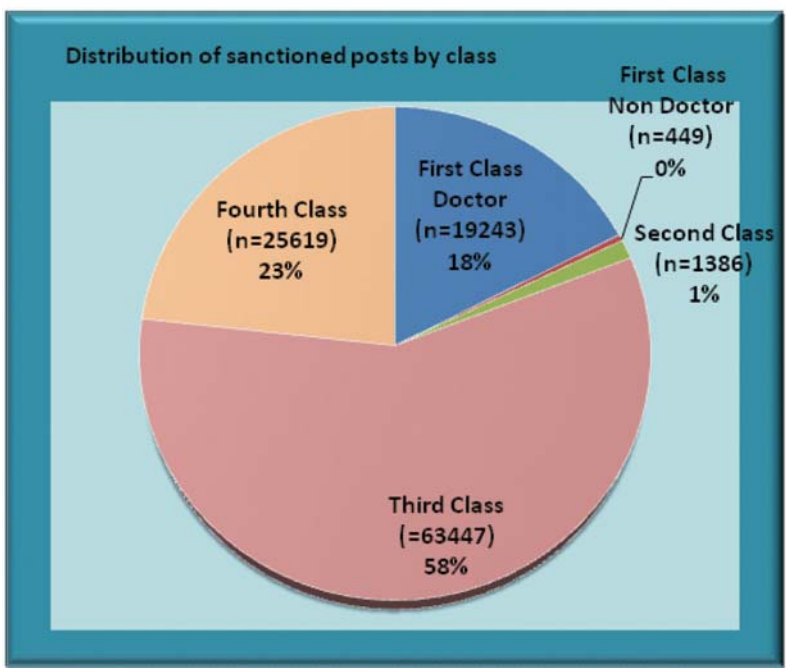
Class		Sanctioned Post	Existing			Vacant
			Male	Female	Total	
Class I	Doctors	19243	9713	2669	12382	6861
	Non-Doctors	449	173	47	220	229
Class II	-	1386	338	703	1041	345
Class III	-	63447	28826	17914	46740	16707
Class IV	-	25619	14601	4912	19513	6106
	Total	110144	53651	26245	79896	30248

Division wise manpower summary under DGHS (No.)

Name of division	Class		Sanctioned Post	Existing			Vacant
				Male	Female	Total	
Dhaka Division	Class I	Doctors	7201	4250	1474	5724	1477
		Non-Doctors	216	93	29	122	94
	Class II		600	196	287	483	117
	Class III		20236	8945	6713	15658	4578
	Class IV		9355	5407	1960	7367	1988
Chittagong Division	Class I	Doctors	3205	1548	389	1937	1268
		Non-Doctors	57	17	4	21	36
	Class II		199	31	108	139	60
	Class III		11727	5247	2628	7875	3852
	Class IV		4175	2318	661	2979	1196
Rajshahi Division	Class I	Doctors	4343	1954	434	2388	1955
		Non-Doctors	86	34	7	41	45
	Class II		272	62	137	199	73
	Class III		15161	7608	3921	11529	3632
	Class IV		6223	3638	1194	4832	1391
Khulna Division	Class I	Doctors	1918	878	161	1039	879
		Non-Doctors	46	20	7	27	19
	Class II		142	30	85	115	27
	Class III		7018	3056	2233	5289	1729
	Class IV		2389	1309	509	1818	571
Barisal Division	Class I	Doctors	1364	537	95	632	732
		Non-Doctors	24	6	0	6	18
	Class II		94	10	63	73	21
	Class III		4938	1955	1634	3589	1349
	Class IV		1775	1002	264	1266	509
Sylhet Division	Class I	Doctors	1212	546	116	662	550
		Non-Doctors	20	3	0	3	17
	Class II		79	9	23	32	47
	Class III		4367	2015	785	2800	1567
	Class IV		1702	927	324	1251	451
		Total	110144	53651	26245	79896	30248


Number of sanctioned, available and vacant posts under DGHS
Sanctioned Post
Available
Vacant
120000
100000
80000
60000
40000
20000
0
19243
12382
6861
449
220
229
1386
1041
345
63447
46740
16707
25619
19513
6106
110144
79896
30248
Class I Doctor
Class I Non Doctor
Class
Class III
Class IV
Total


Gender Distribution of the existing health workforce working in DGHS, 2008
Male
Female
80.00%
70.00%
60.00%
50.00%
40.00%
30.00%
20.00%
10.00%
0.00%
78.45%
21.55%
78.67%
21.33%
32.47%
67.53%
61.67%
38.33%
74.83%
25.17%
67.15%
32.85%
First Class Doctor (n=12382)
First Class Non Doctor (n=220)
Second Class (n=1041)
Third Class (n=46740)
Fourth Class (n=19513)
Total (n=79896)


Distribution of sanctioned posts by class
First Class Non Doctor (n=449) 0%
First Class Doctor (n=19243) 18%
Second Class (n=1386) 1%
Fourth Class (n=25619) 23%
Third Class (=63447) 58%

Number of sanctioned available and vacant posts of doctors

Post	Sanctioned	Available	Vacant
Director, Principal or equivalent	70	64	6
Deputy Director/Equivalent	81	54	27
Assistant Director / Civil Surgeon	191	170	21
DCS / UHFPO	503	483	20
Vice Principal	15	10	5
Professor	419	264	155
Associate Professor	581	355	226
Asstt. Professor	805	612	193
Senior Consultant	360	202	158
Senior Lecturer	8	6	2
Junior Lecturer	32	29	3
Junior Consultant / Equivalent	2,860	1,577	1,283
Assistant Surgeon	13,730	8,747	4,983
Total:	19,655	12,573	7,082

Number of sanctioned, available and vacant posts of medical technologists under DGHS (2008)

Designation	Sanctioned	Available	Vacant Post
Medical Technologist			
Pharmacist	2,834	2,207	627
Medical Technologist(Lab)	1,773	1,408	365
Medical Technologist(Dental)	517	475	42
Medical Technologist(Radiography)	695	646	49
Medical Technologist(Radiotheraphy)	38	30	8
Medical Technologist(Physiotheraphy)	185	32	153
Total	6,042	4,798	1,244

Number of sanctioned, available and vacant posts of medical Assistants and domiciliary workers

Post	Sanctioned	Available	Vacant
Medical Assistan	5,369	3,589	1,780
Domiciliary Workers			
Health Inspector	1,399	932	467
Assistant Health Inspector	4,196	3,613	583
Health Assistant	20,841	14,075	6,766
Total	26,436	18,620	7,816

Alternative Medical Care

Fourty five Medical Officers (15 Unani, 15 Ayurvedic and 15 Homoeopathic) on Alternative Medicine have been appointed in the selected district level hospitals under the work plan of HPNSP, so that the patients of these districts have the option to receive the types of treatment according to their own choice. To assist the medical officers 64 support personnel (compounder) have been appointed. To develop awareness on medicinal plants, 467 herbal gardens.

Personnel and staff of Alternative Medical Care working under DGHS

Post	No.
Medical officers (Unani)	15
Medical officers (Ayurvedic)	15
Medical officers (Homeopathic)	15
Support staff	64
Gardeners	467
Total	**576**

Source: Line Director, Alternative Medical Care, DGHS

Number of sanctioned, available and vacant posts of nurses under Directorate of Nursing Services

Class	Category	Sanctioned	Available	Vacant
I	Nursing	59	06	53
I	Non-nursing	1	0	1
II	Nursing	387	154	233
	Non-nursing	11	8	3
III	Nursing Supervisor	891	668	223
	Nursing	16149	13359	2790
	Non-nursing	263	215	48
IV	Non-nursing	704	635	69
Total		17574	14377	3197

Human Resource Development

As the population of the country is increasing, the country will need large number of physicians, nurses, medical technologists and other paramedical workforces to cope up with the growing need. Along with the national development, the country is experiencing a positive growth in the development of human resources for health. Both the public and private sectors are expanding their respective capacities of developing skilled health personnel that include physicians, nurses, medical technologists and other staffs like pharmacists. Bangladesh health personnel are also working in the different parts of the world and there is a growing need of Bangladeshi personnel globally.

In the subsequent part of this chapter, information on capacity of teaching/training institutes for development of human resources for health will be outlined.

Postgraduate medical degree

There are 33 institutes in Bangladesh which offer postgraduate specialist degrees in medical fields. Of these institutes, 22 are in public sector, 5 are non-profit organizations (these institutes receive financial grants from government to run the institutes or their affiliated hospitals), one is operated by Bangladesh Armed Forces and others are in private and NGO sector. Of the public sector institutes, 19 are under Directorate General of Health Services (DGHS) and one is autonomous medical university. Of the 19 institutes under DGHS, 9 are postgraduate institutes offering onlypostgraduate medical degrees. Others are medical or dental colleges and offer both undergraduate and postgraduate medical degrees.

List of postgraduate medical institutes (total number: 11) under DGHS

1. Institute of Child and Mother Health (ICMH), Matuail, Dhaka
2. Institute of Nuclear Medicine and Hospital, Dhaka
3. National Institute of Cancer Research and Hospital (NICRH), Mohakhali, Dhaka
4. National Institute of Cardiovascular Diseases (NICVD), Sher-E-Bangla Nagar, Dhaka
5. National Institute of Diseases of the Chest and Hospital (NIDCH), Mohakhali, Dhaka
6. National Institute of Kidney Diseases and Urology (NIKDU), Sher-E-Bangla Nagar, Dhaka
7. National Institute of Mental Health and Research (NIMHR), Sher-E-Bangla Nagar, Dhaka
8. National Institute of Ophthalmology (NIO), Sher-E-Bangla Nagar, Dhaka.
9. National Institute of Preventive and Social Medicine (NIPSOM), Mohakhali, Dhaka
10. National Institute of Traumatology, Orthopedic and Rehabilitation (NITOR), Sher-E-Bangla Nagar, Dhaka
11. Center for Medical Education

List of medical colleges under DGHS, which in addition to providing postgraduate medical degrees (total number: 10)

1. Chittagong Medical College, Chittagong
2. Dhaka Dental College, Dhaka
3. Dhaka Medical College, Dhaka
4. MAG Osmani Medical College, Sylhet
5. Mymensingh Medical College, Mymensingh
6. Rajshahi Medical College , Rajshahi
7. Rangpur Medical College , Rangpur

8. Sher-e-Bangla Medical College, Barisal
9. Sir Salimullah Medical College , Dhaka
10. SZR Medical College, Bogra

Medical University under MOHFW (total number: 1)

1. Bangabandhu Sheikh Mujib Medical University (BSMMU), Shahbagh, Dhaka

Postgraduate Institute under Bangladesh Armed Forces (total number: 1)

1. Armed Forces Medical Institute

List of non-profit institutes which offer postgraduate medical degrees (these institutes receive government grants for running their institute or affiliated hospital; total number: 4)

1. Bangladesh College of Physicians and Surgeons (BCPS), Mohakhali, Dhaka
2. Bangladesh Institute of Child Health, Sher-e-Bangla Nagar, Dhaka
3. Bangladesh Institute of Research and Rehabilitation in Diabetes, Endocrine and Metabolic Disorders (BIRDEM), Shahbagh, Dhaka
4. National Heart Foundation Hospital and Research Institute, Mirpur, Dhaka

List of other institutes in private and NGO sector which offer postgraduate medical degrees (total number: 9)

1. Chattagram Maa and Shishu and General Hospital, Chittagong
2. Institute of Child Health and Shishu Hospital, Shishu Sasthya Foundation, Bangladesh, Mirpur-2, Dhaka
3. Lions Eye Institute and Hospital, Lions Bhaban, Agragaon, Dhaka
4. MAI Institute of Ophthalmology and Islamia Hospital, Sher-e-Bangla Nagar, Dhaka
5. Institute of Health Sciences (Under USTC), Foy's Lake, Chittagong
6. Institute of community Opthalmology, Chittagong
7. James P Grant School of Public Health, BRAC University, Mohakhali, Dhaka
8. State University of Bangladesh, Dhanmondi R/A, Dhaka
9. Gono Bisshobidhyaloya (People's University)

Number of seats in the different postgraduate medical courses provided under institutes and colleges

Institute	Name of course with number of seats						
	MS	MD	M.Phil	Diploma	MPH	Other	Total
1. BSMMU	140	150	70	106	-	MTM-10	476
2. BCPS	-	-	-	-	-	-	0
3. Centre for Medical Education (CME)	-	-	-	-	-	MMED-15	15
4. Dhaka Medical College	70	110	36	82	06	-	304
5. Chittagong Medical College	37	35	13	48	03	-	136
6. Mymensingh Medical College	22	40	33	59	-	-	154
7. Rajshahi Medical College	10	19	25	41	05	-	100
8. MAG Osmani Medical College	20	12	28	40	-	-	100
9. Sher-E-Bangla Medical College	04	-	08	22	-	-	34
10. Rangpur Medical College	08	08	08	22	-	-	46
11. Sir Salimullah Medical College	21	36	18	40	05	-	120
12. BIRDEM	10	22	15	14	-	-	61
13. NICVD	20	20	-	14	-	-	54
14. NIDCH	06	15	-	20	-	-	41

Number of seats in the different postgraduate medical courses provided under institutes and colleges (contd...)

Institute	Name of course with number of seats						
	MS	MD	M.Phil	Diploma	MPH	Other	Total
15. Institute of Child Health and Shishu Hospital	-	-	-	06	-	-	06
16. National Institute of Child Health	10	15	-	15	-	-	40
17. National Institute of Cancer Research and Hospital	06	12	-	-	-	-	18
18. (NIPSOM, Dhaka	-	-	07	-	166	-	173
19. National Heart Foundation	05	05	-	-	-	-	10
20. Institute of Nuclear Medicine and Hospital	-	-	-	10	-	-	10
21. Institute of Child & Mother Health (ICMH)	10	10	-	30	-	-	50
22. National Institute of Ophthalmology	10	-	-	10	-	-	20
23. Mirza Ahmed Ispahani Institute of Ophthalmology and Islamia Hospital	-	-	-	10	-	-	10
24. Lions Eye Institute and Hospital	-	-	-	06	-	-	06
25. National Institute of Kidney Disease and Urology (NIKDU)	06	09	-	-	-	-	15
26. National Institute of Traumatology and Orthopedic Rehabilitation (NITOR)	20	-	-	10	-	-	30
27. Dhaka Dental College	22	-	-	-	-	-	22
28. Chittagong Maa-O-Shishu & General Hospital	-	-	-	06	-	-	06
29. National Institute of Medical Health	-	06	-	-	-	-	06
30. Institute of Health Sciences (Under USTC)	-	05	-	45	-	-	50
31. Institute of Community Ophthalmology	-	-	-	08	-	-	08
32. United Hospital Ltd	06	06	-	-	-	-	12
33. Shahid Ziaur Rahman Medical College	-	-	-	10	-	-	10
Total	463	535	261	674	185	25	2143

Undergraduate medical degree

There are 59 medical colleges and 14 dental colleges in the country to offer bachelor's degree in medicine (MBBS) and dentistry (BDS). Of the 59 medical colleges, 17 are under the MOHFW, one under Bangladesh Armed Forces and 41 are in the private sector. Of the 14 dental colleges, 3 are under MOHFW and 11 are in the private sector. The medical colleges currently have capacity of annual admission 5,549 students for MBBS course. These include 2,494 seats in medical colleges under MOHFW, 100 seats in Armed Forces Medical College and 3,055 seats in private medical colleges. Out of these seats in medical colleges under MOHFW, 40 seats are reserved for children of freedom fighters, 20 for tribal students and 84 for foreign students.

Medical colleges under MOHFW and their number of seats

Name (alphabetical order)	Established	No. of Seats
Dhaka Medical College	1948	178
Chittagong Medical College	1962	178
Mymensingh Medical College	1962	178
Rajshahi Medical College	1962	178
MAG Osmani Medical College	1966	178
Sher-E-Bangla Medical College	1968	178
Rangpur Medical College	1972	178
Sir Salimullah Medical College	1972	178
Comilla Medical College	1992	107
Dinajpur Medical College	1992	132
Faridpur Medical College	1992	107
Khulna Medical College	1992	132
Shahid Ziaur Rahman Medical College	1992	132
Shahid Sarwardi Medical College	2005	126
Pabna Medical College	2008	50
Coxe's Bazar Medical College	2008	50
Noakhali Medical College	2008	50
	Total	2310

Note: Out of these total seats, 40 seats are reserved for children of freedom fighters, 20 seats for tribal students and 84 seats for foreign students (54 for SAARC and 30 for non-SAARC countries)
Source: Director, Medical Education and Health Manpower Development, Mohakhali, Dhaka

Medical colleges under Bangladesh Armed Forces and its number of seats

Name	No. of Seats
Armed Forces Medical College, Dhaka	100

Source: Director, Medical Education and Health Manpower Development, Mohakhali, Dhaka

Private medical colleges and their number of seats

Name (alphabetical order)	Estd.	No. of Seats
1. Bangladesh Medical College, Dhaka	1985	110
2. Samaj Vittic Medical College, Dhaka	1989	70
3. Institute of Applied Health Sciences, Chittagong	1990	200
4. Jahurul Islam Medical College, Kishoregonj	1992	100
5. Medical College for Women and Hospital, Dhaka	1992	100
6. Z.H Sikder, Women Medical College, Dhaka	1992	75
7. Community Based Medical College, Mymenshing	1995	100
8. Dhaka National Medical College, Dhaka.	1995	125
9. Jalalabad Ragib Rabea Medical College, Sylhet	1996	160
10. Moulana Bhashani Medical College, Dhaka	1998	75
11. North East Medical College, Sylhet	1998	100
12. East West Medical College, Dhaka	2000	75
13. Holy Family Red Crescent Medical College, Dhaka	2000	110
14. International Medical College, Gazipur	2000	75
15. North Bengal Medical College, Sirajganj	2000	50
16. Kumudini Medical College, Tangail	2001	100
17. Tairunnessa Medical College, Gazipur	2001	50
18. BGC Trust Medical College, Chittagong	2002	100
19. Ibrahim Medical College, Dhaka	2002	100
20. Enam Medical College, Dhaka	2003	100
21. Shahabuddin Medical College, Dhaka	2003	75
22. Islami Bank Medical College, Rajshahi	2004	50
23. Central Medical College, Comilla	2005	50
24. Eastern Medical College, Comilla	2005	50
25. IBN Sina Medical College, Dhaka	2005	50
26. Khawja Yunus Medical College, Sirajganj	2005	50
27. Chottogram, Ma -O- Shishu Medical College, Dhaka	2006	75
28. Sylhet Women Medical College, Sylhet	2006	75
29. Nightangel Medical College, Ashulia, Dhaka	2006	50
30. Southern Medical College, Chittagong	2006	50
31. Northern International Medical College, Dhaka	2006	50
32. Northern Medical College, Rangpur	2006	30
33. Uttara Adhunik Medical College, Dhaka	2007	75
34. Delta Medical College, Dhaka	2008	50
35. Ad-din Women's Medical College, Dhaka	2008	50
36. Community Medical College, Dhaka	2008	50
37. TMSS Medical College, Bogra	2008	50
38. Anwar Khan Modern Medical College, Dhaka	2008	50
39. Prime Medical College, Rangpur	2008	50
40. Rangpur Community Hospital Medical College, Rangpur	2008	50
Total=		3055
Sl. No 8, 13, 18, 29, 31, 32, 33, 35 (Total No. 8) is running by high court rule Sl. No. 2 and 26 (Total No.2) college is running by their own rule (Private University) Sl. No. 30 (Total No. 1) New admission stopped from session 2007-2008 (Previous students transferred to Sylhet Women's Medical College		

Dental colleges under MOHFW and their number of seats

Institute	No. of Seats
Dental Unit, Chittagong Medical College, Chittagong	50
Dental Unit, Rajshahi Medical College, Rajshahi	50
Dhaka Dental College, Dhaka	110
Total	210
*10 seats are reserved for children of freedom fighters and 5 seats are for tribal quota . Source: Director, Medical Education and Health Manpower Development, Mohakhali, Dhaka	

Three dental colleges under MOHFW and 11 in private sector offer bachelor degree in dentistry (Bachelor of Dental Surgery). A total of 910 students can be admitted per year, 210 in government dental colleges and 700 in private dental colleges. Ten seats in government dental colleges are reserved for children of freedom fighters.

Private dental colleges and their number of seats

Name (alphabetical order)	Estd.	No. of Seats
1. Pioneer Dental College, 111 Malibag, DIT Road, Dhaka	1995	100
2. University Dental College, 120 Shiddeswari Outer Circular Road, Century Orchid (4th Floor), Moghbazar, Dhaka	1996	75
3. Bangladesh Dental College, House # 35, Road # 14/A, Dhanmondi R/A, Dhaka	1997	50
4. Samaj Vittik Dental College, Mirzanagar, Savar, Dhaka	1997	50
5. City Dental College, 1085/1, Malibag Chowdhury Para, Dhaka	1998	75
6. Saporro Dental College, Road # 1/B, Sector # 9, Uttara, Dhaka	2000	50
7. Chittagong International Dental College, Chittagong	2005	50
8. Marks Dental College, A/3, Main Road, Section-14, Mirpur, Dhaka	2008	50
9. Rangpur Dental College, Rangpur	2008	100
10.Update Dental College, 162, Atish Dipankar Road, West Mugdha, Dhaka	2008	50
11.Udayan Dental College, Rajshahi	2008	50
	Total	**700**
Source: Director, Medical Education and Health Manpower Development, Mohakhali, Dhaka		

Graduation on alternative medicine

Name (alphabetical order)	No. of Seats
Homoeopathic Degree College, Dhaka	50
Sylhet Govt. Tibbia College, Sylhet[1]	25
Unani and Ayurved College, Dhaka	50
Total	**125**
1. Offers diploma Source: Director, Medical Education and Health Manpower Development, Mohakhali, Dhaka	

It is estimated that there are over 19,000 graduates and/or diploma holders of alternative medicines in the country. Their detail distribution is given in the table below.

Graduates and diploma holders in alternative medicine

Number of qualified personnel in Altenative Medical Care (Year 2008)

Qualified Personnel in AMC	Number
Unani Graduates (BUMS)	373
Ayurvedic Graduates (BAMS)	297
Unani Diploma holders (DUMS)	1331
Ayurvedic Diploma holders (DAMS)	491
Homeopathic Graduates (BHMS)	777
Homeopathic Diploma holders (DHMS)	16222
Total	**19015**

Nursing education

There are 13 nursing colleges and 69 nursing institutes in the country for production of nursing workforce. The nursing colleges offer BSc nursing degree, and the nursing institutes offer diploma in nursing. Of the 13 nursing colleges, 5 are under MOHFW and 7 under private sector and one is manned by Bangladesh Armed Forces

Bangladesh Nursing Council Affiliated Nursing Institute & College with Seat

Type of Institute	Number	Seats
Govt. (Under DNS) Nursing Institute	46	1770
A.F.M.I, Dhaka Cant.	01	50
Private Nursing Institute	22	575
Nursing Institutes Diploma Course (Total)	69	2395
Nursing Colleges		
Govt. Nursing College (Post Basic) Mohakhali (Foreign-5)	1	125
A. F. M. I, Dhaka Cant. (Basic)	1	25
Govt. Nursing College (Basic)	4	400
Private Nursing College (Basic & Post Basic) No. of Seat	07	215
Total	**13**	**765**

Of the 69 nursing institutes, 46 are under MOHFW, 1 under Bangladesh Armed Forces and 22 under private sector. Under the MOHFW, 6 nursing institutes are attached with medical college hospitals, 11 with general hospitals and 17 with district hospitals. The nursing institute under the Bangladesh Armed Forces is attached with the Armed Forces Medical Institute, Dhaka. Besides, 2 institutes produce specialized nurses. These are National Heart Foundation, Mirpur, Dhaka (20 seats; intensive care unit, coronary care unit, cardiac nursing) and Bangladesh Health Professionals Institute, Savar, Dhaka (20 seats; rehabilitation nursing).

Names of Nursing Institute, College & Specialized Nursing Institute

Attached to Medical College-6	
Name of Institute	**seats**
1. S.S.M.C. Mitford Hospital, Dhaka	50
2. M.A.G. Osmani Medical College Hospital, Sylhet	100
3. R.M.C.H, Rangpur	100
4. Sher-e-Bangla Medical College Hospital, Barisal	100
5. Comilla Medical College Hospital, Comilla	50
6. Faridpur Medical College Hospital, Faridpur	50
Total	**450**
ATTACHED TO GENERAL HOSPITAL-11	
7. General Hospital, Khulna	50
8. Mohammad Ali. Hospital, Bogra	50
9. General Hospital, Dinajpur	50
10. General Hospital, Noakhali	50
11. General Hospital, Pabna	50
12. General Hospital, Jessore	50
13. General Hospital, Kushtia	50
14. General Hospital, Tangail	50
15. General Hospital, Rangamati	50
16. General Hospital, Patuakhali	50
17. General Hospital, Sirajgonj	30
Total	**530**

Names of Nursing Institute, College & Specialized Nursing Institute (contd...)

No	Name of Institute	
ATTACHED TO SADAR HOSPITAL -29		**Seats**
18.	Sadar Hospital, Munshigonj	30
19.	Sadar Hospital, Chuadanga	30
20.	Sadar Hospital, Magura	30
21.	Sadar Hospital, Cox's Bazer	30
22.	Sadar Hospital, Moulvibazer	30
23.	Sadar Hospital, Sherpur	30
24.	Sadar Hospital, Chapainowabgonj	30
25.	Sadar Hospital, Joypurhat	30
26.	Sadar Hospital, Satkhira	30
27.	Sadar Hospital, Thakurgaon	30
28.	Sadar Hospital, Rajbari	30
29.	Sadar Hospital, B. Baria	30
30.	Sadar Hospital, Feni	30
31.	Sadar Hospital, Bagerhat	30
32.	Sadar Hospital, Kurigram-01712-657082	30
33.	Sadar Hospital, Bhola	30
34.	Sadar Hospital, Netrokona	30
35.	Sadar Hospital, Gopalgonj	30
36.	Sadar Hospital, Madaripur	30
37.	Sadar Hospital, Pirojpur	30
38.	Sadar Hospital, Baroguna	30
39.	Sadar Hospital, Naugon	30
40.	Sadar Hospital, Nilphamari	30
41.	Sadar Hospital, Panchgar	30
42.	Sadar Hospital, Kishorgonj	30
43.	Sadar Hospital, Jamalpur	30
44.	Sadar Hospital, Jhinaidah	30
45.	Sadar Hospital, Chandpur	30
46.	Sadar Hospital, Hobigonj	30
	Total	870
	Total Govt. (Under DNS) 46 N.I. Seat =	1770
	A.F.M.I. DHAKA CAN'T. DHAKA	50

Private Nursing Institute -22		Seats
1.	B A Siddiki N.I, H.F.R.C. Hospital, Dhaka	50
2.	Kumudini Nursing Institute, Mirzapur, Tangail	50
3.	Zahurul Islam Nursing Institute, Bajitpur, Kishorgonj	20
4.	Christian Mission Hospital, Rajshahi	20
5.	Christian Hospital, Chandraghona	30
6.	N. I, BHPI, (C.R.P) Savar, Dhaka	20
7.	KYAMCH Nursing Institute, Anyatpur, Sirajgonj	50
8.	Daiabatic Association Nursing Institute, Faridpur	25
9	Moulana Bhasani Nursing Institute, Uttara, Dhaka	20

Names of Nursing Institute, College & Specialized Nursing Institute contd...

Private Nursing Institute -22		Seats
10	Fatima Nursing Institute, Ad-din Maternity Hospital, Bara-Moghbazar, Dhaka	50
11	Ad-din Nursing Institute, Ad-din Maternity Hospital, Jessore	30
12	Safina Nursing Institute, Ad-din Maternity Hospital, Kushtia	30
13	Islami Bank M.C.H. Nursing Institute, Rajshahi	40
14	North East Nursing Institute, Talyhawr, Sylhet	25
15.	N.I Shisu Shastha Foundation Hospital, Mirpur, Dhaka	20
16.	N.I, CHP Joyrumkura, Haluaghat, Mymunsing	20
17.	N.I, Medical College for Women & Hospital, Uttara.	25
18.	Ctg. Maa-O-Shisu Hospital N.I, Agrabad, Chittagong	25
19.	N.I, Central Hospital, Green Road, Dhanmondi, Dhaka	20
20.	TMMC Nursing Institute, Targas, Bordbazar, Gazipur	20
21.	Greenlife Hospital Nursing Institute, Green Road, Dhaka	20
22.	TMSS Nursing Institute, Thangamara, Gokul, Bogra	25
	Total	**635**

List of Government Nursing Colleges

Govt. Nursing College – 06		Seats
1.	College of Nursing, Mohakhali, Dhaka (Post Diploma) (Reserved for Foreign Student -5 Seat)	125
2.	College of Nursing, DMCH, Dhaka (Diploma)	100
3.	College of Nursing, MMCH, Mymunsingh (Diploma)	100
4.	College of Nursing, RMCH, Rajshahi (Diploma)	100
5.	College of Nursing, CMCH, Chittagong (Diploma)	100
6.	AFMI, Dhaka Cant. Dhaka (Army Board) (Diplomac)	25

List of Private Nursing Colleges

Private Nursing College - 07	Seats
Kumudini Nursing College, Mirzapur, Tangail (Diploma & Post Diploma)	30 30
State College of Health Science, Dhanmondi, Dhaka (Diploma & Post Diploma)	20 20
International Medical College, Tongi, Gazipur (Diploma)	20
TMSS Nursing College, Bogra (Post Diploma)	30
Moulana Bhashani Nursing College (Diploma)	25
North East Nursing College, S.Surma, Syllhet (Diploma)	20
Jalalabad Ragib-Rabeya N/C. Pathantula, Syllhet (Diploma)	20
Total	**215**

Specialized Nursing Course (Private-Total 2 Nos.)	Seats
National Heart Foundation, Mirpur, Dhaka –Diploma in Cardiac (ICU/CCU/Cathlab) Nursing	20
BHPI, (CRP) Chapain, Savar, Dhaka -Diploma in Rehabilitation Nursing	20
Total	45

Junior Midwifery Institute (Total 8 Nos.)	Seats
Junior Midwifery Institute, Hollyfamily Red Crescent Hospital, Dhaka	25
S.M.U.R. Maternity Hospital, Bangla 1azaar, Dhaka.	20
Mamo n Hospital, City Corporation, Chittagong	25
C. R. Maternity Hospital, Chandpur	20
Fatema Hospital, Jessore	10
Ad-din Maternity Hospital, Jessore	20
Christian Hospital, Bogra	10
Kumudini Hospital, Mirzapur, Tangail.	10
Total	140

Production of community based skilled birth attendants

There are 39 government and 2 private training institutes to produce community based skilled birth attendants. Nearly 5000 C-SBA have been trained and are serving in their own area.

Training institutes for production of community based skilled birth attendants

Type of Institute	Location
CSBA Institute, (Civil Surgeon) With G.Hospital/Sadar Hospital	Narayangonj (WHO), Manikgonj, Kishoregonj, Jamalpur, Habigonj, Gopalgonj, Narsingdi, Nilphamari, Natore, Naogaon, Kurigram, Panchagarh, Gaibandha, Jhinaidah, Bagerhat, Rajbari, Madaripur, Munshigonj, Chandpur
Family Welfare Visitor Training Institute	Tangail (WHO), Barisal, Faridpur, Comilla (WHO), Kushtia, Khulna (WHO)
CSBA Institute (With Nursing Institute)	Noakhali, Jessore, Satkhira, Thakurgaon, Feni, Joypurhat, Pabna, B. Baria, Netrokona, Chuadanga, Cox’s Bazer, Patuakhali, Chapai-Nowabgonj, Sirajgonj
CSBA-Institute (Private)	Kumudini Hospital. Mirzapur, Tangail, Lamb Hospital, Parbatipur, Dinajpur

Production of medical assistants

Medical assistants are the assistants to the doctors working at the upazila health complexes or union sub-centers. Bangladesh has a shortage of graduate medical doctors. In this context, the medical assistants serve as the doctors. Currently there are 7 medical assistant training schools in the country which together have seat capacity of 650 students. Medical assistant's course requires a student to complete a 3 years' course to obtain a medical diploma.

Medical Assistant Training School under DGHS and their number of seats

Name of MATS	No. of seats
Medical Assistant Training School, Bagerhat	150
Medical Assistant Training School, Kushtia	100
Medical Assistant Training School, Noakhali	100
Medical Assistant Training School, Sirajganj	100
Medical Assistant Training School, Tangail	100
Medical Assistant Training School, Comilla	50
Medical Assistant Training School, Faridpur	50

Note: A new MATS is being constructed in Jenidah district
(Admission of 50 students have been started in Faridpur and Comilla MATS in 2008-2009 session, and 50 seats are added in Tangail, Sirajganj, Noakhali, Kushtia and Bagerhat MATS).
Source: Director, Medical Education and Health Manpower Development, Mohakhali, Dhaka

Private medical assistant training school and their number of seats

Name of institute	Established	Total
Sylhet Medical Assistant Training School, Sylhet	2008	40
Comilla IHT, Thakur Para Medical Assistant Training School, Thakur Para, Comilla	2008	50
SIMT Medical Assistant Training School, Kalabagan, Dhaka	2008	100
SIMT Medical Assistant Training School, Nishindhora, Bogra	2008	20
SPKS Medical Assistant Training School, Mirpur, Dhaka	2008	100
Moulobibazar Medical Assistant Training School, Kushumbag, Moulobibazar	2008	80
Advance Medical Assistant Training School, Green road, Dhaka	2008	100
The Medical Assistant Training School, Mirpur, Dhaka	2008	100
New Pilot Medical Assistant Training School, Tangail Sadar	2009	50
Bangladesh Medical Assistant Training School, Uttara, Dhaka	2009	50
AR Medical Assistant Training School, Mohammadpur, Dhaka	2009	75
	Total	1855

Production of medical technologists

Medical technologists are technicians who perform the laboratory tests, take x-ray images, provide physiotherapy or radiotherapy, help making artificial dentures, etc. To produce medical technologists there are currently both graduate and diploma courses in the country.

Sixteen institutes conduct BSc medical technology courses in laboratory medicine, physiotherapy, occupational therapy and dentistry. Total seat capacity is 1065. Of the 16 institutes, 3 are under MOHFW and the rests are in private sector.

To produce diploma medical technologists, the DGHS has 3 institutes of health technology (IHT) in the government sector. They altogether have 1010 seats. There are 47 private IHTs which have total seat capacities of 5696. These IHTs offer 3 years diploma in medical technology in 7 disciplines, viz. laboratory, radiography, physiotherapy, dental technology, radiotherapy and pharmacy.

Currently 50 institutes both in government and private sector have total seat capacity of 6706.

Two institutes (Gonobisshobidhyaloya, Savar and BHPI, Savar: total 25 seats)have started MSc course in medical technology in discipline of physiotherapy.

Institute of Health Technology under DGHS and their number of seats

Name of nursing institute	Estd.	No. of seats
Institute of Health Technology , Dhaka	1963	327
Institute of Health Technology , Rajshahi	1976	326
Institute of Health Technology, Bogra	2006	357
	Total	**1010**

Note: Another 7 Institutes of Health Technologies are being established.
Source: Director, Medical Education and Health Manpower Development, Mohakhali, Dhaka

BSc Institutes of Health Technology under DGHS and their number of seats

Name	Total
NITOR, Sher-e-Bangla Nagar, Dhaka	25
Institute of Health Technology, Dhaka	60
Institute of Health Technology, Rajshahi	60
Total	145

Source: Director, Medical Education and Health Manpower Development, Mohakhali, Dhaka

BSc and MSC degrees provided by Institutes of Health Technology in public and private sector and their number of seats

Name	Institute Type of Teaching	Number of Institute	Number of Seats
BSc in Health Technology	Government	3	110
	Private	13	1065
	Total	16	1175
MSc in Health Technology	Government	-	-
	Private	2	65
	Total	2	65
		Total	1240

Institute of Health Technology in private sector offering diploma in medical technology and their number of seats

Name of institute	Estd.	Total
Bangladesh Medical College, Dhaka	-	25
Bangladesh Institute of Medical and Dental Technology, Dhaka	1996	150
Centre for Rehabilitation of the Paralyzed (CRP), Dhaka	1999	200
Institute of Health Technology, Chittagong	2000	200
Health Wage Institute of Medical Technology, Bogra	2002	181
Janata Institute of Medical Technology, Bogra	2002	115
Marks Institute of Medical Technology, Dhaka	2002	150
Green view Institute of medical technology, Dhaka	2002	105
Rajshahi Institute of Medical Technology, Rajshahi	2002	170
National Institute of Medical Technology, Dhaka	2003	150
Radiant College of Medical Technology, Dhaka	2003	80
Institute of Medical Technology, Chittagong	2003	150
Institute of Medical Technology, Faridpur	2005	100
National Institute of Medical & Dental Technology, Dhaka	2005	65
New lab Institute of Medical Technology, Dhaka	2005	110
Chittagong Institute of Medical Technology, Chittagong	2005	150
Elah College of Medical Technology, Comilla	2005	50
Institute of Community Health Bangladesh, Dhaka	2005	75

Institute of Health Technology in private sector offering diploma in medical technology and their number of seats (contd...)

Name of institute	Estd.	Total
SAIC Institute of Medical Technology, Dhaka	2005 & 2008	160
Gonoshasthya Institute of Health Sciences, Gazipur	2006	100
International Institute of Health Sciences, Dhaka	2006	160
Prime Institute of Medical Technology, Rajshahi.	2006	200
Islami Bank Institute of Health Technology, Rajshahi	2007	150
Millennium Institute of Medical Technology, Dhaka	2007	75
Bangladesh Institute of Medical Technology, Pabna	2007	120
Fortune Institute of Medical Technology, Dhaka	2007	165
Professor Shohrab Uddin Institute of Medical technology, Tangail	2007	190
Shad S. A. Memorial Institute of Medical Technology, Dhaka	2007	75
Sumona Institute of Medical technology, Sadarghat	2007	130
Comilla Institute of Medical Technology, Comilla	2007	100
Comilla Institute of Medical Technology, Comilla	2007	75
Prime Institute of Science and Technology (PRISMET), Rangpur	2007	125
Rumdo Institute of Health Technology, Mymensingh	2007	85
Ad-deen Women's Institute of Health Technology, Jessore	2007	100
TMSS Medical Technology Institute, Thengamara, Bogra	2007	210
Institute of Medical and Dental Technology, Tangail	2007	105
Ahsania Mission Institute of Health Technology, Dhaka	2008	75
Trauma Institute of Medical Technology, Dhaka	2008	100
Dhaka Institute of Health Technology, Dhaka	2008	145
Institute of British Colombia Medical Technology, Dhaka	2008	125
Prince Institute of Medical Technology, Dhaka	2008	115
CSCR Institute of Medical Technology, Chittagong	2008	115
State college of health sciences, Dhaka	2008	50
SAIC Institute of Medical Technology, Bogra	2008	125

Institute of Health Technology in private sector offering diploma in medical technology and their number of seats (contd...)

Name of institute	Estd.	Total
A.R Institute of Medical Technology, Dhaka	2008	225
Fashion Eye Hospital Limited Fashion Tower, Dhaka	2008	20
Bangladesh Jatiyo Andah Kallan Samiti, Comilla	2008	50
Jefri Institute of Health Sxience & Technology, Dhaka	2009	250
Total		5946
Source: Director, Medical Education and Health Manpower Development, Mohakhali, Dhaka		

In-service Training

The Directorate General of Health Services (DGHS) has provision of giving the personnel and staffs on the job training, both local and foreign. In 2008-2009, a total number of 12,371 personnel and staffs were given local training and 129 personnel were given foreign training. The Technical Training Unit (TTU) under the Line Director of In-service Training is responsible for providing local training to the personnel and staffs under DGHS.

Type of training	No. of participants
Local Training	12,305
Foreign Training	129
Total	**12,434**

Training Achievement of In-Service Training (IST), DGHS, Mohakhali, Dhaka for the (year 2008) Local training & Overseas Training	
Name of Activities	**No. of Participants**
Local Training	
6 days ESP orientation for auxiliary service providers. Including curriculum Review	150
6 days ESP refresher training for field service providers including curriculum development.	2000
Curriculum development/Review for Nurses and Paramedics on advanced ESP clinical skills from district, upazila and below on Reproductive health (10 days)	Done
Curriculum development for Nurses and Paramedics on advanced ESP clinical skills from district, upazila and below on Child health care (6 days)	Done
Training for Nurses and Paramedics on advanced ESP clinical skill s from district, upazila and below on Reproductive health (10 days)	58
Breast feeding counselling for health care providers (Doctors and Nurses) (6 days) Including curriculum Review	148
5 days training on Asthma prevention and management for Medical graduates. Including curriculum review	126
5 days training on Cancer awareness, screening and primary detection for doctors including curriculum development.	20
1 day orientation on survical and breast cancer awareness for opinion leaders including curricu lum and teaching aids development.	200

Training Achievement of In-Service Training (IST),DGHS, Mohakhali, Dhaka for the (year 2008) Local training & Overseas Training (contd...)	
6 days training on nutrition for field service providers	1750
2 days Worksop on Medical Biotechnology	50
Breast feeding counseling training for health care providers (HAs/Field service providers.) (3 days)	275
2 weeks training on intensive coronary care for junior doctors workin in the CCU / cardiology department of medical colleges including curriculum review.	37.00
2 weeks training on intensive coronary care for staff nurses workin in the CCU/cardiology department of medical colleges including curriculum review.	18.00
5 days Training on awareness of primary health care doctors on biochemical parameters for prevention and control of cardiovascular diseases	47.00
3 days Training of health technologists on biochemical tests for diagnosis of cardiovascular risks and diseases	
3 days training program on primary health care physicians on mental health including curriculum development	60
2 days training program on primary health workers on mental health	90
Training on primary management & prevention of kidney & urological diseases for primary health care physicians (6 days)	60
Training on kidney & urological diseases for nurses working at p rimary health care level (6 days).	60
Training on Kidney & urological diseases for health workers working at primary health care level (6 days).	120
6 days training for doctors on violence against women and girls.	188
6 days training for nurses on violence against women and girls.	215
6 days training on management & prevention of substance abuse including alcohol for doctors. (including curriculum development)	25
6 days training on management & prevention of substance abuse including alcohol for nurses and medical assistants. (including curriculum development)	50
2 days orientation on medicolegal activities for CS, DCS,RMO etc	150
6 days training on Applied forensic Medicine including post mortem for MOs , RMOs and UH&FPO (including curriculum development)	200
One year Training Course on Diploma in Aneasthesia (DA) and Diploma in Gynae & Obs (DGO) for Doctors	
6 days training on improved financial management for personnel working at Division,District,Upazila and Specialized Institutions,TTU and Others Including curriculum Review	275
15 days basic service management training for doctors.	60
2 days training on monitoring and supportive supervision for supervisors at upazila level and below (HI, AHI, SI, EPI Tech. , MA etc) including curriculum review.	720
Hardware training on computer operation for officer and staff.	30
Computer programming on MS access and SPSS for officer and staff (including Curriculum review).	39
Computer programming on Graphics Design and webpage design for officer and staff	30

28 days basic computer training on operating system , installation, internet etc.for the persons of MOHFW, DGHS and autonomous institute.	**89**
14 days refresher computer training on operating system, installation, internet etc.for the persons of MOHFW, DGHS and autonomous institute.	900

Training Achievement of In-Service Training (IST),DGHS, Mohakhali, Dhaka for the (year 2008)Local training & Overseas Training (contd...)	
2 days PMIS training for PMIS recording & reporting tools.	301
2 days training for service statistics related MIS recording & reporting tools.	301
5 days training on standard operating procedures (SOP) regarding IPD, OPD, OT, emergency, housekeeping, record keeping, nursing services, diagnostic services, etc. for service providers of primary, secondaryand tertiary Hospitals including monitoring and supervision including curriculum review.	180
3 days Traning on SOP for MLSS, aya, attendant, sweper, cleaner, security, guard, etc. from primary, secondary and tertiary level hospitals including monitoring and supervision. And development/review of hand out.	180
15 Days Computer Training on DMIS for Health Personnel from district and Upazilla	120
3 days Women's professional development programme for personnel from district/ directorate/ Secrtariate level manegers.	60
5 days Mid level manegment development programme for personnel from district level Health manegers & UH& FPO.	75
3 days training on technique of developing training media and maintenance of audiovisual equipment for audiovisual operator,audiovisual projectionist,audiovisual helper and audiovisual technician including curriculum development	64
Reporting and Dissemination of Different activities under Inservice training and update training facility and resource inventory of district /Upazilla	Done
Maintenance and Further Development of Training Management Information System (TMIS)	Done
21 days advanced computer training on District management Information System (DMIS).	253
Need assessment of different categories of training	done
Evaluation of different categories of training activities	done
28 days English language course for health personnel	125
Training on updating media and messages in support on HEP for HEO/HE/Other related officers	100
Training on gendrer issue and poverty alliviation for HEO/HE/Other related officers	25
1 day training for doctors , medical assistant, MA, Paramedical Health/ Field Staff, Nurses, RMP, Drug distributor, formal and informal leaders etc on filariasis elimination & morbidity control to be held at ditivional / district / upazilla level with field implementation of HH registration , Mass drug administration and coverage survey .	200
Organization of 2 days joint simulation exercise with BDRCS at most cyclone prone districts. (Multi-sectoral approach) on EPR	50
Conduct vulnerability and Capacity Assessment at 10 (ten) selected hazard prone areas on EPR.	100
2 days traininge for field staff on Diseaseter Mitigition	90
Two days orientation course for Disaster Focal Points on EPR	
Training course on Mass casualty management for hospital level staffs.	40
2 days Orientation on service statistics of MO, MA, MT, HI, AHI and HA	150
HEP Training for the mid level managers (5 days) Health Education officers/ Health Educator	28

Programme/ Administrative Management Training for the HEP Focal Points . (Do) Health Education officers/ Health Educator	38
Implemented by the Following cost Center of In -service Training	
Bangladesh College of Physicians and Surgeons (BCPS)	
TOT Program	199

Training Achievement of In-Service Training (IST),DGHS, Mohakhali, Dhaka for the (year 2008) Local training & Overseas Training (contd...)

Orientation Training program of Participants	404
Attachment of Participants at different Department at different Institution.	186
Workshop, Seminar etc.	0
Workshop on Accredation and assesment Procedure, Dissertation writing, Information technology & Communication Skill development,Curriculumn development, Research methodology, OSPE & OSCE, Examination and assesmentprocedure for 40 subject	141
Institute of Epidemiology Disease control and Research (IEDCR)	
Training on Outbreak Investigation for One Week	136
Training on Laboratory Diagnosis of emerging and reemerging diseases for one week	86
Certificate Course on Clinical Epidemiology for three months	15
Institute of Public Health (IPH)	
Microbiological Lab Training of the Lab personnel	304
Institute of Child and Maternal Health (ICMH)	
TOT for Doctors on advanced ESD clinical skills training course on Reproductive Health	27
TOT for Doctors on advanced ESD clinical skills training course on Child Health care	23
Training on Complementary Feeding	27
Breastfeeding counseling for health care providers	23
TOT on Nutrition Program, Planning & Mangement	12
National Institute of Preventive & Social Medicine (NIPSOM)	
Continuation of ongoing Training on Epidemiology and Bio-Statistics using SPSS with Cambrige University (Data analysis) and linkage with NIPSOM	2
Oversease Training	129
Total	12434

Achievement of In-Service Training (IST) from July,2008 to April,2009 Local Training

Name of Activity	Achievement
Training for Nurses and Paramedics on advanced ESP clinical skills from district, upazila and below on Reproductive health (10 days)	136
Training for Nurses and Paramedics on advanced ESP clinical skills from district, upazila and below on Reproductive health (06 days)	67
1 day orientation on survical and breast cancer awareness for opinion leaders including curriculum and teaching aids development.	550
Breast feeding counseling training for health care providers (HAs/Field service providers.) (3 days)	365
6 days training for doctors on violence against women and girls.	165
6 days training for nurses on violence against women and girls.	410
2 days orientation on medico legal activities for CS, DCS, RMO etc including Curriculum Review	300
1 day Orientation on Continuing Performance Development (CPD) on Medical, Surgical & Management skill for Medical personnel at division level	100

3 days Training on basic management skill with curriculum development and curriculum review	134
6 days Training on Applied forensic Medicine including post mortem for MOs, RMOs and UH&FPO (including curriculum development)	250

Achievement of In-Service Training (IST) from July,2008 to April,2009 Local Training (contd...)

Name of Activity	Achievement
6 days Training on improved financial management for personnel working at Division, District, Upazila and Specialized Institutions, TTU and Others Including curriculum Review	369
21 days basic service management training for newly recruited doctors including TOT & curriculum development/Review.	265
Training on TMIS recording and reporting for personnel of DTCC & DUTT and other related institutions including guide book	200
Advanced training on computer networking (including Curriculum/Guide book Development review (28 days)	56
Hardware training on computer operation for officer and staff (including Curriculum/Guide book Development/ review.) (28 days)	173
Computer programming on MS access and SPSS for officer and staff (including Curriculum/Guide book Development/ review.) (28 days)	77
Computer programming on Graphics Design and webpage design for officer and staff (including Curriculum / Guide book Development/ review.) (28 days)	75
28 days basic computer training on operating system, installation, internet etc. for the persons of MOHFW, DGHS and autonomous institute.	210
14 days refresher computer training on operating system, installation, internet etc.for the persons of MOHFW, DGHS and autonomous institute.	289
2 days PMIS Training for PMIS recording & reporting tools.	370
2 days Training for service statistics related MIS recording & reporting tools.	494
5 days Training on standard operating procedures (SOP) regarding IPD, OPD, OT, emergency, house keeping, record keeping, nursing services, diagnostic services, etc. for service providers of primary, secondary and tertiary Hospitals including monitoring and supervision including curriculum review.	150
15 Days Computer Training on DMIS for Health Personnel from district and Upazilla	377
5 days Mid level management development programme for personnel from district level Health managers & UH& FPO including curriculum development/Review	208
Development of Management Information System (MIS) at Primary, Secondary, tertiary and Specialized Hospitals and developing computer skill of related personnel by providing Basic Computer Training and hands on training on developed software including Curriculum development.	44
21 days advanced computer training on District management Information System (DMIS).	90
28 days English language course for health personnel	25
1 day orientation for awareness building on violence against women for community leaders	723
2 days Training programme on reproductive health for community Gate keeper (UP Chairman, UP member, Imam, School teachers and health volunteers) including curriculum review	201
2 days Training on infection prevention policy and practice for District & Upazilla Health Personnel including Curriculum review	915
3 days Training on basic management skill with curriculum development and curriculum review	60
6 month Training for Doctors on Obs & Gymea (EmOC) including TOT & curriculum development/review	162
6 month Training for Doctors on Anesthesia (EmOC) including TOT & curriculum Development/Review	162

Achievement of In-Service Training (IST) from July 2008 to April 2009 Local Training (contd...)	
Name of Activity	Achievement
Training on updating media and messages in support on HEP for HEO/HE/Other related officers	25
Organization of 2 days joint simulation exercise with BDRCS at most cyclone prone districts. (Multi-sectoral approach) on EPR	175
Conduct vulnerability and Capacity Assessment at 10 (ten) selected hazard prone areas on EPR	87
2 days training for field staff on Disaster mitigation	315
Workshop on Search, Rescue, Evacuation and First Aid for Health Workers & volunteers (2days)	326
2 days Orientation on service statistics of MO, MA, MT, HI, AHI and HA	95
Emergency management Training of Traffic police/Launch driver and Helper/Bus driver and helper/Petrol pump worker	350
Breast feeding counseling training for health care providers (HAs/Field service providers.) (3 days)	125
3 days Women's professional development programme for personnel from district/directorate/ Secretariat level managers including curriculum	75
Advanced programming on visual basic 6, SQL server for officer and staff.	22
3 days training on technique of developing training media and maintenance of audiovisual equipment for audiovisual operator, audiovisual projectionist, audiovisual helper and audiovisual technician including curriculum development	52
Training on gender issue and poverty alleviation for HEO/HE/Other related officers	651
1 day training for doctors , medical assistant, MA, Paramedical Health/ Field Staff, Nurses, RMP, Drug distributor, formal and informal leaders etc on filariasis elimination & morbidity control to be held at divisional/district/upazilla level with fie	396
Orientation Training program on Femily Medicine (BCPS)	525
Total	11188

No. of admissions in different medical and dental colleges (2004-2007)

Medical or Dental College	2004			2005			2006			2007			2008			Total		
	M	F	T	M	F	T	M	F	T	M	F	T	M	F	T	M	F	T
Dhaka Medical College	82	79	161	94	69	163	111	73	184	94	93	177	108	63	190	489	377	876
Sir Salimullah Medical College	80	78	158	93	68	161	104	82	186	86	91	177	97	59	182	460	378	865
Rajshahi Medical College	82	76	158	85	70	155	112	67	179	107	71	178	96	57	182	482	341	852
Rangpur Medical College	68	85	153	67	89	156	91	87	178	88	90	178	88	51	178	402	402	843
Mymensingh Medical College	71	89	160	89	70	159	97	85	182	110	68	178	89	61	171	456	373	850
Chittagong Medical College	80	77	157	96	62	158	100	86	186	101	82	183	82	60	185	459	367	869
MAG Osmani Medical College	93	61	154	89	66	155	104	84	188	87	92	179	96	65	178	469	368	854
Shere Bangla Medical College	90	63	153	84	70	154	105	79	184	110	68	178	100	62	187	489	342	856
Faridpur Medical College	25	24	49	28	21	49	51	51	102	53	50	103	63	21	117	220	167	420
Sahid Ziaur Rahman Medical College	20	37	57	56	45	101	65	45	110	51	51	102	21	23	50	213	201	420
Dinajpur Medical College	23	28	51	21	33	54	56	53	109	66	57	123	55	56	121	166	171	337

No. of admissions in different medical and dental colleges (2004-2007) (contd...)

Medical or Dental College	2004			2005			2006			2007			2008			Total		
	M	F	T	M	F	T	M	F	T	M	F	T	M	F	T	M	F	T
Khulna Medical College	27	29	56	26	30	56	66	42	108	69	63	132	69	23	132	257	187	484
Comilla Medical College	36	21	57	44	63	107	44	63	107	48	59	107	47	29	107	219	235	485
Dhaka Dental College	35	33	68	58	40	98	61	83	144	55	83	138	39	35	88	248	274	536
Noakhali Medical College	-	-	-	-	-	-	-	-	-	-	-	-	20	30	50	20	30	50
Chittagong Dental College	5	13	18	19	23	42	-	-	-	-	-	-	-	-	-	24	36	60
Rajshahi Dental College	7	10	17	18	23	41	-	-	-	-	-	-	-	-	-	25	33	58

No. of students passed from different medical and dental colleges (2004-2008)

Medical or Dental College	2004			2005			2006			2007			2008			Total		
	M	F	T	M	F	T	M	F	T	M	F	T	M	F	T	M	F	Total
Dhaka Medical College	58	50	108	75	83	158	69	63	132	98	72	170	144	63	207	444	331	775
Sir Salimullah Medical College	109	88	197	72	76	148	102	72	174	69	73	142	94	59	153	446	368	814
Rajshahi Medical College	181	103	284	97	79	176	102	58	160	91	59	130	90	57	147	561	356	897
Rangpur Medical College	176	70	246	115	77	192	72	79	151	82	76	158	32	51	83	477	353	830
Mymensingh Medical College	102	95	197	102	79	181	88	62	150	71	78	149	63	61	124	426	375	801
Chittagong Medical College	140	68	208	171	108	279	116	82	198	100	74	174	65	60	125	592	392	984
MAG Osmani Medical College	79	69	148	88	77	165	136	160	296	112	85	197	62	65	127	477	456	933
Shere Bangla Medical College	157	96	253	112	76	188	96	74	170	71	78	149	74	62	136	510	386	896
Faridpur Medical College	37	31	68	22	16	38	39	39	78	19	21	40	21	21	42	138	128	266
SZMC, Bogra	34	34	68	35	32	67	21	38	59	39	39	78	23	23	46	152	166	318
Dinajpur Medical College	23	24	47	21	18	39	23	26	49	35	27	62	21	20	41	123	115	238
Khulna Medical College	35	26	61	29	25	54	31	17	48	24	19	43	24	23	47	143	110	253
Comilla Medical College	31	37	68	16	36	52	24	33	57	28	26	54	11	29	40	110	161	271
Dhaka Dental College	50	31	81	157	150	307	158	151	309	160	166	326	34	35	69	559	533	1092
Chittagong Dental College	12	1	13	6	3	9	-	-	-	-	-	-	-	-	-	18	4	22
Rajshahi Dental College	9	4	13	9	6	15	-	-	-	-	-	-	-	-	-	18	10	28

Health MIS in Bangladesh

Introduction

Despite obstacles, MIS (health) made commendable successes recently. Table-7 lists those successes. The speed may be boosted up tremendously by urgently placing few young, dynamic, computer literate and capable doctors preferably of public health background. MIS (health) also desperately needs full time server maintenance engineers to ensure 24h uptime of Internet database servers and mirror (backup) servers and also full time database programmers for quick development of web-based database programs and fixing database problems.

The routine jobs of MIS (health) Bangladesh

In the MIS (health) there is mismatch between "the workload of the department" and "its systems and infrastructure". One frequently used statement can be quoted here to understand the reason of this failure, which says, "MIS (health) is building the systems now, which were needed to build 10 years back".

Following are the routine responsibilities of MIS (health):

a. Population information (viz. population size, age-sex distribution, birth rate, death rate, growth rate, age-specific mortality rates, maternal mortality rate, etc.);

b. Health service statistics (viz. number, type and location of all health facilities, number of patients treated by each facility in outdoor, indoor and emergency departments, bed utilization rates, average length of stay, volume of services by category, disease profile, causes of deaths, etc. from public and private health facilities;

c. Emergency Obstetric Care (EmOC) statistics (in-depth obstetric care related information) from over 500 health facilities

d. Integrated Management of Childhood Illness (IMCI) statistics (age and sex-disaggregated information of out-patient, emergency and in-patient children, availability and quality of services) from 275 sub-districts

e. Health workforce statistics (viz. staffing pattern of health facilities, vacancy statement, LPR list, personal data sheets, health workforce: population ratio, health workforce distribution, etc.;

f. Logistics statistics (viz. number and condition of vehicles and major equipment, etc.);

g. Financial statistics: not done before.

Looking deeper into the problems

The following paragraphs describe the prevailing system for routine information collection:

a. **Population information:** MIS (health) conducts annual household survey to collect demographic information from each household of the country. The system is well known as GR or Geographical Reconnaissance. For GR, the rural health workers of DGHS (health assistant, Assistant Health Inspector and Health Inspector) use to visit the households in January and February each year. GR form is complicated. Each form covers 18 families with 47 different type of information (variables or fields) per household. No provision exists for giving incentives to the health workers. The allocation for GR per union (lowest administrative unit, average number of households: ~5,000) is only Tk. 2,000 (US$ 29), which covers all expenses excluding GR Forms. Having considered that all posts of health workers are filled up, each health worker needs to visit 60 to 70 households daily (considering 25 working days per month) to complete GR in January and February. It is both theoretically and practically impossible. But, nobody cared to recognize this weakness of the system. Monitoring and supervision system was weak. Therefore, there was doubt about the data quality. The data forms were complied at union level and submitted to sub-district (subdistrict) health office where sub-district form is constructed putting individual union data and compute sub-district total and then sent to Dhaka via district health office. All sub-district forms were brought to MIS (health), DGHS office for data entry. Only about 12 persons, already over burdened with other responsibilities, were supposed to enter the data in computer. But, due to poor motivation about the data quality they never cared to do the job. Yet GR was conducted year after year with no report published in any year, except in 2004 when the last report was published using GR data of 2002. GR data could be good source for estimation of, amongst others, child and maternal mortality rates, cause-specific death rates, age-and sex-specific population sizes, for local as well as national level planning and decision making. The idea of GR is great, which can be synonym of annual health census and a strong demographic health surveillance system. But, this dream was never fulfilled although time, money and efforts of health managers and health workers all over the country were heavily engaged.

b. **Health service statistics:** It is estimated that only about 20% of all patients in Bangladesh, who seek health care, go to public health facilities. The rest seek health care from any of the different type of non-public providers, viz. informal healers, drug retailers, private practitioners, private clinics and hospitals, through out of pocket payments. Despite this fact, health service statistics of MIS (health) were primarily based on public health facilities. The number of functioning public health facilities other than community clinics under the DGHS of Bangladesh is 1951, of which 1,362 are union sub centers having only out-patient day services, and 589 hospitals having out-patient, in-patient and emergency services. School health clinics and urban dispensaries are not included in this estimate.

Outpatient data from union sub-centers come to the respective sub-district health office where the data are compiled. In subdistrict health complex, in-patient and emergency patient profiles are also prepared. Data from sub-district are sent to civil surgeon's office in district. The district and general hospitals also prepare their own out-patient, in-patient and emergency patient profiles and send to respective civil surgeon. The civil surgeons send the subdistrict and district data to MIS (health), DGHS. All other public hospitals send their patient profiles directly to MIS (health), DGHS.

What are the types of data the health facilities produce? They produce the hospital utilization data (number of outpatient and emergency visits, number of admissions, bed occupancy rate, average length of stay and in-patient death rate), disease profile (age and sex disaggregated morbidity data both for out-patient and inpatient based on common diseases and diagnoses translated in ICD-10 codes). Each hospital needs to send mortality data for each in-patient death on case by case basis. Compliance of monthly reporting is reasonably good for district levels and below; but for medical college hospitals, and for tertiary care and private hospitals are unsatisfactory. However, things are improving. How reliable the data are? In a rural sub-center, a medical assistant sees patients without any investigation support.

S/he can provide symptomatic treatment. How much reliable his/her diagnosis can be? In an out-patient department (OPD) of any hospital, a doctor may have to see on average 100 to 300 patients in OPD hours per day. S/he is not in a position to completely recording all information in the register. The health facilities also have to prepare other health service statistics, viz. emergency obstetric care statistics and IMCI statistics.

Type of health facility statistics received by MIS (health), DGHS

Type of health facility statistics	OPD & emergency	In-patient	Source
Hospital utilization data	No. of OPD and emergency visits age and sex disaggregated	• No. of admissions age and sex disaggregated • Bed occupancy rate • Average length of stay • In-patient death rate	• All health facilities • All hospitals
Disease profile on common morbidities	Age and sex disaggregated translated in ICD-10 code	Age and sex disaggregated translated in ICD-10 code	• All health facilities • All hospitals
Causes of death	-	Case by case basis	All hospitals
Emergency Obstetric Care data	-	Obstetric care facilities	Around 600 facilities
Integrated Management of Childhood Illness data	275 sub-districts	275 sub-districts	275 sub-districts

Who really does the data handling job? In each sub-district, there is a class III staff called statistician, who is responsible for all the data gathering, compilation and transmission jobs. In some districts there are two posts of statistical staff, one class I post of Statistician and one class II post of Statistical assistant. In other districts, there is only one post of Statistician. But, most of the class I posts are vacant. In the divisional health offices, there are 2 to 4 posts, one class I (Assistant Chief) and one to three class III (Statistical assistant). Sylhet divisional health office does not have any statistical staff. Follwing table shows the detail. Experience shows that workloads are more in sub-district and district than in divisional health offices. But, number of staff in sub-district and district health offices is less than in divisional health offices. The data generated from the hospitals are huge in volume. But, there is no post of statistical staff in any hospital beginning from district hospital as high as up to top level tertiary or specialty hospital. This ambiguity creates situation for irregularity, inadequacy and poor quality in data flow.

Statistical staffs in the peripheral health offices of Bangladesh

Level	Class	Name of post	Sanctioned posts in each		Total posts (N o.)	Vacant posts	Remarks
			Place	No.			
Directorate	1	Statistician/ Statistical Officer	MBDC	1	1	0	MBDC
Directorate	2	Statistical Officer	EPI	1	1	0	EPI
Directorate	3	Statistical Assistant	CDC	2	2	1	CDC
Directorate	3	Statistical Assistant	Hospital	1	1	1	Hosp
Directorate	3	Statistical Assistant	ARI	1	1	0	ARI
Directorate	3	Statistical Assistant	EPI	1	1	0	EPI
Directorate	3	Statistical Assistant	MBDC	2	2	0	MBDC
Division	1	Assistant Chief	All division	1	5	0	No post in Sylhet

Level	Class	Name of post	Sanctioned posts in each		Total posts (N o.)	Vacant posts	Remarks
			Place	No.			
Division	3	Data entry opera tor	5 divisions	1	5	0	None in Sylhet
Division	3	Statistical assistant	Dhaka	3	3	0	Nil in Sylhet
Division	3	Statistical assistant	Chittagong	3	3	0	
Division	3	Statistical assistant	Rajshahi	3	3	0	
Division	3	Statistical assistant	Khulna	3	3	0	
Division	3	Statistical assistant	Barisal	1	1	0	

Statistical staffs in the peripheral health offices of Bangladesh

Level	Class	Name of post	Sanctioned posts in each		Total posts (No.)	Vacant posts	Remarks
			Place	No.			
Institute	3	Assistant Statistician	NIKDU	1	1	0	NIKDU
Civil Surgeon Office	1	Statistician	All districts	1	64	55	Most Class I posts are vacant
Civil Surgeon Office	3	Statistical assistant	56 districts	1	56	0	
Sub-district level	2	Statistician	31-bed hosp	1	1	1	Haragach
Sub-district level	3	Statistical assistant	31-bed hosp	1	5	0	-
Sub-district level	3	Statistician	SUB-DISTRICT HOSPITAL	1	476	85	-
200-bed Hosp	3	Statistical Assistant	Narayanganj	1	1	0	Narayanganj
250-bed hosp	3	Statistical Assistant	Khulna, Noakhali	1	2	2	Khulna, Noakhali
Medical College Hospital	1	Statistician/ Statistical Officer	10 MCH	1	10	8	8 old MCH, SSH, SZMCH
Institute	1	Statistician/ Statistical Officer	5 Institutes (IPHN, NIPSOM, NICVD, NICRH, NCCRFH)	1	5	3	
Institute	2	Statistical Officer	NIDCH	1	1	1	NIDCH
Instiute	2	Statistician	IPHN, NIKDU	1	2	1	
Institute	3	Statistical Assistant	IPHN, IEDCR, NIPSOM, NCCRFH	1	4	1	
TB clinic	3	Statistical Assistant	TB Clinic (Shaymoli)	1	1	1	TB Clinic
			Total=		661	160	

The data handling environment is still almost manual and duplication of same work at different levels causes just wastage of time of the limited statistical workforce. Normally information of each patient in a health facility is recorded in a register, from where a compiled hard copy-based report is produced. The hard copy is sent to hire level, where again data are entered in computer and compiled with similar data from other facilities. A new hard copy is produced and sent to next higher level. This repetition takes place in sub-district level and then at district level finally at MIS (health), DGHS. If the data could be entered in a computer database or simply in a spread sheet program at the source and the soft copy could be transmitted to the next higher level, much of the valuable time of the limited statistical workforce could be saved. Again, health facility statistics for utilization rates, disease profiles, causes of death, emergency obstetric care and IMCI could be collected in integrated format on case by case basis and entered in computer database or spreadsheet program to auto generate reports.

c. **Health workforce statistics:** MIS (health) currently deals with statistics for personnel and staffs working under DGHS only. The DGHS has over 100,000 health workforces, comprising doctors, nurses, technologists and other non-technical staffs. Until recently personal data sheets (PDS) were maintained only for doctors (<20% of all staffs). Other staffs were uncovered. PDS database was maintained in a standalone computer without being available through network or Internet. Offline PDS system was not successful due to poor compliance of the doctors. Online PDS database has been launched for all classes of staffs. This is building upbut would require administrative measures to get full compliance. Experience shows that an initial stock taking through active-staff census may be required. Institute-wise staffing pattern, existing staff list and institute-wise list of staffs leaving job are updated periodically through collection of hard copy reports from the institutions. The responsibilities and workloads go on the same existing staffs who are already over burdened with population and hospital statistics. The MIS (health) staffs at DGHS, few in number and lagging behind in capacity and tools, cannot cope with the data entry for reports they receive in hard copies.

Health workforce statistics prepared by MIS (health)

Type of statistics	Description
Institute-wise staffing pattern	Designation, type of post, number of sanctioned post, number of existing staff by sex in each type of post, number of vacancies by each type of post, pay scale
Existing staff list(Class I)	ID number, code number,name of person,, father's name, mother's name, designation, status, joining date.
Institute-wise list of staffs leaving job	Class, designation, name of person, father's name, permanent address, date of joining date of LPR, death, dismissal, ID number, code number
Personal Data Sheet	Detail resume of individual health staff

So, they cannot engage sufficient time in tracking whether all reports were received in time with complete and correct information. The design of the reports is also faulty which necessitates duplication of same work. Other than the PDS, all the remaining three reports (staffing pattern, existing staff list and list of staffs leavingjob) can be integrated in a single format and put in easily updatable online database. There is weakness in coordination. MIS (health) often remains in darkness about the number of new posts created, number of posts abolished, number of posts upgraded, staffs transferred, promoted, joined in new position, left job, given lien or deputation, etc. A strong and integrated coordination mechanism must be in place to improve the health workforce information system. To compute "doctor: population ratio" or "nurse: population ratio", estimates of private health workforce would also be needed. Mechanism must be found out to get information on private health workforce. Perhaps involvement of the local government can be of great help in this regard. Geographic distribution of health workforce is not difficult to work out across public facilities; but computation of overall distribution would require information on private health workforce.

d. Logistic statistics: Paper-based logistics reports are collected from public health facilities on quarterly basis. Reports describe functional status of ambulance, vehicles and major equipment. Problems related to collection of logistic status report are similar to that of collection of staffing pattern data. Compliance is not satisfactory. An integrated database system available and updatable online can make the system efficient and capable to provide real time update.

Name of item	Model	No.	Source	Year received	Present status			Remarks
					Functioning	Non-functioning	Repairable out non-functioning	

e. Financial statistics: Developing Financial MIS has not been attempted seriously by MIS (health) before. However, it would be better to consolidate the other routine MIS areas, before going for development of financial MIS. Currently the Planning wing of MOHFW monitors the progress of implementation of HNPSP. On the other hand, the ministry's Health Economics Unit undertakes different economic evaluations pertaining to health. By working with them in close collaboration, the current need of financial MIS can be met.

Is the existing workforce of MIS (health) capable?

The simple answer is only partially capable. MIS (health) does not have a regular post of director. The current director is working as OSD. The total number of sanctioned posts is 66 of which 26 is vacant. The vacancy rate is 39%. The nomenclature of the positions may mislead one about the qualifications and capacity of the personnel and staff. The director due to his previous exposure and personal interest about ICT, databases, biostatistics, and biomedical research, is capable to lead a team for development of reasonablygood health information system. Other administrators, being deployed from purely managerial positions in health administration on consideration of seniority, lack sufficient knowledge and skill about the MIS (health) system. The ICT personnel are really few in number and lack advanced hardware, software and programming knowledge and skills, which are required to run a centralized network and web based information system of government's health services. The statistical staffs do not know statistical analysis techniques. There are no knowledgeable and skilled persons who can support director to design survey or research protocol, administer and manage information system, administer and promote web portal and information dissemination system, generate and interpret summary data from statistical software, and write and publish quality health reports on various issues. Scopes for staff development could not be materialized earlier because of heavy dependence on manual systems of data management that killed all of their time without benefitting with success. The relatively few intelligent and capable staffs are needed to engage in implementation of tightly scheduled HNPSP and bilateral donor (WHO and UNICEF) supported projects. The director needs to remain engaged lot of his time in attending meetings and workshops.

The recommended actions for solution are:

a. Freeing up MIS (health) office from manual system of data management as quickly as possible through ensuring all data feeding from sources where data are generated;

b. Engaging MIS (health) staff on monitoring data quality (viz. coverage, completeness, accuracy, timeliness, etc.), analysis and report writing;

c. Improving local capacity for data feeding: (i) establishing accountability of local health managers to send complete set of accurate and timely data online; (ii)making available computers and Internet connections in all remote points as quickly as possible; (iv) enhancing computer skills of as many persons as possible in remote points through local arrangements;

d. Improving capacity of MIS (health): (i) providing short duration and continued training to existing staffs; (ii) deploying capable young and dynamic persons to empty positions without looking to seniority; (iii) deploying few additional personnel on deputation to support data analysis and report writing; (iv) hiring immediately full time skilled persons to support quick and immediate hardware and software development and trouble shooting.

Weaknesses of MIS identified by World Bank Mission

The Annual Project Implementation Review (APIR 2008) of HNPSP done by World Bank Mission identified following problems of MIS (health):

a. Too many reporting formats;
b. Lack of timely collection of program data;
c. Lack of training plan on ICT to modernize MIS;
d. Lack of use of information in decision making process.

The observations truly reflect the current situation. Both in the fields as well as in the facilities there are multiple forms often on same subject. When a new form was introduced, it was not considered that an earlier form existed. So, both forms continued causing burden over the field workers, hospital staffs and managers. Line Directors lack coordination in setting schedule for data collection. LDs with field program have own separate data collection form(s). While the field workers could get all data in a single go to the particular family, due to different calendars sometimes imposed on ad hoc basis, they need to visit multiple times the same household in the same month. So, they become de-motivated and often produce data without visiting homes. Effort has been made to reduce number of data collection and reporting forms. As the data communication system is not automated and there always remains a human factor, the data sources need remainders, and monitoring and supervision to get data in time.

Staff profile of MIS (health), DGHS

Section	Post	Sancti oned	No. oCategory f p	Existin g osts	Vacant	Remarks
Personnel MIS	Assistant chief (medical)	1	1	1	0	
Personnel MIS	Assistant chief (statistician)	1	1	0	1	
Personnel MIS	Assistant director	1	1	1	0	
IT	Assistant programmer	2	2	2	0	
Health Information Unit	Assistant statistician	2	4	1	3	SSMIS
Personnel MIS	Assistant statistician	2				SSMIS
Health Information Unit	Chief, Health Information Unit	1	1	1	0	SSMIS
IT	Data entry operator	4	4	4	0	
Health Information Unit	Deputy chief (medical)	1	1	0	1	SSMIS
Health Information Unit	Deputy chief (non-medical)	1	1	0	1	SSMIS
Health Information Unit	Deputy chief (statistical)	1	1	1	0	SSMIS
Personnel MIS	Deputy director	1	1	1	0	
Logistic MIS	Deputy Program Manager	1	2	1	0	
Population MIS (new)	Deputy Program Manager	1		0	1	
Health Information Unit	Deputy Program Manager (SSMIS)	1	1	1	0	SSMIS
Administration	Director	0	1	1	0	As OSD
IT	Draftsman	1	1	0	1	
Administration	Driver	2	2	2	0	
Administration	Duplicating machine operator	1	1	1	0	
Health Information Unit	Investigator	2	3	1	2	SSMIS
Personnel MIS	Investigator	1		0		
Health Information Unit	Machine room clerk	1	3	0	3	SSMIS
IT	Machine room clerk	1		0		
Personnel MIS	Machine room clerk	1		0		
Administration	MLSS	5	5	2	3	
Administration	Office assistant	2	5	1	2	
Health Information Unit	Office assistant	1		1		SSMIS
IT	Office assistant	1		1		
Personnel MIS	Office assistant	2		1		
Administration	Office peon	1	1	0	1	

Staff profile of MIS (health), DGHS (contd...)

Section	Post	No. of posts				Remarks
		Sanctioned	Category	Existing	Vacant	
Logistic MIS	Program Manager	1	2	0	2	
Population MIS (new)	Program Manager	1		0		
IT	Programmer	1	1	1	0	
Administration	Security Guard	1	1	1	0	
Health Information Unit	Statistical assistant	2	8	2	0	SSMIS
IT	Statistical assistant	4		4		
Personnel MIS	Statistical assistant	2		2		
Health Information Unit	Statistician	2	4	2	0	SSMIS
IT	Statistician	1		0	1	
Personnel MIS	Statistician	1		0	1	
Administration	Steno-typist	1	3	1	0	
Health Information Unit	Steno-typist	2		0	2	SSMIS
Administration	Sweeper	1	1	1	0	
IT	System analyst	1	1	1	0	
Administration	UDA	2	2	1	1	
	Total =	66	66	41	28 (39%)	

Due to engagement of critical workforce into data entry jobs, the reminder systems were not in place before. Truly speaking MIS (health) lacked for long period right leadership to foresee the need for ICTs and appropriately skilled personnel. So the existing training plan and deployment of manpower and system could not satisfy the need expected from MIS (health).

Use of information in policy making process

It is not entirely true that information is not used in policy making process. Personal Data Sheets and vacancy statements are widely and routinely used by the personnel departments. Logistic information is also routinely used. The Cabinet Division of the Government gets data on monthly and yearly forms from MIS (health) that cover wide range of health service matters. However, there are more opportunities for using health information in policy making process. Unavailability of relevant and reliable health information on time is one of the reasons why policy makers can not use information for decision making. MIS departments will have to make their systems efficient and effective to be able to undertake a stewardship role through providing right information at the right time. They will have to sensitize the policy makers to use information for better decision making. MIS (health) has some routine publications, such as, Health Bulletin, Year Book, MIS Newsletters, etc. These are contributing to give impression to the policy makers about the health situations of the country influencing their decision. Now it is also important to build an effective feed back system with the data senders and working with them to improve data quality, schedule and completeness.

Important changes in MIS (health) made during last one year

From March 2008, an attempt is in motion to continuously revisit and streamline the systems and needs of MIS (health) to perfectly match each other. It is well-known that Government systems have limitations and bottlenecks and everything has to be done within the line permissible by existing rules and guidelines. Many important changes have been made creating scopes within the limitations -but others could not be done as barriers were absolute.

Important changes made in MIS (health) during last one year

Action	Purpose	How solved
1. Computerization of all health facilities as low as up to sub-district level	• Done as a first step towards decentralization of data entry job to data sources and avoid duplication of data entry at each point of hierarchy	Computers procured for unplanned distribution were redistributed to go as low as up to sub-district level, to important points, and to points where computer is needed but there was none

Important changes made in MIS (health) during last one year (contd...)

Action	Purpose	How solved
2. Internet connection	· To provide Internet connectivity to all major health information points	· 24-h unlimited Internet connections using wireless broadband EDGE modems have been provided to about 800 health data points as low as sub-district hospitals or health offices. No major data points were excluded
3. Establishment of web-based Internet server at MIS (health)	· Up gradation of LAN server to Internet based web server to enable remote data sources directly feed and retrieve health data interactively or through batch files · To host web portal of MIS health · To establish full control on data · To make the Internet bandwidth more cost-effective · To increase number of servers for data security and uninterrupted service	· LAN server of MIS (health) upgraded to serve dual purposes, LAN server and Internet server · Static website hosted in commercial server transferred in own server to save money, apply full control and dynamicity · Centralized web-based databases stored in own server guarantying security and control on data · The same Internet bandwidth purchased for local users is also used for web servers · LAN servers purchased for divisional health offices relocated at MIS (health) for better utilization and ensuring back up of data and services
4 Launch of innovative participatory dynamic web portal	To make for DGHS a vibrant one-stop information sharing center where all stakeholders can put content through own control and information gets accessible throughout the world	· Static web site of DGHS has been converted to dynamic web portal and hosted in own server · All health stakeholders (LDs, institutes, hospitals, health NGOs) given opportunity to take user name and password to put and edit content interactively · Interface easy and user friendly · New content is uploaded almost everyday
5. Launch of online Personal Data Sheet	· To ensure hassle free updating of personal data sheets by staff members by themselves	· Web based software database created and hosted in own server · Old offline database migrated to new online database

	· To enable MIS (health) staffs to concentrate on other important jobs · To expand scope of PDS to all classes of health staffs, to all MOHFW personnel and staffs and to private health workforce	· Newspaper advertisement given asking DGHS doctors to update individual PDS online · The way forward: more inputs will be required; local health managers should take active role; significant progress is expected when Internet connectivity will be established in all data points
6. Internet mail server under own domain	To give all health offices under DGHS email addresses under own domain, viz. dg@dghs.gov.bd; keraniganj@uhfpo.dghs.gov.bd	Robust email server with own domain name and accessible throughout the world has been created free of charge in collaboration with Google

Important changes made in MIS (health) during last one year

Action	Purpose	How solved
7. Establishment of MIS Data Center	MIS (health) staff had to perform data entry jobs in tiny, unhealthy, poorly ventilated and poorly illuminated space	· New comfortable space has been created as MIS data center (data collection, analysis, report generation) converting a verandah into room with Internet connectivity, good illumination, 24 hour power back up, air conditioning, toilet and drinking water provisions · Data center is contributing to improving data collection system
8. Establishment of MIS IT Lab	To enable data management staff at MIS (health) and all over health services quickly acquire ICT and data analysis skills	· On request, PWD (civil) renovated earlier data entry room into IT Lab and MIS Server room · New computer tables procured · Computers and computer chairs arranged under redistribution plan
9. Establishment of MIS Resource Center	To create an efficient center for quick communication and information dissemination	A room has been re-designated as MIS resource center where fax, phone, email, internet, computer, scanner, color printer, etc. backed by trained communication persons have been placed
10. Placement of generator to ensure 24 hour power backup	To ensure 24 hour run time of Internet servers and keep staffs of MIS at work with all computers, printers and equipment in operation in the face of frequent power failure during hot seasons	· A 60 KW generator sitting idle in Panchagar district hospital brought to MIS (health) · Physical structure to house the generator built with own arrangement · Through tireless negotiations, generator installed with help of PWD (electrical)
11. Introduction of Electronic Office Attendance System	To improve staff punctuality through electronically monitor and control staff attendance of MIS (health)	· Electronic office attendance system placed · Staff members issued machine readable ID cards and circular to use the office attendance system · Deviation informed back to concerned staff members for improving punctuality
12. Revision of Operational Plan (OP)	MIS (health) OP 2003-2010 appeared unsuitable for achieving goals of MIS (health) properly	· OP revised for 2008-2011 period to rationalize activities for building effective health MIS · Approval on revision taken · Communication is under way to get approval for revised Annual Development Plan based on revised OP
13. Publications of health reports	To disseminate health information to policy makers and stakeholders	MIS (health) published following health reports: · Year book 2007 on HNPSP (2006-07) activities · Health Bulletin 2008 on country's health situation · Voice of MIS newsletter on EOC activities Manuscripts of: · Year book 2008 on HNPSP (2007-08) is ready for printing · Health Bulletin 2009 under process · Manuscript of IMCI newsletter is ready

Action	Purpose	How solved
14. Health workers' diary	To reduce community-based data collection forms, harmonize data collection system, reduce work burden of health workers and improve data archiving	• Initial draft done by MIS (health) • Opinion taken from users • Line Directors and experts of ICDDR,B and UNICEF consulted to finalize draft • Bureau of Health Education printed & distributed
15. Development of more online database software	To minimize duplication in data collection and compilation, reduce number of data collection forms and speed up data transmission system	• A number of web-based database software has been developed to collect data online • New software are being developed • Old software are being further improved
16. Capacity building and motivation of staffs	• To improve capacity of data management staff both at MIS (health) and all over health services • To improve motivation of staffs on newer possibilities	• The new IT Lab enhanced training capacity of MIS (health) manifolds • Data management staffs at all levels have been brought under repeated training and motivation sessions
17. More coverage of data providers	To encourage tertiary hospitals in public sector, and hospitals and clinics in private sector to provide hospital statistics to MIS (health)	• Workshops organized with support from WHO where relevant stakeholders at mid- to top-levels of the organizations were invited • They are gradually coming up to send reports
18. Formation of organization level MIS Committees	To encourage local initiative to achieve MIS (health) goals for health information system improvement	• MIS (health) committees framed at MOHFW, DGHS, and in all tertiary hospitals, districts and sub-districts to start activities soon • Terms of reference developed • Workshops designed to improve capacity of the committees
19. Sub-district mobile phone health care service	To introduce mobile phone based tele-health service based on each sub-district health office to enable citizens of the respective catchments area get health advice calling an assigned mobile phone	• One mobile phone has been provided to each sub-district health office • The phone number has been circulated locally through different local communication channels • A doctor remains available to receive phone calls to listen health problems of people, advice instant treatment or welcome to health center
20. Video conferencing system with civil surgeons	To enable top policy makers in the Ministry of Health and DGHS hold real time video conversations with civil surgeons on urgent and important health issues	• Web cams have been provided to each civil surgeon of 64 districts • Existing Internet connections are used for video conferencing • At the initial stage, video conferencing platform like Skype are being used • The system will be made sophisticated gradually • Two way seminars, monitoring office attendance and telemedicine services are being provided
21. Extension of MIS (health) building	To create a specially designed conference room for MIS (health) to enable it demonstrate cheaper and effective information communication solutions	• Work is progressing rapidly

How MIS (health) plans to redesign its routine jobs

Population information

The advantage of Geographical Reconnaissance (GR) will be heavily used. A computerized database of permanent citizens' health registry will be prepared for use of Directorate General of Health Services, Directorate General of Family Planning, National Nutrition Program and even other ministries. The GR form will be simplified to introduce one form for one family. Each family member will be given a unique identification number in line with national ID scheme. Basic demographic data, viz. date of birth, sex, educational status, marital status, chronic disease, mobile phone, etc. will be collected. Health workers will visit every household through a period of 6 to 8 months for data collection. Family planning workers will also be invited to assist in data collection. Database software and financial assistance will be provided to sub-district or district level to do data entry locally. Alternatively Integelligent Character Reconition Technology through outsourcing can be used or data entry. After completion of data entry, hard copies will be produced for data authentication at community level. Finally all sub-district data will be assembled to generate national citizens' registry which will be hosted in secured data server available online to appropriate authorities. After preparation of the first complete GR dataset, no more annual GR will be required. Instead, health field workers will collect updated information during their routine household visit using the health worker's diary and submit data to subdistrict statistical staff to update family information online. On further development of the system, health workers will be able to update information through their mobile phones right from the family. If implemented successfully, new GR system will provide following benefits: (i) availability of current profile of local and national figures for total and age-sex disaggregated population size; and age and sex disaggregated death rate, NMR, IMR, U5MR, MMR; (ii) easy to differentiate old and new patients in hospitals for statistical purpose; (iii) no need for recording name, age, sex, etc. of respondents as identification number will provide those information from national database. The ID scheme will be such that a person will not need to carry the ID card. Responding to few simple questions will enable service providers get his or her ID number.

Health service statistics

Emphasis will be given to collect complete set of in-patient data on case by case basis both from public and private hospitals. Desktop based software "capable to connect to centeralized server via Internet on instruction given by the user" will be provided to all data sources free of costs. The closed patient files on discharge, death, referral, DORB, etc. of patients will be sent to hospital's computer room to enter summary data into the software which will be submitted from time to time to central server. ICD-10 classification will be autogenerated by the software. Patient data will be available both locally and at MIS (health). All current patient information system, viz. disease profile, in-patient utilization statistics, causes of death, emergency obstetric care, integrated management of childhood illness wil be combined in common platform to avoid duplication of same job. Once entered in computer and submitted to central server, the hospitals will not require to compile data any more or submit report. Reports will be automatically generated. To assist in development of this system, computers and Internet connections are being provided to all hospitals as low as up to sub-district level. However, more computers, at least one per ward, need to be provided gradually to get full benefit of the system. Private facilities will need to create their own data communication system. For out-patient and emergency departments, it is not wise at this moment to demand detail patient data. Due to huge patient load in short OPD hours, it is hard to record data maintaining quality. Until finding a right solution, it is decided to keep data need at minimum. Only number of daily visits in age and sex-disagregated forms may be reliably obtained.

Health workforce statistics

The online PDS database system will be further improved. After launching of Internet, local authorities will further be communicated to mobilize all health staffs to update individual PDS. Support from MOHFW will be required. A baseline staff-census through active process may be required. An integrated online system will be developed to update staffing status, viz. sanctioned posts, existing staffs, staff leaving job, and vacancy statement. Initially a complete listing of all posts, all existing staffs, all staffs in LPR will be done institute by institute, then each institute will be asked to update online the specific areas of changes as and when they occur. Automatic electronically generated reminder system will be developed and tracking of compliance will be logged. Incorporation of private health workforce will remain as a challenge. Assistance of BMDC, professional bodies, clinic associations and pharmaceutical industries may appear effective to find solution.

Logistic statistics

Like the health workforce, initially a complete listing of the major equipment available in each hospital along with their functional status will be prepared. Then, the information will be provided in an online database. Respective hospitals will update the database from time to time when any change in equipment profile will occur. Similar reminder and logging systems as for health workforce will be developed.

Financial statistic

Perhaps it would be wise to consolidate the activities which are already in implementation phase before starting too much for financial statistics. It is being thought that an online software be prepared to track procurement process and item by item unit price spent by organizations.

Development and maintenance of DGHS web portal as a health information warehouse

The web portal of DGHS maintained by MIS (health) is playing key role in information dissemination. However, it has more opportunities of collaboration and further development. A dedicated smart person should be engaged to improve its information content and promotion.

Publications

MIS (health) started to publish a Year Book to report progress on different operational plans by Line Directors of DGHS. MIS (health) also publishes reports on emeregncy obstetric care and integrated management of childhood illness. The health bulletin is the yearly publication which reflects the country's health situation. There are ample opportunities to improve the contents of these publications with in-depth analysis and interpretation. Experts in research background and writing ability should be engaged immediately to make the publications better.

Introduction of new technology or piloting advanced technology on lay staffs

New technology, such as, Service Availability Mapping (SAM) and Geographic Information System (GIS) enables to spot geographic location of a place and generate computer-aided instant maps on distribution of health facilities, health services, health resources, health manpower, morbidities, disease epidemics or endemics, mortalities, etc. across the country. MIS (health) plans to try SAM or GIS on pilot basis. On success, this new technology will be scaled up. Video-conferencing across the civil surgeons' offices will also be tried.

Monitoring, supervision and feed back system

One of the slowest processes which MIS (health), DGHS currently faces is communication. The existing land phone, mobile phone and fax communication with many people across health service is time consuming. It is expected that on introduction of Internet connection, communication will be faster with field managers. MIS (health) is also introducing mass SMS system to quickly broadcast announcement. To further improve monitoring and supervision, MIS Committees have been constituted at MOHFW, DGHS, and at different managerial levels/hospitals with specific terms of references (TORs). The committees will periodically meet to check whether MIS (health) reporting requirements are met as per TOR.

Physical infrastructure and logistics

Currently there is acute shortage of space in MIS (health). So, new physical space will be built through vertical extension of MIS (health) building with provisions of meeting/training room. Servers, Internet connection, phone-fax communication, mobile phones, printers, full time power supply, etc. will be ensured. Trouble shooting arrangements will be kept available round the clock. Also repair and maintenance of ICT and communication equipments will be given extra emphasis. To facilitate the remote offices to solve local problems financial assistance should be provided to them.

Sub-district Mobile Phone Health Care Service

This is an innovative technology-based health service proposed by MOHFW to respond to interest of E-Governance Cell of Prime Minister's Office. It is a Tele-Health service for community people based on each SUB-DISTRICT HOSPITAL of the country. Doctors working in the health center answer calls of people for medical advice. The Subdistrict hospitals have been supplied mobile phones for dedicated use (hotline). The phone numbers have been circulated locally through local government, educational and religious institutions, NGOs, social, religious and other communication channels. In the service hours, people living in the catchments area call the hotline and get at least one doctor to listen to their calls, discuss health problems, advice instant treatment or for welcoming them to the health center on particular date and time or to refer them to other right centre. It is expected that this Tele-Health service will help building confidence of the local people on the near by health center, improve doctor-patient relationship, encourage patients to visits more to public health facilities, and indirectly influence doctors to remain available in health centers. The idea of this technology-based service solution has been appreciated due to certain unique characteristics, viz., wider coverage of the population; reaching people living in the remote areas; availability to economically deprived community; and simple to use by the technologically lagging people.

Recommendations that need immediate implementation

1. Place few young, dynamic, computer literate and capable doctors preferably of public health background to lead teams for

a. designing database forms
b. expediting data collection
c. data analysis, report writing and publication
d. web portal development and promotion

2. Provide full time server maintenance engineers to ensure 24h uptime of Internet database servers and mirror (backup) servers
3. Provide full time database programmers for quick development of web-based database programs and fixing database problems

4. Conduct active staff-census across ministry of health to capture basic information of all personnel and staffs with a view to prepare the first complete staff database

5. Fill up vacancies in the statistical staff posts across country

6. Ensure compliance of health managers for quick implementation of Internet backbone, use of email and Internet framework and quick response on request for information

7. Set priority and consolidate the ongoing activities than including new activities

8. Incorporate policies that require less movement from office through minimizing meetings, workshops, and more reliance on email discussion, web based information communication, video conferencing, etc.

9. Improve monitoring, supervision and feed back: Involve in addition to MIS (health), offices of other directors in the monitoring system. Other directors must also be compliant to send their own information, and monitor and supervise subordinate officesand staff to comply with the information requirement through more reliance on ICT based information flow and sending reliable, timely and complete dataset.

10. Ensure availability of adequate amount of fund in time.

Conclusion

MIS (health), DGHS currently most requires assistance to improve system and compliance across the data sources and data processing lines throughout the country both in public, private and NGO communities. While money and technology are crucial; equally important are human beings (managers, staffs and stakeholders) who can drive the movement forward. MIS (health), DGHS seriously needs assistance to mobilize all these three factors.

Financial Health Services

Allocation, Expenditures and Progress of Operational Plans of HNPSP 2008-2009under DGHS (in lakh taka)

Name of the Programme	Fund Allocation (RADP 2009-2010)				Expenditure (2009-2010)				Progress Against Allocation %
	Total	GOB & JDCF	RPA		Total	GOB & JDCF	RPA		
			GOB	Other			GOB	Other	
Essential Service Delivery (ESD)	39600.00	7500.00	9557.00	0.00	13151.14	6259.40	6891.74	0.00	80.25
Communicable disease control (CDC)	9413.00	2125.00	7000.00	288.00	6427.93	1790.81	4172.63	464.49	68.29
TB and Leprosy Control (TLC)	11991.00	600.00	495.00	10031.00	7387.99	294.00	264.97	6829.02	62.92
Health Education and Promotion (HEP)	2306.00	716.00	1540.00	0.00	2215.87	695.68	1520.19	0.00	98.26
Improved Hospital Service Management (IHSM)	15100.00	5100.00	9900.00	0.00	11786.96	4069.25	7717.71	0.00	78.67
Alternative Medical Care (AMC)	995.00	800.00	195.00	0.00	894.08	760.91	133.17	0.00	89.86
Non-Communicable Disease and Other Public Health Interventions (NCD)	5066.00	204.00	4862.00	0.00	3769.47	17.84	3751.63	0.00	74.41
National AIDS/STD Program (NASP)	11865.00	131.00	10469.00	0.00	3792.40	18.26	3774.14	0.00	42.62
In-service Training (IST)	4683.00	600.00	4083.00	0.00	2880.72	335.17	2545.55	0.00	88.38
Pre-service Education (PSE)	7500.00	2500.00	5000.00	0.00	7235.21	2412.70	4822.51	0.00	96.47
Procurement Storage and Supplies (PSS)	4007.00	3907.00	100.00	0.00	4002.69	3904.00	98.69	0.00	99.89
Research and Development (Health)	423.00	23.00	400.00	0.00	252.75	8.23	244.52	0.00	59.75
MIS (Health)	1549.00	169.00	1300.00	0.00	1187.39	124.51	1062.88	0.00	81.82
Quality Assurance (QA)	160.00	17.00	143.00	0.00	152.46	12.82	139.64	0.00	95.29
Sector-wide Program Management (SWPM)	390.00	20.00	370.00	0.00	193.32	10.32	183.00	0.00	49.57
Human Resource Management (HRM)	150.00	50.00	100.00	0.00	107.05	24.43	82.62	0.00	71.37
Improved Financial Management (IFM)	50.00	12.00	38.00	0.00	25.00	6.00	19.00	0.00	50.00
Micronutrient Supplementation (MS)	1945.00	540.00	1255.00	0.00	1424.08	210.29	1213.79	0.00	87.72
National Eye Care (NEC)	500.00	273.00	123.00	0.00	232.88	168.00	64.88	0.00	67.38
Total=	117693.00	25287.00	56930.00	10319.00	67119.39	21122.62	38703.26	7293.51	75.65

Allocation, Expenditures and Progress of 6 Investment Project of HNPSP 2008-2009 DGHS (in lakh taka)

Name of the Programme	Project cost	ADP Allocation (2009-2010)				Expenditure (2009-2010)				Progress Against Allocation %
		Total	GOB	RPA		Total	GOB	RPA		
				GOB	Other			GOB	Other	
Establishment of 250 bedeed National Institute of Ophthalmology & Hospital (01/7/03-30/6/09) 1st phase	13369.00	1469.00	469.00	1000.00	0.00	370.99	370.99	0.00	0.00	25.25
Upgradetion of 50 beded National Cencer Research Institute & Hospital to 300 Bed (2nd Revised -01/07/03-30/06/2010)	29552.00	2220.00	1220.00	1000.00	0.00	8.80	8.80	0.00	0.00	0.40
Establishment of 250 bedeed National Institute of Neuro Science (01/7/07-30/6/09)	10848.00	1125.00	1125.00	0.00	0.00	1124.96	1124.96	0.00	0.00	100.00
Establishment of 100 bedeed Sharkari Karmochari Hospital, Fulbaria, Dhaka (01/7/07-30/6/09)	4239.00	710.00	710.00	-	-	562.46	562.46	0.00	0.00	79.22
Establishment of National Institute of ENT at Dhaka (01/7/07-30/6/09)	4127.00	300.00	300.00	-	-	285.59	285.59	0.00	0.00	95.20
Expension & Modernization of Dhaka Medical College Hospital (01/07/08-30/06/2011)	6000.00	700.00	700.00	-	-	617.40	617.40	0.00	0.00	88.20
Total=	68135.00	6524.00	4524.00	2000.00	0.00	2970.20	2970.20	0.00	0.00	25.84

Midterm budget framework of Ministry and Family Welfare

Description	Budget	Estimated		Projected	
	2008-09	Revised 2008-09	Budget 2009-10	2010-11	2011-12
Health Nutrition & Population Sector Program	228674.00	251448.00	292853.00	298633.69	291759.69
Administration (Directorate Ge neral of Health Services, Office of the civil surgeons, Upazila health office ,Drug Directorate Nursing directorate)	282671.55	302399.03	346339.74	354723.51	350616.25
Medical Education (Medical colleges , Paramedical Institute, Medical Assistant Training School, TB Control & Training Institute ,Dental colleges , Nursing colleges Sylhet ayurved & Tibbia college, Government unani & ayurvedic degree college & hospital, Government Homeopathic Degree college & hospital , Center for Medical Education)	12351.18	11491.12	14435.14	15818.57	17152.46
Hospitals (Medical College Hospitals, District Hospitals Other District Hospitals , Upazila Health Complex & Rural Health Center, Dental College Hospitals, Specialized Hospital & Institutes ,TB Center, School Health Center)	114958.05	117466.01	131770.29	141477.5	158831.13
IEDCR	211.29	184.75	236.95	239.32	245.57
Directorate General for Family Planning	75218.01	70083.42	79194.19	82548.14	85512.19
Alternative Medical Care	225.55	229.7	248.15	272.91	300.22
Medical post-graduation and research (Bangladesh Medical Research Council, Dhaka National Medical Institute Hospital, Bangladesh College of Physicians & Surgeons, Dhaka Shishu Hospital, Bangladesh institute of child health, Bangladesh National Nut rition Council, BSMMU)	5895.28	6100.83	6886.5	7134.3	9873.59
Non-Government Institute and organization (CRP, ICDDR,B and Others)	4813.86	5292.75	4129.5	6146.5	6482
Total- Revenue	366569.52	359346.20	392502.97	415176.73	466918.00
Total- Development	243948.00	261504.00	307533.00	315186.82	310110.69
Total- Program	0.00	0.00	0.00	0.00	0.00
Grand total-MOHFW	**610517.52**	**620850.20**	**700035.97**	**730363.55**	**777028.69**

Allocation of fund for Annual Development Program under Health, Nutrition and Population Program (HNPSP) 2009-2010 (in Lakh Taka)

Name of Program	Allocation in ADP of 2009 -2010			
	Total	Taka (Revenue)	Expenditure Capital (RPA) Revenue	Project Aid (RPA)
Essential Service Delivery (ESD)	54000	7000(6500)	7400(4300) 46600	47000(9900)
Communicable disease control (CDC)	9500	1500(1200)	700(400) 8800	8000(7000)
TB and Leprosy Control (TLC)	8010	220(195)	258 7752	7790(750)
Health Education and Promotion (HEP)	2500	100(12)	244(156) 2256	2400(1900)
Improved Hospital Service Management (IHSM)	21100	7000(6500)	13600(12900) 7500	14100(13900)
Alternative Medical Care (AMC)	716	572(500)	96(24) 620	144(144)
Non-Communicable Disease and Other Public Health Interventions (NCD)	3500	300(250)	1100(1050) 2400	3200(3200)
National Aids-Std Program (NASP)	9500	300(150)	4125(3975) 5375	9200(9200)
Pre-service Education (PSE)	2500	500(400)	1900(1800) 600	2000(2000)
In-service Training (IST)	5008	800(218)	582 4426	4208(4016)
Management for Procurement Logistics and Supplies	4066	3794(525)	3579(10) 487	272(100)
Research and Development (Health)	502	25	47(22) 455	477(477)
MIS (Health)	2900	300(290)	1100(1090) 1800	2600(2300)
Quality Assurance (QA)	181	16(16)	-181	165(150)
Sector-wide Program Management (SWPM)	430	120(120)	76(76) 354	310(300)
Human Resource Management (HRM)	220	50(15)	35 185	170(120)
Improved Financial Management (IFM)	74	20(20)	2(2) 72	54(34)
Micronutrient Supplementation (MS)	3100	400(300)	140(40) 2960	2700(2200)
National Eye Care (NEC)	480	200(76)	124 356	280(178)

Annex

Division and district wise no. of different types of hospitals under DGHS with bed capacity as of December 2008

Barisal Division

Division	District	Name of Institute	Type of Institute	No. of Institute	Total	Revenue	Development	Proposed Bed	Bed will increase
Barisal	Barguna	District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex (No.-4)	UHC	4	181	181	0	0	0
		Taltoli 20 Beded Hospital	20 bed	1	20	20	0	0	0
		Kukua 10 Beded Hospital(RHC)	10 bed	1	10	10	0	0	0
				Sub-Total :	311	311	0	0	0
	Barisal	Medical College Hospital	MCH	1	600	600	0	1000	400
		General Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	9	336	336	0	0	0
		Chest Hospital	Chest Hospital	1	20	20	0	0	0
		Babuganj 20 Beded Hospital	20 bed	1	20	0	20	0	0
		Chhakhar 10 Beded Hospital(RHC)	10 bed	1	10	10	0	0	0
				Sub-Total :	1086	1066	20	1000	400
	Bhola	District Hospital	District Level	1	100	100	0	150	50
		Upazila Health Complex	UHC	6	262	262	0	0	0
		Char Aicha 20 Beded Hospital(Devlop)	20 bed	1	20	20	0	0	0
		Khairhat 10 Beded Hospital(RHC)	10 bed	1	10	10	0	0	0
				Sub-Total :	392	392	0	150	50
	Jhalokathi	District Hospital	District Level	1	100	100	0	150	50
		Upazila Health Complex	UHC	3	112	112	0	0	0
		Kirtipasha 10 Beded Hospital(RHC)	10 bed	1	10	10	0	0	0
				Sub-Total :	222	222	0	150	50
	Patuakhali	General Hospital	District Level	1	250	150	100	0	0
		Upazila Health Complex	UHC	6	224	224	0	0	0
		Kuakata 20 Beded Hospital	20 bed	1	20	20	0	0	0
		Kathaltoli 20 Beded Hospital	20 bed	1	20	20	0	0	0
				Sub-Total :	514	414	100	0	0
	Pirojpur	District Hospital	District Level	1	100	100	0	150	50
		Upazila Health Complex	UHC	6	224	224	0	0	0
				Sub-Total :	324	324	0	150	50
				Total (Barisal Division) :	2849	2729	120	1450	550

Division and district wise no. of different types of hospitals under DGHS with bed capacity as of December 2008 (cont...)

Chittagong Division

Division	District	Name of Institute	Type of Institute	No. of Institute	Total	Revenue	Development	Proposed Bed	Bed will Increase
Chittagong	Bandarban	District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	6	186	186	0	0	0
		Rowanchari 10 Beded Hospital(RHC)	10 bed	1	10	10	0	0	0
			SubTotal :		296	296	0	0	0
	Brahmanbaria	District Hospital	District Level	1	100	100	0	250	150
		Upazila Health Complex	UHC	7	255	255	0	0	0
		Chest Hospital	Chest Hospital	1	20	20	0	0	0
		Gulnak 10 Beded Hospital(RHC)	10 bed	1	10	10	0	0	0
			SubTotal :		385	385	0	250	150
	Chandpur	District Hospital	District Level	1	200	100	100	0	0
		Upazila Health Complex	UHC	7	274	274	0	0	0
			SubTotal :		474	374	100	0	0
	Chittagong	Chittagong Medical College Hospital	MCH	1	1010	1010	0	0	0
		General Hospital	District Level	1	150	150	0	0	0
		Upazila Health Complex	UHC	14	624	624	0	0	0
		Chest Hospital	Chest Hospital	1	150	150	0	0	0
		Sultanpur 31 Bed Hospital, Rawjan	31 bed	1	31	0	31	0	0
		Infectious Diseases Hospital, Chittagong	IDH	1	20	20	0	0	0
		Bibirhat 20 Baded Hospital	20 bed	1	20	20	0	0	0
		Sandwip 10 Beded Hospital(RHC)	10 bed	1	10	10	0	0	0
		General Hospital(Proposed Port Area	Proposed	1	0	0	0	100	100
		Institute of Tropical Medicin Chittagong	Prposed	1	0	0	0	100	100
			SubTotal :		2015	1984	31	200	200
	Comilla	Medical College Hospital	MCH	1	250	250	0	500	250
		General Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	15	560	560	0	0	0
		20 Beded Hospital	20 bed	4	80	80	0	0	0
		Kalikapur 10 Beded	10 bed	1	10	10	0	0	0

Division and district wise no. of different types of hospitals under DGHS with bed capacity as of December 2008 (contd...)

Chittagong Division (contd...)

Division	District	Name of Institute	Type of Institute	No. of Institute	Total	Revenue	Development	Proposed Bed	Bed will Increase
	Cox's Bazar	District Hospital	District Level	1	250	100	150	0	0
		Upazila Health Complex	UHC	7	274	274	0	0	0
		St. Martin 10 Beded Hospital	10 bed	1	10	10	0	0	0
				Sub-Total :	**534**	**384**	**150**	**0**	**0**
	Feni	District Hospital	District Level	1	100	100	0	250	150
		Upazila Health Complex	UHC	5	193	193	0	0	0
		Chest Hospital	Chest Hospital	1	20	20	0	0	0
				Sub-Total :	**313**	**313**	**0**	**250**	**150**
	Khagrachari	District Hospital	District Level	1	50	50	0	100	50
		Upazila Health Complex	UHC	7	217	217	0	0	0
		Talbalchhari 10 Beded Hospital(RHC)	10 bed	1	10	10	0	0	0
				Sub-Total :	**277**	**277**	**0**	**100**	**50**
	Lakshmipur	District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	4	143	143	0	0	0
		Char Alekgender 20 Baded Hospital	20 bed	1	20	20	0	0	0
		Sub-Total :			**263**	**263**	**0**	**0**	**0**
	Noakhali	District Hospital	District Level	1	250	150	100	0	0
		Upazila Health Complex	UHC	8	362	362	0	0	0
		Char Agli 20 Beded Hospital(Devlop)	20 bed	1	20	0	20	0	0
		Begumganj 10 Baded Hospital	10 bed	1	10	10		0	0
				Sub-Total :	**642**	**522**	**120**	**0**	**0**
	Rangamati	General Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	9	279	279	0	0	0
		Kaptai 10 Beded Hospital	10 bed	1	10	10	0	0	0
		Sub-Total :			**389**	**389**	**0**	**0**	**0**
		Total (Chittagong Division)			**6588**	**6187**	**401**	**1300**	**800**

Division and district wise no. of different types of hospitals under DGHS with bed capacity as of December 2008 (contd...)

Dhaka Division

Division	District	Name of Institute	Type of Institute	No. of Institute	Total	Revenue	Development	Proposed Bed	Bed will Increase
Dhaka	Dhaka	National Institute of Chest Disease and Hospital(NIDCH)	PGIH	1	600	600	0	0	0
		National Institute of Traumatology and Rehabilitation (NITOR)	PGIH	1	500	500	0	0	0
		National Institute of Cardiovascular Disease (NICVD)	PGIH	1	414	250	164	0	0
		National Institute of Ophthalmology (NIO)	PGIH	1	250	250	0	0	0
		National Institute of Kidney Disease and Urology (NIKD&U)	PGIH	1	100	0	100	0	0
		National Institute of Mental Health & Research (NIMH&R)	PGIH	1	100	50	50	0	0
		National Institute of Cancer Research and Hospital (NICR&H)	PGIH	1	50	50	0	250	200
		Medical College Hospital	MCH	3	2675	2675	0	2000	300
		Homoeopathic Degree College & Hospital	AMH	1	100	100	0	0	0
		Unani & Ayurvadic College & Hospital	AMH	1	100	100	0	0	0
		Dental College and Hospital	Dental College	1	20	20	0	200	180
		Sarkari Karmochari Hospital	SKH	1	100	100	0	0	0
		National Asthma Center	Center	1	100	0	100	0	0
		Infectious Diseases Hospital	IDH	1	100	100	0	0	0
		Burn Unit at DMCH	Center	1	50	0	50	200	150
		Upazila Health Complex	UHC	5	231	231	0	0	0
		Leprosy Hospital	Leprosy Hospital	1	30	30	0	0	0
		Bangladesh Korea Moitree Hospital	Friendship Hospital	1	20	0	20	0	0
		20 Baded Hospital	20 bed	3	60	60	0	0	0
		10 Beded Hospital(RHC)	10 bed	3	30	30	0	0	0
		National TB Control Project	Proposed	1	0	0	0	250	250

Division and district wise no. of different types of hospitals under DGHS with bed capacity as of December 2008 (contd...)

Dhaka Division (contd...)

Division	District	Name of Institute	Type of Institute	No. of Institute	Total	Revenue	Development	Proposed Bed	Bed will Increase
Dhaka		General Hospital(Proposed), Khilgaon	Proposed	1	0	0	0	500	500
		General Hospital(Proposed), Kurmitolla	Proposed	1	0	0	0	500	500
		General Hospital(Proposed), Mirpur	Proposed	1	0	0	0	500	500
		Institute of Neuro Science, Dhaka	Proposed	1	0	0	0	300	300
		Sub-Total :			**5630**	**5146**	**484**	**4700**	**2880**
	Faridpur	Medical College Hospital	MCH	1	250	250	0	0	0
		General Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	7	274	274	0	0	0
		Chest Hospital	Chest Hospital	1	20	20	0	0	0
		Sub-Total :			**644**	**644**	**0**	**0**	**0**
	Gazipur	District Hospital	District Level	1	100	100	0	0	0
		Tongi 50 beded Hospital	50 bed	1	50	50	0	0	0
		Upazila Health Complex	UHC	4	162	162	0	0	0
		Sub-Total :			**312**	**312**	**0**	**0**	**0**
	Gopalganj	District Hospital	District Level	1	250	100	150	0	0
		Upazila Health Complex	UHC	4	124	124	0	0	0
		Gopinathpur 10 Beded Hospital(RHC)	10 bed Hospial	1	10	10	0	0	0
		Sub-Total :			**384**	**234**	**150**	**0**	**0**
	Jamalpur	General Hospital	District Level	1	250	100	150	0	0
		Upazila Health Complex	UHC	6	243	243	0	0	0
		Sub-Total :			**493**	**343**	**150**	**0**	**0**
	Kishoreganj	District Hospital	District Level	1	250	100	150	0	0
		Upazila Health Complex	UHC	12	429	429	0	0	
		Sub-Total :			**679**	**529**	**150**	**0**	**0**

Division and district wise no. of different types of hospitals under DGHS with bed capacity as of December 2008 (contd...)

Dhaka Division (contd...)

Division	District	Name of Institute	Type of Institute	No. of Institute	Total	Revenue	Development	Proposed Bed	Bed will Increase
Dhaka	Madaripur	District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	3	112	112	0	0	0
		Sub-Total :			**212**	**212**	**0**	**0**	**0**
	Manikganj	District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	6	186	186	0	0	0
		Sub-Total :			**286**	**286**	**0**	**0**	**0**
	Munshiganj	District Hospital	District Level	1	100	100	0	250	150
		Upazila Health Complex	UHC	5	250	250	0	0	0
		Sub-Total :			**350**	**350**	**0**	**250**	**150**
	Mymensingh	Medical College Hospital	MCH	1	800	800	0	1000	200
		Upazila Health Complex	UHC	11	436	436	0	0	0
		20 Baded Hospital	20 bed	1	20	20	0	0	0
		Sub-Total :			**1256**	**1256**	**0**	**1000**	**200**
	Narayanganj	200 bed Hospital	200 bed	1	200	200	0	250	50
		General Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	4	124	124	0	0	0
		20 Baded Hospital	20 bed	3	60	60	0	0	0
		Sub-Total :			**484**	**484**	**0**	**250**	**50**
	Narsingdi	100 Bed Hospital (Development)	100 bed	1	100	0	100	0	0
		District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	5	174	174	0	0	0
		Sub-Total :			**374**	**274**	**100**	**0**	**0**
	Netrokona	District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	9	374	374	0	0	0
		Sub-Total :			**474**	**474**	**0**	**0**	**0**

Division and district wise no. of different types of hospitals under DGHS with bed capacity as of December 2008 (contd...)

Dhaka Division (contd...)

Division	District	Name of Institute	Type of Institute	No. of Institute	Total	Revenue	Development	Proposed Bed	Bed will Increase
	Rajbari	District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Comple	UHC	4	162	162	0	0	0
		Sub-Total :			262	262	0	0	0
	Shariatpur	District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	5	155	155	0	0	0
		Kodalpur 20 Baded Hospital	20 bed	1	20	20	0	0	0
		Sub -Total :			275	275	0	0	0
	Sherpur	District Hospital	District Level	1	100	50	50	0	0
		Upazila Health Comple	UHC	4	124	124	0	0	0
		Sub -Total :			224	174	50	0	0
	Tangail	General Hospital	District Level	1	250	100	150	0	0
		Upazila Health Comple	UHC	11	436	436	0	0	0
		Sub- Total :			686	536	150	0	0
	Total (Dhaka Division)				13025	11791	1234	6200	3280

Division and district wise no. of different types of hospitals under DGHS with bed capacity as of December 2008 (contd...)

Khulna Division

Division	District	Name of Institute	Type of Institute	No. of Institute	Total	Revenue	Development	Proposed Bed	Bed will Increase
Khulna	Bagerhat	District Hospital	District Level	1	100	100	0	150	50
		Upazila Health Complex	UHC	8	305	305	0	0	0
		Sub-Total :			405	405	0	150	50
	Chuadanga	District Hospital	District Level	1	100	100	0	150	50
		Upazila Health Complex	UHC	3	93	93	0	0	0
		Sub-Total :			193	193	0	150	50

Division and district wise no. of different types of hospitals under DGHS with bed capacity as of December 2008 (contd...)

Khulna Division (contd…)

Division	District	Name of Institute	Type of Institute	No. of Institute	Total	Revenue	Development	Proposed Bed	Bed will Increase
	Jessore	General Hospital	District Level	1	250	250	0	0	0
		Upazila Health Complex	UHC	7	274	274	0	0	0
		Chest Hospital	Chest Hospital	1	20	20	0	0	0
		Sub-Total :			**544**	**544**	**0**	**0**	**0**
	Jhenaidah	District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	5	231	231	0	0	0
		25-bed Shishu Hospital	25-bed Shishu Hospital	1	25	0	25	0	0
		Sub-Total :			**356**	**331**	**25**	**0**	**0**
	Khulna	Medical College Hospital	MCH	1	250	250	0	500	250
		Shekh Abu Naser Specialized Hospital	Specialized Hospital	1	250	0	250	0	0
		General Hospital	District Level	1	150	150	0	0	0
		Chest Hospital	Chest Hospital	1	100	100	0	0	0
		Upazila Health Complex	UHC	9	317	317	0	0	0
		Infectious Diseases Hospital	IDH	1	20	20	0	0	0
		Kopilmoni 10 Beded Hospital(RHC)	10 bed Hospial	1	10	10	0	0	0
		Sub-Total :			**1097**	**847**	**250**	**500**	**250**
	Kushtia	General Hospital	District Level	1	250	150	100	0	0
		Upazila Health Complex	UHC	5	212	212	0	0	0
		Sub-Total :			**462**	**362**	**100**	**0**	**0**
	Magura	District Hospital	District Level	1	100	100	0	150	50
		Upazila Health Complex	UHC	3	93	93	0	0	0
		Binodpur 20 Baded Hospital	20 bed	2	40	40	0	0	0
		Sub-Total :			**233**	**233**	**0**	**150**	**50**

Khulna Division (contd...)

Division	District	Name of Institute	Type of Institute	No. of Institute	Total	Revenue	Development	Proposed Bed	Bed will Increase
Khulna	Meherpur	District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	2	81	81	0	0	0
		Sub-Total :			181	181	0	0	0
	Narail	District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	2	81	81	0	0	0
		Sub-Total :			181	181	0	0	0
	Satkhira	District Hospital	District Level	1	100	100	0	150	50
		Upazila Health Complex	UHC	6	243	243	0	0	0
		Sub-Total :			343	343	0	150	50
		Total (Khulna Division)			3995	3620	375	1100	450

Division and district wise no. of different types of hospitals under DGHS with bed capacity as of December 2008 (contd...)

Rajshahi Division

Division	District	Name of Institute	Type of Institute	No. of Institute	Total	Revenue	Development	Proposed Bed	Bed will Increase
Rajshahi	Bogra	Medical College Hospital	MCH	1	500	500	0	0	0
		Mohammad Ali Hospital	District Level	1	250	250	0	0	0
		Upazila Health Complex	UHC	11	474	474	0	0	0
		Chest Hospital	Chest Hospital	1	20	20	0	0	0
		20 Baded Hospital	20 bed	3	60	60	0	0	0
		Sub-Total :			**1304**	**1304**	**0**	**0**	**0**
	Chapai Nawabganj	District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	4	143	143	0	0	0
		Sub-Total :			**243**	**243**	**0**	**0**	**0**

Division and district wise no. of different types of hospitals under DGHS with bed capacity as of December 2008 (contd...)

Rajshahi Division (contd...)

Division	District	Name of Institute	Type of Institute	No. of Institute	Total	Revenue	Development	Proposed Bed	Bed will Increase
Rajshahi	Dinajpur	Dinajpur Medical College Hospital	MCH	1	250	250	0	500	250
		General Hospital	District Level	1	250	250	0	0	0
		Upazila Health Complex	UHC	12	410	410	0	0	0
		Sub-Total :			**910**	**910**	**0**	**500**	**250**
	Gaibandha	District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	6	205	205	0	0	0
		Ramchandpur 10 Beded Hospital(RHC)	10 bed	1	10	10	0	0	0
		Sub-Total :			**315**	**315**	**0**	**0**	**0**
	Joypurhat	District Hospital	District Level	1	100	100	0	150	50
		Upazila Health Complex	UHC	4	162	162	0	0	0
		Sub-Total :			**262**	**262**	**0**	**150**	**50**
	Kurigram	District Hospital	District Level	1	100	100	0	150	50
		Upazila Health Complex	UHC	8	286	286	0	0	0
		Raiganj 10 Beded Hospital(RHC)	10 bed	1	10	10	0	0	0
		Sub-Total :			**396**	**396**	**0**	**150**	**50**
	Lalmonirhat	District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	4	143	143	0	0	0
		Dahogram 10 Beded Hospital(RHC)	10 bed Hospial	1	10	10	0	0	0
		Sub-Total :			**253**	**253**	**0**	**0**	**0**

Division and district wise no. of different types of hospitals under DGHS with bed capacity as of December 2008 (contd...)

Rajshahi Division (contd...)

Division	District	Name of Institute	Type of Institute	No. of Institute	Total	Revenue	Development	Proposed Bed	Bed will Increase
Rajshahi	Naogaon	District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	10	424	424	0	0	0
		Sub-Total :			**524**	**524**	**0**	**0**	**0**
	Natore	District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	5	174	174	0	0	0
		Sub-Total :			**274**	**274**	**0**	**0**	**0**
	Nilphamari	District Hospital	District Level	1	100	100	0	150	50
		Upazila Health Complex	UHC	5	193	193	0	0	0
		Leprosy Hospital	Leprosy Hospital	1	80	80	0	0	0
		Saidpur 50 beded Hospital	50 bed	1	50	50	0	0	0
		Sub-Total :			**423**	**423**	**0**	**150**	**50**
	Pabna	Pabna Mental Hospital	Mental Health	1	500	400	100	0	0
		General Hospital	District Level	1	250	120	130	250	0
		Upazila Health Complex	UHC	8	305	305	0	0	0
		Chest Hospital	Chest Hospital	1	20	20	0	0	0
		Sub-Total :			**1075**	**845**	**230**	**250**	**0**
	Panchagarh	District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	4	181	181	0	0	0
		Sub-Total :			**281**	**281**	**0**	**0**	**0**

Division and district wise no. of different types of hospitals under DGHS with bed capacity as of December 2008 (contd...)

Rajshahi Division (contd...)

Division	District	Name of Institute	Type of Institute	No. of Institute	Total	Revenue	Development	Proposed Bed	Bed will Increase
	Rajshahi	Medical College Hospital	MCH	1	600	600	0	0	0
		Chest Hospital	Chest Hospital	1	100	100	0	0	0
		Upazila Health Complex	UHC	9	355	355	0	0	0
		Godagari 31 Bed Hospital	31 bed	1	31	31	0	0	0
		Infectious Diseases Hospital	IDH	1	20	20	0	0	0
		Sub-Total :			**1106**	**1106**	**0**	**0**	**0**
	Rangpur	Medical College Hospital	MCH	1	600	600	0	1000	400
		Upazila Health Complex	UHC	7	274	274	0	0	0
		Haragach 31 Bed Hospital	31 bed	1	31	31	0	0	0
		Chest Hospital	Chest Hospital	1	20	20	0	0	0
		Sub-Total :			**925**	**925**	**0**	**1000**	**400**
	Serajganj	General Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	8	248	248	0	0	0
		Ullapara 20 Beded Hospital	20 bed Hospial	1	20	20	0	0	0
		Khokshabari 10 Beded Hospital(RHC)	10 bed Hospial	1	10	10	0	0	0
		Sub-Total :			**378**	**378**	**0**	**0**	**0**
	Thakurgaon	District Hospital	District Level	1	100	100	0	0	0
		Upazila Health Complex	UHC	4	181	181	0	0	0
		Sub-Total :			**281**	**281**	**0**	**0**	**0**
Total (Rajshahi Division					**8950**	**8720**	**230**	**2200**	**800**

Division and district wise no. of different types of hospitals under DGHS with bed capacity as of December 2008 (contd...)

Sylhet Division

Division	District	Name of Institute	Type of Institute	No. of Institute	Total	Revenue	Development	Proposed Bed	Bed will Increase
Sylhet	Habiganj	District Hospital	District Level	1	100	100	0	150	50
		Upazila Health Complex	UHC	7	236	236	0	0	0
		Sub-Total :			**336**	**336**	**0**	**150**	**50**
	Moulavibazar	District Hospital	District Level	1	100	100	0	250	150
		Upazila Health Complex	UHC	6	205	205	0	0	0
		Sub-Total :			**305**	**305**	**0**	**250**	**150**
	Sunamganj	District Hospital	District Level	1	250	100	150	0	0
		Upazila Health Complex	UHC	10	329	329	0	0	0
		20 Baded Hospital	20 bed	2	40	40	0	0	0
		Kaitak 10 Beded Hospital(RHC)	10 bed Hospial		10	10	0	0	0
		Sub-Total :			**629**	**479**	**150**	**0**	**0**
	Sylhet	Medical College Hospital	MCH	1	900	900	0	1000	100
		Shahid Shamsuddin Hospital	District Level	1	100	100	0	0	0
		Chest Hospital	Chest Hospital	1	56	56	0	0	0
		Upazila Health Complex	UHC	11	398	398	0	0	0
		Leprosy Hospital	Leprosy Hospital	1	20	20	0	0	0
		Infectious Diseases Hospital, Sylhet	IDH	1	20	20	0	0	0
		Sub-Total :			**1494**	**1494**	**0**	**1000**	**100**
Total (Sylhet Division) :					**2764**	**2614**	**150**	**1400**	**300**
Grand Total (6 Division) :					**38171**	**35661**	**2510**	**13650**	**6180**